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Introduction

Research in computational conformal mappings has lately taken two major directions. One direction involves the conformal mapping from a standard region, like the unit disk or the upper half-plane, onto the problem region, whereas in the other it is from the problem region onto a standard region. In the former case one solves a nonlinear integral equation involving the conjugate operator (e.g., Theodorsen's integral equation), by fast Fourier transform, polynomial approximation, iteration, or Newton's method. In the latter case the integral equation, derived from the Dirichlet problem, is linear or singular linear if it is derived from potential theory (e.g., Symm's integral equation). Depending on the nature of the problem region, these methods often use the Schwarz-Christoffel transformations.

Various kinds of applications of conformal mappings have been and are being studied. It is an important area and we will provide relevant information from different publications.

The historical development of different methods for computational conformal mapping of simply and multiply connected regions is sketched below.

1.1 Historical Background

The oldest transformation, known as the *stereographic projection* of the sphere, was used by Claudius Ptolemy (ca. 150 A.D.) to represent the celestial sphere. Later, in a totally different mapping of a sphere onto a plane, known as Mercator's projection, the spherical earth is cut along a meridian circle and conformally mapped onto a plane strip. Gerhard Kremer published the first world map in 1569 using this technique, and ever since all sea maps are generated by this method. These two projections, however, do not provide similar maps of the same region, which shows that conformal mapping does not imply similarity of figures. Lambert [1772] was the first mathematician who studied the mathematical projection of maps. These and other similar considerations enabled Joseph-Louis Lagrange in 1779 to obtain all conformal representations of a portion of the earth's surface onto a plane where all circles of longitudes and latitudes are represented by circular arcs.

Gauss [1822] was the first to state and completely solve the general problem of determining all conformal mappings that transform a very small neighborhood of a point on an

arbitrary analytic surface onto a plane area. However, Gauss' work did open the harder problem of finding the way whereby a given finite portion of a surface can be mapped onto a portion of the plane. A breakthrough came in 1851 when Riemann gave the fundamental result, known as the *Riemann mapping theorem*, which has since been a turning point for all subsequent developments in the theory of conformal mapping. In the proof of this theorem he assumed that a variational problem, now known as the Dirichlet problem, possesses a solution. It was fifty years later that Hilbert [1901] proved the existence of the solution of the Dirichlet problem. In the mean time the validity of Riemann's result was established rigorously by Schwarz [1890] by using a number of theorems from the theory of logarithmic potential. A detailed proof of the Riemann mapping theorem and the Riemann-Hilbert problem is available in Appendix C.

After the basic theoretical aspects of the theory of functions of a complex variable and conformal mapping were established by Cauchy, Riemann, Schwarz, Christoffel, Bieberbach, Carathéodory, Goursat, Koebe, and others in the nineteenth and early decades of the twentieth century, the first numerical research into developing a method for mapping a region bounded by finitely many Jordan curves Γ_i onto an n -sheeted Riemann surface where the curves Γ_i correspond to rectilinear slits was done by Burnside [1891]. A minimizing principle was established by Bieberbach [1914], namely, that among all suitably normed conformal maps of a given simply connected region the one with the least area is the conformal map onto a circle. In particular, this principle evolved as a result of minimizing the integral $\iint_D \|f'(z)\|^2 dx dy$, where f is regular in D and normalized by $f(0) = 0$, $f'(0) = 1$, and the area theorem. Bieberbach used the Ritz method to find an approximate solution in the form of a polynomial for the above integral and used it to construct the conformal map of a simply connected region onto a circle. An exposition of the Ritz method can be found in Kantorovich and Krylov [1936]. The estimates obtained by this method were later improved by Höhndorf [1926] and Müller [1938] in the problem of a conformal map of a nearly circular region onto a circle. Other improvements on the Bieberbach method were produced by Landau [1926], Julia [1926] and Kantorovich and Krylov [1936] who also developed a graphical method of conformal mapping due to Melent'ev [1937].

The first integral equation method was developed by Lichtenstein [1917] who solved the problem of conformally mapping a simply connected region bounded by a Jordan contour onto a circle by reducing it to the solution of an integral equation. Other attempts in this direction were made by Krylov and Bogolyubov [1929] who reduced the Dirichlet problem to an integral equation which is approximately solved by the Fredholm method with error estimates. Nyström [1930] gave a method for an approximate solution of integral equations, which is useful in conformal mapping. For conformal mappings of nearly circular regions the first numerical work was done by Fock [1929] who determined numerically the mapping function for a circular quadrilateral with angles $\pi/2, \pi/2, \pi/2, \alpha\pi$ ($0 < \alpha < 1$). An important result, now known as Theodorsen's integral equation, was developed by Theodorsen [1931] and improved upon by Theodorsen and Garrick [1933] for conformally mapping nearly circular regions onto a circle. Another integral equation, known as Gershgorin's equation, was developed by Gershgorin [1933] as a result of conformally mapping a simply connected region bounded by a Jordan contour onto a circle, which is solved by the Nyström method. An exposition on Gershgorin's integral equation and its application to the mapping of a simply connected region onto a circle, of a doubly connected region onto an annulus, and of a multiply connected region, in general, onto slit planes is available in Kantorovich and Krylov [1936]. Various applications of conformal mapping are available in the monograph by Kantorovich and Krylov [1936], where both harmonic and biharmonic functions are studied. The Dirichlet and Neumann problems in $\mathbb{R}^2$ are solved by conformal mapping for

different boundary conditions, and the general Hilbert problem is also investigated. Later Banin [1943] developed a method of approximately replacing Gershgorin's integral equation by a system of linear differential equations. Krylov [1938] also reduced the problem of conformally mapping an n -connected region bounded by n Jordan contours onto various canonical regions to the problem of solving a system of simultaneous integral equations.

Successive approximations in the integral equation method for simply and multiply connected regions were used by Kantorovich [1933; 1937]. Following Theodorsen's integral equation method, Warschawski [1945] reduced the problem of conformal mapping of a simply connected region bounded by a Jordan contour onto the unit circle to that of solving a nonlinear integral equation which is then solved by the method of successive approximations where the precise estimates for the convergence are also provided. Goluzin [1934] investigated the region exterior to the circles $C_1, \dots, C_n$ and determined a harmonic function $u(x, y)$ for this region such that it takes preassigned values on C_i , $i = 1, \dots, n$, and $u(\infty) < \infty$. The problem is then reduced to that of solving a finite system of functional equations which are solved by successive approximations. He applied this method to solve the Neumann problem for Laplace's equation for such regions and determined the Green's function for these regions and the conformal maps that carry these regions onto slit planes. The Schwarz method is used to develop an integral equation which is solved by successive approximations in Nevanlinna [1939] and Epstein [1948].

Julia [1927] determined a sequence of polynomials which converges to a suitably normed mapping of a simply connected region. An application of orthogonal polynomials to conformal mapping of simply connected regions was first developed by Szegő [1921]. He has defined a set of polynomials $\sigma_0(z), \sigma_1(z), \dots, \sigma_n(z), \dots$ (subsequently known as the Szegő polynomials) for a closed Jordan contour Γ in the z -plane, satisfying the following two properties: (i) $\sigma_n(z)$ is a polynomial of degree n , i.e., $\sigma_n(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$, such that $a_n > 0$, and (ii) $\frac{1}{l} \int_{\gamma} \sigma_m(z) \overline{\sigma_n(z)} ds = \delta_{m,n}$, where l is the length of Γ , s its arc length, and $\delta_{m,n}$ the Kronecker delta. Then the series $S(z, a) = \sum_{n=0}^{\infty} \sigma_n(z) \overline{\sigma_n(a)}$ converges uniformly and absolutely in every closed subregion of the interior of Γ . A formula is then obtained which defines the mapping function of the interior of Γ onto the unit disk in terms of $S(z, a)$. Szegő also gave a mapping function for the exterior of Γ by the polynomials $\sigma_n(z)$. Further developments in the Szegő polynomial method were produced by Smirnov [1928] and later by Kantorovich and Krylov [1936] who investigated the Szegő and Bochner-Bergman type polynomials and applied them to the minimizing problem. Some results about approximate mapping functions for a simply connected region by means of certain polynomials were given by Schaginyan [1944].

The problem of minimizing a functional was first solved by Hadamard [1908] which later became known as the Hadamard variational method. Löwner [1923] also developed an important variational method, whereas Julia [1926] provided another characterization of the mapping function by a minimum principle. The problem of reducing the mapping problem to that of minimizing of a functional was solved by Douglas [1931] who used the Riemann mapping theorem and the Osgood-Carathéodory theorem. Kufarev [1935-1937] investigated a minimal problem for a single-valued analytic function in an annulus and discussed the mapping of the minimizing function. Later, in 1947 he used the Löwner method to study a polygonal problem region which consists of the whole plane cut by a broken polygonal line with finitely many sides, one of which extends to infinity.

The problem of mapping a doubly connected region onto an annulus was reduced to that of minimizing an area integral by Khajalia [1940], who also showed that if the region

is accessible from without, then there exists a sequence of minimal rational functions that converges uniformly to the desired mapping function. The problem of mapping a simply connected region bounded by a Jordan contour is reduced by Shiffman [1939] to that of minimizing a functional, almost similar to that of Douglas. This problem deals with the Plateau problem, and the electrostatic characterization of the resulting functional provides an effective method for determining the conformal maps. The Hadamard formula and the variation of domain functions were used by Schiffer [1946] to derive a new variational method.

In 1931, Grötsch solved the problem of conformally mapping a multiply connected region onto some canonical regions based on the assumption that the solution of a similar problem for a simply connected region is known. Iterative methods were established by Goluzin [1939] who used the mapping of a multiply connected region conformally onto some canonical regions, thereby reducing the problem to a sequence of conformal maps of simply connected regions. Heinhold [1954] investigated the problem of conformally mapping the simply connected region lying in the exterior of the unit circle onto the exterior of the unit circle.

Green's function method for an arbitrary region was established by Leja [1934; 1936] where the approximating functions were found closely related to Lagrange polynomials. A set of polynomials was also obtained which were used as mapping function for a region D , containing the point at infinity, onto the exterior of the unit circle. It was determined that the map is univalent if D is simply connected.

The Schwarz-Christoffel formula was used by Bergman [1923-24] to map a half-plane onto a particular polygon where a method for determining the parameters in the formula from the lengths of the sides of the polygon is given. In 1925 he investigated the conformal map of a special polygon onto a rectangle and computed level curves and their orthogonal trajectories which were presented in tabular and graphical forms. Bergman also gave the first punch-card machine method in 1947 to solve the torsion problem where the orthogonal polynomials were applied to solve Laplace's equation numerically. The notion of the kernel functions was developed by Bergman [1922] where the existence of a complete orthonormal system with respect to a region is established and the kernel of the system is related to the conformal map of the region. Based on Bergman's kernel function and complete orthonormal sets of functions, Zarankiewicz [1934] found a method for effectively constructing the conformal map of a doubly connected region onto an annulus; in 1934 he published details of this method. Schiffer [1946] found an expression for Bergman's kernel function $K(z, a)$ of a region in terms of Green's function for the region and gave formulas for the variation of $K(z, a)$. In 1948 he extended the concept of kernel functions and orthonormal sets of functions to a wider class of functions. Further study of the relationship between the kernel function and conformal mappings of regions was done by Bergman and Schiffer [1948-1951].

Hodgkinson and Poole [1924] used elliptic functions to map doubly connected regions of certain types onto the whole plane with two slits on the real axis. Using hyperelliptic integrals, a generalization of the solution by Hodgkinson and Poole was given by Vladimírsky [1941] for the problem of conformal mapping of a doubly connected region bounded by rectangular segments or circular arcs, who also extended the solution to n -tuply connected regions ($n > 2$). Hodgkinson [1930] also established the relationship between the theory of Lamé differential equations and the Schwarz theory of conformal mapping. An up-to-date survey of conformal mapping of multiply connected regions onto canonical domains was published by Keldyš [1939]. Important formulas for various conformal maps of n -tuply connected regions in terms of the kernel function of the region were established by Nehari in 1949. These formulas served as tools for numerical computation of conformal maps since

the kernel functions are constructed more easily than the mapping function. Gerabedian and Schiffer [1949] obtained many significant relations between various domain functions of an n -tuply connected region and solved some minimal problems.

The first systematic construction of conformal maps by the method of networks was done by Liebmann [1918]. The Dirichlet problem in $\mathbb{R}^2$ and $\mathbb{R}^3$ was first solved by this method by Phillips and Wiener [1923]. The boundary problems in $\mathbb{R}^2$ with Laplace's equation were solved numerically by this method by Luysternik [1947].

Relaxation methods were first applied to the problems of conformal mapping by Gandy and Southwell [1940] and Southwell [1946]. Several examples of technical interest were given, in which regions of arbitrary shape are mapped onto the interior and exterior of circles and onto rectangles.

The small parameters method developed by Kantorovich [1933] was later used by Rosenblatt and Turski [1936] for conformal mapping of a special type of region. Rosenblatt [1943] constructed the conformal maps of regions onto the unit disk by the Kantorovich method of small parameters and applied it to the dynamic problems of airfoils.

Special conformal mappings were studied by Rothe [1908] who constructed the mapping function for a circular quadrilateral with angles $\pi/2, \pi/2, \pi/2, \rho\pi$ ($\rho = 1/\sqrt{20}$). Hodgkinson's work [1924] on a similar problem has been mentioned above. Wirtinger [1927] derived an explicit formula for computing the conformal map of a triangle with circular arcs and arbitrary angles. The mapping of an ellipse onto a circle and of a region bounded by two symmetrically placed ellipses onto an annulus were investigated by Zmorovich [1935] who also gave an approximate solution in the latter problem. Catalogs of various types of explicit mappings are available in von Koppenfels' papers [1937; 1939]. Other authors on special conformal mappings during this period were Muratov [1937], Krylov [1937], Melent'ev [1937], and Goluzin [1937]. Later Jeffreys and Jeffreys [1946] and Wittich [1947] produced special conformal maps. A very useful dictionary of conformal mappings, which is still a source work in this area, was produced by Kober [1945-1948].

The theory of homogeneous boundary problems for analytic functions was presented by Hilbert [1924] for a boundary that is a single Jordan contour. This problem was then transformed into a Fredholm integral equation, for which Green's functions of the Neumann problem must be determined. Hilbert did not give a complete solution of this integral equation. A complete solution by means of Cauchy integrals for a particular case had already been given by Plemelj [1908]. The solution for the general case, based on the Plemelj method, was later provided by Khvedelidze [1941]. Earlier, Picard [1927] had studied the homogeneous Hilbert problem.

The nonhomogeneous Hilbert problem which is solved to determine a sectionally analytic function $\Phi(z)$ of finite degree at infinity such that $\Phi^+(\zeta) = G(\zeta)\Phi^-(\zeta) + g(\zeta)$ on the boundary Γ , where $G(\zeta)$ and $g(\zeta)$ are defined on the boundary Γ and satisfy the Hölder condition there, was first considered by Carleman [1922]. Later Privalov [1934] investigated it for the case when the boundary Γ is a rectilinear contour, $G(\zeta)$ and $g(\zeta)$ are Lebesgue-integrable, $G(\zeta)$ is bounded, and the limiting value of the solution is taken along a nontangential path. He used Picard's method but did not find a complete solution. A complete solution was first given by Gakhov [1937] for the case when Γ is a single contour. Later Khvedilidze [1941] gave a complete solution of the nonhomogeneous Hilbert problem. Carleman's and Gakhov's methods are essentially alike. In 1851 Riemann considered a very general boundary problem which is now known as the Riemann boundary problem. This problem was also studied and solved by Hilbert in 1904, and therefore it is also called the Riemann-Hilbert problem. It deals with determining an analytic function in a given region with boundary

values involving a relation between its real and imaginary parts. Hilbert reduced this problem to a singular equation and then applied it to the solution of two Dirichlet problems. Later, Noether [1921] used this solution to study the subject of singular integral equations.

One of the widely investigated practical aspects of conformal mapping was the development of the airfoil theory which started with the pioneering work by Joukowski [1890]. He studied the flows around a variety of so-called Joukowski airfoils. The developments in this area are very pertinent for computational aerodynamic flows. Extensive research to develop computational methods for solving the direct and inverse problems of airfoil theory has been done by Theodorsen [1931], Glauert [1948], Andersen, Christiansen, Møller and Tornehave [1962], Timman [1951], James [1971], Ives [1976], and Halsey [1979; 1982]. In single and multi-element airfoils, Theodorsen's integral equation has been solved with the von Karman-Trefftz transformation and the FFT by Garrick's method of conjugate functions (Garrick [1949]) by Ives [1976]. James's method (James [1971]) is another approach in the conformal mapping of single and multiple-element airfoils (Halsey 1979; 1982).

1.2 Modern Developments

During the past decades several methods have evolved for numerical and computational evaluation of mapping functions. The results of some of these methods provide us with an explicit form of a function which approximately evaluates the mapping function for a certain source region. This is possible because of an important result which states that **an analytic function defined on a simply connected and bounded region $D \subset \mathbb{C}$ can always be uniformly approximated on every compact subset of D with any preassigned accuracy by means of a polynomial. In most applications the mapping function is continuous in $\bar{D}$, and it can be uniformly approximated in D by a polynomial.**

One major approach in developing methods for numerical conformal mapping is based on the following interpretation of the Riemann mapping theorem: there exists a conformal mapping $f : D \mapsto U$ with $f(z_0) = 0$ and $f'(z_0)$ nonzero real, where $z_0 \in D$, and this function has a power series expansion $f(z) = c_1 (z - z_0) + \sum_{n=2}^{\infty} c_n (z - z_0)^n$, with c_1 nonzero real and $z_0 \in D$, which converges uniformly in every closed disk with center z_0 and contained in D . However, a polynomial which is a good approximation of f in D is not the same as a truncated power series. If a polynomial $p(z) = c'_1 (z - z_0) + \sum_{n=2}^N c'_n (z - z_0)^n$ approximates f with accuracy $\varepsilon > 0$, then it is necessary that every term of $p(z)$ must approximate the corresponding term of the power series with accuracy $\varepsilon > 0$ on the set $D \cap B(z_0, R)$, where $R = |z - z_0|$ is the radius of convergence of the power series. All this means is that a polynomial p which is a good approximation of the power series starts in the same way as the power series, but the relative error in the coefficients increases with increasing n .

Another direction for developing numerical methods for conformal mapping is based on computing a table of values for the mapping function f at several points of D . The mapping function f always maps the boundary of the source region D in a simple way onto the boundary of the unit disk U . Once the mapping of the boundary is known, the mapping of D itself is determined by the Cauchy formula with positive orientation of the boundary curve. Thus, it is sufficient to determine the mapping of the boundary of D onto the boundary of U , and once we obtain a parametric representation of the boundary of D , we determine only one real-valued function of a real variable t . But in this approach it so happens that this unknown real-valued function satisfies an integral equation. Hence, solving the problem of conformal mapping reduces to that of solving an integral equation.

The widely used current computational techniques are based on the integral equation methods where an integral equation is developed to relate the boundaries of the problem region and the standard region like the unit disk. Once the boundaries are discretized at n points, the integral equation reduces to an algebraic system of equations. The majority of ongoing research in computational conformal mapping is divided basically into two groups: one where the maps are constructed from a standard region such as the unit disk onto the problem region, and the other where the maps are constructed the other way around. In the first group the integral equation is nonlinear and involves the conjugation operator, which can be solved by FFT on a discrete mesh in $O(n \log n)$ operations. This method evolves with the numerical solution of Theodorsen's integral equation, or a related equation, which is solved by using the fixed-point iteration method. The recent development of Newton's method is faster for sufficiently smooth boundaries and the choice of a good initial guess. The basic work in this group has been produced by Wegmann [1978; 1986], Hübner [1986], and Gutknecht [1986]. The first quadratically convergent algorithm for Theodorsen's equation was presented by Wegmann [1978] based on the following two ideas: an induction scheme along tangents to the boundary of the problem region, and a computation scheme similar in formulation to the Riemann-Hilbert problem for the unit disk at each iteration. This method is efficient because the solution of the Riemann-Hilbert problem can be represented by Cauchy integrals. A generalization of this algorithm to the conformal mapping of an annulus onto a doubly connected region was published by Wegmann in 1986. This algorithm is also based on the Riemann-Hilbert problem and is so far the fastest known for this problem. Hübner [1979; 1986] studied Newton's method for the solution of Theodorsen's integral equation. His method is also based on the solution of the Riemann-Hilbert problem. He established the quadratic convergence of Newton's method and obtained a quadratically convergent conformal mapping. An extensive survey of almost all known methods for numerical conformal mapping of the unit disk onto a simply connected region is available in Gutknecht's work [1986]. He has derived integral and integro-differential equations involving the conjugation operator for the boundary correspondence function. Then various iterative schemes for solving these equations are presented in this work. The general theory is described by specific methods, especially the successive conjugation methods of Theodorsen [1931], Timman [1951], Freiberg [1951], the projection method of Bergström [1958], Newton's method of Vertgeim [1958], Wegmann [1978], and Hübner [1979].

In the other group where the maps are constructed from the problem region onto the unit disk, the integral equations, mostly derived from the Dirichlet problem, are generally linear, and require $O(n^2 \log n)$ operations. In cases where the geometry is simpler, they may require a smaller number of operations. The methods in this group are based on Symm's equation which is a singular integral equation of the first kind derived by using a single-layer potential as the basis of conformal mapping. Symm [1966; 1969] investigated an integral equation method, like the one he developed for the boundary integral equation method, for computing the conformal mapping of a simply connected region onto the unit disk. Berrut [1976; 1985; 1986] solved Symm's equation numerically by a Fourier method. This equation is a Fredholm integral equation of the second kind for the derivative of the boundary correspondence function for the conformal mapping of a Jordan region with a piecewise twice differentiable boundary onto the unit disk. Kerzman and Trummer [1986] presented a new method to compute the Riemann mapping function numerically. The solution of the integral equation of the second kind is expressed in terms of the Szegő kernel and is based on an earlier work of Kerzman and Stein [1978] on the Cauchy kernel, Szegő kernel and Riemann mapping function, and of Kerzman and Trummer [1986] on a method for the numerical solution of the conformal mapping problem.

A variant of Symm's integral equation which is suitable for conformal mapping of both simply and multiply connected regions was presented by Reichel [1985]. It uses a Fourier-Galerkin technique to produce an extremely fast iterative solution of $O(n^2 \log n)$ which is due to the singularity of the kernel so that the linear algebraic system becomes block diagonally dominant. Another fast method for solving Symm's equation for multiply connected regions was presented by Mayo [1986]. This integral equation formulation is based on a similar formulation in Mikhlin [1957] and has the advantage of reducing the problems to integral equations of the second kind with unique solutions and boundary kernels. Because the solutions are periodic, the trapezoidal rule can be applied effectively. Once the integral equation is solved, a rapid method is available to determine the mapping function in the interior of the region. This method uses a fast Poisson solver for the Laplacian, thus avoiding the time-consuming computation of integrals at singular points near the boundary.

Other significant modern research includes work by Hoidn [1982] which deals with conformal mapping of simply connected regions where the singularity problem near and on the boundary is solved by reparameterization of the boundary curve. Then this method is applied to Symm's equations which are solved by using spline functions. The singularity problem has been solved by Levin, Papamichael and Sideridis [1978], Hough and Papamichael [1981; 1983], Papamichael and Kokkinos [1981; 1982], Papamichael and Warby [1984], and Papamichael, Warby and Hough [1983; 1986] by integral equation methods as well as expansion methods based on the Bergman kernel or on the Ritz approximation. No research is available in this area for multiply connected regions of higher connectivity. The Chebyshev approximation in conjunction with linear programming has been used by Hartman and Opfer [1986] for conformal mapping of simply connected regions onto the unit disk. A simple approximation formula has been derived by Zemach [1986] for the boundary mapping function which gives a remarkably good fit for mappings of regions with highly distorted boundaries. This method is based on reducing the nonlocal integral equation for the mapping function to a local equation depending on the nature of the distorted regions. The Schwarz-Christoffel formula has always been used in mappings of polygons and related regions, but the treatment of the singularity problem in these cases has been only recently investigated by Barnard and Pearce [1986], Elcrat and Trefethen [1986], and Trefethen and Williams [1986]. Boundary problems for analytic functions and integral equations with transformations have been discussed by Lu [1994], and a comprehensive theoretical account on conformal mapping and boundary problems can be found in Wen [1992].

The advantages of integral equation formulation in conformal mapping can be summarized as follows: All integral equations obtained in any conformal mapping problem (except Arbenz's integral equation, see §13.3) are Fredholm integral equations of the second kind with bounded kernel, except for Symm's integral equation which is of the first kind with a kernel that has a logarithmic singularity. The Fredholm integral equations of second kind are never ill-conditioned, and there are reliable error estimates available for them. However, the drawback with the equations of second kind is that the kernel has singularity at points in the neighborhood of the boundary (but not on the boundary itself unless it has a corner singularity) where computational difficulties often arise. This situation, on the other hand, does not occur with the equations of the first kind.

The best strategy for developing a computational method based on an integral equation formulation is to make sure that the solution is periodic and unique. This permits an effective use of the trapezoid rule which is highly accurate on smooth contours. Another feature to look for is that the mapping onto canonical regions (unit disk, annulus, or slit disks) produces systems of linear equations and avoids solving systems of nonlinear equations as in Fornberg's, Guteknecht's, or Wegman's methods.

Some classical applications of conformal mappings to steady-state problems of mathematical physics and especially for the solution of Laplace's equation can be traced to the beginning of the twentieth century. A noteworthy contribution to the theory of elasticity is by Muskhelishvili [1963]. Modern contributions can be found in areas of fluid flow, heat conduction, solidification, electromagnetics, ion optics, acoustics, vibrations, wave guides, and grid generation, to name a few; a detailed review and biography of the applications through 1972 is available in Laura [1975]. The problem of flow and heat transfer in conduits of arbitrary shape in space vehicles was investigated by Sparrow and Haji-Sheikh [1966]. This study was extended to noncircular conduits with uniform wall temperature by Casarella et al. [1971]. Unsteady heat conduction problems in bars of arbitrary cross section were investigated by Laura et al. [1964; 1965; 1968]. Ives [1976] analyzed the incompressible flow between two concentric circles and computed the streamlines by using Garrick's method of conjugate functions. The problem of solidification of steady-state and transient frozen layers in rectangular channels has been solved by Siegel, Goldstein and Savino [1970]. In transient solidification the shape of a frozen region is determined by mapping it onto a potential plane and then computing the time-dependent conformal map between the potential and the physical plane. The thermoelastic problem of uniform heat flow distributed by an isolated hole of ovaloid form was investigated by Florence and Goodier [1960] and extended by Deresiewicz [1961] to holes which are mapped onto the unit circle and approximated by polynomials.

Wilson [1963] and Richardson [1965] used conformal mapping to determine the stresses in solid propellant rocket grains by solving an integral equation of the Fredholm type and a system of coupled integral equations. The method of computing the conformal mapping function and the related eigenvalue problem for plane regions with irregular boundaries was published by Laura [1968]. The ion problem connected with the trajectory of a charged particle in a plane electric field and a normal magnetic field was solved by Naidu and Westphal [1966] by using the Schwarz-Christoffel transformation. Conformal mapping techniques of simply and doubly connected regions have been used, among others, by Kasin and Merkulov [1966], Laura [1967], Laura et al. [1972], and Schinzinger and Laura [2003].

Conformal mapping has been applied to acoustic waveguides of complicated cross section where the Galerkin method is applied to obtain a functional approximation for the solution of the boundary value problem. The grain of a solid propellant rocket motor with a starlike internal propagation is in the form of a circular cylinder bounded by a thin case. To solve any boundary value or eigenvalue problem, the grain cross section is conformally mapped onto a circle or an annulus. Studies on the shear vibrations of such rocket motors were done by Baltrukonis et al. [1965] and Laura and Shahady [1966], and Schinzinger and Laura [2003]. Conformal mapping techniques are used in a study on the Rayleigh-Taylor instability for ideal fluid by Menikoff and Zemach [1980; 1983]. Grid generation for cascades of blades and inlet flows has been investigated by Inoue [1983; 1985]. The Kirchhoff flow problem past a polygonal obstacle was solved by Trefethen [1986]. Other contemporary applications can be found in the book on numerical conformal mapping edited by Trefethen [1986].

Computational conformal mapping in the present decade is still progressing steadily. Fast algorithms are being developed. Computational conformal mapping is being used in engineering problems that require grid generation and related domain simplification. Although the use of conformal mapping in reducing or eliminating singularities in solutions of integral and integro-differential equations is one important issue, the choice of an initial good guess in Newton's method is another. Much has developed since the evolution of computer technology around 1955, and this subject has reached maturity.

1.3 In Retrospect

Finally, the following quote from Bernhard Riemann's Inaugural Dissertation, Göttingen, 1851, is worthy of presentation.

“Die Ausführung dieser Theorie, welche, wie bemerkt, einfache durch Grössenoperationen bedingte Abhängigkeitsgesetze ins Licht zu setzen bestimmt ist, unterlassen wir indess jetzt, da wir die Betrachtung des Ausdrucken einer Function gegenwärtig ausschliessen.

Aus demselben Grunde befassen wir uns hier auch nicht damit, die Brauchbarkeit unserer Sätze als Grundlagen einer allgemeinen Theorie dieser Abhängigkeits-gesetze darzuthun, wozu der Beweis erfordert wird, dass der hier zu Grunde gelegte Begriff einer Function einer veränderlichen complexen Grösse mit dem einer durch Grössenoperationen ausdrückbaren Abhängigkeit völlig zusammenfällt.”

This book starts with the Riemann mapping theorem and its implications developed since its acceptance as “one of the most important theorems of complex analysis” (Ahlfors [1953: 172]).

REFERENCES USED: Ahlfors [1953], Andersen et al. [1962], Barnard and Pearce [1986], Berrut [1986], Elcrat and Trefethen [1986], Freiberg [1951], Gaier [1964; 1983], Goluzin [1969], Gutknecht [1986], Hartman and Opfer [1986], Hoidn [1986], Hübner [1979; 1986], Ives [1982], James [1971], Jawson [1963], Jawson and Symm [1977], Kantorovich and Krylov [1958], Kerzman and Stein [1978], Kerzman and Trummer [1986], Kober [1957], Laura [1975], Lawrentjew and Schabat [1967], Lu [1994], Mayo [1986], Mikhlin [1957], Nehari [1949; 1952], Papamichael et al. [1981; 1983; 1984; 1986], Schinzinger and Laura [2003], Seidel [1952], Symm [1966], Trefethen [1980], Trefethen and Williams [1986], Wegmann [1979; 1986], Wen [1992], Zemach [1986].

2

Conformal Mapping

Some basic concepts and results from complex analysis are presented. They include harmonic functions, Cauchy's theorem, Cauchy kernel, Riemann mapping theorem, analytic continuation, and the Schwarz reflection principle. Proofs for most of the results can be found in textbooks.

2.1 Definitions

Let $\mathbb{R}^n$ denote the Euclidean n -space, and $\mathbb{R}^+$ the set of nonnegative real numbers. The complement of a set B with respect to a set A is denoted by $A \setminus B$ (or $\text{compl}(B)$ if the reference to set A is obvious), the product of the sets A and B by $A \times B$, and the closure of a set A by $\bar{A}$.

A complex-valued function f is said to belong to the class $C^k(D)$ if it is continuous together with its k th derivatives, in a domain D , $0 \leq k < \infty$. In this case we often say that f is a $C^k(D)$ -function, or that f is a C^k -function on D , where a C^0 -function is written simply as a C -function. The function f in the class $C^k(D)$, for which all k th derivatives admit continuous continuations in the closure $\bar{D}$, form the class of functions $C^k(\bar{D})$. The class $C^\infty(D)$ consists of functions f which are infinitely differentiable on D , i.e., continuous partial derivative of all orders exist. These classes are linear sets. Thus, every linear combination $\lambda f + \mu g$, where λ and μ are arbitrary complex numbers, also belongs to the respective class.

Let $\mathbb{C}$ denote the complex plane. If $a \in \mathbb{C}$ and $r > 0$, then

$$\begin{aligned} B(a, r) &= \{z \in \mathbb{C} : |z - a| < r\}, \\ \bar{B}(a, r) &= \{z \in \mathbb{C} : |z - a| \leq r\}, \\ \partial B(a, r) &= \{z \in \mathbb{C} : |z - a| = r\}, \end{aligned} \tag{2.1.1}$$

denotes, respectively, an open disk, a closed disk, and a circle, each of radius r and centered at a . The open unit disk $B(0, 1)$ is sometimes denoted by U . A connected open set $A \subseteq \mathbb{C}$ is called a *region* (or *domain*). The extended complex plane is denoted by $\mathbb{C}_\infty$. Then $\partial_\infty D$

is the boundary of a set D in $\mathbb{C}_\infty$, i.e.,

$$\partial_\infty D = \begin{cases} \partial D & \text{if } D \text{ is bounded,} \\ \partial D \cup \{\infty\} & \text{if } D \text{ is unbounded.} \end{cases}$$

If D is a region in $\mathbb{C}_\infty$, then the following statements are equivalent: (a) D is simply connected, (b) $\mathbb{C}_\infty \setminus D$ is connected, and (c) $\partial_\infty D$ is connected. **Regions that have more than one layer over the complex plane are called Riemann surfaces.**

Let $z = x + iy$ be the complex variable in the z -plane, where x and y are arbitrary real numbers. Similarly, let $w = u + iv$ be the complex variable in the w -plane, where u and v are arbitrary real numbers. Then a function $w = f(z)$ defined on a domain D in the z -plane and taking values in the w -plane is called a (complex) *function of a complex variable*. Let $z_0 \in D$ be a nonisolated point of D , so that every neighborhood of z_0 contains a point of D other than z_0 itself. Then a complex number A is said to be the *derivative* of the function $w = f(z)$ at a point $z = z_0$ relative to the domain D , denoted by $f'_D(z_0)$, if, given any $\varepsilon > 0$, there exists a $\delta > 0$ such that $0 < |z - z_0| < \delta$, $z \in D$, implies

$$\left| A - \frac{f(z) - f(z_0)}{z - z_0} \right| < \varepsilon. \quad (2.1.2)$$

In this case we say that $w = f(z)$ is *differentiable* at $z = z_0$ with derivative $A = f'_D(z_0)$, where A is the limit of the differential quotient $\frac{f(z) - f(z_0)}{z - z_0}$ in the direction $z \rightarrow z_0$ in the deleted neighborhood $0 < |z - z_0| < \delta$. Note that above differential quotient is bounded by some constant C , and thus, $|f(z) - f(z_0)| \leq C|z - z_0|$ for sufficiently small $|z - z_0|$, $z \in D$. Hence,

$$\lim_{z \rightarrow z_0} f(z) = f(z_0).$$

Thus, $f(z)$ is continuous at $z = z_0$ relative to the domain D .

2.1.1 Analytic Functions. **A function is said to be *analytic* (synonymously, *holomorphic* or *regular*) on a domain D in the z -plane if $f(z)$ is differentiable on D ,** i.e., it is differentiable at every point of D . The derivative of $f(z)$ is denoted by $f'(z)$, without explicit reference to the domain D . Thus, a function $f(z)$ is said to be *analytic* on a domain D if $f(z)$ is analytic at a point $z = z_0$, where $z_0 \in D$.

The increment of a complex function $w = f(z)$ differentiable relative to a domain D can be written as

$$\Delta w = f(z_0 + h) - f(z) = f'_D(z_0)h + \varepsilon(h)h, \quad h = z - z_0, \quad (2.1.3)$$

where $\varepsilon(h) \rightarrow 0$ as $h \rightarrow 0$. Suppose we denote the quantity $f'_D(z_0)h$, called the *principal linear part* of the increment Δw , by $dw(z_0)$, or briefly by dw or df , while writing the increment h as an independent variable $d_D z$ or simply dz . Then the quantities dw, df, dz are called the *differentials*, and the derivative can be written as the ratio of differentials

$$f'(z_0) = \frac{dw}{dz} = \frac{df}{dz}. \quad (2.1.4)$$

Some of the properties of the differential are:

- (i) $d(kf) = kdf$;
- (ii) $d(f \pm g) = df \pm dg$;
- (iii) $d(fg) = f dg + g df$;
- (iv) $d\left(\frac{f}{g}\right) = \frac{g df - f dg}{g^2(z_0)}$, $g(z_0) \neq 0$.

Let $z = x + iy$ be a complex number. Then $\bar{z} = x - iy$, $x = \frac{z + \bar{z}}{2}$, and $y = \frac{z - \bar{z}}{2i}$. Also

$$\partial f = \frac{\partial f}{\partial z} = \frac{1}{2}(f_x - i f_y), \quad \bar{\partial} f = \frac{\partial f}{\partial \bar{z}} = \frac{1}{2}(f_x + i f_y). \quad (2.1.5)$$

A function $f : D \mapsto \mathbb{C}$ is analytic on D iff $\bar{\partial} f = 0$, which is equivalent to the *Cauchy-Riemann equations* for the function $w = f(z) = u(x, y) + iv(x, y)$:

$$u_x = v_y, \quad u_y = -v_x, \quad (2.1.6)$$

or, in polar form ($z = re^{i\theta}$),

$$u_r = \frac{1}{r} v_\theta, \quad v_r = -\frac{1}{r} u_\theta. \quad (2.1.7)$$

Thus,

$$f'(z) = u_x + i v_x = v_y - i u_y. \quad (2.1.8)$$

The Cauchy-Riemann equations are necessary conditions for $f(z)$ to be analytic on D . However, merely satisfying the Cauchy-Riemann equations alone is not sufficient to ensure the differentiability of $f(z)$ at a point in D . For example, the function $f(z) = \frac{x^4 - y^4}{x^3 + y^3} + i \frac{x^4 + y^4}{x^3 + y^3}$, $f(0) = 0$, $z \in \mathbb{C}$, is not differentiable at the origin, although it satisfies the Cauchy-Riemann equations there: $u_x(0, 0) = 1 = v_x(0, 0) = v_y(0, 0)$, $u_y(0, 0) = -1$, but by limit definition, $f'(0) = (1 + i)/2$ along the line $y = x$.

The following results are obvious: $\partial(\log|z|) = \frac{1}{2z}$, $\bar{\partial}(\log|z|) = \frac{1}{2\bar{z}}$, $\overline{\partial f} = \bar{\partial} \bar{f}$, and the Laplacian

$$\nabla^2 \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} = 4\bar{\partial}\partial = 4\partial\bar{\partial}. \quad (2.1.9)$$

The Cauchy-Riemann equations (2.1.6) for the function $f(z) = u(x, y) + iv(x, y)$ satisfy the partial differential equations

$$\begin{aligned} u_x v_x + u_y v_y &= 0, \\ \nabla^2 u &= 0, \quad \nabla^2 v = 0. \end{aligned} \quad (2.1.10)$$

Using the gradient vector $\nabla \equiv \mathbf{i} \frac{\partial}{\partial x} + \mathbf{j} \frac{\partial}{\partial y}$, the first equation in (2.1.10) can be written as the inner (scalar) product:

$$\langle \nabla u, \nabla v \rangle = 0. \quad (2.1.11)$$

Then the Cauchy-Riemann equations yield $|\nabla u| = |\nabla v| = |f'(z)|$. Eq (2.1.11) also signifies the orthogonality condition for the families of level curves defined by $u(x, y) = \text{const}$ and $v(x, y) = \text{const}$.

If $w = u + iv$, then

$$\Re\{z\} \cdot \Re\{w\} = \Re\left\{\frac{z^2 + |z|^2}{2z} w\right\}, \quad \Im\{z\} \cdot \Im\{w\} = \Im\left\{\frac{z^2 - |z|^2}{2iz} w\right\}. \quad (2.1.12)$$

Example 2.1. Let $f(z) = \bar{z} = x - iy$. Its real and imaginary parts do not satisfy the Cauchy-Riemann equations, and hence $\bar{z}$ does not have a complex derivative. In general, the function $f(z, \bar{z})$ that depends on $\bar{z}$ is not complex-differentiable. ■

Let E denote a closed bounded infinite set of points in the z -plane. For the points $z_1, z_2, \dots, z_n \in E$ the Vandermonde determinant is defined by

$$V(z_1, z_2, \dots, z_n) = \prod_{\substack{i,j=1 \\ i \neq j}}^n (z_i - z_j), \quad n \geq 2. \quad (2.1.13)$$

Let us define the numbers

$$d_n = V_n^{2/n(n-1)}, \quad (2.1.14)$$

where $d_{n+1} \leq d_n$. Since d_n does not exceed the diameter¹ of the set E for any n , it follows that the sequence $\{d_n\}$ approaches a finite limit as $n \rightarrow \infty$. This limit is called the *transfinite diameter* of the set E and is denoted by $\text{diam}(E)$, i.e., $\text{diam}(E) = \lim_{n \rightarrow \infty} d_n$. When the set E has finitely many points, we take $\text{diam}(E) = 0$. ■

Example 2.2. If $f(z) = z\bar{z}$, then $f'(z)$ exists only at $z = 0$. To prove, note that $f(z) = |z|^2 = x^2 + y^2$, so that $u = x^2 + y^2$ and $v = 0$. Then $\frac{\partial u}{\partial x} = 2x$, $\frac{\partial u}{\partial y} = 2y$, $\frac{\partial v}{\partial x} = 0 = \frac{\partial v}{\partial y}$. Hence, the Cauchy-Riemann equations are satisfied only when $x = 0 = y$, i.e., only when $z = 0$. This function not analytic anywhere else. ■

2.1.2 Integration. Let f be a C -function on an open set $D \subset \mathbb{C}$, and let Γ be a piecewise C^1 -continuous curve in D . If $|f(z)| \leq M$ for all points $z \in \Gamma$, i.e., for all $z = \gamma(t)$ for $t \in [a, b]$, where $M > 0$ is a constant, then

$$\left| \int_{\Gamma} f dz \right| \leq M l(\Gamma), \quad (2.1.15)$$

where

$$l(\Gamma) = \int_a^b |\gamma'(t)| dt = \int_a^b \sqrt{x'(t)^2 + y'(t)^2} dt \quad (2.1.16)$$

is the arc length of the path Γ . In general, by applying the triangle inequality, we get

$$\left| \int_{\Gamma} f dz \right| \leq \int_{\Gamma} |f| |dz| = \int_a^b |f(\gamma(t))| |\gamma'(t)| dt. \quad (2.1.17)$$

If $\gamma : [a, b] \mapsto \mathbb{C}$ defines a piecewise smooth contour Γ and F is a function defined and analytic on a region containing Γ , then

$$\int_{\Gamma} F'(z) dz = F(\gamma(b)) - F(\gamma(a)). \quad (2.1.18)$$

¹The diameter of the set E is defined as $\sup_E \{|z_i - z_j| : z_i, z_j \in E, i, j = 1, 2, \dots, n (i \neq j)\}$.

If $\gamma(a) = \gamma(b)$, then

$$\int_{\Gamma} F'(z) dz = 0. \quad (2.1.19)$$

This result is known as the **fundamental theorem for line integrals (contour integration)** in the complex plane. Thus, if a function f is defined and analytic on a region $D \subset \mathbb{C}$ and if $f'(z) = 0$ for all points $z \in D$, then f is a constant on D . If f is a C -function on a region D , then the following three statements are equivalent:

- (i) If Γ_1 and Γ_2 are two paths in D from a point $z_1 \in D$ to a point $z_2 \in D$, then $\int_{\Gamma_1} f(z) dz = \int_{\Gamma_2} f(z) dz$, i.e., the **integrals are path-independent**.
- (ii) If Γ is a Jordan contour lying in D , then $\int_{\Gamma} f(z) dz = 0$, i.e., the **integrals on a closed contour are zero**.
- (iii) There exists a function F defined and analytic on D such that $F'(z) = f(z)$ for all $z \in D$, i.e., there exists a global antiderivative of f on D .

A function $\Phi(z)$, analytic inside a region with boundary $\Gamma = \cup_{k=1}^n \Gamma_k$, is said to be *sectionally analytic* on Γ if $\Phi(z)$ is continuous on each Γ_k from both left and right except at the end points where it satisfies the condition $|\Phi(z)| \leq \frac{C}{|z - c|^\alpha}$, where c is the corresponding end point of Γ_k , and C and α are real constant with $\alpha < 1$.

Let $w = f(z)$ be an analytic function, regular in a simply connected region D , and α be an interior point of D and a simple zero of $f(z)$ such that $f(\alpha) = 0$ and $f'(\alpha) \neq 0$. Moreover, let $z_0 \in D$ be a first approximation of the zero α , i.e., z_0 is close to α . Then

$$\begin{aligned} \alpha &= \sum_{n=0}^{\infty} (-1)^n \frac{f(z_0)^n}{n!} \left[\frac{d^n f^{-1}(w)}{dw^n} \right]_{w=f(z_0)} \\ &= \exp \left\{ -f(z_0) \left[\frac{df^{-1}(w)}{dw} \right]_{w=f(z_0)} \right\}, \end{aligned}$$

where $z = f^{-1}(w)$ denotes the inverse function of $f(z)$ and the exponential function operates symbolically on the differential symbol (Blaskett and Scherdtfeger [1945-1946:266]).

2.1.3 Fatou's Lemma. Let $f_n : I \mapsto \mathbb{R}$ be nonnegative, extended real-valued, measurable functions defined in an interval I and such that the sequence $\{f_n\}$ converges pointwise to the function $f : I \mapsto \mathbb{R}$. If $\varliminf_{n \rightarrow \infty} \int_{\Gamma} f_n < \infty$, then f is integrable and

$$\int_I f \leq \varliminf_{n \rightarrow \infty} \int_{\Gamma} f_n. \quad (2.1.20)$$

The integral of the pointwise liminf is always the liminf of the integrals.

2.2 Jordan Contour

A simple closed curve Γ in $\mathbb{C}$ is a path $\gamma : [a, b] \mapsto \mathbb{C}$ such that $\gamma(t) = \gamma(s)$ iff $t = s$ or $|t - s| = b - a$. In what follows, a simple closed curve will be called a *Jordan contour*. The *Jordan curve theorem* states that if Γ is a simple contour, then $\mathbb{C} \setminus \Gamma$ has two components, one called the interior of Γ , denoted by $\text{Int}(\Gamma)$, and the other called the exterior of Γ ,

denoted by $\text{Ext}(\Gamma)$, each of which has Γ as its boundary. Thus, if Γ is a Jordan contour, then $\text{Int}(\Gamma)$ and $\text{Ext}(\Gamma) \cup \{\infty\}$ are simply connected regions.

Let a continuous curve Γ , defined by $\gamma(t) = \alpha(t) + i\beta(t)$, be divided into n arcs $\sigma_k = \widehat{z_{k-1} z_k}$, $k = 1, \dots, n$, where $z_k = \gamma(t_k)$ for $k = 0, 1, \dots, n$, such that the end point of each arc, except the last one, overlaps the initial point of the next arc. If we join each segment $[z_{k-1}, z_k]$, $k = 1, \dots, n$, by straight line segments (see Figure 2.1, where $n = 5$), we obtain a polygonal line L inscribed in Γ . The segments of L are the chords joining the end points of the arcs σ_k , and

$$l = \text{length of } L = \sum_{k=1}^n |z_k - z_{k-1}|. \quad (2.2.1)$$

The curve Γ is said to be **rectifiable** if

$$\sup_{\mathcal{P}} \sum_{k=1}^n |z_k - z_{k-1}| = l < +\infty,$$

where the least upper bound is taken over all partitions $\mathcal{P} = \{a = t_0, t_1, \dots, t_n = b\}$ of the interval $[a, b]$, $a \leq t \leq b$. The nonnegative number l is called the length of the curve Γ . The curve is said to be *nonrectifiable* if the sums (2.1.13) become arbitrarily large for suitably chosen partitions.

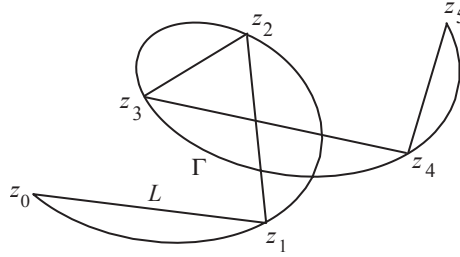


Figure 2.1 Rectifiable curve.

A piecewise smooth curve $\tilde{\Gamma}$, defined by $\tilde{\gamma} = [\tilde{a}, \tilde{b}] \mapsto \mathbb{C}$, is called a **reparameterization** of a curve Γ , defined by $\gamma = [a, b] \mapsto \mathbb{C}$ if there exists a function $\alpha \in C^1$, $\alpha : [a, b] \mapsto [\tilde{a}, \tilde{b}]$ with $\alpha'(t) > 0$, $\alpha(a) = \tilde{a}$ and $\alpha(b) = \tilde{b}$, such that $\gamma(t) = \tilde{\gamma}(\alpha(t))$, $t \in [a, b]$. Then

$$\int_{\Gamma} f = \int_{\tilde{\Gamma}} f$$

for any C -function f defined on an open set containing the image of γ (which is equal to the image of $\tilde{\gamma}$).

2.2.1 Hölder Condition. Let $f(t)$ be defined on a Jordan curve Γ (open or closed). If

$$|f(t_1) - f(t_2)| \leq A |t_1 - t_2|^\alpha, \quad 0 < \alpha \leq 1, \quad (2.2.2)$$

for arbitrary points $t_1, t_2 \in \Gamma$ ($t_1 \neq t_2$), where $A > 0$ and α are real constants, then $f(t)$ is said to satisfy the *Hölder condition of order α* , or simply $f(t)$ satisfies the condition H^α , denoted by $f(t) \in H^\alpha$. The condition H^1 is known as the *Lipschitz condition*. If $f(t) \in C(\Gamma)$ and $f(t) \in H^\alpha$, then we say that $f(t)$ is *Hölder-continuous* on Γ . If $f \in C(\Gamma)$ and $f \in H^1$, then $f(t)$ is said to be *Lipschitz-continuous*.

Criterion describing how “regular” or “smooth” a function is.
Smaller α allows “less smooth” functions; larger α means smoother.

2.3 Metric Spaces

Let S denote the set of all real-valued sequences. Then S is a vector space if addition and scalar multiplication of vectors $s_i, t_i \in S$ for $i = 1, 2, \dots$ are defined coordinatewise, i.e., $\{s_i\} + \{t_i\} = \{s_i + t_i\}$, and $\lambda \{s_i\} = \{\lambda s_i\}$ for a scalar λ . The *Fréchet metric* (distance) $d(\{s_i\}, \{t_i\})$ is defined for $\{s_i\}, \{t_i\} \in S$ by

$$d(\{s_i\}, \{t_i\}) = \sum_{i=1}^{\infty} \frac{\|s_i - t_i\|}{(1 + \|s_i - t_i\|)^{2^i}}. \quad (2.3.1)$$

The space ℓ^p , $1 \leq p < \infty$, is defined by $\ell^p = \left\{ \{s_i\} : \sum_{i=1}^{\infty} |s_i|^p < \infty \right\}$. Since the sum of any two elements in ℓ^p is also in ℓ^p , then ℓ^p is a vector subspace of S . Define a norm $\|\cdot\|_p$ on ℓ^p by

$$\|\{s_i\}\|_p = \left(\sum_{i=1}^{\infty} |s_i|^p \right)^{1/p}. \quad (2.3.2)$$

Then it can be shown that ℓ^p is closed under vector addition and $\|\cdot\|_p$ satisfies the triangle inequality. Let $\{s_i\}, \{t_i\} \in \ell^p$ be such that $\sum |s_i|^p = 1$ and $\sum |t_i|^q = 1$, where p and q are called conjugate exponents such that $p > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$. Then

Hölder inequality:

$$\sum_{i=1}^{\infty} |s_i t_i| \leq \left(\sum_{i=1}^{\infty} |s_i|^p \right)^{1/p} \left(\sum_{i=1}^{\infty} |t_i|^q \right)^{1/q}. \quad (2.3.3)$$

Cauchy-Schwarz inequality:

$$\sum_{i=1}^{\infty} |s_i t_i| \leq \sqrt{\sum_{i=1}^{\infty} |s_i|^2} \sqrt{\sum_{i=1}^{\infty} |t_i|^2}. \quad (2.3.4)$$

Minkowsky's inequality:

$$\left(\sum_{i=1}^{\infty} |s_i + t_i|^p \right)^{1/p} \leq \left(\sum_{i=1}^{\infty} |s_i|^p \right)^{1/p} + \left(\sum_{i=1}^{\infty} |t_i|^p \right)^{1/p}, \quad p \geq 1. \quad (2.3.5)$$

Note that $p = 1$ gives the *triangle inequality*.

Let L^2 denote the Hilbert space of all square-integrable analytic functions f in a simply connected region D with boundary Γ . A function $f(z)$ regular in D is said to belong to the class $L^2(D)$ (we say, $f \in L^2(D)$) if the integral $\iint_D |f(z)|^2 dS_z < +\infty$, where $dS_z = dx dy$ denotes an area element in D . The (surface)-norm $\|f\|_{2,D}$ of $f(z)$ is defined by

$$\|f\|_{2,D}^2 = \iint_D |f(z)|^2 dS_z. \quad (2.3.6)$$

If two functions $f, g \in L^2(D)$, then their inner product is defined by

$$\langle f, g \rangle = \iint_D f(z) \overline{g(z)} dS_z. \quad (2.3.7)$$

Let Γ be the rectifiable Jordan boundary of the region D , of length l , and let Γ_ρ denote the image of the circle $|w| = \rho$ under the mapping $z = g(w) = f^{-1}(w)$, $0 < \rho < R$. If $f(z)$ is regular in D , then the integral

$$\int_{\Gamma_\rho} |f(z)|^2 ds_z = \rho \int_{\phi=0}^{2\pi} \left| f(g(w)) \sqrt{g'(w)} \right|^2 d\phi, \quad w = \rho e^{i\phi}, \quad (2.3.8)$$

where ds_z denotes a line element on Γ , is a monotone increasing function. We say that $f \in L^2(\Gamma)$ if this integral remains bounded as $\rho \rightarrow R$, i.e.,

$$\lim_{\rho \rightarrow R} \int_{\Gamma_\rho} |f(z)|^2 ds_z = \int_{\Gamma} |f(z)|^2 ds_z = \|f\|_{2,\Gamma}^2,$$

where $\|f\|_{2,\Gamma}^2$ is the (line)-norm of f . For any two functions $f, g \in L^2(\Gamma)$, we define their inner product as

$$\langle f, g \rangle = \int_{\Gamma} f(z) \overline{g(z)} ds_z. \quad (2.3.9)$$

If the disk $B(z_0, r) \subset D$, then

$$|f(z_0)|^2 = \begin{cases} \frac{1}{\pi r^2} \iint_D |f(z)|^2 dS_z, & \text{if } f \in L^2(D), \\ \frac{1}{2\pi r} \int_{\Gamma} |f(z)|^2 ds_z, & \text{if } f \in L^2(\Gamma). \end{cases} \quad (2.3.10)$$

Let a region have a piecewise continuous boundary Γ , and let $f(z), g(z)$ be regular in D , and $f'(z)$ and $g'(z)$ continuous in $\bar{D}$. Then

$$\langle f, g' \rangle = \iint_D f(z) \overline{g'(z)} dS_z = \frac{1}{2i} \int_{\Gamma} f(z) \overline{g(z)} ds_z. \quad (2.3.11)$$

This is known as *Green's formula*. It is useful in converting a surface integral into a line integral. This formula also holds for multiply connected regions. The integral along Γ is taken in the positive sense, i.e., the region D remains to the left as one traverses the contour Γ . The following inequality is also useful: If $f \in L^2(D)$, then $f \in L^2(\Gamma)$, and

$$\iint_D |f(z)|^2 dS_z \leq \frac{l}{2} \int_{\Gamma} |f(z)|^2 ds_z. \quad (2.3.12)$$

We will denote by L^∞ the Hilbert space of 2π -periodic and bounded functions f with the norm

$$\|f\|_\infty = \max_{[0, 2\pi]} |f(x)|, \quad (2.3.13)$$

and by W the Sobolev space of 2π -periodic and absolutely continuous functions f , with $f' \in L^2[0, 2\pi]$. Then

$$\|f\| = \max (\|f\|_\infty, \|f'\|_2),$$

and $(W, \|\cdot\|)$ is a Banach space. Thus, if $f, g \in (W, \|\cdot\|)$, then (i) $\|f\| \geq 0$; (ii) $\|f\| = 0$ implies that $f = 0$; (iii) $\|\alpha f\| = |\alpha| \|f\|$ for a scalar α ; and $\|f + g\| \leq \|f\| + \|g\|$ which is known as the triangle inequality.

2.4 Basic Theorems

Some basic theorems from the theory of functions of a complex variable are presented without proofs which can be found in standard textbooks on the subject.

Theorem 2.1. (Cauchy's theorem) *Let f be analytic on a region D , and let Γ be a closed contour which is homotopic to a point in D . Then $\int_{\Gamma} f = 0$.*

Note that a set is said to be simply connected if every closed contour $\Gamma \subset D$ is homotopic (as a closed curve) to a point in D , i.e., to some constant curve. Some local versions of Cauchy's theorem are as follows:

Theorem 2.2. (Cauchy-Goursat theorem for a disk) *Let $f : B \mapsto \mathbb{C}$ be analytic on a disk $B = B(z_0, r) \subset D$. Then (i) there exists a function $F : B \mapsto \mathbb{C}$ which is analytic on B and is such that $F'(z) = f(z)$ for all $z \in B$ (i.e., f has an antiderivative on B); and (ii) If Γ is a closed contour in B , then $\int_{\Gamma} f = 0$.*

This theorem also holds if f is continuous on B and analytic on $D \setminus \{z_1\}$ for some fixed $z_1 \in D$.

Theorem 2.3. (Cauchy-Goursat theorem for a rectangle) *Let R denote a rectangle with sides parallel to the coordinate axes, and let f be a function defined and analytic on an open set D containing R . Then $\int_{\Gamma} f = 0$.*

Even if the function f is analytic on D except at some fixed point $z_1 \in D$ which does not lie on the contour R , and if $\lim_{z \rightarrow z_1} (z - z_1)f(z) = 0$, then also $\int_{\Gamma} f = 0$. The two theorems 1.2.2 and 1.2.3 hold (a) if f is bounded on a deleted neighborhood of z_1 , or (b) if f is continuous on D , or (c) if $\lim_{z \rightarrow z_1} f(z)$ exists.

The *index* of a curve Γ with respect to a point $z_0 \in \mathbb{C}$ is the integer n that expresses how many times Γ winds around z_0 . This index is denoted by $I(\Gamma, z_0)$ and is called the *winding number* of Γ with respect to z_0 . Thus,

$$I(\Gamma, z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{dz}{z - z_0}. \quad (2.4.1)$$

In fact,

$$I(\Gamma, z_0) = \begin{cases} \pm n & \text{if } z_0 \in \text{Int}(\Gamma), \\ 0 & \text{if } z_0 \in \text{Ext}(\Gamma). \end{cases}$$

Theorem 2.4. (Cauchy's integral formula) *Let f be analytic on a region D , and let Γ be a simple closed contour in D that is homotopic to a point in D . Let $z_0 \in D$ be a point not on Γ . Then*

$$f(z_0) \cdot I(\Gamma, z_0) = \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z_0} d\zeta. \quad (2.4.2)$$

The integrand in (2.4.2) is known as the *Cauchy kernel* defined by

$$H(z, z_0) = \frac{1}{2\pi i} \frac{f(z)}{z - z_0}. \quad (2.4.3)$$

The formula (2.4.2) is a special case of integrals of the Cauchy type. If we set

$$F(z) = \frac{1}{2\pi i} \int_{\Gamma} \frac{g(\zeta)}{\zeta - z} d\zeta, \quad (2.4.4)$$

where $\Gamma : [a, b] \mapsto \mathbb{C}$ is a simple contour and g a C -function defined on the image $\Gamma([a, b])$, then F is analytic on $\mathbb{C} \setminus \Gamma([a, b])$ and is infinitely differentiable, such that its k th derivative is given by

$$F^{(k)}(z) = \frac{k!}{2\pi i} \int_{\Gamma} \frac{g(\zeta)}{(\zeta - z)^{k+1}} d\zeta, \quad k = 1, 2, \dots \quad (2.4.5)$$

Then Cauchy's integral formula for the derivatives is

$$f^{(k)}(z) \cdot I(\Gamma, z_0) = \frac{k!}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{(\zeta - z)^{k+1}} d\zeta, \quad k = 1, 2, \dots \quad (2.4.6)$$

Let f be analytic on a region D and let $\Gamma = \partial B(z_0, R)$ be a circle lying in D . If $|f| \leq M$ for all $z \in \Gamma$, then for $k = 0, 1, \dots$, we have

$$|f^{(k)}(z_0)| \leq \frac{k!}{R^k} M. \quad (2.4.7)$$

This result is known as *Cauchy's inequality*. A corollary is the *Liouville theorem* which states that the only bounded entire functions are constants. A partial converse of Cauchy's theorem is known as *Morera's theorem* which states that if f is continuous on a region D and if $\int_{\Gamma} f dz = 0$ for every closed contour Γ in D , then f is analytic on D , and $f = F'$ for some analytic function F on D .

Two very useful corollaries of Cauchy's integral formula (2.4.2) are:

- (i) The *maximum modulus theorem* which states that if f is a nonconstant analytic function on a region D with a simple boundary Γ , then $|f|$ cannot have a local maximum anywhere in $\text{Int}(\Gamma)$.
- (ii) The *mean value property* of an analytic function f defined on the circle $\partial B(z_0, r)$ states that

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{i\theta}) d\theta. \quad (2.4.8)$$

An application of the maximum modulus theorem is

Lemma 2.1. (Schwarz lemma) *Let f be analytic on the open unit disk U , and suppose that $|f(z)| \leq 1$ for all $z \in U$ and $f(0) = 0$. Then $|f(z)| \leq |z|$ for all $z \in U$ and $|f'(0)| \leq 1$. If $|f(z_0)| = |z_0|$ for some $z_0 \in U$, $z_0 \neq 0$, then $f(z) = cz$ for all $z \in U$, where c is some constant such that $|c| = 1$.*

A consequence of the Schwarz lemma is: If $f(z)$ is analytic on the disk $|z| < R$ such that $f(0) = 0$, and $|f(z)| \leq M$, where $M > 0$ is a constant, then $|f(z)| \leq \frac{M}{R} |z|$, $|f'(0)| \leq \frac{M}{R}$ on this disk, where the equality holds only for $f(z) = e^{i\alpha} \frac{M}{R} z$, α real (Wen [1992:26-27]).

2.4.1 Singularities. Let f be analytic on a deleted neighborhood of z_0 : $\{z : 0 < |z - z_0| < r\}$. In such a case we say that z_0 is an *isolated singularity* of f , and the Laurent series expansion of f at z_0 is

$$f(z) = \dots + \frac{b_n}{(z - z_0)^n} + \dots + \frac{b_1}{z - z_0} + a_0 + a_1(z - z_0) + \dots, \quad 0 < |z - z_0| < r. \quad (2.4.9)$$

If z_0 is an isolated singularity of f and if all but a finite number of b_n in the expansion (2.4.9) are zeros, then z_0 is called a *pole* of f , and if $k \neq 0$ is the highest integer such that

$b_k \neq 0$, then z_0 is called a *pole of order k* . In this case $f(z) = \frac{g(z)}{(z - z_0)^n}$, where $g(z)$ is analytic at $z = z_0$ and $g(z_0) \neq 0$. If z_0 is a pole of order 1, it is called a *simple pole*. If infinitely many b_n are nonzero, then z_0 is called an *essential singularity*. A singularity is *essential* if it is not a pole or a branch point; for example, $e^{1/z}$ at $z_0 = 0$.

If f has a removable singularity at z_0 , then $f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$ is a convergent power series, where $f(z_0) = a_0$ which is analytic at z_0 .

Cauchy's Formula. The deleted neighborhood $0 < |z - a| < r$ is a multiply connected domain, and the function $\frac{1}{(z - a)^n}$, $n = 1, 2, \dots$, is analytic on this domain, but not on the whole disk $|z - a| < r$. Then we have

$$\oint_C \frac{dz}{(z - a)^n} = ir^{1-n} \int_0^{2\pi} e^{i(1-n)\theta} d\theta = \begin{cases} ir^{1-n} \frac{e^{i(1-n)\theta}}{i(1-n)} \Big|_0^{2\pi} = 0 & \text{if } n \neq 1, \\ i\theta \Big|_0^{2\pi} = 2\pi i & \text{if } n = 1. \end{cases} \quad (2.4.10)$$

Branch Points. There are *algebraic branch points* of degree n as in $f(z) = \sqrt[n]{z}$ at $z = z_0$; and there are *logarithmic branch points* of degree ∞ such as $\log z$ at $z = z_0$. In general, the power function $z^a = e^{a \log z}$ is analytic at $z = z_0$ if $a \in \mathbb{Z}$ is an integer; it has an *algebraic branch point* of degree q at the origin if $a = p/q$ is rational, non-integral with $0 \neq p \in \mathbb{Z}$ and $2 \leq q \in \mathbb{Z}$ with no common factor, and is a *logarithmic branch point* of degree ∞ at $z = 0$ for a not rational.

2.4.2 Residues. If f has an isolated singularity at z_0 , then the Laurent series expansion of f in a deleted neighborhood of z_0 is given by (2.4.9). The coefficient b_1 is called the *residue* of f at z_0 , written as $b_1 = \text{Res}(f, z_0)$. If f is analytic in a region $D \in \mathbb{C}$, and if Γ is any circle around z_0 whose interior lies in D , then

$$\int_{\Gamma} f(z) dz = 2\pi i b_1. \quad (2.4.11)$$

As an example, using the Laurent series expansion $e^{1/z} = 1 + \frac{1}{z} + \frac{1}{z^2} + \dots + \frac{1}{n!z^n} + \dots$, we have $\text{Res}\{e^{1/z}\} = 1$ at $z_0 = 0$.

Theorem 2.5. If f is analytic in a region $D \in \mathbb{C}$ and has an isolated singularity at $z_0 \in D$, then

(a) z_0 is an isolated singularity if any one of the following conditions hold: (i) f is bounded in a deleted neighborhood of z_0 ; (ii) $\lim_{z \rightarrow z_0} f(z)$ exists; or (iii) $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$.

(b) z_0 is a simple pole iff $\lim_{z \rightarrow z_0}$ exists and is nonzero; this limit is equal to the residue of f at z_0 .

(c) z_0 is a pole of order $\leq k$ (or possibly a removable singularity) iff any one of the following conditions hold: (i) There is a constant $M > 0$ and an integer $k \geq 1$ such that $|f(z)| \leq \frac{M}{|z - z_0|^k}$ in a deleted neighborhood of z_0 ; (ii) $\lim_{z \rightarrow z_0}^{k+1} f(z) = 0$; or (iii) $\lim_{z \rightarrow z_0}^k f(z)$ exists.

(d) z_0 is a pole of order $k \geq 1$ iff there is an analytic function g defined on a neighborhood A of z_0 defined by $A \setminus \{z_0\} \subset D$, such that $g(z_0) \neq 0$, and such that $f(z) = \frac{g(z)}{(z - z_0)^k}$ for $z \in A, z \neq z_0$.

Let f be analytic in a region D and let $z_0 \in D$. Then f has a zero of order k at z_0 iff $f(z_0) = 0, \dots, f^{(k-1)}(z_0) = 0, f^{(k)}(z_0) \neq 0$. It is obvious from Taylor's series of f at z_0 that f can be written as $f(z) = (z - z_0)^k g(z)$, where $g(z)$ is analytic at z_0 , and $g(z_0) = f^{(k)}(z_0)/k! \neq 0$.

It follows from Theorem 2.5(d) that if $f(z)$ has a pole (of finite order k) at z_0 , then $|f(z)| \rightarrow \infty$ as $z \rightarrow z_0$. However, if f has an essential singularity at z_0 , then $|f|$ will not, in general, approach ∞ as $z \rightarrow z_0$. In fact, by Picard's theorem (Picard [1879]), if f has an essential singularity at z_0 and if A is any arbitrary small deleted neighborhood of z_0 , then for all $\zeta \in \mathbb{C}, \zeta \neq z_0$, the equation $f(z) = \zeta$ has infinitely many solutions in A except possibly for one value.

Theorem 2.6. (Casorati-Weierstrass theorem) *Let f have an essential singularity at z_0 and let $w \in \mathbb{C}$. Then there exist $z_1, z_2, \dots, z_n \rightarrow z_0$, such that $f(z_n) \rightarrow w$.*

Example 2.2. (i) The function $f(z) = \frac{\sin z}{z}$ has removable singularity at $z = 0$, because $\lim_{z \rightarrow 0} z \cdot (\sin z)/z = \lim_{z \rightarrow 0} \sin z = 0$. (ii) The function $f(z) = e^z/z$ has a simple pole at $z = 0$, because $\lim_{z \rightarrow 0} z \cdot e^z/z = 1$; this singularity is not removable. (iii) The function $f(z) = (e^z - 1)^2/z$ has a removable singularity at $z = 0$, because $\lim_{z \rightarrow 0} z \cdot (e^z - 1)^2/z = 1$. (iv) The function $f(z) = z/(e^z - 1)$ has a removable singularity, because $(e^z - 1)/z = 1 + z/2! + z^2/3! + \dots \rightarrow 1$ as $z \rightarrow 0$. (v) If $f(z)$ and $g(z)$ are analytic, both having zeros of order k at z_0 , then $f(z)/g(z)$ has a removable singularity at z_0 and $\lim_{z \rightarrow z_0} \frac{f(z)}{g(z)} = \frac{f^{(k)}(z_0)}{g^{(k)}(z_0)}$. (vi) If f is analytic in a region containing a circle Γ and its interior and has a simple zero at z_0 inside or on Γ , then $z_0 = \frac{1}{2\pi i} \int_{\Gamma} \frac{zf'(z)}{f(z)} dz$.

Also, $f(z) = \frac{e^z(z-3)}{(z-1)(z-5)}$ has simple pole at $z = 1$ and $z = 5$.

$f(z) = \frac{\cos z}{1-z}$ has simple pole at $z = 1$. ■

Since it is not convenient to develop the Laurent series expansion of all functions $f(z)$ with all types of singularities, it is desirable to directly develop techniques to determine residues of f .

Case 1. Since f has a removable singularity at z_0 iff $\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$, we have

Theorem 2.7. *If $g(z)$ and $h(z)$ are analytic functions, both having zeros at z_0 of the same order, then $f(z) = \frac{g(z)}{h(z)}$ has a removable singularity at z_0 .*

Some examples are: (i) $e^z/(z-1)$ has no singularity at $z_0 = 0$; (ii) $(e^z - 1)/z$ has a removable singularity at 0 because both $e^z - 1$ and z have zeros of order 1 (although each

of then vanishes at $z = 0$ but their derivatives do not); and (iii) $z^2/\sin z$ has a removable singularity at $z_0 = 0$ because both numerator and denominator have zeros of order 2.

Case 2. Pole of order k .

Theorem 2.8. Let f have an isolated singularity at z_0 and let k be the smallest non-negative integer such that $\lim_{z \rightarrow z_0} (z - z_0)^k f(z)$ exists. Then $f(z)$ has a pole of order k at z_0 . Let $\phi(z) = (z - z_0)^k f(z)$, such that ϕ is defined uniquely at z_0 and analytic at z_0 . Then

$$\text{Res}(f, z_0) = \frac{\phi^{(k-1)}(z_0)}{(k-1)!}. \quad (2.4.12)$$

Theorem 2.9. Let g and h be analytic at z_0 , with $g(z_0) \neq 0$, and let $h(z_0) = 0 = \dots = h^{(k-1)}(z_0)$, $h^{(k)}(z_0) \neq 0$. Then the function $g(z)/h(z)$ has a pole of order k and the residue at z_0 is given by

$$\text{Res}(g/h, z_0) = \left[\frac{k!}{h^{(k)}(z_0)} \right]^k \times \begin{vmatrix} \frac{h^{(k)}(z_0)}{k!} & 0 & 0 & \dots & 0 & g(z_0) \\ \frac{h^{(k+1)}(z_0)}{(k+1)!} & \frac{h^{(k)}(z_0)}{k!} & 0 & \dots & 0 & g'(z_0) \\ \frac{h^{(k+2)}(z_0)}{(k+2)!} & \frac{h^{(k+1)}(z_0)}{(k+1)!} & \frac{h^{(k)}(z_0)}{k!} & \dots & 0 & \frac{g''(z_0)}{2!} \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ \frac{h^{(2k-1)}(z_0)}{(2k-1)!} & \frac{h^{(2k-2)}(z_0)}{(2k-2)!} & \frac{h^{(2k-3)}(z_0)}{(2k-3)!} & \dots & \frac{h^{(k+1)}(z_0)}{(k+1)!} & \frac{g^{(k-1)}(z_0)}{(k-1)!} \end{vmatrix}. \quad (2.4.13)$$

Example 2.3. Let $f(z) = \frac{e^z}{\sin^3 z}$. Here $g(z) = e^z$, $h(z) = \sin^3 z$, $k = 3$. Then by formula (2.4.13),

$$\text{Res}(e^z/\sin^3 z, 0) = \frac{3!}{6} \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ -1 & 0 & \frac{1}{2} \end{vmatrix} = 1.$$

Case 3. (Essential singularity at z_0) In the case of essential singularity at z_0 there is no simple formula except finding the Laurent series expansion for $f(z)$.

Example 2.4. Since

$$\begin{aligned} f(z) &= e^{z+1/z} = e^z \cdot e^{1/z} = \left(1 + z + \frac{z^2}{2!} + \dots\right) \left(1 + \frac{1}{z} + \frac{1}{2!z^2} + \dots\right) \\ &= \frac{1}{z} \left\{ 1 + \frac{1}{2!} + \frac{1}{2!3!} + \frac{1}{3!4!} + \dots \right\}, \end{aligned}$$

which gives

$$\text{Res}(f, 0) = 1 + \frac{1}{2!} + \frac{1}{2!3!} + \frac{1}{3!4!} + \dots,$$

where the resulting series cannot be summed explicitly. ■

A summary of methods to determine residues is given in Table 2.1.

Table 2.1: Summary of methods to determine residues.

Function	Test	Singularity Type	Residue at z_0
1. $f(z)$	$\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0$	removable	0
2. $\frac{g(z)}{h(z)}$	g and h have zeros of same order	removable	0
3. $f(z)$	$\lim_{z \rightarrow z_0} (z - z_0)f(z)$ exists and is not zero	simple pole	$\lim_{z \rightarrow z_0} (z - z_0)f(z)$
4. $\frac{g(z)}{h(z)}$	$g(z_0) \neq 0, h(z_0) = 0, h'(z_0) \neq 0$	simple pole	$\frac{g(z_0)}{h'(z_0)}$
5. $\frac{g(z)}{h(z)}$	g has zero of order k , h has zero of order $k + 1$	simple pole	$(k + 1) \frac{g^{(k)}(z_0)}{h^{(k+1)}(z_0)}$
6. $\frac{g(z)}{h(z)}$	$g(z_0) \neq 0, h(z_0) = 0 = h'(z_0), h''(z_0) \neq 0$	pole of order 2	$2 \frac{g'(z_0)}{h''(z_0)} - \frac{2}{3} \frac{g(z_0)h'''(z_0)}{[h''(z_0)]^2}$
7. $\frac{g(z)}{(z - z_0)^2}$	$g(z_0) \neq 0$	pole of order 2	$g'(z_0)$
8. $\frac{g(z)}{h(z)}$	$g(z_0) = 0, g'(z_0) \neq 0, h(z_0) = 0, h'(z_0) = h''(z_0) = 0, h'''(z_0) \neq 0$	pole of order 2	$3 \frac{g''(z_0)}{h'''(z_0)} - \frac{3}{2} \frac{g'(z_0)h^{(iv)}(z_0)}{[h'''(z_0)]^2}$
9. $f(z)$	k is the smallest integer such that $\lim_{z \rightarrow z_0} \phi(z_0)$ exists, where $\phi(z) = (z - z_0)^k f(z)$	pole of order k	$\lim_{z \rightarrow z_0} \frac{\phi^{(k-1)}(z)}{(k - 1)!}$
10. $\frac{g(z)}{h(z)}$	g has zero of order l ; and h has zero of order $(k + l)$	pole of order k	$\lim_{z \rightarrow z_0} \frac{\phi^{(k-1)}(z)}{(k - 1)!}$, where $\phi(z) = (z - z_0)^k (f(z)/h(z))$
11. $\frac{g(z)}{h(z)}$	$g(z_0) \neq 0, h(z_0) = \dots = h^{(k-1)}(z_0) = 0, h^{(k)}(z_0) \neq 0$	pole of order k	residue given by formula (2.4.13) in Theorem 2.9

2.4.3 Boundary Values for Cauchy Integral. If the boundary Γ of a simply connected region D is rectifiable and if $f(z)$ is regular in D and continuous on $\bar{D}$, then Cauchy's integral formula (2.4.2) holds for every $z_0 \in D$. For the boundary values we have the following result: Let the boundary Γ of D be smooth, and let $f(z)$ be regular in D and continuous on $\bar{D}$. Then for a point $z_0 \in \Gamma$ which is not a corner point

$$f(z_0) = \frac{1}{\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - z_0} d\zeta. \quad (2.4.14)$$

In the case when $z_0 \in \Gamma$ is a corner point with inner angle $\alpha\pi$, $0 < \alpha < 2$, the boundary

value $f(z_0)$ is given by

$$f(z_0) = \frac{1}{\alpha\pi i} - \int_{\Gamma} \frac{f(\zeta)}{\zeta - z_0} d\zeta, \quad (2.4.15)$$

where the integral is taken as a Cauchy p.-v. integral. If D is the region exterior to the Jordan curve Γ which is traversed in the positive sense and if $f(z)$ is regular at $z = \infty$, then the boundary value at a point $z_0 \in \Gamma$ which is not a corner point is given by

$$f(z_0) = -\frac{1}{\pi i} - \int_{\Gamma} \frac{f(\zeta)}{\zeta - z_0} d\zeta + 2f(\infty), \quad (2.4.16)$$

and at a corner point $z_0 \in \Gamma$ with the inner angle α , $0 < \alpha < 2$, by

$$f(z_0) = \frac{1}{(2 - \alpha)\pi i} - \int_{\Gamma} \frac{f(\zeta)}{\zeta - z_0} d\zeta + 2f(\infty). \quad (2.4.17)$$

The integrals in (2.4.14)-(2.4.17) are taken as Cauchy principal-values (see Appendix B for details on p.-v. integrals)

2.4.4 Argument Principle. (Cauchy's argument principle) This principle relates to the difference between the number of zeros and poles of a meromorphic function to a contour integral of the function's logarithmic derivative, and is defined by

$$\oint_{\Gamma} \frac{f'(z)}{f(z)} dz = 2\pi i(N - P), \quad (2.4.18)$$

where N is the number of zeros and P the number of poles, each counted according to its multiplicity. In other words, it is the number of windings around the point $w = 0$ made by the point $w = f(z)$ as z traverses Γ once in the counterclockwise direction equals $N - P$.

2.4.5 Plemelj Formulas. If Γ is a contour, then

$$\begin{aligned} F^+(\zeta_0) &= \frac{1}{2} f(\zeta_0) + \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta, \\ F^-(\zeta_0) &= -\frac{1}{2} f(\zeta_0) + \frac{1}{2\pi i} \int_{\Gamma} \frac{f(\zeta)}{\zeta - \zeta_0} d\zeta, \end{aligned} \quad (2.4.19)$$

where $F^+(\zeta_0)$ and $F^-(\zeta_0)$ are the limiting values from the right and left of Γ , respectively, $f(\zeta)$ satisfies the Hölder condition on Γ , and ζ_0 does not coincide with those end points where $f(\zeta_0) \neq 0$. If ζ_0 coincides with an end point where $f(\zeta_0) = 0$, then $F^+(\zeta_0) = F^-(\zeta_0) = f(\zeta_0)$. A proof for these formulas can be found in Muskhelishvili [1953/1992].

Another set of useful formulas is as follows:

$$\sin \pi z = \pi z \prod_{n=1}^{\infty} \left(1 - \frac{z^2}{n^2}\right), \quad (2.4.20)$$

$$\pi \cot \pi z = \sum_{n=-\infty}^{\infty} \frac{1}{z - n} = \frac{1}{z} + \sum_{n=1}^{\infty} \left(\frac{1}{z - n} + \frac{1}{z + n} \right), \quad (2.4.21)$$

where the series converges in the Cauchy sense.

2.5 Harmonic Functions

The functions whose Laplacian is zero are known as harmonic functions. Thus, a real-valued function $u(x, y) \in C^2(D)$ is said to be *harmonic* in a region D if $\nabla^2 u = 0$. Some properties of harmonic functions in $\mathbb{R}^2$ are as follows:

(i) The function

$$\frac{1}{r} = \frac{1}{\sqrt{(x - x_0)^2 + (y - y_0)^2}} \quad (2.5.1)$$

is harmonic in a region that does not contain the point (x_0, y_0) .

(ii) If $u(x, y)$ is a harmonic function in a simply connected region D , then there exists a conjugate harmonic function $v(x, y)$ in D such that $u(x, y) + i v(x, y)$ is an analytic function of $z = x + iy = (x, y)$ in D . In view of the Cauchy-Riemann equations (2.1.6),

$$v(x, y) - v(x_0, y_0) = \int_{x_0, y_0}^{x, y} (-u_y dx + u_x dy), \quad (2.5.2)$$

where $(x_0, y_0) = z_0$ is a given point in D . This property is also true if D is multiply connected. However, in that case the conjugate function, $v(x, y)$ can be multiple-valued, as we see by considering $u(x, y) = \log r = \log \sqrt{x^2 + y^2}$ defined on a region D containing the origin which has been indented by a small circle centered at the origin. Then, in view of (2.5.2),

$$v(x, y) - v(x_0, y_0) = \tan^{-1} \frac{y}{x} \pm 2n\pi + \text{const}, \quad n = 1, 2, \dots,$$

which is multiple-valued.

(iii) Since derivatives of all orders of an analytic function exist and are themselves analytic, any harmonic function will have continuous partial derivatives of all orders, i.e., a harmonic function belongs to the class $C^\infty(D)$, and a partial derivative of any order is again harmonic.

(iv) A harmonic function must satisfy the mean-value theorem, where the mean value at a point is evaluated for the circumference or the area of the circle around that point. If u is harmonic on a region containing the closed disk $\bar{B}(z_0, r)$, where $z_0 = x_0 + iy_0$, then

$$u(x_0, y_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + r e^{i\theta}) d\theta. \quad (2.5.3)$$

(v) In view of the maximum modulus theorem (§1.2), the maximum (and also the minimum) of a harmonic function u in a region D occurs only on the boundary of D . This result is known as

Theorem 2.10. (Maximum Principle) *A nonconstant function which is harmonic inside a bounded region D with boundary Γ and continuous in the closed region $\bar{D} = D \cup \Gamma$ attains its maximum and minimum values only on the boundary of the region.*

Thus, u has a maximum (or minimum) at $z_0 \in D$, i.e., if $u(z) \leq u(z_0)$ (or $u(z) \geq u(z_0)$) for z in a neighborhood $B(z_0, \varepsilon)$ of z_0 , then $u = \text{const}$ in $B(z_0, \varepsilon)$.

(vi) The value of a harmonic function u at an interior point in terms of the boundary values u and $\frac{\partial u}{\partial n}$ is given by Green's third identity (A.8).

(vii) If u and U are continuous in $\bar{D}$ and harmonic in D such that $u \leq U$ on Γ , then $u \leq U$ also at all points inside D . In fact, the function $U - u$ is continuous and harmonic in D .

Hence $U - u \geq 0$ on Γ . Then, in view of the maximum principle (Theorem 2.10), we require that $U - u \geq 0$ at all points inside D .

(viii) If u and U are continuous in $\bar{D}$ and harmonic in D for which $|u| \leq U$ on Γ , then $|u| \leq U$ also at all points inside D . In fact, the three harmonic functions $-U$, u , and U satisfy the relation $-U \leq u \leq U$ on Γ . Then, by (vii), $-U \leq u \leq U$ at all points inside D , or $|u| \leq U$ inside D .

Theorem 2.11. (Harnack theorem) *Suppose that $\{u_n(z)\}$ is a monotone increasing sequence of harmonic functions on a region D , which is convergent at a point $z_0 \in D$. Then $\{u_n(z)\}$ is uniformly convergent on closed sets in D .*

Let $z_0 \neq \infty$ be any point inside D , and let K denote the closed disk $\bar{B}(z_0, R)$ such that $K \subset D$. Then, if $r = |z - z_0| \leq R$, the Harnack inequality

$$u_n(z_0) \frac{R-r}{R+r} \leq u_n(z) \leq u_n(z_0) \frac{R+r}{R-r} \quad (2.5.4)$$

holds for any annulus $0 < r < R$ with center at z_0 , provided that $u_n(z)$ are harmonic and nonnegative on the disk $B(z_0, R)$.

Theorem 2.12. (Identity theorem) *If $f(z)$ is analytic on a region D , $z_0 \in D$, and $\{z_k\}_{k=1}^\infty$ is a sequence of distinct points in D such that $z_k \rightarrow z_0$, and $f(z_k) = 0$ for $k = 1, 2, \dots$, then $f(z) \equiv 0$ on D .*

This theorem is useful in establishing certain identities, including the concept of analytic continuation.

Example 2.5. (i) The function $f(z) = \frac{1}{z} = \frac{x}{x^2 + y^2} - i \frac{y}{x^2 + y^2}$ is harmonic everywhere except at the singularity at $x = 0 = y$.

(ii) The function

$$\begin{aligned} f(z) &= \frac{z-1}{z+1} = \frac{(z-1)(\bar{z}+1)}{(z+1)(\bar{z}+1)} = \frac{|z|^2 + z - \bar{z} - 1}{|z+1|^2} \\ &= \frac{x^2 + y^2 - 1}{(x+1)^2 + y^2} = i \frac{2y}{(x+1)^2 + y^2} \end{aligned}$$

is harmonic everywhere except at the singularity at $x = -1, y = 0$.

(iii) The function $f(z) = \log z = \log(r e^{i\theta}) = \log r + i\theta$ has real and imaginary parts as follows: $\Re\{\log z\} = \log r = \log|z| = \frac{1}{2} \log(x^2 + y^2)$, which is harmonic everywhere except at $x = 0 = y$; and $\Im\{\log z\} = \theta = \arg\{z\}$, which is not defined at $x = 0 = y$ and is a multiple-valued harmonic function everywhere else; it is only specified up to integral multiples of 2π . If $z = x > 0$ is real and positive, then $\log z = \log x$ is the real logarithm provided we choose $\arg\{x\} = 0$. However, if $z = x < 0$, then $\log z = \log|x| + (2k+1)\pi i$, $k \in \mathbb{Z}$, is complex for all $\arg\{x\}$. also, $z = 0$ is called the *logarithmic branch point*. ■

2.5.1 Harmonic Conjugate. Write the Cauchy-Riemann equations (2.1.6) as $\frac{\partial v}{\partial x} = -\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial y} = \frac{\partial u}{\partial x}$. These equations can be rewritten in vector form as an equation of $\text{grad } v$, known as the *skew gradient of u* , i.e.,

$$\nabla v = \nabla^\perp u, \quad \text{where} \quad \nabla^\perp = \begin{bmatrix} -u_y \\ u_x \end{bmatrix}. \quad (2.5.5)$$

It is orthogonal to $\text{grad } u$ everywhere and is of same length, i.e.,

$$\nabla u \cdot \nabla^\perp u = 0, \quad \|\nabla u\| = \|\nabla^\perp u\|.$$

This means that the gradient of a harmonic function and that of its harmonic conjugate are mutually orthogonal vector fields having the same Euclidean length:

$$\nabla u \cdot \nabla v = 0, \quad \|\nabla u\| = \|\nabla v\|. \quad (2.5.6)$$

Next, $\nabla^\perp u$ has a potential function v iff the corresponding line integral is independent of path, i.e., for every closed curve $C \subset D$, where D is simply connected, and every harmonic function u ,

$$\begin{aligned} 0 &= \int_C \nabla v \cdot d\mathbf{x} = \int_C \nabla^\perp u \cdot d\mathbf{x} = \int_C \nabla u \cdot \mathbf{n} \, ds \\ &= \iint_D \nabla \cdot \nabla u \, dx \, dy = \iint_D \nabla^2 u \, dx \, dy, \end{aligned}$$

by divergence theorem (Appendix A). This proves the existence of harmonic conjugate functions.

The geometrical significance of Eq (2.5.5) is as follows: The gradient ∇u is normal to the level curves of $u(x, y)$, which are the sets $\{u(x, y) = c\}$ where ∇u assumes a fixed constant value. Also, ∇v is orthogonal to ∇u , which is tangent to the level curves of u . Similarly, ∇v is normal to its level curves. Since their tangent directions ∇u and ∇v are orthogonal, the level curves of u and v form a orthogonal system of plane curves, except at a critical point where $\nabla u = 0$. Then $\nabla u = \nabla^\perp u = 0$. Thus, orthogonality of level curves does not necessarily hold at critical points, although critical points of u are the same as those of v and also the same as critical points of $f(z)$ where $f'(z) = 0$.

Note that the harmonic conjugate does not exist only on the punctured plane $\Omega = \mathbb{C} \setminus \{0\}$ because of the logarithmic potential. In fact, if $u(x, y)$ is a harmonic function on $\Omega_R = \{0 < |z| < R\}$, where $0 < R \leq \infty$, then there exists a constant c such that $\tilde{u}(x, y) = u(x, y) - c \log \sqrt{x^2 + y^2}$ is also harmonic and possesses a single-valued harmonic conjugate $\tilde{v}(x, y)$. Thus, the function $\tilde{f} = \tilde{u} + \tilde{v}$ is analytic on Ω_R and the original function $u(x, y)$ is the real part of the multiple-valued analytic function $f(z) = \tilde{f}(z) + c \log z$. This fact is useful in the analysis of airfoils.

2.5.2 Capacity. Let D^* denote the complement of the region D that includes the point $z = \infty$. Then the region D^* can be covered by regions $D^{*(n)}$ with boundaries $\Gamma^{*(n)}$, $n = 1, 2, \dots$. The Green's function $G_n(z, \infty)$ of the regions $D^{*(n)}$ is a harmonic function in $D^{*(n)}$ except at the point $z = \infty$ which assumes the value zero on $\Gamma^{*(n)}$ and in a neighborhood of ∞ behaves such that $\lim_{z \rightarrow \infty} [G_n(z, \infty) - \log |z|] = \gamma_n$ exists (and is finite). The quantity γ_n is called *Robin's constant* for the regions $D^{*(n)}$. Hence, in a neighborhood of $z = \infty$

$$G_n(z, \infty) = \log |z| + u_n(z) + \gamma_n, \quad (2.5.7)$$

where $u_n(z)$ is a harmonic function in $D^{*(n)}$, including the point $z = \infty$, and $u_n(z) \rightarrow 0$ as $z \rightarrow \infty$. Since $\overline{D^{*(n)}} \subset D^{*(n+1)}$, the function $G_{n+1}(z, \infty) - G_n(z, \infty)$ is harmonic in $D^{*(n)}$, including the point ∞ . In view of the maximum principle (Theorem 2.10), the last statement is true everywhere in the region $D^{*(n)}$, i.e., for all $z \in D^{*(n)}$

$$G_{n+1}(z, \infty) \geq G_n(z, \infty), \quad \gamma_{n+1} \geq \gamma_n. \quad (2.5.8)$$

In view of the Harnack theorem, the sequence of functions $\{u_n + \gamma_n\}$, where $u_n(\infty) = 0$, either converges to ∞ in D^* or converges to a harmonic function $u(z) + \gamma$ such that $u(\infty) = 0$. In the latter case, the quantity γ is called Robin's constant for the region D^* , the quantity $C = e^{-\gamma}$ is called the *capacity* of the region D^* (denoted by $\text{cap}(D^*)$), and the function $G(z, \infty)$ is called Green's function for the region D^* , which assumes nonnegative values everywhere in D^* , but these values need not be zero. For example, at isolated boundary points of D^* the value of $G(z, \infty)$ is positive. In the case when $u_n(z) + \gamma_n \rightarrow \infty$ for $z \in D^*$, we have $\gamma_n \rightarrow \infty$ because $u_n(\infty) = 0$. In this case the capacity of the region D is taken as zero. The following results are useful:

Theorem 2.13. *For the region D^* , containing the point at infinity, to have a Green's function $G(z, \infty)$, it is necessary and sufficient that the capacity of its boundary Γ be positive.*

Theorem 2.14. *The capacity of an arbitrary closed and bounded region D is equal to its transfinite diameter, i.e., $\text{cap}(D) = \text{diam}(D)$.*

2.6 Univalent Functions

If we place a sphere such that the complex plane is tangent to it and the origin coincides with the south pole, then we can transfer all points of $\mathbb{C}$ to the sphere by projection from the north pole. This is called a *stereographic projection* which is a one-to-one map of $\mathbb{C}$ onto the sphere, such that its image is the whole sphere except the north pole which then corresponds to the point at infinity $z = \infty$.

A mapping f of a region D onto a region G is called *analytic* iff it is differentiable. The mapping f is called *conformal* if it is bijective and analytic. The *conformal mapping theorem* states that if a mapping $f : D \mapsto G$ is analytic and $f'(z_0) \neq 0$ for each $z \in D$, then f is conformal. Thus, f rotates tangent vectors to curves through z_0 by a definite angle θ and magnifies (or contracts) them by a factor r . The mapping f is *conformal* if it is analytic with a nonzero derivative. Two important properties are the following:

- (i) If $f : D \mapsto G$ is conformal and bijective (i.e., one-to-one and onto), then $f^{-1} : G \mapsto D$ is also conformal and bijective.
- (ii) If $f : D \mapsto G$ and $g : G \mapsto E$ are conformal, then the composition $f \circ g : D \mapsto E$ is conformal and bijective.

Property (i) is useful in solving boundary value problems (e.g., the Dirichlet problem) for a region D . The method involves finding a map $f : D \mapsto G$ such that G is a simpler region on which the problem can be first solved, and then the result for the original problem is provided by f^{-1} . Since the Dirichlet problem involves harmonic functions (see §1.3), the following result on the composition of a harmonic function with a conformal map is useful: If u is harmonic on a region G and if $f : D \mapsto G$ is conformal, then $u \circ f$ is harmonic on D . In fact, let $z \in D$ and $w = f(z) \in G$. Then there is an analytic function g on the open disk $B(w, \rho) \subset G$ such that $u = \Re\{g\}$. Thus, $u \circ g = \Re\{g \circ f\}$, which is harmonic since $g \circ f$ is analytic. A mapping in which both the magnitude and the sense of the angles between the curves and their images are preserved is said to be a conformal mapping of the first kind, but if the sense of the angles is reversed, then it is called a conformal mapping of the second kind.

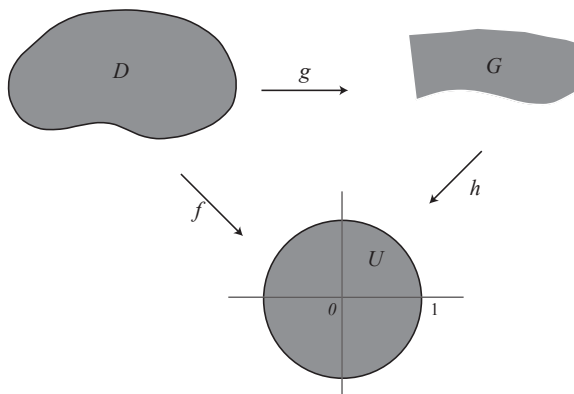


Figure 2.2 Mappings.

Theorem 2.15. (Riemann mapping theorem) *Let $D \subset \mathbb{C}$ be a simply connected region. Then there exists a bijective conformal map $f : D \mapsto U$, where U is the open unit disk. Moreover, the map f is unique provided that $f(z_0) = 0$ and $f'(z_0) > 0$ for $z_0 \in D$.*

A proof of this theorem is available in Appendix C. This theorem implies that if D and G , both contained in $\mathbb{C}$, are any two simply connected regions, then there exists a bijective conformal map $g : D \mapsto G$. If $f : D \mapsto U$ and $h : G \mapsto U$, then $g = h^{-1} \circ f$ is bijective conformal (Figure 2.2). Thus, the two regions D and G are said to be *conformal* if there exists a bijective conformal map between them.

A bijective conformal map is also called a *univalent map*. A function $w = f(z)$ defining a univalent mapping is called a univalent function on a domain D if it is one-to-one and analytic on D . Its inverse image is also a univalent function defined on the image region. In the study of univalent mappings the first question asked is whether a given region can be mapped univalently onto another region. In the case of simply connected regions it is necessary that the two regions have the same connectivity. Once this condition is met, we can inquire about the possibility of univalent conformal mapping of various regions onto a given simply connected region. To determine how a simply connected region is mapped onto a simply connected region, we must know their mappings onto a standard region, such as the unit disk. This enables us to obtain the required mapping first by mapping the given region onto the unit disk and then mapping the unit disk onto the other region. Now the question arises whether an arbitrary simply connected region can be mapped onto the unit disk. It turns out that this is always possible except for two cases, namely, when the region is the entire plane and when it has a single boundary point. Note that the requirement that the boundary orientation be preserved even in the mapping of regions with non-Jordan boundaries is important. Consider, for example, the map $w = \sin z$ which is analytic on the strip $-\pi/4 < x \leq 3\pi/4$ and maps bijectively the two boundaries of this strip onto the two boundaries of the hyperbola $u^2 - v^2 = 1/2$, which make the boundary of a simply connected region (curved strip). But the boundary of this curved strip is not traversed in the positive sense. Other simple examples of functions that are not univalent in the lower or the upper half-plane are $w = z^2$, and $w = \sinh z$, although they are analytic in $\mathbb{C}$ and map bijectively the real axis $\Im\{z\} = 0$ onto the real axis $\Im\{w\} = 0$. Thus, the function $w = f(z)$, analytic on the half-plane $\Im\{z\} > 0$ and continuous on the closed region $\Im\{z\} \geq 0$, grows as $|z| \rightarrow \infty$ at most as fast as Cz^2 , i.e., the ratio $\frac{f(z)}{z^2}$ is bounded for $0 \leq \arg\{z\} \leq \pi$ and a sufficiently large $|z|$.

The Riemann mapping theorem also implies that there exists a unique function $w = F(z)$ that is regular in D , that is normalized at a finite point $z_0 \in D$ by the conditions $F(z_0) = 0$ and $F'(z_0) = 1$, and that maps the region D univalently onto the disk $|w| < 1$. In fact, the function $F(z) = \frac{f(z)}{f(z_0)}$ is such a function, where $f(z)$, with $f(z_0) = 0$ and $f'(z_0) > 0$, is the function mentioned in the Riemann mapping theorem, and the radius of the disk onto which the function $w = F(z)$ maps the region D is $R = \frac{1}{f'(z_0)}$. If there exists another function $w = F_1(z)$, with $F_1(z_0) = 0$ and $F'_1(z_0) = 1$, that maps D onto a disk $|w| < R_1$, then, by the Riemann mapping theorem, we could have $\frac{F_1(z)}{R} = f(z)$, and hence, $\frac{1}{R_1} = f'(z_0)$, i.e., $F_1(z) = \frac{f(z)}{f'(z_0)} = F(z)$, which proves the uniqueness of the mapping function $w = F(z)$. The quantity $R = \frac{1}{f'(z_0)}$ is called the *conformal radius* of the region D at the point $z_0 \in D$.

Since $D^* = \text{Ext}(\Gamma)$ is simply connected, Green's function $G(z, \infty)$ for D^* coincides with $\log |f(z)|$, where the function $w = f(z)$ maps D^* univalently onto $|w| > 1$ such that $f(\infty) = \infty$. Then, Robin's constant γ for the region D^* is equal to $\log |f'(\infty)|$, and $\cap(D^*) = \frac{1}{|f'(\infty)|} = R$, where R is the conformal radius of the region D^* (with respect to ∞), i.e., the number R is such that the region D^* is mapped univalently onto $|w| > R$ by a normalized function $w = F(z)$ with $F(\infty) = \infty$ and $F'(\infty) = 1$. Thus, in view of Theorems 2.13 and 2.14, we have the following theorem.

Theorem 2.16. *The capacity, and hence the transfinite diameter of a bounded simply connected region D , is equal to the conformal radius of the region D^* which is the complement of the region D in $\mathbb{C}_\infty$ and contains the point at infinity.*

2.6.1 Conformality and Uniqueness. Conformality is a local phenomenon; it applies to small neighborhoods D of a point z_0 which is mapped from the z -plane onto the w -plane by the function $w = f(z)$ (see Theorem 2.17 below). Suppose that $f(z)$ is analytic at the point z_0 and that $f'(z_0) \neq 0$. Let $f'(z_0) = m e^{i\alpha}$, and let $z - z_0 = \Delta r_z e^{i\Delta\theta_z}$. Then $f(z) = f(z_0) + f'(z_0)(z - z_0)$, which in the w -plane yields to $w - w_0 = \Delta r_w e^{i\Delta\theta_w}$. As z_0 approaches z , where $z, z_0 \in D$, the point w_0 approaches w , which gives $w - w_0 = f'(z_0)(z - z_0)$, or

$$\Delta r_w e^{i\Delta\theta_w} = f'(z_0) \Delta z e^{i\Delta\theta_z} = m \Delta r_z E^{i(\alpha + \Delta\theta_z)}. \quad (2.6.1)$$

Then

$$\begin{aligned} \Delta r_w &= m \Delta r_z, & m &= \text{mod } f'(z_0), \\ \Delta\theta_w &= \alpha + \Delta\theta_z, & \alpha &= \arg\{f'(z_0)\}. \end{aligned} \quad (2.6.2)$$

Note that α depends only on z_0 , and not on z . Thus, the angle α that appears in $\Delta\theta_w$ as a rotation remains the same at every $z \in D$. Let us consider a curve C_z in terms of the points z_i , $i = 1, 2, \dots, k$, shown in Figure 2.3. If we map this curve C_z from the z -plane onto the w -plane, it will undergo a small rotation by α and a magnification (compression) by a factor shown in Figure 2.3. Hence, this limiting process with infinitesimally small increments of

area elements ΔS_z and ΔS_w leads to

$$\begin{aligned} \lim_{\Delta z \rightarrow 0} \frac{\Delta S_w}{\Delta S_z} &= |f'(z)|^2 = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial u}{\partial y} \right)^2 \\ &= \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = J, \end{aligned} \quad (2.6.3)$$

where J is the Jacobian of the transformation $w = f(z)$. This transformation is one-to-one when $f'(z) \neq 0$ or $j \neq 0$.

This transformation of incremental length and areas can be summarized, for later use, by the following results:

$$\text{Transformation of incremental length: } |dw| = |f'(z)| |dz|; \quad (2.6.4)$$

$$\begin{aligned} \text{Transformation of incremental areas } \Delta S_z \mapsto \Delta S_w : dw d\bar{w} &= f'(z) \overline{f'(z)} dz d\bar{z}, \\ \text{or } du dv &= |f'(z)|^2 dx dy; \end{aligned} \quad (2.6.5)$$

$$\text{Area bounded} = \iint_D |f'(z)|^2 dx dy; \quad (2.6.6)$$

$$\text{Magnification factor : } m = \frac{dw}{dz} = f'(z). \quad (2.6.7)$$

where

$$|f'(z)|^2 = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 = \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2. \quad (2.6.8)$$

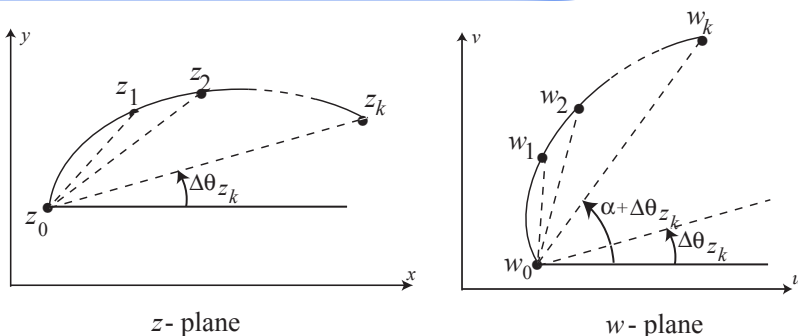


Figure 2.3 Limiting process.

The condition for a well-defined one-to-one mapping is that there must be no critical points, i.e., there must be no point z_0 at which $f'(z_0) = 0$, or $f'(z_0) = \infty$, because otherwise the inverse relationship $\frac{dz}{dw} = \frac{1}{dw/dz}$ as well as $\alpha = \arg\{f'(z_0)\}$ would become indeterminate.

A domain of transformed points in the w -plane is called *schlicht* or *simple* if none of its points originates from more than one point each in the z -plane.

However, the following example shows that the conformal map guaranteed by the Riemann mapping theorem is not unique, in the sense that in certain cases one can get a choice of more than one conformal map.

Example 2.6. In order to construct a mapping that maps the half-disk $C(0, \frac{1}{2}) = \{|z| < 1, \Im\{z\} > 0\}$ onto the entire unit disk $C(0, 1) = \{|z| < 1\}$, the map $w = z^2$ will not work because the image of $B(0, \frac{1}{2})$ omits the positive real axis, thus yielding a disk with a slit $\{|w| < 1, 0 < \arg\{w\} < 2\pi\}$. In order to get the entire unit disk, we may consider the mapping $z = \frac{w-1}{w+1}$ (see Map 3.17) which maps the right half-plane $\{\Re\{w\} > 0\}$ onto the unit disk. It also maps the upper right quadrant $\{0 < \arg\{w\} < \pi/2\}$ onto the half unit disk $C(0, \frac{1}{2})$. Then its inverse $w = \frac{z+1}{z-1}$ will then map the half-disk $C(0, \frac{1}{2})$ onto the upper right quadrant.

However, the mapping (3.3.39), $w = \frac{iz^2 + 1}{iz^2 - 1}$ (see Map 3.44), maps the upper right quadrant onto the unit disk $C(0, 1)$. Since

$$w = \frac{iw^2 + 1}{iw^2 - 1} = \frac{i\left(\frac{z+1}{z-1}\right) + 1}{i\left(\frac{z+1}{z-1}\right) - 1} = \frac{(i+1)(z^2+1) + 2(i-1)z}{(i-1)(z^2+1) + 2(i+1)z} = -i \frac{z^2 + 2iz + 1}{z^2 - 2iz + 1},$$

we find that, omitting the factor $-i$ which merely rotates the disk by $-\pi/2$, the mapping

$$w = \frac{z^2 + 2iz + 1}{z^2 - 2iz + 1} \quad (2.6.9)$$

is another choice for the solution of the problem. ■

Theorem 2.17. (Local inverse theorem) *If $w = f(z)$ is conformal at z_0 , there exists a function f^{-1} such that $z_0 = f^{-1}(w_0)$. Then $w = f(z) = f[f^{-1}(w)]$, where $z = f^{-1}(w)$, and*

$$\frac{df^{-1}(w)}{dw} = \frac{1}{\frac{df(z)}{dz}} = \left(\frac{df(z)}{dz}\right)^{-1}, \quad \text{provided } \frac{df(z)}{dz} \neq 0 \text{ or } f \text{ is conformal.}$$

PROOF. Using $z = x + iy$ and $w = u + iv$, the mapping $w = f(z)$ corresponds to a pair of simultaneous equations $u = u(x, y)$, $v = v(x, y)$ with $du = u_x dx + u_y dy$ and $= v_x dx + v_y dy$. Now, if the inverse mapping f^{-1} exists, then we have $x = x(u, y)$, $y = y(u, v)$ with $dx = x_u du + x_v dv$ and $dy = y_u du + y_v dv$. Hence,

$$dx = x_u (u_x dx + u_y dy) + x_v (v_x dx + v_y dy).$$

Since dx and dy are independent, their coefficients must vanish, i.e.,

$$1 = x_u u_x + x_v v_x, \quad 0 = x_u u_y + x_v v_y, \quad (2.6.10)$$

These equations admit a solution only if

$$\begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} \neq 0.$$

The determinant is known as the *Jacobian* defined by

$$J = \frac{\partial(u, v)}{\partial(x, y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = u_x v_y - v_x u_y.$$

Thus, f^{-1} exists if $J \neq 0$. Since f is analytic, the Cauchy-Riemann equations (2.1.6) give

$$J = u_x^2 + v_x^2 = \left| \frac{\partial f}{\partial x} \right|^2 = \left| \frac{df}{dz} \right|^2 \neq 0.$$

Returning to the solution of (2.6.10), we have

$$x_u = \frac{1}{J} \begin{vmatrix} 1 & v_x \\ 0 & v_y \end{vmatrix} = \frac{v_y}{J}, \quad x_v = \frac{1}{J} \begin{vmatrix} u_x & 1 \\ u_y & 0 \end{vmatrix} = -\frac{u_y}{J}.$$

Similarly

$$y_u = \frac{v_x}{J}, \quad y_v = \frac{u_x}{J}.$$

Then since f is analytic, we have $x_u = \frac{u_x}{J} = y_v$ and $x_v = \frac{v_x}{J} = -y_u$. Hence f^{-1} is analytic. ■

Note that this theorem proves only the existence of a local inverse. Nothing global is mentioned. Thus, if f is a many-to-one mapping, then f^{-1} exists only in a neighborhood of one of the branches of the multiple valued function f^{-1} .

2.6.2 Conformal and Isogonal Mappings. Let C_1 and C_2 be two curves in the z -plane that intersect at the point z_0 , and let G_1 and G_2 be the images of C_1 and C_2 , respectively, under the mapping $w = f(z)$. Then G_1 and G_2 intersect at w_0 which is the image of z_0 . Assume that $f(z)$ is analytic at all points of C_1 and C_2 and that z_0 is not a critical point. Let λ_k , $k = 1, 2$, be the parameters that define C_k and G_k . Then the tangents to the two curves at the point of intersection are given by

$$\tau_1(w_0) = f'(z_0) \frac{dz}{d\lambda_1} \Big|_{z_0} = f'(z_0) t_1(z_0), \quad (2.6.11)$$

$$\tau_2(w_0) = f'(z_0) \frac{dz}{d\lambda_2} \Big|_{z_0} = f'(z_0) t_2(z_0). \quad (2.6.12)$$

Thus, $|\tau_k(w_0)| = |f'(z_0)| |t_k(z_0)|$ for $i = 1, 2$ where the angles $\psi_k(w_0) = \theta_k(z_0) + \phi(z_0)$, and thus,

$$\Delta\psi(w_0) = \psi_1(w_0) - \psi_2(w_0) = \theta_1(z_0) - \theta_2(z_0) = \Delta\theta(z_0). \quad (2.6.13)$$

This result states (see Figure 2.4) that the magnitude and the sign of the angle subtended by the tangents to the two curves in the z -plane is the same as the magnitude and sign of the angle subtended by the tangents to the images of these two curves in the w -plane. **A mapping under which the magnitude and sign of the angle between the tangents to two curves remains unchanged is called a conformal mapping.** The sign of the difference between the angles of Eq (2.6.13) is called the *sense* of the angle subtended by the tangents. A transformation for which

$$|\psi_1(w_0) - \psi_2(w_0)| = |\theta_1(z_0) - \theta_2(z_0)|, \quad (2.6.14)$$

but for which the **sense of the angle may not be preserved is called an isogonal mapping.** Hence, a conformal mapping is isogonal but the converse may not be true. An application

of the isogonal transformation is given in §21.6.

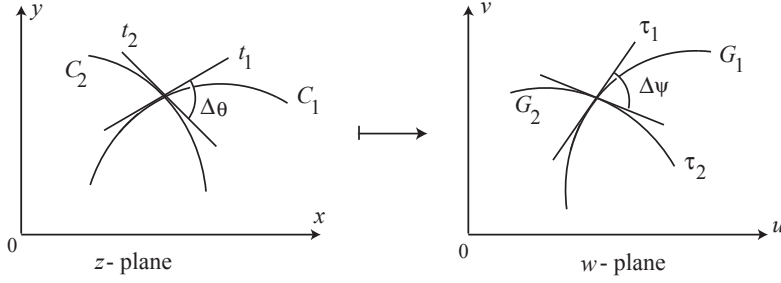


Figure 2.4 Conformal and isogonal mappings.

2.6.3 Conformal Mapping of an Area Element. A differential line element in a region D in the z -plane from the point (x, y) to a point $(x + dx, y + dy)$ is defined by $dz = dx + i dy$. Similarly the differential line element in a region G in the w -plane from (u, v) to $(u + du, v + dv)$ is defined by $dw = du + i dv$. Consider a rectangular area element A with sides $dz_1 = dx$ and $dz_2 = i dy$. Then the images of dz_1 and dz_2 are differential curves in the w -plane defined by $dw_k = du_k + i dv_k$, $k = 1, 2$. The area of a rectangle with sides z_1 and z_2 is given by $A = |\Im\{z_1^* z_2\}|$, and the area element in the z -plane is given by $dA_z = dx dy$ and that in the w -plane by $dA_w = du_1 dv_2 - du_2 dv_1$, where dz_1 is along a line defined by $y = \text{const}$, and dz_2 is along a line defined by $x = \text{const}$, the images of these two lines in the w -plane are

$$dw_1 = \left. \frac{df}{dz} \right|_{y=\text{const}} \cdot dz_1 = \frac{\partial f}{\partial x} dx, \quad dw_2 = \left. \frac{df}{dz} \right|_{x=\text{const}} \cdot dz_2 = \frac{1}{i} \frac{\partial f}{\partial y} i dy = \frac{\partial f}{\partial y} dy,$$

where $w = f(z) = u(x, y) + i v(x, y)$. Since $\frac{\partial f}{\partial x} = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x}$ and $\frac{\partial f}{\partial y} = \frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y}$, we have

$$dw_1 = \left(\frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right) dx, \quad dw_2 = \left(\frac{\partial u}{\partial y} + i \frac{\partial v}{\partial y} \right) dy.$$

Hence, the differential area element in the w plane is given by

$$dA_w = |\Im\{dw_1^* dw_2\}| = \left| \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right| dx dy = J(u, v; x, y) A_w, \quad (2.6.15)$$

where

$$J(u, v; x, y) \equiv J(w; z) = \left| \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x} \right| = \left| \frac{\partial u}{\partial x} \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \frac{\partial v}{\partial y} \right| = \frac{\partial(u, v)}{\partial(x, y)}$$

is the *Jacobian* of the transformation, which in view of the Cauchy-Riemann equations, reduces to

$$J(w; z) = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 = \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2. \quad (2.6.16)$$

Note that $J(w; z)$ is positive for all x and y . Since f is analytic in D , we get

$$J(w; z) = |f'(z)|^2 = \left| \frac{dw}{dz} \right|^2. \quad (2.6.17)$$

2.6.4 Analytic Continuation. Analytic continuation is an important aspect of conformal mapping. Let a function $f_1(z)$ be defined on a domain D_1 and another function $f_2(z)$ on D_2 . If D_1 and D_2 have a common subregion (i.e., if $D_1 \cap D_2 \neq \emptyset$) in which $f_1(z) = f_2(z)$, then $f_2(z)$ is said to be the *analytic continuation* of $f_1(z)$. This concept is closely related to Taylor’s series. Suppose a function $f(z)$ has a Taylor’s series expansion $f_1(z)$ at $z = z_1$. If there exists a singularity at z_s , then the circle of convergence or a domain D_1 within which $f_1(z)$ is analytic is given by the radius $|z_s - z_1|$ centered at z_1 . Within D_1 there exists a point z_2 for which the series expansion of $f(z)$ is $f_2(z)$ with a circle of convergence or domain D_2 which does not include the singular point z_s . If D_2 is close to the border of D_1 and the circle of convergence of $f_2(z)$ extends beyond D_1 , the $f_2(z)$ is the analytic continuation of $f_1(z)$ into D_2 , while $f_1(z) = f_2(z)$ in $D_1 \cap D_2$.

The analytic continuation is unique. In fact, suppose that g_1 and g_2 are analytic continuations of f from D_1 into D_2 . Then $g_1(z) \equiv g_2(z)$ on $D_1 \cap D_2$, and thus, by identity theorem (§1.3), $g_1 = g_2$ throughout D_2 . Thus, in practice we will determine a single function $F(z)$ that is analytic on $D_1 \cup D_2$ and is given by $F(z) = f(z)$ on D_1 , and $F(z) = g(z)$ on D_2 . For example, let $f(z) = \int_0^\infty e^{-zt} dt$. Notice that $f(z)$ is analytic on $\Re\{z\} > 0$ because after evaluating the improper integral we find that $f(z) = 1/z$ is analytic on $\Re\{z\} > 0$. So we take $g(z) = 1/z$. Since $g(z)$ is analytic on $\mathbb{C} \setminus \{0\}$, $g(z)$ becomes the analytic continuation of $f(z)$ from the right half-plane into the whole plane indented at the origin. Similarly, the Laplace transform of $\cos at$ is analytic for $\Re\{z\} > 0$; but the function $\frac{z}{z^2 + a^2}$ is its analytic continuation from the right half-plane into the whole plane indented at the points $\pm ia$.

2.6.5 Chain Property. Let $D_0, \dots, D_n$ be regions in $\mathbb{C}$, and let $f_k : D_{k-1} \mapsto D_k$ denote conformal mappings for $k = 1, \dots, n$. Then the mapping $g = f_n \circ \dots \circ f_1$, defined by

$$g(z) = f_n(f_{n-1}(\dots f_2(f_1(z))\dots)) \quad (2.6.18)$$

is a conformal mapping of D_0 onto D_n . The mapping g is said to be composed of a *chain* of mappings $f_1, \dots, f_n$ and is represented by the scheme

$$D_0 \xrightarrow{f_1} D_1 \xrightarrow{f_2} D_2 \xrightarrow{f_3} \dots \xrightarrow{f_{n-1}} D_{n-1} \xrightarrow{f_n} D_n. \quad (2.6.19)$$

Thus, the set of regions on $\mathbb{C}$ can be divided into *mapping classes* such that two regions can be mapped conformally onto each other iff they belong to the same mapping class. Figure 2.5, which represents an example of this chain property, is a composite of the following six maps, in that order:

Map 2.1. $w = -iz$ (rotation by $-\pi/2$), which maps the upper-half of the z -plane with a vertical slit from $z = 0$ to $z = i$ onto the right half-plane with a slit from $w = 0$ to $w = 1$.

Map 2.2. $w = z^2$, which maps the right half-plane with a slit from $z = 0$ to $z = 1$ onto the entire w -plane with a slit from $-\infty < u < 1$.

Map 2.3. $w = z^2 - 1$ (translation), which maps the entire z -plane with a slit from $-\infty < x < 1$ onto the entire w -plane with the slit $-\infty < u < 0$ (negative real axis).

Map 2.4. $w = \sqrt{z}$ (see Map 4.26), which maps the entire z -plane with a slit $-\infty < x < 0$ (negative real axis) onto the right-half w -plane.

Map 2.5. $w = iz$ (rotation by $\pi/2$), which maps the right-half z -plane onto the upper half-plane $\Im\{w\} > 0$.

Map 2.6. $w = \frac{1-z}{1+z}$ (see Map 3.21), which maps the right-half z -plane onto the unit disk $|w| < 1$.

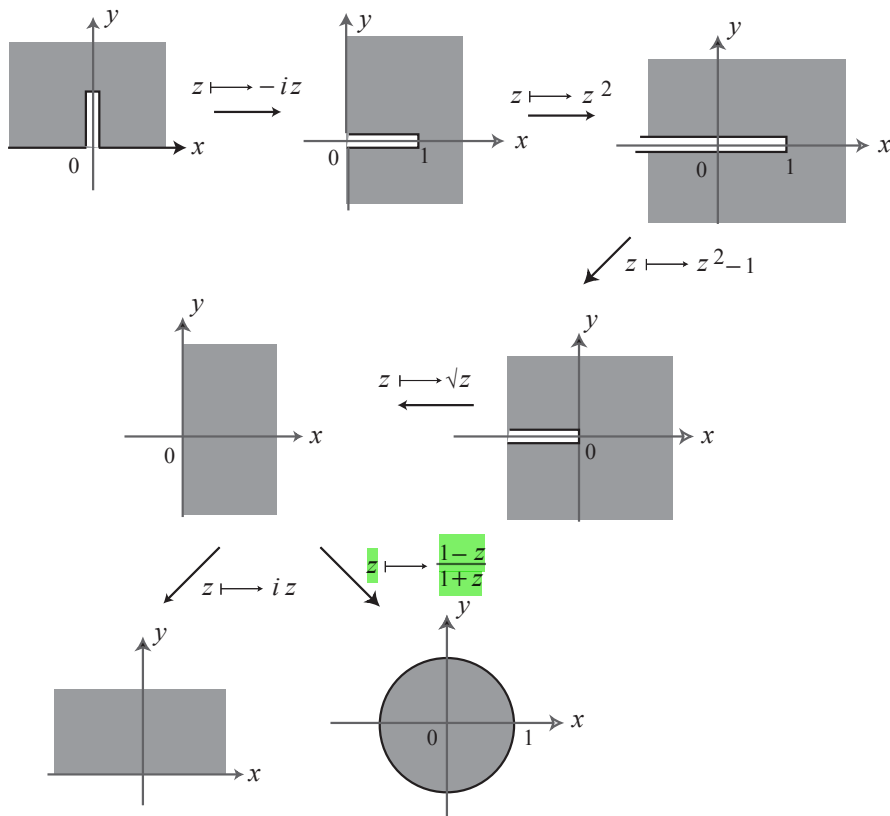


Figure 2.5 Chain property of conformal mappings.

The practical applications of conformal mappings are related to the problem of constructing a function which maps a given region onto a given region. Often we find an explicit expression for the mapping function and determine it by applying the chain property.

Another example of the chain property is that the chain of mappings $f_1 : z \mapsto z - 1$, $f_2 : z \mapsto 2z$, $f_3 : z \mapsto z + 3i/2$ maps the circle $|z - 1| = 1$ onto the circle $|w - 3i/2| = 2$, being composite of the following maps, in that order:

Map 2.7. $w(z) = z - 1$ (translation); **Map 2.8** $w(z) = 2z$ (magnification); and **Map 2.9.** $w(z) = z + \frac{3i}{2}$ (translation).

2.6.6 Schwarz Reflection Principle. If $f(z)$ is analytic on a region containing a segment of the real axis and is real-valued on this segment, then $\overline{f(z)} = f(\bar{z})$. This leads to the concept of analytic continuation across the real axis when the function is known only on one side. Let I denote a segment $a < x < b$, and let Γ be a Jordan arc joining a and b and lying in $\Re\{z\} > 0$. Then $C = I \cup \Gamma$ is a contour that encloses a region D lying entirely in $\Re\{z\} > 0$. If we reflect this region into the real axis, we get a region D' that lies in the lower half-plane and is bounded by I and Γ' which is the reflection of Γ into the real axis.

Then the region $I \cup D'$ is symmetric about the real axis.

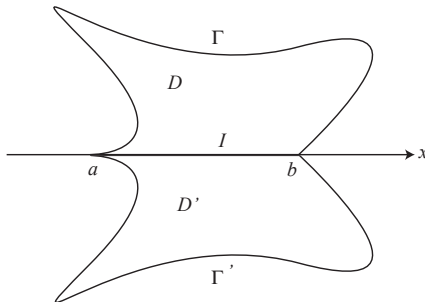


Figure 2.6 Schwarz reflection principle.

If $f(z)$ is continuous on $D \cup I$, analytic on D , and real-valued on I , then the function $F(z)$ defined by

$$F(z) = \begin{cases} f(z), & \text{if } z \in D \cup I, \\ \overline{f(\bar{z})}, & \text{if } z \in D' \end{cases}$$

is analytic on $D \cup I \cup D'$ (see Figure 2.6). If f is analytic on one side of a curve Γ and cannot be continued analytically across Γ , then Γ is called a *natural boundary* for f . For example, the unit circle $|z| = 1$ is a natural boundary for the function $f(z) = \sum_{n=0}^{\infty} z^{2^n}$.

Let S^+ (S^-) denote the upper (lower) half-plane $\Re\{z\} > 0$ ($\Re\{z\} < 0$), respectively, or vice-versa, with L as their common boundary (i.e., the real axis, Figure 2.7). Let $f(z)$ be a function defined at $z \in S^+$, and let it be connected with the function $f^*(z)$ defined in S^- by the relation

$$f^*(z) = \overline{f(\bar{z})}, \quad (2.6.20)$$

i.e., $f(z)$ and $f^*(z)$ take conjugate complex values at points symmetric with respect to the real axis, since the points z and $\bar{z}$ are reflections of each other in L . We can rewrite the relation (2.6.20) as $f^*(z) = \bar{f}(\bar{z})$, i.e., if $f(z) = u(x, y) + iv(x, y)$, then $\bar{f}(\bar{z}) = u(x, -y) - iv(x, -y)$. If $f(z)$ is regular (or meromorphic) in S^+ , then $f^*(z) = \bar{f}(\bar{z})$ is regular (or meromorphic) in S^- , and the relation (2.6.20) is symmetric as regards f and f^* , i.e.,

$$f(z) = \overline{f^*(\bar{z})}, \quad (f^*(z))^* = f(z).$$

If $f(z)$ is a rational function

$$f(z) = \frac{a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0}{b_m z^m + b_{m-1} z^{m-1} + \cdots + b_0}, \quad (2.6.21)$$

then $f^*(z) = \bar{f}(\bar{z})$ is obtained by simply replacing the coefficients by their conjugate complex values. Let t be a real number, and assume that $f(z)$ takes a definite limit value $f^+(t)$ as $z \rightarrow t$ from S^+ . Then $f^{*-}(t)$ exists and

$$f^{*-}(t) = \bar{f}^-(t) = \overline{f^+(t)}, \quad (2.6.22)$$

because $\bar{z} \rightarrow t$ from S^- as $z \rightarrow t$ from S^+ , and hence, $f^*(z) = \overline{f(\bar{z})} \rightarrow \overline{f^+(t)}$.

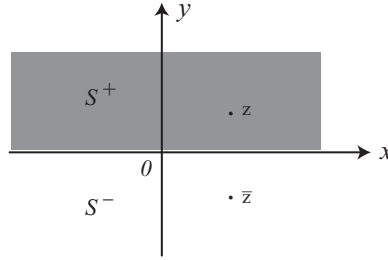


Figure 2.7 Reflections in the real axis.

We will assume that $f(z)$ is regular on S^+ , except possibly at infinity, and continuous on L from the left. Define a sectionally regular function $F(z)$ by

$$F(z) = \begin{cases} f(z) & \text{for } z \in S^+, \\ f^*(z) & \text{for } z \in S^-. \end{cases} \quad (2.6.23)$$

Then, in view of (2.6.22)

$$F^-(t) = \overline{F^+(t)}, \quad F^+(t) = \overline{F^-(t)}. \quad (2.6.24)$$

These relations are useful when transforming the boundary conditions in any boundary problem containing $f^+(t)$ and $\overline{f^+(t)}$, or $f^-(t)$ and $\overline{f^-(t)}$ into those involving $F^+(t)$ and $F^-(t)$.

In view of the Schwarz reflection principle, another property is that of extending $f(z)$. If $\Im\{f^*(t)\} = 0$ in any interval I of the real axis, then the function $f^*(z)$ is the analytic continuation of $f(z)$ through the interval I because $f^{*-}(t) = f^+(t)$ on this interval.

2.6.7 Conformal Equivalence. A *conformal equivalence* between two regions D and G in the complex plane is a one-to-one analytic function f with $f(D) = G$. Thus, $f'(z) \neq 0$ for all z in D . Conversely, if $f : D \mapsto \mathbb{C}$ is analytic such that f' never vanishes, then f is not necessarily a conformal equivalence. As an example, consider $f(z) = e^z$. However, if $f'(z) \neq 0$ on D , then f is locally one-to-one and conformal. The Riemann mapping theorem (Theorem 2.15) establishes the conformal equivalence of two regions. If f is a conformal equivalence between the open sets D and G , then

$$\text{Area}(G) = \iint_D |f'|^2 dz. \quad (2.6.25)$$

Note that if f is an analytic function, then the Jacobian of f , regarded as a mapping from $\mathbb{R}^2$ into $\mathbb{R}^2$, is $|f'|^2$. Hence, if G is a simply connected region, $g : D \mapsto G$ is a Riemann map, and $g(z) = \sum_n a_n z^n$ in D , then from (2.6.25)

$$\text{Area}(G) = \iint_D |g'|^2 dz = \pi \sum_n n |a_n|^2. \quad (2.6.26)$$

In fact, since $g'(z) = \sum_n na_n z^{n-1}$, then, for $r < 1$,

$$\begin{aligned} |g'(re^{i\theta})|^2 &= \left(\sum_n na_n r^{n-1} e^{i(n-1)\theta} \right) \overline{\left(\sum_m ma_m r^{m-1} e^{i(m-1)\theta} \right)} \\ &= \sum_{n,m} nma_n \bar{a}_m r^{n+m-2} e^{i(n-m)\theta}, \end{aligned}$$

which converges uniformly in θ . Since $\int_0^{2\pi} e^{i(n-m)\theta} d\theta = 0$ for $n \neq m$,

$$\begin{aligned} \iint_D |g'|^2 &= \sum_n n^2 |a_n|^2 2\pi \int_0^1 r^{2n-1} dr \\ &= 2\pi \sum_n n^2 |a_n|^2 \frac{1}{2n} \\ &= \pi \sum_n n |a_n|^2. \end{aligned}$$

This relation is useful in solving the minimum area problem in conformal mapping.

If a function $w = f(z)$ maps conformally a region D onto another region G , and if z_0 is an isolated boundary point of D , then z_0 is a removable singularity or a simple pole of $f(z)$. That is the reason to assume the regions to be without isolated boundary points (Goluzin [1969:205]; Wen [1992:95]).

2.6.8 Riemann Sphere. Riemann, using the stereographic projection from the North Pole, assumed this Pole on the sphere to be mapped onto the point at infinity. If a is a pole of f , then, as $z \rightarrow a$, the function $f(z)$ composed with the inverse of stereographic projection tend to the North Pole, so that a meromorphic function in a domain $D \in \mathbb{C}$ is regarded as a continuous image of D onto the sphere. The map at the North Pole is continuous and behaves just like at any other point on the sphere. Thus, at a simple pole where $1/(z-a)$ occurs, the mapping onto the sphere is conformal at a . If higher powers of $1/(z-a)$ occur, the same is true as at any point a where f is analytic but $f' = 0$. The unit sphere in this sense as the extended complex plane via stereographic projection is called the *Riemann sphere*. If a is a pole, we write $f(a) = \infty$. Thus, this mapping technique defined the true mapping significance of the point at infinity.

Theorem 2.18. *Let $f(z)$ be a meromorphic function defined in the entire plane. If $f(z)$ maps the entire plane onto the Riemann sphere, the image of f covers the entire sphere with at most two exceptions. Also, if f is not a rational function, then the same is true for $f(z)$ with $|z| > M$ for $M > 0$.*

Let $f(z)$ be an analytic or meromorphic function in a domain $D \in \mathbb{C}$. A value c of $f(z)$ is said to be *totally ramified*, or *branched*, if, whenever $f(b) = c$, we have $f'(b) = 0$. This means that if f is a mapping from the z -plane onto the w -plane, then f does not define a locally one-to-one conformal map at those points that map onto c . In this case f behaves like the function z^n , $n > 1$ in the neighborhood of the origin, and the image is ramified or branched in a neighborhood of c .

Theorem 2.19. (Nevanlinna [1925]) *An entire function in the z -plane can have at most*

meromorphic = holomorphic except "allowed" singularities(poles).

two totally ramified values, whereas a meromorphic function in the z -plane can have at most four totally ramified values.

Examples that both 2 and 4 numbers are sharp: (i) Complex sine and cosine functions are entire functions defined by $2 \cos z = e^{iz} + e^{-iz}$, $2i \sin z = e^{iz} - e^{-iz}$; thus, $\sin^2 z + \cos^2 z = 1$ and $\cos z = \frac{d \sin z}{dz}$. Thus, $\sin z = \pm 1$ iff $\cos z = 0$, which shows that the two values ± 1 are totally ramified for $f(z) = \sin z$. Similarly, as Nevanlinna [1925] has shown, Weierstrass's elliptic function $\wp(z)$ has four totally ramified values.

Theorem 2.20. (Bloch [1925]) *Let $w = f(z)$ be analytic in the unit disk and be normalized so that $f'(0) = 0$. Then there exists a constant $B > 0$ with the following property: For every $r < B$ there exists a disk of radius r in the w -plane that is the one-to-one conformal image under f of a domain inside the unit disk in the z -plane.*

The largest value of B is known as *Bloch's constant*; its precise value is unknown.

Corollary 2.1. *Let an arbitrary nonconstant entire function $w = f(z)$ define the mapping of the z -plane onto the w -plane such that for every $R > 0$ there is a disk of radius R in the w -plane that is the one-to-one image under f of some domain in the z -plane.*

2.6.9 Bieberbach Conjecture. It was 99 years ago that Ludwig Bieberbach [1916] mentioned this conjecture in a footnote. This conjecture deals with the bound on the coefficients of the class $\mathcal{S}$ of univalent (schlicht) functions $f : E \mapsto f(E)$, where E is the unit disk, with the series expansion $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$, and normalized by $f(0) = 0$ and $f'(0) = 1$. This normalization is a scaling that also implies that the boundary of $f(E)$ cannot get close to the origin. The Bieberbach conjecture states that $|a_n| \leq n$ for all $n = 2, 3, \dots$.

The historical development shows that there was a sustained and broad interest in the Bieberbach conjecture and its impact on creative thinking of mathematicians of the last century. This single phenomenon has been responsible for development of certain beautiful aspects of complex analysis, especially in the geometric-function theory of univalent functions. Although de Branges theorem is deep and meritorious, the work of Fitzgerald [1985], Fitzgerald and Pommerenke [1985], Pommerenke [1985; 1992a, b], and all previous researchers provided significant progress.

Let $\mathcal{S}$ denote the family of univalent functions analytic on the unit disk $|z| < 1$, which are of the form $f(z) = z + \sum_{n=2}^{\infty} a_n z^n$. The coefficients a_n satisfy the inequality $|a_n| \leq n$ for $n = 2, 3, \dots$, and this bound, known as de Branges estimate (see de Branges [1985]), is sharp for the Koebe function $f(z) = \frac{z}{(1-z)^2}$. (Wen [1992:54], Kythe [2016]). This result is true only in $\mathbb{C}$, and there is no such conjecture for $\mathbb{C}^n$, $n \geq 2$.

2.6.10 Mercator's Projection. In 1569 Mercator chose to abandon size in favor of shape while constructing his famous map. He used the following definitions for a particular class of maps known in cartography as *cylindrical projections*.

The *equator* is the great circle equidistant from North and South Poles.

The *meridians* are the circular arcs joining the North and South Poles. They are perpendicular to the equator, and are the curves one travels when traveling due north or south from any point.

The *parallels of latitude*, or *parallels* in short, are the circles to the meridians. They are also the circles at fixed distance from the North or South Pole, and they are the curves one traverses when traveling due east or west from any point other than the two Poles.

A *cylindrical projection*² is a map constructed as follows: The equator is represented by a horizontal line segment drawn in the center. The length of the segment determines the *scale* of the map along the equator. In other words, if L is the length of the Earth's equator and w the width of the map (which is the length of the horizontal line-segment representing the equator), then all distances along the equator are represented by a fixed factor w/L . The meridians are represented by vertical lines of finite or infinite length,³ and the parallels are represented by horizontal line segments of the same fixed length w as the equator. Hence, all cylindrical projections have the following two properties which illustrate the above Euler's theorem (Euler [1775]):

(i) The map is in the form of a rectangle or infinite vertical strip representing all the Earth except the Poles, with two vertical sides corresponding to a single meridian, such that every other meridian corresponds to a unique vertical line.

(ii) The quarter circle along a meridian from the equator to the North or South Pole is divided into 90 degrees, and the latitude of any point is the number of degrees along the meridian north or south of the equator. Thus, for example, at latitude ϕ (radian) north or south, the parallel is a circle of radius $R \cos \phi$, where R is the radius of the Earth so that the length of the equator is $L = 2\pi R$. Thus, at latitude ϕ , the parallel is mapped with fixed scale $\frac{w}{2\pi R \cos \phi} = \frac{w}{L \cos \phi} = s \sec \phi$, where $s = w/L$ is the scale of the map along the equator (see Figure 2.8).

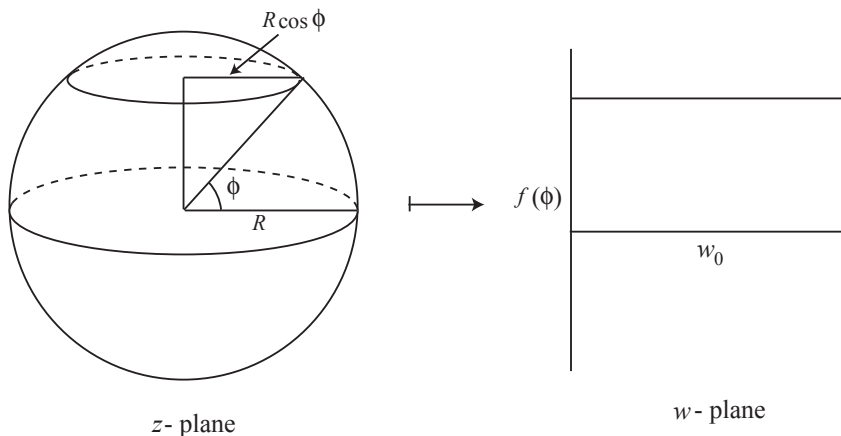


Figure 2.8 Cylindrical projection.

The mathematically true cylindrical projection is described as follows: Let S denote the globe that is a sphere representing the surface of the Earth, and let C denote a cylinder tangent to S along the equator. Now, project S onto C along rays from the center of S (Figure 2.9). Then each meridian on the sphere will map onto a vertical line, and each

² The word ‘projection’ used in map-making means any systematic representation in a broad sense; it should not be confused with the narrow mathematical definition, or the colloquial word ‘projection’ or ‘projection’ using a slide projector.

³ The infinite version is a purely theoretical map; the actual finite map is cut off to represent the portion of the Earth between two fixed parallels.

parallel onto a circle on the cylinder parallel to the equator. Next, cut the cylinder along a vertical line and unroll it onto a vertical strip of the plane, thus yielding the so-called true cylindrical projection or central cylindrical projection.

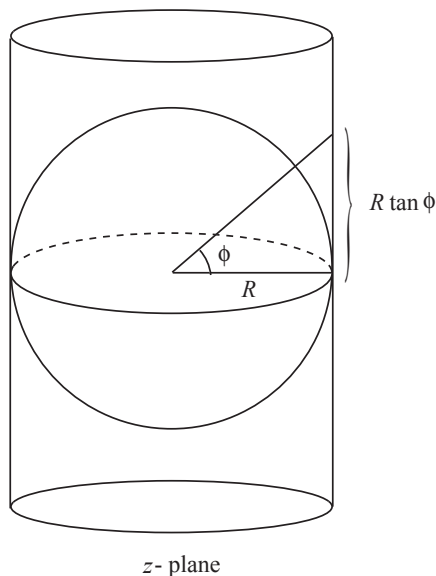


Figure 2.9 Central cylindrical projection.

Much later, Euler [1775] proved the following theorem:

Theorem 2.21. *It is impossible to make an exact scale map of any part of a spherical surface.*

All maps in which the North is the vertical direction on the map use a fixed scale along East-West line; they are the result of the cylindrical projection and cannot have a fixed scale for the entire map. This is because the so-called scale is the factor $\sec \phi$ in the expression $s \sec \phi$ for the scale, which will remain approximately the same over a small portion of the earth's surface and will make very negligible difference. Thus, each meridian on the sphere is mapped onto a vertical line, and each parallel of latitude onto a circle on the cylinder parallel to the equator. Once we cut the cylinder along a suitable vertical line, the resulting flat map becomes a practical central cylindrical projection (Figure 2.9).

If $x = au$ and $y = af(v)$, where $z = x + iy$ and $w = u + iv$, then the meridians and parallels of the earth will be mapped onto the coordinate lines in the (x, y) -plane. The equator will be mapped onto the x -axis if $f(0) = 0$. To make a rhumb line on the sphere, which cuts all the meridians at the same angle α map onto a straight line in the z -plane cutting the lines $x = \text{const}$ at the same angle α , we must have $\cot \alpha = \frac{dv}{\cos v du} = \frac{dy}{dx} =$

$\frac{f'(v) dv}{du}$, which yields $f(v) = \int_0^v \sec v dv = \log \tan \left(\frac{v}{2} + \frac{\pi}{4} \right)$.

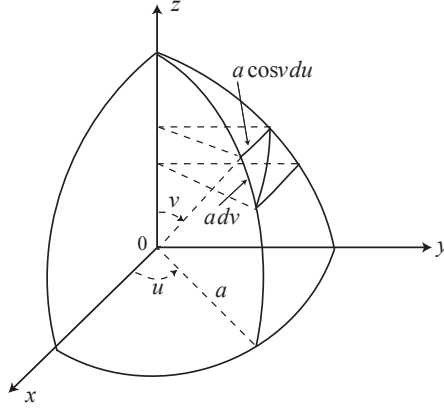


Figure 2.10 Mercator's projection.

The resulting mapping is called *Mercator's projection*, given by

$$z = a \left[u + \log \tan \left(\frac{v}{2} + \frac{\pi}{4} \right) \right]. \quad (2.6.27)$$

This mapping is conformal in the sense that angles are preserved and lengths are multiplied by a factor depending on the latitude. Mercator's projection can also be written as $z = a [u + \text{gd}^{-1}(v)]$, where $\text{gd}(u)$ denotes the Gudermannian of u (Franklin [1944:140-141]). For definition and other results on Gudermannian, see Appendix D. For interested readers the original work of Mercator is available in Mercator [1569], Lambert [1772], and Osserman [2004].

2.7 Taylor Series Approximations

Let Γ denote the boundary of a domain D in the z -plane, and let $C \equiv B(0, 1)$ be the unit circle $|w| = 1$ (see Figure 2.11). To determine a mapping function $z(w)$ that maps the boundary C onto the boundary Γ , we will use the Cauchy integration formula (2.4.2) (with $I(\gamma, z_j) = 1$) to translate any interior point z_j of Γ to the w -plane. Thus, we have

$$w(z_j) = \frac{1}{2\pi i} \int_{\Gamma} \frac{w(z)}{z - z_j} dz, \quad (2.7.1)$$

where $z = r e^{i\theta}$, so that Eq (2.7.1) reduces to

$$w(z_j) = \frac{1}{2\pi} \int_{\Gamma} \frac{w(r e^{i\theta})}{r e^{i\theta} - w_j} r e^{i\theta} d\theta. \quad (2.7.2)$$

Thus, using the Cauchy integral formula to expand $w(z)$ in a Taylor series about any z_j , the n th derivative with respect to z_j is given by

$$w^{(n)}(z_j) = \frac{n!}{2\pi i} \int_{\Gamma} \frac{w(z)}{(z - z_j)^{n+1}} dz. \quad (2.7.3)$$

This will give us the Taylor series expansion of $w(z)$ about a point $z = a \in \Gamma$ as

$$w(z) = w(a) + (z - a)w'(a) + \frac{(z - a)^2}{2!}w''(a) + \cdots, \quad (2.7.4)$$

or about $w = 0$ as

$$w(z) = c_0 + c_1 z + c_2 z^2 + \cdots, \quad (2.7.5)$$

where $c_j = zw^{(j)}(0)$, $j = 0, 1, 2, \dots$.

The inverse function $z = f(w)$ can be approximated as a power series (or a Taylor series) using the expansion (2.7.4) about the point $w = 0$ by reversing the role of w and z . Thus, to map the boundary Γ and its *interior* $\text{Int}(\Gamma)$, which must include the point $z = 0$, onto the unit disk $C \cup \text{Int}(C)$, we can write the approximate series as

$$z \equiv p_n(w) = a_0 + a_1 w + a_2 w^2 + \cdots + a_n w^n. \quad (2.7.6)$$

Similarly, the mapping of the *exterior* region $\text{Ext}(\Gamma)$ that does not include the point $z = 0$ can be approximated by the approximate power series as

$$z \equiv q_m(w) = a_0 + a_{-1} w^{-1} + a_{-2} w^{-2} + \cdots + a_{-m} w^{-m}. \quad (2.7.7)$$

Combining these two power series, we obtain the Laurent series approximation as

$$z = \sum_{j=-m}^m a_j w^j. \quad (2.7.8)$$

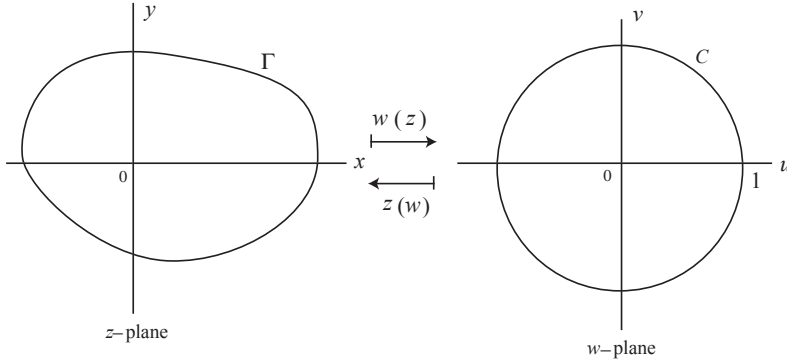


Figure 2.11 Boundary Γ in z -plane and the boundary C in the w -plane.

2.7.1 Interior of the Unit Circle. Consider the inverse transformation $z = z(w)$, where $w = e^{i\Theta}$. Then in view of (2.7.3)

$$z = x + iy = c_0 + c_1 w + c_2 w^2 + \cdots = \sum_{j=0}^{\infty} (x_j + iy_j) = \sum_{j=0}^{\infty} c_j e^{ij\Theta}. \quad (2.7.9)$$

Separating the real and imaginary parts in Eq (2.7.9) we have for the j th term

$$\begin{aligned} c_j &= a_j + ib_j; \\ x_j &= a_j \cos j\Theta - b_j \sin j\Theta; \\ y_j &= b_j \cos j\Theta + a_j \sin j\Theta. \end{aligned} \quad (2.7.10)$$

Map 2.10. The function $w = f(z)$ that maps the unit disk U onto itself such that the point $z_0 \in U$ goes into the origin of the w -plane is given by

$$w = \frac{z - z_0}{1 - \bar{z}_0 z}. \quad (2.7.11)$$

Accordingly, both $f(z)$ and the associated Bergman kernel function $K(z, z_0)$ have a pole at $z_0 = 1/\bar{z}_0$, and the kernel $K(z, z_0)$ can be represented by the polynomial series

$$K(z, z_0) = \frac{1}{\pi} \sum_{k=1}^{\infty} k (z \bar{z}_0)^{k-1}, \quad (2.7.12)$$

which converges rapidly when $|z_0|$ is small; however, the convergence becomes considerably slower the faster $|z_0| \rightarrow 1$, i.e., the closer the pole $1/\bar{z}_0$ gets to the boundary of U . For Bergman kernel, see §8.4.1.

REFERENCES USED: Ahlfors [1966], Betz [1964], Bieberbach [1916], Blaskett and Schwerdtfeger [1945-1946], Boas [1987], Carrier, Krook and Pearson [1966], de Branges [1985], Fitzgerald [1985], Franklin [1944], Gaier [1964], Goluzin [1969], Jeffrey [1992], Kythe [1998; 2016], Lawrentjew and Schabat [1967], Marsden and Hoffman [1987], Muskhelishvili [1953/1992], Nehari [1952], Osserman [2004], Schinzinger and Laura [2003], Wen [1992].

3

Linear and Bilinear Transformations

The central problem in the theory of conformal mapping is to determine a function f which maps a given region $D \subset \mathbb{C}$ conformally onto another region $G \subset \mathbb{C}$. The function f does not always exist, and it is not always uniquely determined. The Riemann mapping theorem (§2.6, Theorem 2.15) guarantees the existence and uniqueness of a conformal map of D onto the unit disk U under certain specific conditions. First, we will introduce definitions of certain curves, and some elementary mappings, before we will study linear and bilinear transformations.

3.1 Definition of Certain Curves

Some curves are defined as follows:

3.1.1 Line, or straight line, has the equation $lx + my + p = 0$, where l, m, p are real, and $l^2 + m^2 > 0$. Then $\Re\{\bar{\lambda}z\} = p$, or $\Im\{\mu z\} = p$, where $\lambda = l + im \neq 0$, and $\mu = m + il = i\bar{\lambda}$.

The distance of a point z_0 from the line $lx + my + p = 0$ is given by $\left| \frac{\Re\{\bar{\lambda}z_0\} - p}{\lambda} \right|$.

The angle between the x -axis and the line $lx + my + p = 0$ is given by $\arg\{\lambda + \frac{\pi}{2} + n\pi\}$, $n = 0, \pm 1, \pm 2, \dots$

3.1.2 Circle in the extended sense means a circle or a straight line. There is no distinction between circles and lines in the theory of bilinear transformations. However, a circle in the z -plane is defined by $|z - z_0| = r$ with center at z_0 and radius r ; in the w -plane it is defined by $|w - w_0| = R$ with center at w_0 and radius R .

3.1.3 Ellipse has the equation $|z - z_1| + |z - z_2| = k$, $k > |z_1 - z_2|$; its foci are z_1 and z_2 ; major axis is k ; eccentricity is $\frac{|z_1 - z_2|}{k}$; exterior of the ellipse is defined as $|z - z_1| + |z - z_2| > k$, and interior as $|z - z_1| + |z - z_2| < k$.

3.1.4 Hyperbola has the equation $|z - z_1| + |z - z_2| = \pm k$, $0 < k < |z_1 - z_2|$, where the plus or minus sign is used according to whether the branch is nearer to the focus z_2 or it is nearer to the focus z_1 . The real axis is k ; eccentricity $\frac{|z_1 - z_2|}{k}$; equations of asymptotes

are

$$\left\{ (\bar{z}_1 - \bar{z}_2) e^{\pm\theta} \left(z - \frac{z_1 + z_2}{2} \right) \right\} = 0, \quad \cos \theta = \frac{k}{|z_1 - z_2|};$$

the exterior, not containing foci is defined by $||z - z_1| - |z - z_2|| < k$.

3.1.5 Rectangular Hyperbola is the hyperbola of §3.1.4 if $k = \frac{|z_1 - z_2|}{\sqrt{2}}$. Moreover, if $z_2 = -z_1$ (i.e., the origin is the center of the curve), then the equation of the rectangular hyperbola is

$$\Re \left\{ \frac{z^2}{z_1^2} \right\} = \frac{1}{2},$$

with foci $\pm z_1$. The exterior of the hyperbola is defined by

$$-\infty \leq \Re \left\{ \frac{z^2}{z_1^2} \right\} < \frac{1}{2}.$$

Equations of asymptotes are (three forms):

$$\Re \left\{ \frac{z^2}{z_1^2} \right\} = 0; \quad \Re \left\{ \frac{z}{z_1} \right\} = \pm \Im \left\{ \frac{z}{z_1} \right\} = \frac{1}{2}; \quad \Re \left\{ \frac{z(1 \pm i)}{z_1} \right\} = 0.$$

3.1.6 Parabola has the equation $|z - z_0| = |k - \Re\{\lambda z\}| |\lambda|^{-1}$, $\lambda \neq 0$, $k \geq \Re\{\lambda z_0\}$. It has focus at $z = z_0$; directrix $\Re\{\lambda z\} = k$; latus rectum $2|k - \Re\{\lambda z_0\}| |\lambda|^{-1}$; and vertex $z = z_0 + \frac{k - \Re\{\lambda z_0\}}{2\lambda}$.

3.1.7 Cassini's ovals or Cassinians, presented in Figure 3.1, are defined by $|z - z_1||z - z_2| = \alpha$, $\alpha > 0$; its foci are z_1, z_2 . For $\alpha = \frac{1}{4}|z_1 - z_2|^2 = 1$, the curve is a *lemniscate*.

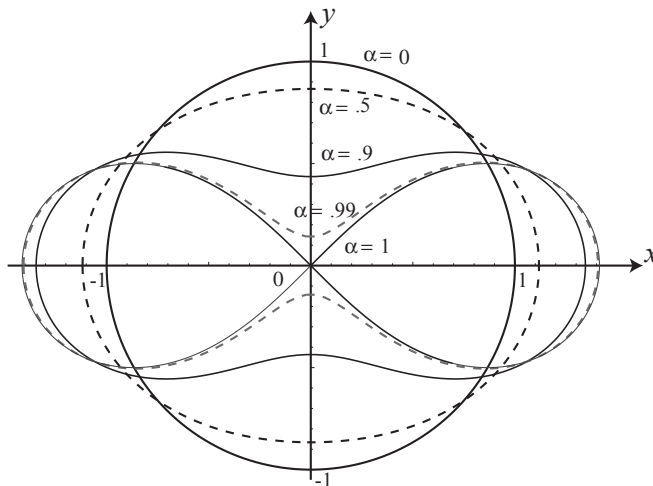


Figure 3.1 Cassini's ovals and lemniscate.

Cassini's ovals are plotted using the definition with rectangular coordinates:

$$F(x, y, \alpha) = [(x + \alpha)^2 + y^2] [(x - \alpha)^2 + y^2] = 1$$

for values of $\alpha = 0, 0.5, 0.9, 0.99$, and 1. For $\alpha = 0$ the curve becomes the unit circle. The region does not remain simply connected for $\alpha = 1$; it is a doubly connected region and its boundary curve is called the *lemniscate*.

3.1.8 Cardioid and Limaçons. The cardioid, defined by $r = 2a(1 + \cos \theta)$, is the curve in Figure 3.2(a) that is described by a point P of a circle of radius a as it rolls on the outside of a fixed circle of radius a . It is a special case of the limaçon of Pascal, which has the equation $r = b + a \cos \theta$. As described in Figure 3.2(b), the line OQ joining the origin to any point Q on a circle of diameter a passes through O ; then the limaçon is the locus of all points P such that $PQ = b$. Figure 3.2(b) represents the case when $2a > b$, and Figure 3.2(c) the case when $2a < b$. If $b \geq 2a$, the curve is convex. If $b = a$, the limaçon becomes the cardioid.

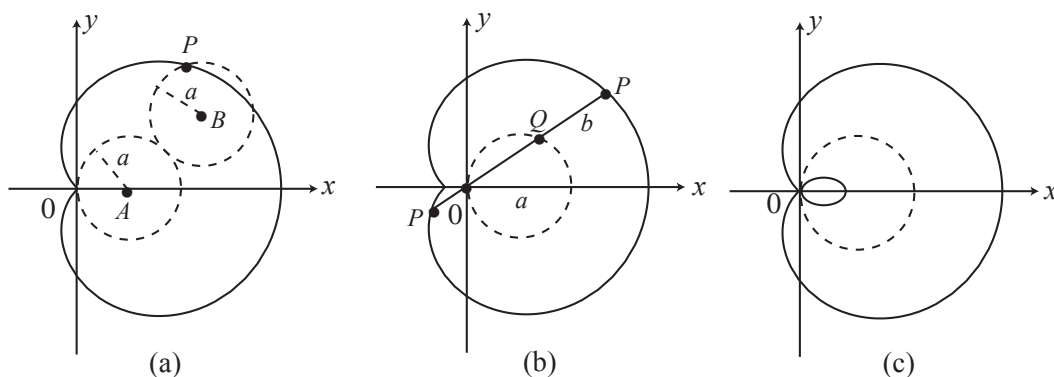


Figure 3.2 Cardioid and limaçons.

3.2 Bilinear Transformations

The **bilinear (linear-fractional, or Möbius)** transformation is of the form

$$w = f(z) = \frac{az + b}{cz + d}, \quad (3.2.1)$$

where a, b, c, d are complex constants such that $ad - bc \neq 0$ (otherwise the function $f(z)$ would be identically constant). If $c = 0$ and $d = 1$, or if $a = 0$, $d = 0$ and $b = c$, then the function (3.2.1) reduces to a linear transformation $w = az + b$, or an inversion $w = \frac{1}{z}$, respectively. The transformation (3.2.1) is also written as

$$w = \frac{a}{c} + \frac{bc - ad}{c(cz + d)}, \quad (3.2.2)$$

that can be viewed as composed of the following three successive functions:

$$z_1 = cz + d, \quad z_2 = \frac{1}{z_1}, \quad w = \frac{a}{c} + \frac{bc - ad}{c} z_2,$$

which shows that the mapping (3.2.1) is a linear transformation, followed by an inversion which is followed by another linear transformation. The bilinear transformation (3.2.1) maps the extended z -plane conformally onto the extended w -plane such that the pole at

$z = -d/c$ is mapped into the point $w = \infty$. The inverse transformation

$$z = f^{-1}(w) = \frac{b - dw}{-a + cw} \quad (3.2.3)$$

is also bilinear defined on the extended w -plane, and maps it conformally onto the extended z -plane such that the pole at $w = a/c$ is mapped into the point $z = \infty$. Note that $f'(z) = \frac{ad - bc}{(cz + d)^2} \neq 0$; also $[f^{-1}(w)]' = \frac{-ad + bc}{(cw - a)^2} \neq 0$. A bilinear transformation carries circles into circles (in the extended sense), as explained in §3.1.

Example 3.1. The bilinear transformation that maps the square of side 2, centered at the origin of the z -plane onto the parallelogram shown in Figure 3.3 is given by $w = \frac{az + b}{cz + d}$ such that the points $1 + i$, $1 - i$, $-1 - i$, and $-1 + i$ are mapped onto the points $1 - 2i$, $1 + 2i$, $\frac{1 + 2i}{5}$, and $-\frac{1 + 2i}{5}$, respectively, as shown in Figure 3.3. ■

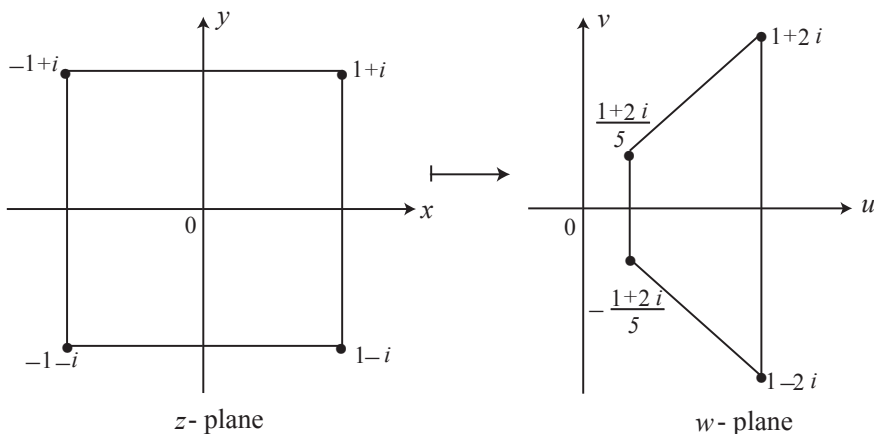


Figure 3.3 Conformal and isogonal mappings.

Example 3.2. The transformation that maps the half-plane $\Im\{z\} > \Re\{z\}$ onto the interior of the circle $|w - 1| = 3$ is

$$w = \frac{(1 + 3i)z + 4i}{z + i}.$$

To prove it, let us regard the given half-plane as the ‘interior’ of the ‘circle’ through ∞ defined by the line $\Im\{z\} = \Re\{z\}$. The three points in clockwise order are $z_1 = \infty$, $z_2 = 0$, and $z_3 = -1 - i$. The three points on the circle $|w - 1| = 3$ in clockwise order are $w_1 = 1 + 3i$, $w_2 = 4$, and $w_3 = -2$. The linear fractional transformation that maps three points in order is defined by Eq (3.2.1), where from the images of ∞ and 0 we have $a = 1 + 3i$ and $b/c = 4$, i.e., $b = 4i$ and $c = i$. Substituting these into Eq (3.2.1), we get the desired mapping.

Since $z_0 = i$ is in the half-plane $\Im\{z\} > \Re\{z\}$, its image under the above map is $w_0 = 2.5 + 1.5i$; thus, as $w_0 - 1 = 1.5 + 1.5i$ has a modulus $1.5 \times \sqrt{2} \approx 2.12132 < 3$, we find that w_0 is in the interior of $|w - 1| = 3$ (see Figure 3.4). ■

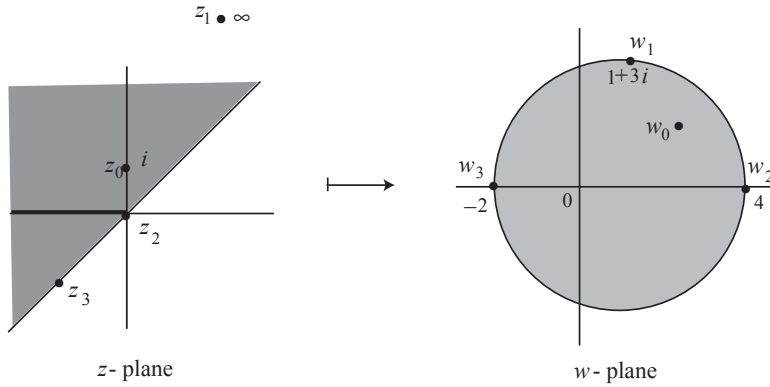


Figure 3.4 $w = \frac{(1+3i)z + 4i}{z + i}$.

3.2.1 Fixed Points. The **fixed points of a mapping are those points at which $w = z$** . For example, the mapping $w = z^3$ has a fixed point $z = w = 1$. Let F_1 and F_2 be the **fixed points of the bilinear transformation (3.2.1), $ad - bc \neq 0$. Then F_1 and F_2 are the roots of the equation**

$$cF^2 + (d - a)F - b = 0.$$

We will ignore the *identical transformation* $w = z$, and consider the following cases:

(i) If $F_1 = F_2 = \infty$, we have $w = z + b$, $b \neq 0$, p, q real. Then (i) any line $\Im\{\bar{b}z\} = p$ is mapped onto itself, but onto no other ‘circle’ (in the extended sense); and (ii) the line $\Re\{\bar{b}z\} = q$ is mapped onto $\Re\{\bar{b}w\} = q - |b|^2$.

(ii) If $F_1 = F_2 \neq \infty$, we have

$$\frac{1}{w - F} = \frac{1}{z - F} + k,$$

where

$$c \neq 0, F = \frac{a - d}{2c} = \frac{2b}{d - a}, k = \frac{2c}{a + d}.$$

The transformation is ‘parabolic’. The straight line $\Im\{k(z - r)\} = 0$ and every circle touching this straight line at F is mapped onto itself; no other circle has this property. The set of ‘circles’ orthogonal to the above set is, as a whole, mapped onto itself.

(iii) If $F_1 \neq F_2 = \infty$, we have $w = F_1 = \alpha(z - F_1)$, where $\alpha = a/d$, $\alpha \neq 0, 1$, $c = 0$, and $F_1 = \frac{b}{d - a}$. Let D denote the set of concentric circles centered at F_1 , and E denote the set of straight lines through F_1 . Then the entire set D is mapped onto itself; so is E . None of the ‘circles’ other than those stated below is mapped onto itself:

- (a) α real, but $\alpha \neq -1$: then every line of E is mapped onto itself;
- (b) $\alpha = -1$: then every circle of D , and every line of E , is mapped onto itself;
- (c) $|\alpha| = 1$, but $\alpha \neq -1$: then every circle of D is mapped onto itself;
- (d) α not real, $|\alpha| \neq 1$: then no circle or line is mapped onto itself (this is known as the *loxodromic transformation*).

(iv) If $F_1 \neq F_2$, $F_1 \neq \infty$, $F_2 \neq \infty$, we have

$$\frac{w - F_1}{w - F_2} = \alpha \frac{z - F_1}{z - F_2},$$

where

$$\alpha = \frac{cF_2 + d}{cF_1 + d} = \frac{(a + d - \sqrt{(a - d)^2 + 4bc})^2}{4(ad - bc)}.$$

Let E denote a pencil of ‘circles’ through F_1 and F_2 , and let D denote the set of ‘circles’ orthogonal to the circles of E (co-axial system). Then the results are the same as in (iii) above, but replace circle or straight line there by ‘circle’.

The fixed points should, however, not be confused with the *involution* transformation for which the mapping $w = f(z)$ has the same form as the mapping $z = f^{-1}(w)$. An example is the mapping $w = \frac{1}{z}$, or the mapping $w = \frac{z + 1}{z - 1}$.

3.2.2 Linear Transformation. $w = az + b$, $a \neq 0$. The special cases are as follows:

(a) $w = az$: (i) if $a > 0$, then there is *magnification* by ratio a centered at 0; (ii) if $a = e^{i\tau}$, τ real, then there is *rotation* by the angle τ about 0; and (iii) if $a = me^{i\tau}$, $m > 0$, and τ real, then there is *rotation* by the angle τ about 0, followed by a *magnification* of ratio m centered at 0.

(b) $w = z + b$: There is a *translation* by the vector $\overline{0b}$.

(c) $w = az + b$, $a \neq 1$: Then $F_0 = \frac{b}{1 - a}$, $a = me^{i\tau}$, $m > 0$, τ real. There is *rotation* about F_0 by the angle τ , followed by *magnification* by ratio m centered at F_0 .

The correspondence of the above linear transformations between the z -plane and the w -plane is as follows:

- (i) The line $x = p$ is mapped onto the line $\Re\left\{\frac{w - b}{a}\right\} = p$.
- (ii) The line $y = q$ is mapped onto the line $\Im\left\{\frac{w - b}{a}\right\} = q$.
- (iii) The line $lx + my = p$ is mapped onto the line $\Re\left\{(l - im)\frac{w - b}{a}\right\} = p$.
- (iv) The line $\Re\{az + b\} = p$ is mapped onto the line $u = P$.
- (v) The line $\Im\{az + b\} = q$ is mapped onto the line $v = Q$.
- (vi) The line $\Re\{(l - im)(az + b)\} = p$ is mapped onto the line $lu + mv = P$.
- (vii) The circle $|z - z_0| = r$ is mapped onto the circle $|w - (aw_0 + b)| = |a|R$.
- (vii) The circle $\left|z - \frac{w_0 - b}{a}\right| = \frac{r}{|a|}$ is mapped onto the circle $|w - w_0| = R$.

3.2.3 Composition of Bilinear Transformations. We will consider the general bilinear transformation (3.2.1) with $ad - bc \neq 0$. Note that if $ad - bc = 0$, then we have $w = \text{const}$. The general bilinear transformation (3.2.1) consists of the following three successive transformations:

- (i) translation of z : $z_1 = z + d/c$;
- (ii) inversion of z_1 : $z_2 = 1/z_1$; and
- (iii) translation, rotation, and scale change of z_2 : $w = \frac{a}{c} - \frac{ad - bc}{c^2} z_2$.

Map 3.1. If $a = -d$, the transformation is *involutory*, i.e., w and z may be interchanged.

The magnification is $\left| \frac{dw}{dz} \right| = \left| \frac{ad - bc}{(cz + d)^2} \right|$, or

$$|f'(z)|^2 = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 = \left(\frac{\partial u}{\partial y} \right)^2 + \left(\frac{\partial v}{\partial y} \right)^2.$$

The scale or magnification factor m is then defined as $m = |dw/dz| = |f'(z)|$, where $m = |m|$, and $M = 1/m$. The critical points are $z = -d/c$ and $z = \infty$; at these points the transformation is *not* conformal in the usual sense. The details of the transformation are as follows:

- (i) The points $z = 0$; $-b/a$; $z_\infty = -d/c$ are mapped onto the points $w = b/d$; 0 ; ∞ .
- (ii) The point at infinity $z = \infty$ is mapped onto the point $w_\infty = a/c$.

Map 3.2. The bilinear transformation $w = f(z)$ that transforms the three points z_k into the points w_k , $k = 1, 2, 3$, can be expressed as (Saff and Snider [1976:329])

$$\begin{vmatrix} 1 & z & w & zw \\ 1 & z_1 & w_1 & z_1 w_1 \\ 1 & z_2 & w_2 & z_2 w_2 \\ 1 & z_3 & w_3 & z_3 w_3 \end{vmatrix} = 0.$$

Map 3.3. Let z be fixed and let $\Re\{z\} \geq 0$. Define a sequence of bilinear transformations

$$T_0(w) = \frac{a_0}{z + a_0 + b_1 + w}, \quad T_k(w) = \frac{a_k}{z + b_{k+1} + w}, \quad k = 1, \dots, n-1,$$

such that each $a_j > 0$ is real, and each b_j is zero or purely imaginary for $j = 0, 1, 2, \dots, n-1$. We use induction to prove that the chain of transformations defined by $\zeta = S(w) = (T_0 \circ \dots \circ T_{n-2} \circ T_{n-1})(w)$ maps the half-plane $\Re\{w\} > 0$ onto a region contained in the disk $|\zeta - 1/2| < 1/2$. Thus, using the *Wallis criterion*, if $P(z) = z^n + c_1 z^{n-1} + c_2 z^{n-2} + \dots + c_n$ is a polynomial of degree $n > 0$ with complex coefficients $c_k = p_k + i q_k$, $k = 1, 2, \dots, n$, if $Q(z) = p_1 z^{n-1} + i q_2 z^{n-2} + p_3 z^{n-3} + i q_4 z^{n-4} + \dots$, and if $Q(z)/P(z)$ can be written as a continued fraction, then we have $Q(z)/P(z) = (T_0 \circ T_{n-2} \circ \dots \circ T_{n-1})(z)$ (Saff and Snider [1976:330]).

Theorem 3.1. (Schwarzian derivative) (i) *The Schwarzian derivative (also called the Schwarz differential operator), defined by*

$$\{w, z\} = \left(\frac{w''}{w'} \right)' - \frac{1}{2} \left(\frac{w''}{w'} \right)^2 = \frac{w'''}{w'} - \frac{3}{2} \left(\frac{w''}{w'} \right)^2,$$

is invariant under bilinear transformations. (Nehari [1952:199]; Wen [1992:76]).

(ii) The Schwarzian derivative of the function $w = w(z)$ that maps the upper half-plane $\Im\{z\} > 0$ conformally onto a circular triangle with interior angles $\alpha\pi$, $\beta\pi$ and $\gamma\pi$ is given by

$$\{w, z\} = \frac{1 - \alpha^2}{2z^2} + \frac{1 - \beta^2}{2(1 - z)^2} + \frac{1 - \alpha^2 - \beta^2 + \gamma^2}{2z(1 - z)}.$$

(Nevanlinna and Paatero [1969:342]).

3.3 Cross-Ratio

A cross-ratio between four distinct finite points z_1, z_2, z_3, z_4 is defined by

$$(z_1, z_2, z_3, z_4) = \frac{z_1 - z_2}{z_1 - z_4} \cdot \frac{z_3 - z_4}{z_3 - z_2}. \quad (3.3.1)$$

If z_2, z_3 , or z_4 is a point at infinity, then (3.3.1) reduces to

$$\frac{z_3 - z_4}{z_1 - z_4}, \quad \frac{z_1 - z_2}{z_1 - z_4}, \quad \text{or} \quad \frac{z_1 - z_2}{z_3 - z_2},$$

respectively. The cross-ratio (z, z_1, z_2, z_3) is invariant under bilinear transformations.

Theorem 3.2. *A bilinear transformation is uniquely defined by a correspondence of the cross-ratios, i.e.,*

$$(w, w_1, w_2, w_3) = (z, z_1, z_2, z_3), \quad (3.3.2)$$

which maps any three distinct points z_1, z_2, z_3 in the extended z -plane into three prescribed points w_1, w_2, w_3 in the extended w -plane. The cross-ratio (z, z_1, z_2, z_3) is the image of z under a bilinear transformation that maps three distinct points z_1, z_2, z_3 into $0, 1, \infty$.

3.3.1 Symmetric Points. The points z and z^* are said to be *symmetric* with respect to a circle C (in the extended sense) through three distinct points z_1, z_2, z_3 iff

$$(z^*, z_1, z_2, z_3) = \overline{(z, z_1, z_2, z_3)}. \quad (3.3.3)$$

The mapping that carries z into z^* is called a *reflection* with respect to C . Two reflections obviously yield a bilinear transformation.

If C is a straight line, then we choose $z_3 = \infty$, and the condition for symmetry (3.3.3) gives

$$\frac{z^* - z_1}{z_1 - z_2} = \frac{\bar{z} - \bar{z}_1}{\bar{z}_1 - \bar{z}_2}. \quad (3.3.4)$$

Let z_2 be any finite point on the line C . Then, since $|z^* - z_1| = |z - z_1|$, the points z and z^* are equidistant from the line C . Moreover, since $\Im\left\{\frac{z^* - z_1}{z_1 - z_2}\right\} = -\Im\left\{\frac{z - z_1}{z_1 - z_2}\right\}$, the line C is the perpendicular bisector of the line segment joining z and z^* . If C is the circle $|z - a| = R$, then

$$\begin{aligned} \overline{(z, z_1, z_2, z_3)} &= \overline{(z - a, z_1 - a, z_2 - a, z_3 - a)} \\ &= \left(\bar{z} - \bar{a}, \frac{R^2}{z_1 - a}, \frac{R^2}{z_2 - a}, \frac{R^2}{z_3 - a} \right) \\ &= \left(\frac{R^2}{\bar{z} - \bar{a}}, z_1 - a, z_2 - a, z_3 - a \right) \\ &= \left(\frac{R^2}{\bar{z} - \bar{a}} + a, z_1, z_2, z_3 \right). \end{aligned}$$

Hence, in view of (3.3.3), we find that the points z and $z^* = \frac{R^2}{\bar{z} - \bar{a}} + a$ are symmetric, i.e.,

$$(z^* - a)(\bar{z} - \bar{a}) = R^2. \quad (3.3.5)$$

Note that $|z^* - a||z - a| = R^2$; also, since $\frac{z^* - a}{z - a} > 0$, the points z and z^* are on the same ray from the point a (Figure 3.5). Also, the point symmetric to a is ∞ .

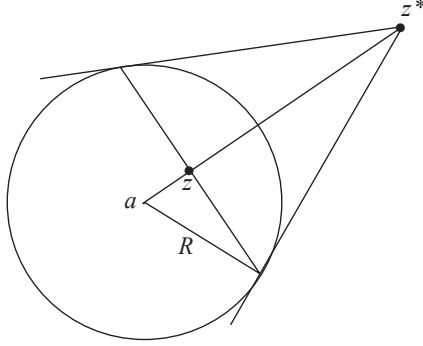


Figure 3.5 Symmetry with respect to a circle.

A GENERALIZATION of this result is as follows: If Γ denotes an analytic Jordan curve with parametric equation $z = \gamma(s)$, $s_1 < s < s_2$, then for any point z sufficiently close to Γ , the point

$$z^* = \gamma(\overline{\gamma^{-1}(z)}) \quad (3.3.6)$$

defines a symmetric point of z with respect to Γ (Sansone and Gerretsen, [1960: 103]; Papamichael, Warby and Hough [1986]). Some examples are as follows:

(i) If Γ is the circle $x^2 + y^2 = a^2/9$, then $z^* = a^2/9\bar{z}$.

(ii) If Γ is the ellipse $\frac{(x + a/2)^2}{a^2} + y^2 = 1$, then

$$z^* = -\frac{a}{2} + \frac{(a^2 + 1)(\bar{z} + a/2) + 2ia\sqrt{a^2 - 1 - (\bar{z} + a/2)}}{a^2 - a}.$$

(iii) If Γ_1 is a cardioid defined by $z = \gamma(s) = \left(\frac{1}{2} + \cos \frac{s}{2}\right) e^{is}$, $-\pi < s \leq \pi$, then from (3.3.6) we cannot write an explicit expression for symmetric points with respect to Γ_1 . However, for any real t ,

$$\gamma(\pm it) = \left(\frac{1}{2} + \cosh \frac{t}{2}\right) e^{\mp t}$$

defines two real symmetric points with respect to Γ_1 , provided the parameter t satisfies the equation $\gamma(it)\gamma(-it) = a^2$, i.e., $\frac{1}{2} + \cosh \frac{t}{2} = a$ which has the roots

$$t = 2 \cosh^{-1}(a - 1/2) = \pm 2 \log \rho, \quad (3.3.7)$$

where $\rho = a - \frac{1}{2} + \sqrt{\left(a - \frac{1}{2}\right)^2 - 1}$.

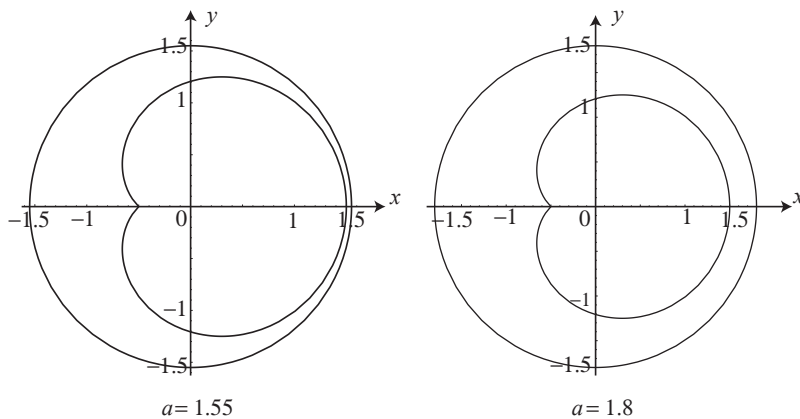


Figure 3.6 Cardioid inside a circle.

(iv) If a doubly connected region Ω is bounded outside by the circle $\Gamma_2 = \{z : |z| = a, a > 1.5\}$ and inside by the cardioid Γ_1 defined above in (iii) (see Figure 3.6), then it follows from (3.3.7) that there is one pair of real common symmetric points $\zeta_1 \in \text{Int}(\Gamma_1)$ and $\zeta_2 \in \text{Ext}(\Gamma_2)$ such that $\zeta_1 = a/\rho^2$ and $\zeta_2 = a\rho^2$, where ρ is defined in (3.3.7).

3.3.2 Symmetry Principle. The *symmetry principle* states that if a bilinear transformation maps a circle C_1 onto a circle C_2 , then it maps any pair of symmetric points with respect to C_1 into a pair of symmetric points with respect to C_2 . This means that bilinear transformations preserve symmetry.

A practical application of the symmetry principle is to find bilinear transformations which map a circle C_1 onto a circle C_2 . We already know that the transformation (3.3.2) can always be determined by requiring that the three points $z_1, z_2, z_3 \in C_1$ map onto three points $w_1, w_2, w_3 \in C_2$. But a bilinear transformation is also determined if a point $z_1 \in C_1$ should map into a point $w_1 \in C_2$ and a point $z_2 \notin C_1$ should map into a point $w_2 \notin C_2$. Then, by the symmetry principle, the point z_2^* which is symmetric to z_2 with respect to C_1 is mapped into the point w_2^* which is symmetric to w_2 with respect to C_2 , and then the desired bilinear transformation is given by

$$(w, w_1, w_2, w_2^*) = (z, z_1, z_2, z_2^*). \quad (3.3.8)$$

Let U^+ denote the region $|z| < 1$, U^- the region $|z| > 1$, and let $C = \{|z| = 1\}$ be their common boundary (Figure 3.7). Let $f(z)$ be a function defined on U^+ . Then this function can be related to the function $f^*(z)$ defined in U^- in the same manner as in (2.6.3) for half-planes, except that now the conjugate complex points are replaced by points inverse with respect to the circle C according to the relation (3.3.5). Thus,

$$f^*(z) = \overline{f\left(\frac{1}{\bar{z}}\right)} = \bar{f}\left(\frac{1}{z}\right). \quad (3.3.9)$$

This relation is symmetrical, i.e.,

$$f(z) = \overline{f^*\left(\frac{1}{\bar{z}}\right)}, \quad (f^*(z))^* = f(z). \quad (3.3.10)$$

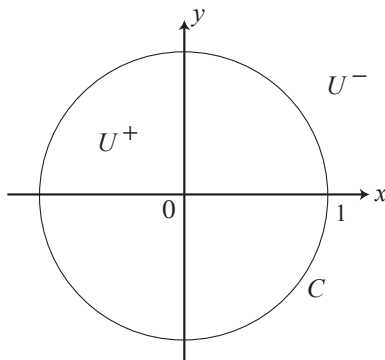


Figure 3.7 Unit disk.

If $f(z)$ is regular or meromorphic in U^+ , then $f^*(z)$ is regular or meromorphic in U^- . Also, if $f(z)$ is a rational function defined by (2.6.4), then

$$f^*(z) = \frac{\bar{a}_n z^{-n} + \bar{a}_{n-1} z^{-n+1} + \cdots + \bar{a}_0}{\bar{b}_m z^{-m} + \bar{b}_{-m+1} z^{m-1} + \cdots + \bar{b}_0}. \quad (3.3.11)$$

Moreover, if $f(z)$ has the power series expansion

$$f(z) = \sum_{k=-\infty}^{\infty} a_k z^k \quad z \in U^+,$$

then

$$f^*(z) = \sum_{k=-\infty}^{\infty} \bar{a}_k z^{-k}, \quad z \in U^-. \quad (3.3.12)$$

If $f(z)$ has a zero (pole) of order k at $z = \infty$ ($z = 0$), so does $f^*(z)$. Let us assume that $f(z)$ approaches a definite limit value $f^+(t)$ as $z \rightarrow t \in C$ from U^+ . Then $f^{*-}(t)$ exists and

$$f^{*-}(t) = \bar{f}^- \left(\frac{1}{t} \right) = \overline{f^+(t)}, \quad (3.3.13)$$

because $1/\bar{z} \rightarrow t$ from U^+ as $z \rightarrow t$ from U^- , and hence $f^*(z) = \bar{f} \left(\frac{1}{z} \right) = \overline{f \left(\frac{1}{\bar{z}} \right)} \rightarrow \overline{f^+(t)}$.

If $f(z)$ is regular in U^+ except possibly at infinity and continuous on C from the left, then let $F(z)$ be sectionally regular and be defined by

$$F(z) = \begin{cases} f(z) & \text{for } z \in U^+, \\ f^*(z) & \text{for } z \in U^-. \end{cases} \quad (3.3.14)$$

Then, $F^*(z) = F(z)$, and, as in (2.6.7),

$$F^-(t) = \overline{F^+(t)}, \quad F^+(t) = \overline{F^-(t)}. \quad (3.3.15)$$

Moreover, in view of the Schwarz reflection principle, if $\Im\{f^+(t)\} = 0$ on some part of the circle C , then $f^*(z)$ is the analytic continuation of $f(z)$ through this part of C .

3.3.3 Special Cases. We will discuss the following special cases of the transformations (3.2.1) and (3.2.3).

Map 3.4. (Linear transformation $w = az + b$, where $a = a_1 + ia_2$ and $b = b_1 + ib_2$. Then $w = u + iv = (\alpha_1 + i\alpha_2)(x + iy) + (b_1 + b_2) = (a_1x - a_2y + b_1) + i(a_1y + a_2x + b_2)$. The magnification factor and the clockwise rotation are

$$m = \frac{|dw|}{|dx|} = |a| = \sqrt{a_1^2 + a_2^2}, \quad (3.3.16)$$

$$\alpha = \arg \left\{ \frac{dw}{dz} \right\} = \arg \{m\} = \arctan \left(\frac{a_2}{a_1} \right). \quad (3.3.17)$$

Let $a = 1 + i$ and $b = -\frac{1}{2} - i\frac{3}{2}$. Then $m = \sqrt{2}$ and $\alpha = \pi/4$. The rectangular lines $u = k_u$ and $v = v_k$ are represented in the z -plane by the rectangular lines $y = x - (\frac{1}{2} + k_u)$ and $y = -x + (\frac{3}{2} + k_v)$, respectively. The transformation is presented in Figure 3.8.

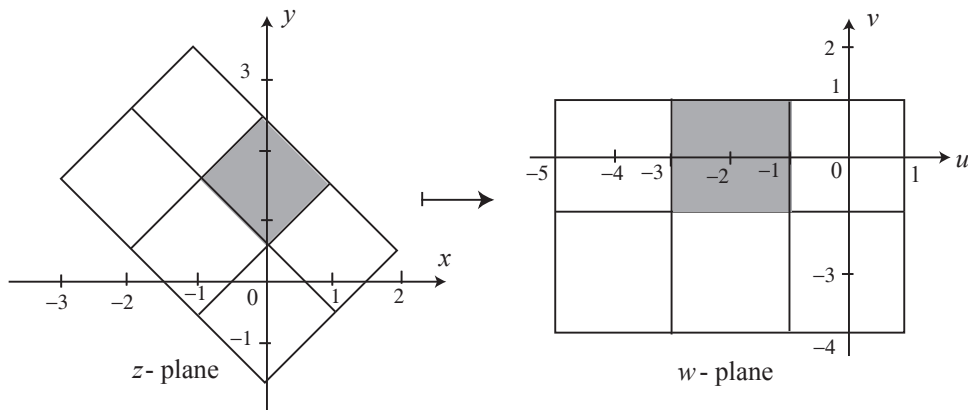


Figure 3.8 Linear transformation $w = az + b$.

In general, the mapping $w = az + b$ translates by b , rotates by $\arg\{a\}$ and magnifies (or contracts) by $|a|$.

Example 3.3. Consider the transformation $w = (1+i)z + (1+2i) = \sqrt{2} e^{i\pi/4} z + 1+2i$ in the z -plane. It maps the rectangle bounded by the lines AB: $x = 0$; BC: $y = 0$; CD: $x = 2$; and DA: $y = 1$ in the z -plane onto the rectangle bounded by the lines A'B': $u + v = 3$; B'C': $u - v = -1$; C'D': $u + v = 7$; and D'A': $u - v = -3$, respectively, such that the points $z = (0, 1), (0, 0), (2, 0), (2, 2)$ are mapped onto the points $w = (0, 3), (1, 2), (3, 4), (2, 5)$, respectively. The rectangle ABCD is translated by $(1 + 2i)$, rotated by an angle $\pi/4$ in the counterclockwise direction, and contracted by $\sqrt{2}$. ■

Map 3.5. The transformation $w = 2z - 2i$ maps the circle $|z - i| = 1$ onto the circle $|w| = 2$. The transformation that maps the disk $|z - i| = 1$ onto the exterior of the circle $|w| = 2$ is $w = \frac{2}{z - i}$. In fact, let $w = g(z) = \frac{az + b}{cz + d}$. Choose $g(i) = \infty$, so $g(z) = \frac{az + b}{z - i}$ without loss of generality. Take three points on the circle $|z - i| = 1$ as $0, 1 + i$, and $2i$; then $g(0) = ib, g(1 + i) = a(1 + i)$, and $g(2i) = 2a - ib$. Since all these three points must lie on $|w| = 2$, the simplest choice is $a = 0$ and $b = 2$. Then $g(z) = \frac{2}{z - i}$.

Map 3.6. The inversion $w = \frac{1}{z}$ is an involutory transformation with fixed points $F_1 = 1$ and $F_2 = -1$. The mapping from z -plane to w -plane, or inversely, is presented in Figure 3.9, where the radius $\overline{OA}$ is 1; the angle $\angle z0\bar{z}$ is bisected by the real axis; $a\bar{z} : \bar{z}B = Aw : Bw$; p, q and r are real; and the points $\bar{z}$ and w are inverse with respect to the unit circle.

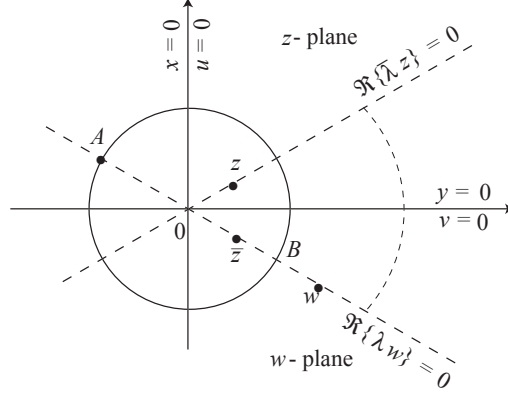


Figure 3.9 Transformation $w = \frac{1}{z}$.

There are five special cases under the inversion $w = \frac{1}{z}$. They are:

- (a) The interior (exterior) of the circle of D in the z -plane is mapped onto the interior (exterior) of the same circle, with points $z_\infty = 0; \infty; i; -i$ in the z -plane mapped into the points $\infty; w_\infty = 0; -i; i$, respectively, in the w -plane.
- (b) The line $\Re\{\bar{\lambda}z\} = 0$ in the z -plane is mapped onto the line $\Re\{\lambda w\} = 0$ in the w -plane; and the circle $|z| = r$ in the z -plane is mapped onto the circle $|w| = 1/|z|$ in the w -plane. These two cases are geometrically obvious and no figures are needed for them.

The remaining three cases are presented in Figure 3.10.; details, with $z = p, q, r, r_p, r_q$ real, and $w = P, Q, R, R_p, R_q$ real, are as follows:

- (c) The line $\Re\{\bar{\lambda}z\} = 0$ is mapped onto the circle

$$\left| w - \frac{\bar{\lambda}}{2p} \right| = \left| \frac{\lambda}{2p} \right|.$$

See Figure 3.10(c).

- (d) The circle $|z - z_0| = |z_0| \neq 0$ is mapped onto the line $\Re\{z_0\} = \frac{1}{2}$.

See Figure 3.10(d).

- (e) The circle $|z - z_0| = r \neq |z_0|$ is mapped onto the circle

$$\left| w - \frac{\bar{z}_0}{|z_0|^2 - r^2} \right| = \frac{r}{||z_0|^2 - r^2|}.$$

See Figure 3.10(e).

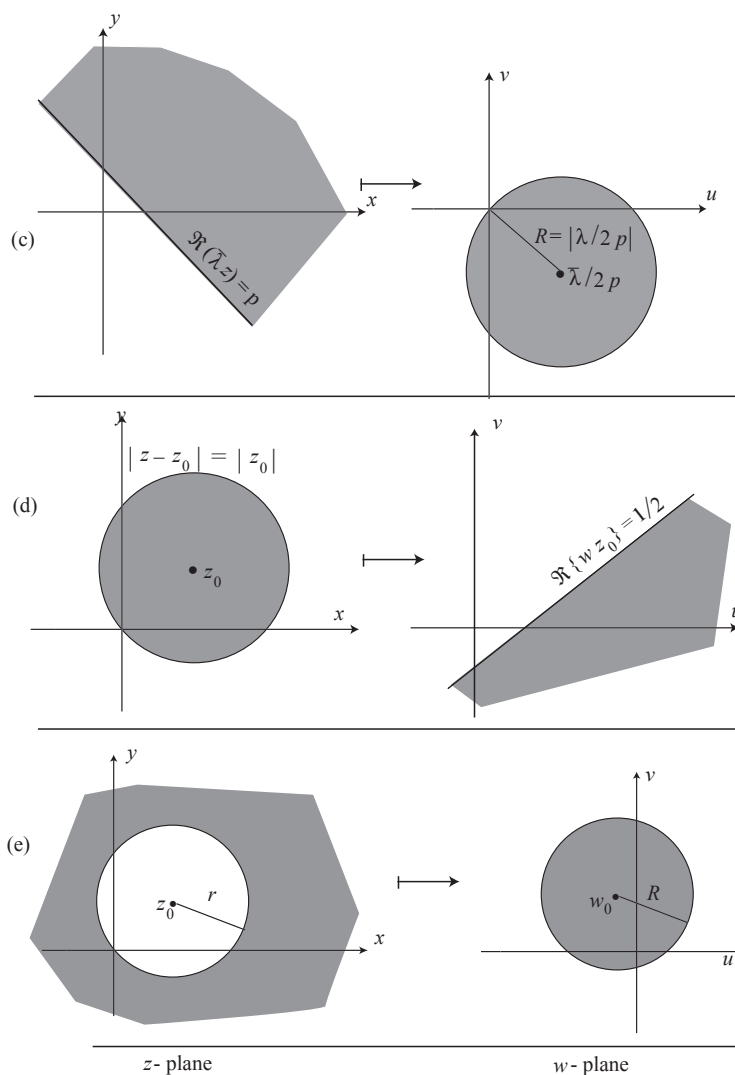
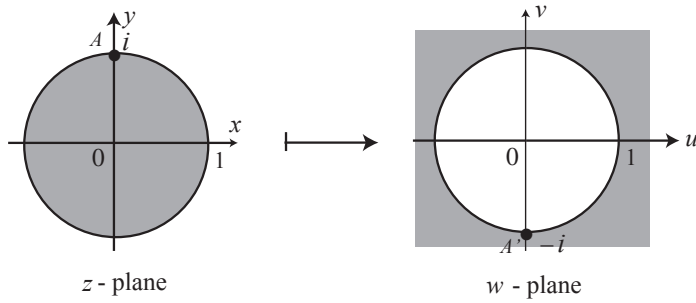


Figure 3.10 Transformation $w = \frac{1}{z}$.

Map 3.7. The transformation $w = \frac{1}{z}$, or $z = \frac{1}{w}$, which maps the interior of a circle in the z -plane onto the exterior of the circle in the w -plane and conversely, is shown in Figure 3.11.

Figure 3.11 Circle in the z -plane onto the exterior of a circle.

Since

$$w = u + iv = \frac{x}{x^2 + y^2} - \frac{y}{x^2 + y^2}i, \quad z = x + iy = \frac{u}{u^2 + v^2} - \frac{v}{u^2 + v^2}i,$$

we get

$$u = \frac{x}{x^2 + y^2}, \quad v = -\frac{y}{x^2 + y^2}.$$

A line through the origin in the z -plane is mapped onto a line through the origin in the w -plane. However, a line not passing through the origin in the z -plane is mapped onto a circle in the w -plane which passes through the origin, as the following example shows.

Example 3.4. The equation $ax + by + c$, $c \neq 0$ represents a straight line in the z -plane. Under the inversion map $w = 1/z$, this line is mapped onto a circle in the w -plane which passes through the origin. The details are as follows: The mapped curve is

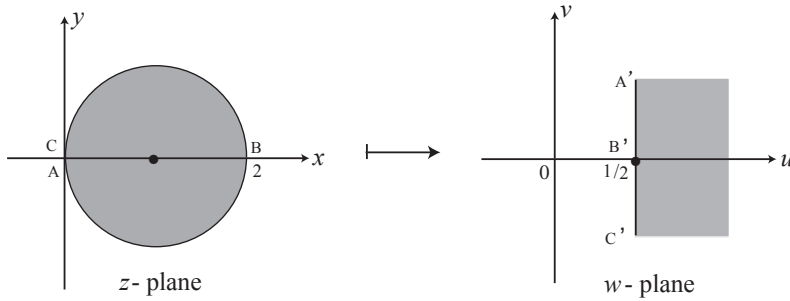
$$\frac{au}{u^2 + v^2} - \frac{bv}{u^2 + v^2} + c = 0.$$

Thus, $au - bv + c(u^2 + v^2) = 0$. Dividing by c , we obtain

$$u^2 + v^2 + \frac{a}{c}u = \frac{b}{c}v = 0,$$

which is the equation of the required circle. ■

Map 3.8. The transformation $w = \frac{1}{z}$, which maps the interior of the circle $|z - 1| = 1$ in the z -plane onto the right plane $u \geq \frac{1}{2}$, $-\infty < v < \infty$ in the w -plane, is shown in Figure 3.12.

Figure 3.12 Circle in the z -plane onto a rectangle.

Map 3.9. The transformation that maps the exterior of the unit circle in the z -plane onto the interior of the unit circle in the w -plane is defined by

$$w = 1/z,$$

and is presented in Figure 3.13.

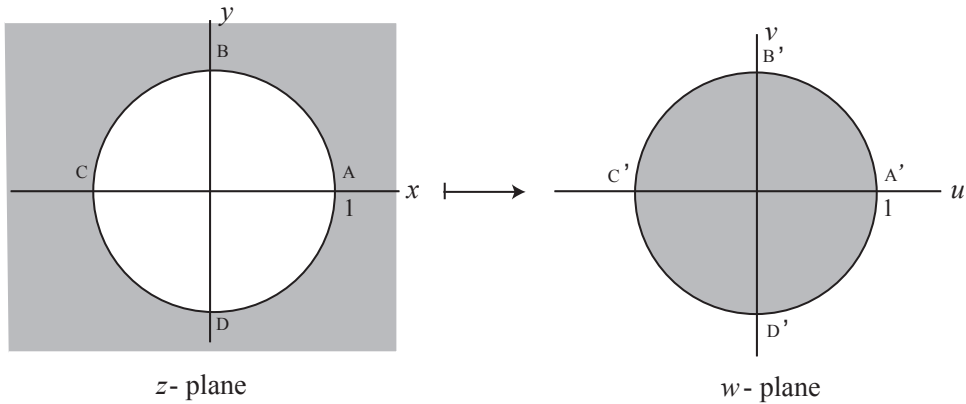


Figure 3.13 Exterior of the unit circle onto the interior of the unit circle.

Map 3.10. The transformation $w = k/z$ is known as the *reciprocal* or the *inverse transformation* for real k . We have $w = R e^{i\Theta} = k(r e^{i\theta})^{-1} = \frac{k}{r} e^{-i\theta}$, which gives $R = k/r$, and $\Theta = -\theta$. The fixed points are: $F_{1,2} = \pm\sqrt{k}$, and the critical points are $z = 0, \infty$. It is a useful transformation in electric transmission through a single wire, discussed in Example 22.4 with $k = 3600$. Two particular cases for $k = -\alpha$, $\alpha > 0$, and $k = 1$ are given in Maps 3.6, 3.7, 3.8, and 3.9, respectively.

Map 3.11. The transformation $w = a/z$ maps the exterior of a circle of radius a : $x^2 + y^2 \geq a^2$, in the z -plane onto the unit disk $|w| \leq 1$ in the w -plane.

Map 3.12. The transformation $w = z + \frac{1}{z}$, which maps the exterior of the upper

semi-circle in the z -plane onto the upper-half of the w -plane, is presented in Figure 3.14.

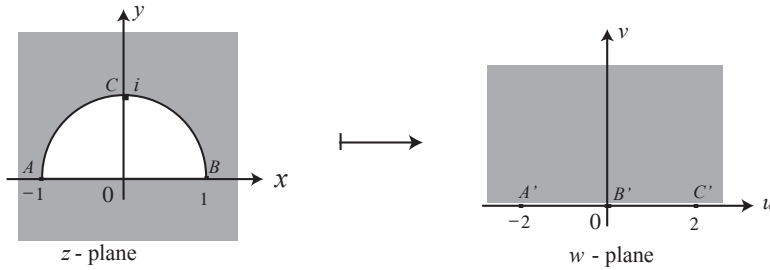


Figure 3.14 Exterior of a circle in the z -plane onto the upper-half of the w -plane.

Map 3.13. The transformation $w = z + \frac{1}{z}$, or $z = \frac{w + \sqrt{w^2 - 1}}{2}$, which maps the interior of the upper semi-circle in the z -plane onto the lower-half of the w -plane, is presented in Figure 3.15.

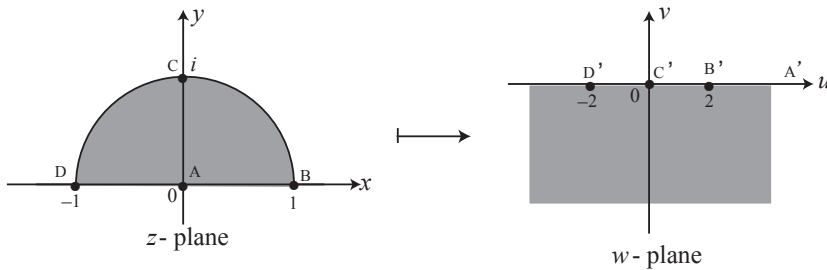


Figure 3.15 Interior of the upper half circle onto the lower-half of the w -plane.

Map 3.14. The transformation $w = z + \frac{1}{z}$, which maps the annular region between two upper semi-circles in the z -plane onto the upper-half of the ellipse in the w -plane, where the ellipse ($B'C'D'$) is defined by

$$\left(\frac{ku}{k^2 + 1}\right)^2 + \left(\frac{kv}{k^2 - 1}\right)^2 = 1,$$

is presented in Figure 3.16.

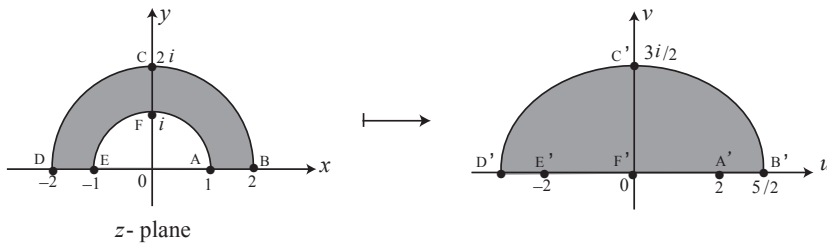


Figure 3.16 Annulus between two semi-circles onto the upper-half of the ellipse.

Map 3.15. If $a = c = 1, d = -b = \alpha$, then $w = \frac{z - \alpha}{z + \alpha}$. For this bilinear transformation

the above three successive cases are presented in Figure 3.17; these three cases may be treated as separate transformations (see cases 3.11 and 3.44 also).

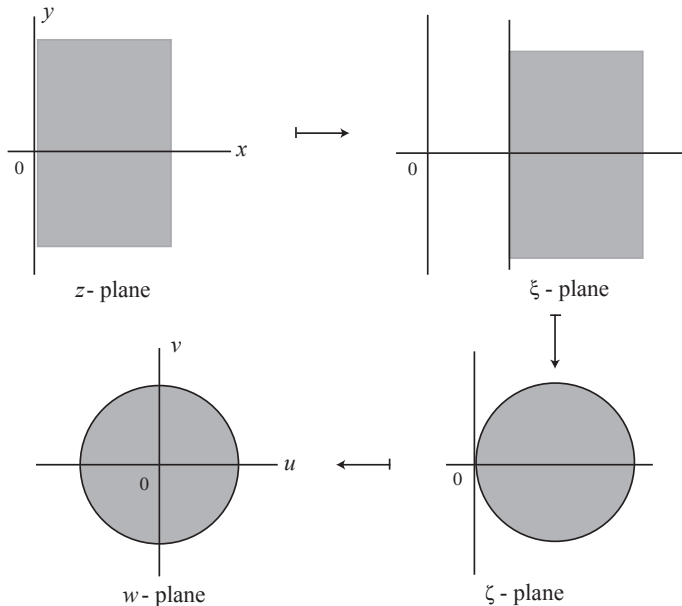


Figure 3.17 $w = \frac{z - \alpha}{z + \alpha}$.

In Figure 3.17, we start in the z -plane and translate the right half-plane by α in the ξ -plane so that $\xi = z + \alpha$; the right half-plane in the ξ -plane is mapped onto the circle in the ζ -plane by $\zeta = 1/\xi = 1/(z + \alpha)$; finally, the circle in the ζ -plane is mapped onto the circle in the w -plane so that $w = 1 + 2\alpha\zeta = 1 - \frac{2\alpha}{z + \alpha} = \frac{z - \alpha}{z + \alpha}$.

We can also select suitable parameters for this transformation; for example, given two circles, marked I with radius r' and marked II with radius r'' , such that the distance between their centers is h , and the distance of their centers as c_1 and c_2 from the origin of the coordinates (see Figure 3.18), we have the case of translation along the real axis, since

$$w = \frac{z - \alpha}{z + \alpha} = \frac{x + iy - \alpha}{x + iy + \alpha} = Re^{i\Theta}, \quad (3.3.18)$$

where

$$R = \left| \frac{z - \alpha}{z + \alpha} \right|, \quad R^2 = \frac{(x - \alpha)^2 + y^2}{(x + \alpha)^2 + y^2} = \frac{x^2 + y^2 + \alpha^2 - 2\alpha x}{x^2 + y^2 + \alpha^2 + 2\alpha x}. \quad (3.3.19)$$

The equation of circle I from Figure 3.18 is $(x - c_1)^2 + y^2 = (r')^2$, or $x^2 + y^2 = (r')^2 - c_1^2 + 2xc_1$. Then the radius R' of the map of circle I is

$$R'^2 = \frac{r'^2 - c_1^2 + 2xc_1 + \alpha^2 - 2\alpha x}{r'^2 - c_1^2 + 2xc_1 + \alpha^2 + 2\alpha x} = \frac{r'^2 - c_1^2 + \alpha^2 + 2x(c_1 - \alpha)}{r'^2 - c_1^2 + \alpha^2 + 2x(c_1 + \alpha)}. \quad (3.3.20)$$

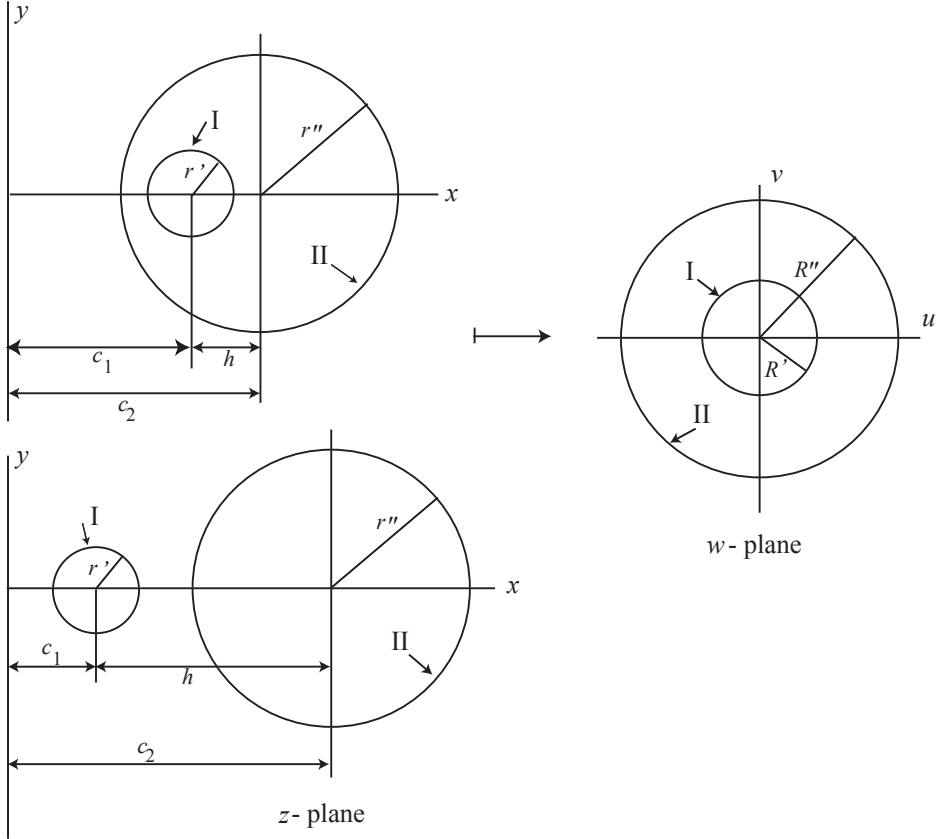


Figure 3.18 $w = \frac{w - \alpha}{w + \alpha}$, Two circles.

If we select α and c_1 such that the relation $\alpha^2 = c_1^2 - r'^2$ is satisfied, then R' becomes independent of x , i.e.,

$$R' = \left(\frac{c_1 - \alpha}{c_1 + \alpha} \right)^{1/2} = \frac{r'}{c_1 + \alpha}, \quad r' = \{(c_1 - \alpha)(c_1 + \alpha)\}^{1/2}. \quad (3.3.21)$$

The circle II can be treated similarly by selecting $\alpha_2^2 = c_2^2 - r''^2$, thus finally giving

$$R'' = \frac{r''}{c_2 + \alpha}. \quad (3.3.22)$$

If we substitute $h = c_2 - c_1$, then this transformation can be written as

$$w = \frac{z - \alpha}{z + \alpha} \quad \text{where } \alpha^2 = c_1^2 - r'^2, \quad (3.3.23)$$

and

$$c_1 = \frac{r''^2 - r'^2 - h^2}{2h}, \quad c_2 = \frac{r''^2 - r'^2 + h^2}{2h} = c_1 + h, \quad R' = \frac{r'}{c_1 + \alpha}, \quad R'' = \frac{r''}{c_2 + \alpha}. \quad \blacksquare$$

A critical point for this transformation is $z = -\alpha$, and the inverse of this transformation is of the same type, i.e., $z = \alpha \frac{1+w}{1-w}$.

The general form of the bilinear transformation (3.2.1) includes as special cases all the intermediate transformations which are discussed below.

Map 3.16. The bilinear transformations $w = \frac{z+1}{z-1}$, $z = \frac{w+1}{w-1}$, are involutory, with p, q real, and the fixed-points $F_{1,2} = 1 \pm \sqrt{2}$. The interior of the circle of D is mapped onto the interior of the same circle; however, the exterior of the circle of E is mapped onto the interior of the same circle (see Figure 3.19).

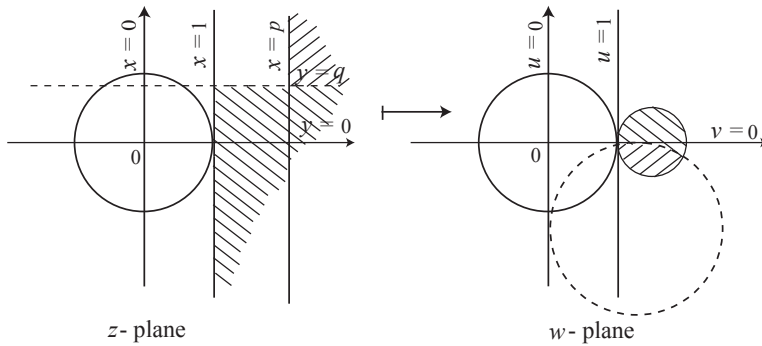


Figure 3.19 Transformation $w = \frac{z+1}{z-1}$.

Other details for some particular cases, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The points $z = 0; z_\infty = 1; \infty$ are mapped onto the points $w = -1; \infty; w_\infty = 1$.
- (b) The half-plane $x \geq 0$ is mapped onto the region $|w| \geq 1$.
- (c) The half-plane $y \geq 0$ is mapped onto the half-plane $v \leq 0$.
- (d) The region $|z| \leq 1$ is mapped onto the half-plane $u \leq 0$.
- (e) The line $x = 1$ in D is mapped onto the line $u = 1$.
- (f) The line $x = p, p \neq 1$ is mapped onto the circle $\left|w - \frac{p}{p-1}\right| = \frac{1}{|p-1|}$.
- (g) The line $y = q, q \neq 0$ is mapped onto the circle $\left|w - \frac{q-1}{q}\right| = \frac{1}{|q|}$.

Map 3.17. The transformation $w = \frac{1+z}{1-z} = e^{i\pi} \frac{z+1}{z-1}$, is the inverse of the Map 3.16. It maps the upper half-plane $\Im\{z\} > 0$ onto the unit disk $B(0, 1)$.

Map 3.18. The transformation $w = \frac{1+z}{1-z}$, or $z = -\frac{1-w}{1+w}$ maps the unit upper semi-circle in the z -plane onto the first quadrant of the w -plane, such that the points $z = -1, 0, 1$

map into the points $w = 0, 1, -1$, respectively, as shown in Figure 3.20.

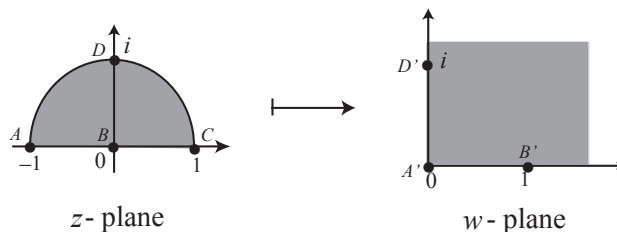


Figure 3.20 Upper semi-circle in the z -plane onto the first quadrant of the w -plane.

Map 3.19. The transformation $w = \frac{z-1}{z+1}$, or $z = \frac{1+w}{1-w}$ maps the right-half of the z -plane onto the unit circle, such that the points $z = 0, i, -i$ map into the points $w = -1, i, -i$, respectively. This map is shown in Figure 3.21(a).

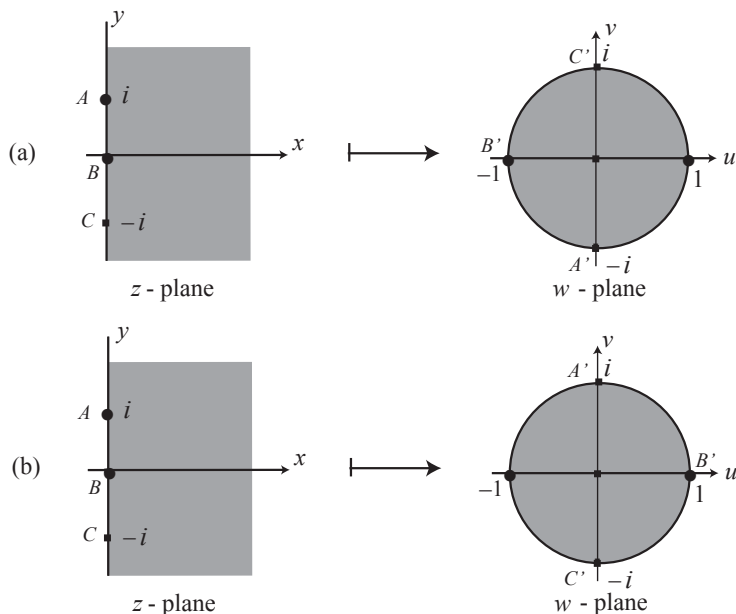


Figure 3.21 Right-half z -plane onto the unit circle.

Example 3.5. To find the linear fractional transformation that maps the interior of the circle $|z - i| = 2$ onto the exterior of the circle $|w - 1| = 3$, we need only three points on the first circle in clockwise order and three on the second circle in counterclockwise order. Let $z_1 = -i$, $z_2 = -2 + i$, and $z_3 = 3i$ and $w_1 = 4$, $w_2 = 1 + 3i$, and $w_3 = -2$. Then we get

$$\frac{(w-4)(3+3i)}{(w+2)(-3+3i)} = \frac{(z+i)(-2-2i)}{(z-3i)(-2+2i)},$$

or

$$w = \frac{z-7i}{z-i}.$$

The center of the first circle is $z = i$, which, under this transformation, is mapped onto the

point $w = \infty$, and thus it lies outside the second circle. ■

Map 3.20. The transformation $w = \frac{1-z}{1+z} = e^{i\pi} \frac{z-1}{z+1}$, is the inverse of the Map 3.18. Thus, this transformation, or its inverse $z = \frac{1-w}{1+w}$, also maps the right-half of the z -plane onto the unit circle such that the points $z = 0, i, -i$ map into the points $w = 1, -i, i$, respectively, as in Figure 3.21(b).

Map 3.21. The transformation $w = \frac{1-z}{1+z}$ maps the exterior of the unit circle $D : |z| > 1$ onto the region $G : \Re\{w\} < 0$, which is the left-half of the w -plane. To see this, first consider the left of the regions D and G , and choose any three points on the unit circle that give negative (clockwise) orientation, say, $z_1 = 1, z_2 = -i, z_3 = -1$. Similarly the three points $w_1 = 0, w_2 = i, w_3 = \infty$ in the left region of G . Then the bilinear transformation is given by

$$\frac{w-0}{i-0} = \frac{(z-1)(-i+1)}{(z+1)(-i-1)},$$

which simplifies to the above transformation.

Map 3.22. The bilinear transformation $w = \frac{z-i}{z+i}, z = i \frac{1+w}{1-w}$, has the fixed-points $F_{1,2} = \frac{1}{2}(1-i)(1 \pm \sqrt{3})$, and p, q, P, Q real. Two particular cases (A) and (B) are presented in Figure 3.22 and Figure 3.23, respectively.

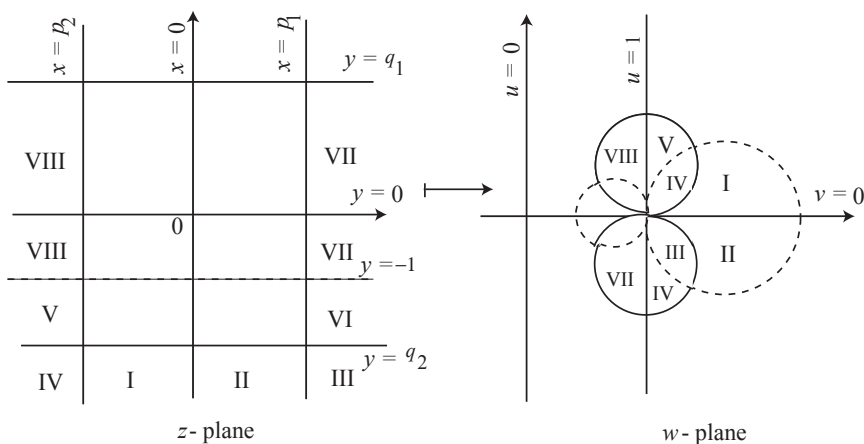


Figure 3.22 Transformation $w = \frac{z-i}{z+1}$ (Case A).

The details of the case A, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- The points $z = 0, i, -i; z_\infty = \infty$ are mapped onto the points $w = -1, 0, \infty; w_\infty = 1$.
- The half-plane $x > 0$ is mapped onto the half-plane $v < 0$.
- The half-plane $y > 0$ is mapped onto the region $|w| < 1$.
- The region $|z| < 1$ is mapped onto the half-plane $u < 0$.
- The line $x - y = 1$ is mapped onto the line $u - v = 1$.

- (f) The line $x = p, p \neq 0$ in D is mapped onto the circle $\left|w - \left(1 - \frac{1}{p}\right)\right| = \frac{1}{|p|}$.
- (g) The line $y = -1$ is mapped onto the line $u = 1$.
- (h) The line $y = q, q \neq -1$ is mapped onto the circle $\left|w - \frac{q}{q+1}\right| = \frac{1}{|q+1|}$.

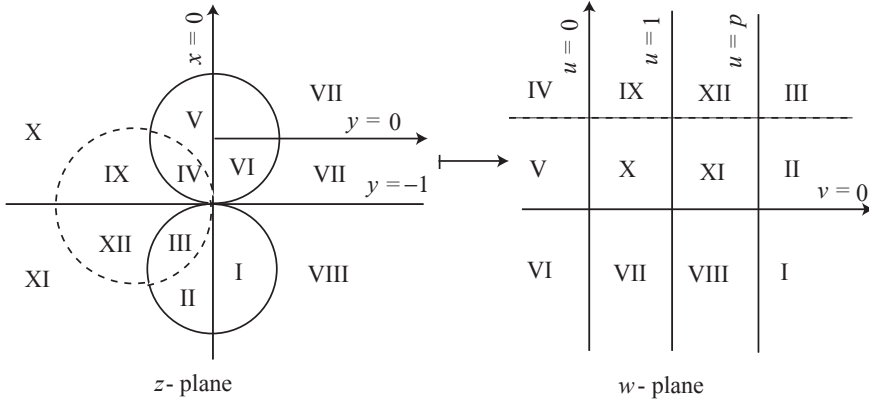


Figure 3.23 Transformation $w = \frac{z-1}{z+1}$ (Case B).

The details of the case B, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The line $y = -1$ is mapped onto the line $u = 1$.
- (b) The circle $\left|z - \frac{ip}{1-p}\right| = \frac{1}{|1-p|}$ is mapped onto the line $u = P, P \neq 1$.
- (c) The circle $\left|z - \left(-i - \frac{1}{q}\right)\right| = \frac{1}{|q|}$ is mapped onto the line $v = Q, Q \neq 0$.

Map 3.23. The transformation $w = \frac{z-i}{z+i}$, or $z = i\frac{1+w}{1-w}$, which maps the upper-half z -plane onto the circle, is presented in Figure 3.24.

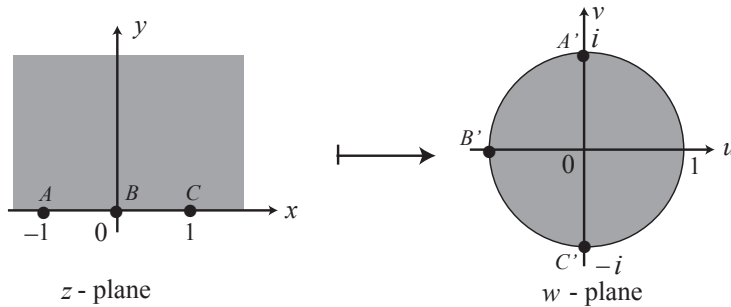


Figure 3.24 Upper-half z -plane onto a circle.

Map 3.24. The bilinear transformation $w = \frac{i+z}{i-z}$ is a composition of the following three transformations: (i) $w = -\frac{1}{s}$ (Map 3.6); (ii) $s = \frac{\zeta-i}{\zeta+i}$ (Map 3.23); and (iii) $\zeta = z$

(translation). It maps the upper half-plane $\Im\{z\} > 0$ in the z -plane onto the unit disk $|w| < 1$ in the w -plane (see Figure 3.25).

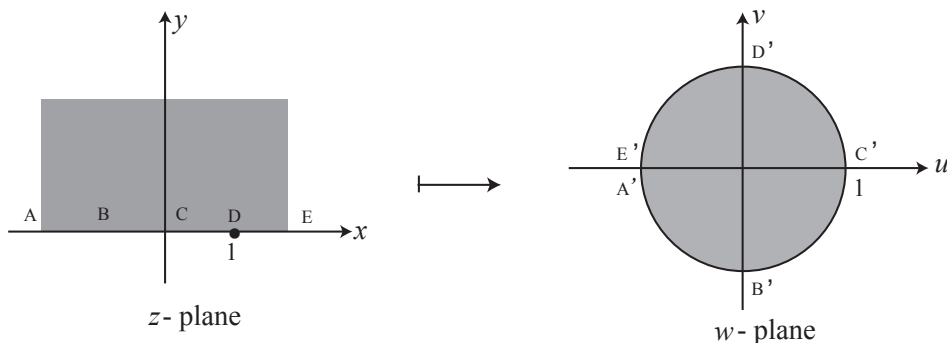


Figure 3.25 Upper half-plane onto the unit disk.

Map 3.25. The transformation

$$w(z) = i \frac{1 - z}{1 + z}$$

maps the unit disk onto the upper half-plane $\Im\{w\} > 0$.

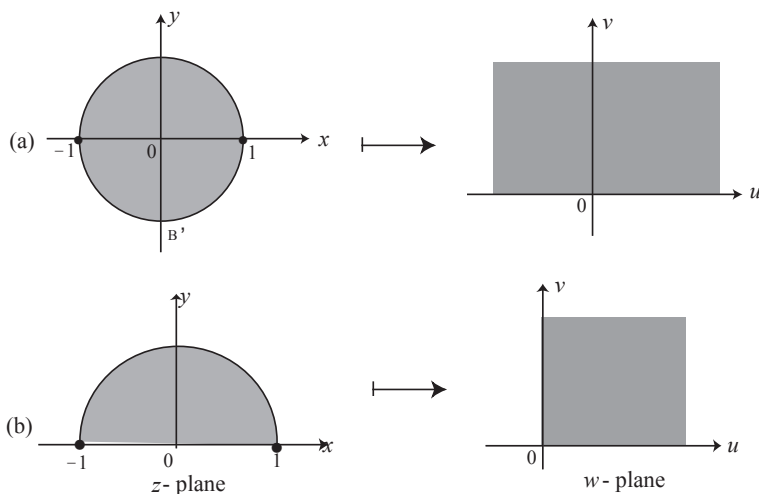


Figure 3.26 $w = i \frac{1 - z}{1 + z}$.

Since the lower boundary of the semi-disk, i.e., the interval $[-1, 1]$, is perpendicular to the upper semi-circle at the point 1, and since w is conformal at $z = 1$, the images of the interval $[-1, 1]$ and the semi-circle intersect at right angle. Since they both pass through the point -1 , which is mapped onto ∞ , we conclude that these images are perpendicular lines. Thus, using $w(0) = i$ and $w(i) = 1$, the interval $[-1, 1]$ in the z -plane is mapped onto the upper half of the imaginary v -axis, and the semi-circle is mapped on the right half of the real u -axis. Finally, testing the image of an interior point, say $i/2$, we find that $w(i/2) = 4/5 + 3i/5$, which is a point in the first quadrant. Hence, the upper semi-disk in the z -plane is mapped onto the first quadrant $\{u > 0, v > 0\}$ of the w -plane, since the

boundary is mapped onto boundary and interior points onto interior points (Figure 3.26).

Map 3.26. The transformation

$$w = -i \frac{z+i}{z-i}, \quad \text{or} \quad z = i \frac{w-i}{w+i},$$

maps the points $z = i, -i, 0$, respectively, onto the points $w = \infty, 0$, and a point with $v > 0$. Thus, this map guarantees that the circle in the z -plane transforms into a straight line, which passes through the origin $w = 0$ and the interior of the circle is mapped onto the upper half of the w -plane.

Map 3.27. The bilinear transformation $w = f(z) = \frac{z}{2z-8}$ maps the region $\text{Int}(\Gamma)$, where $\Gamma = \{|z-2| = 2\}$, conformally onto the region $\Re\{w\} < 0$, such that the point $z = 2$ goes into $w = -1/2$ (Saff and Snider [1976:317]).

Map 3.28. The bilinear transformations $w = \frac{z}{z-1}$, $z = \frac{w}{w-1}$ are involutory, with $z_\infty = 1, w_\infty = 1$, the fixed-points $F_1 = 0, F_2 = 2$, and p, q real. The interior (exterior) of circles of D and E map onto exterior (interior) of circles. The details of different particular cases, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The half-plane $y \geq 0$ is mapped onto the half-plane $v \leq 0$.
- (b) The half-plane $x \geq 0$ is mapped onto the region $|w - \frac{1}{2}| \geq \frac{1}{2}$.
- (c) The region $|x| \leq 1$ is mapped onto the half-plane $u \leq \frac{1}{2}$.
- (d) The half-plane $x \leq \frac{1}{2}$ is mapped onto the region $|w| \leq 1$.
- (e) The line $x = 1$ is mapped onto the line $u = 1$, passing through $w = 1$.
- (f) The line $x = p, p \neq 1$ is mapped onto the circle $|w - \frac{2p-1}{2(p-1)}| = \frac{1}{|2p-2|}$ passing through $w = 1$.
- (g) The line $y = q$ is mapped onto the circle $|w - (1 - \frac{1}{2q})| = \frac{1}{2|q|}$ passing through $w = 1$.

Map 3.29. The function $w = f(z) = \frac{z}{z-(1+i)}$ maps the lens-shaped region bounded by the circles $|z-1| = 1$ and $|z-i| = 1$ onto the region bounded by the rays $\arg\{w\} = 3\pi/4$ and $\arg\{w\} = 5\pi/4$ such that $f(0) = 0$ and $f(1+i) = \infty$.

Map 3.30. The transformation $w = \frac{z}{iz+2}$ maps the disk $|z-i| = 1$ onto the real line. To see this, since the points $z = 0, 1+i, 2i$ lie on the given circle in the z -plane and since the real line in the w -plane passes through the points $w = 0, 1, \infty$, so we choose $h(z)$ such that $h(0) = 0, h(1+i) = 1$, and $h(2i) = \infty$. Then $h(z) = \frac{z}{iz+2}$. Since $h(i) = i$, so h maps the disk $|z-i| < 1$ onto the upper half-plane.

Map 3.31. CIRCLE ONTO CIRCLE. The transformation is

$$w - w_0 = \Re \left\{ e^{i\tau} \frac{z - z_0 - r\alpha}{\bar{\alpha}z - \alpha z_0 - r} \right\}, \quad (3.3.24)$$

where τ is real, $|\alpha| \neq 1$, and is presented in Figure 3.27.

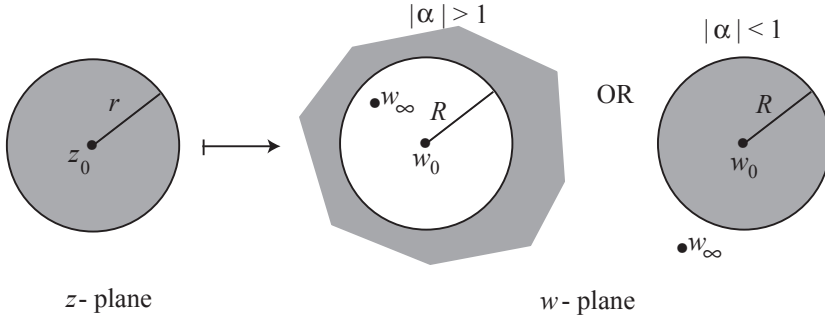


Figure 3.27 Circle onto circle.

Map 3.32. Mapping of the unit circle onto itself:

$$w = e^{i\tau} \frac{z - \alpha}{\bar{\alpha}z - 1}, \quad \tau \text{ real and } |\tau| \neq 1. \quad (3.3.25)$$

Map 3.33. The transformation $w = e^{i\theta} \frac{z - z_0}{1 - \bar{z}_0 z}$, which maps a circle onto the circle in the w -plane such that the point z_0 maps into the point $w = 0$, is shown in Figure 3.28.

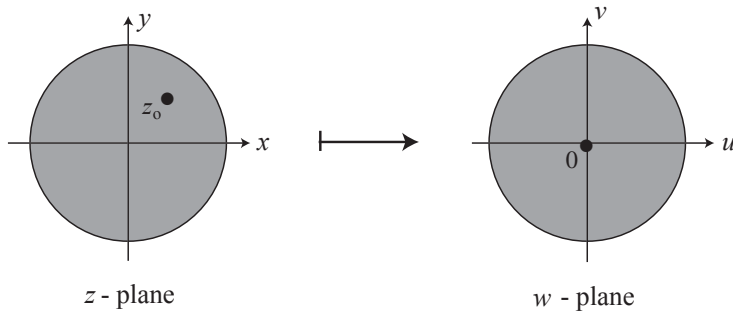


Figure 3.28 Circle onto a circle.

Map 3.34. The function $f(z) = \rho e^{i\alpha} \frac{z - z_1}{z - z_2}$ maps the region enclosed by two arcs onto the region enclosed by the sector shown in Figure 3.29, such that a fixed point $\zeta \neq z_{1,2}$ on Γ_1 goes into a point ω on the u -axis, where $\alpha = \arg \left\{ \frac{\zeta - z_1}{\zeta - z_2} \right\}$ and $\rho = \omega \left| \frac{\zeta - z_1}{\zeta - z_2} \right|$ (Pennisi [1963:321]).

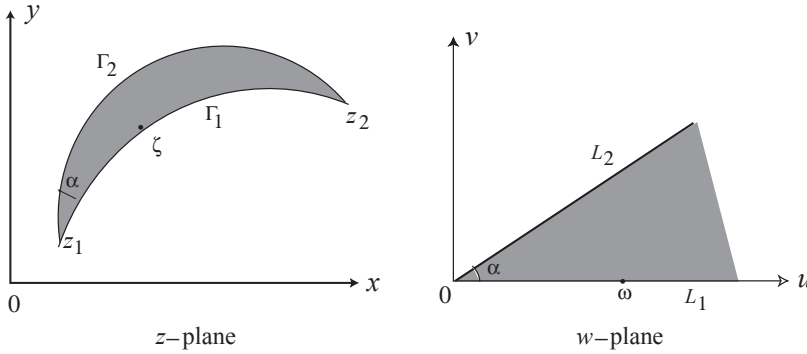
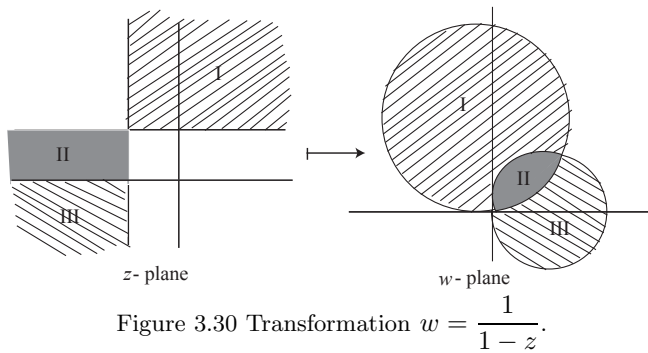


Figure 3.29 Crescent-shaped region.

Map 3.35. The transformation $w = \frac{1}{1-z}$, $z = \frac{w-1}{w}$, with $z_\infty = 1$; $w_\infty = 0$; the fixed points $F_1 = \frac{1}{2} + \frac{i}{2}\sqrt{3}$, and $F_2 = \frac{1}{F_1}$, and p, q, P, Q are real. The interior of the circle in the z -plane is mapped onto the interior of the circle in the w -plane. The details of the transformation, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The half-plane $y \geq 0$ is mapped onto the half-plane $v \geq 0$.
- (b) The half-plane $x \geq 0$ is mapped onto the region $|w - \frac{1}{2}| \geq \frac{1}{2}$.
- (c) The region $|z| \leq 1$ is mapped onto the half-plane $u \geq \frac{1}{2}$.
- (d) The region $|z - 1| \leq 1$ is mapped onto the region $|w| \geq 1$.

Figure 3.30 Transformation $w = \frac{1}{1-z}$.

There are two special cases, one presented in Figure 3.30, and the other in Figure 3.31; their respective details, with $z = p, q$ real and $w = P, Q$ real, are as follows:

For Figure 3.30:

- (a) The line $x = 1$ is mapped onto the line $u = 0$.
- (b) The line $x = p \neq 1$ is mapped onto the circle $\left|w - \frac{1}{2(1-p)}\right| = \frac{1}{2|1-p|}$.

- (c) The line $y = q \neq 0$ is mapped onto the circle $\left|w - \frac{1}{2q}\right| = \frac{1}{2|q|}$.

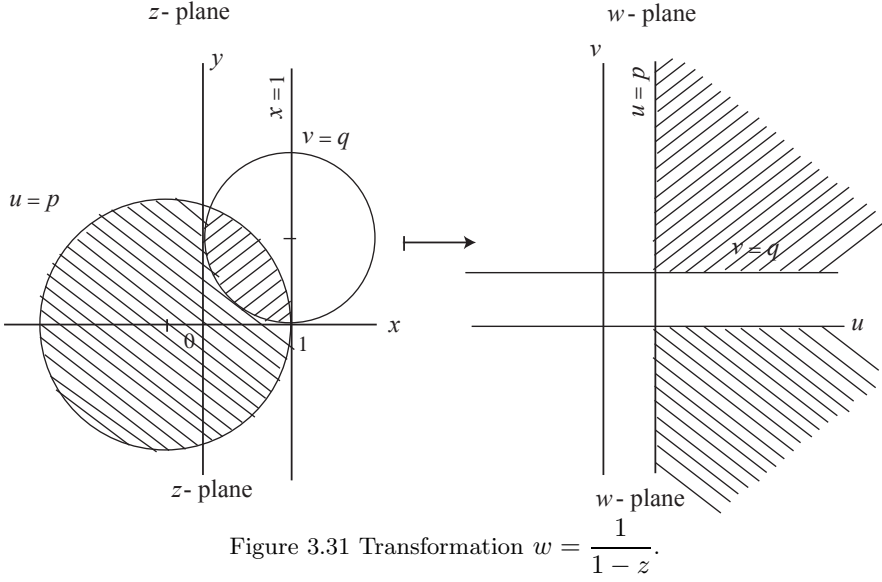


Figure 3.31 Transformation $w = \frac{1}{1-z}$.

For Figure 3.31:

- (a) The circle $\left|z - \left(1 - \frac{1}{2p}\right)\right| = \frac{1}{2|p|}$ is mapped onto the line $u = P \neq 0$.
 (ib) The circle $\left|z - \left(1 + \frac{1}{2q}\right)\right| = \frac{1}{2|q|}$ is mapped onto the line $v = Q \neq 0$.

Map 3.36. The transformation $w = \frac{1}{z} - a$, a real, maps the line $y = 0$ and the circle $|z + \frac{1}{2}i| = \frac{1}{2}$ onto the lines $v = 0$ and $v = 1$, respectively.

Map 3.37. The transformation $w = \frac{2ir}{z} - a$, a real, maps the line $x = 0$ and the circle $|z - r| = r > 0$ onto the lines $v = 1$ and $v = 1$, respectively. ■

Map 3.38. The transformation is

$$w = \frac{r_2}{r_2 - r_1} \left(\frac{2r_1}{z} + i \right). \quad (3.3.26)$$

It is presented in Figure 3.32.

The details of the transformation for this case with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The points $z = 0; \infty; 2ir_1$ are mapped onto the points $w = \infty; \frac{ir_2}{r_2 - r_1}; 0$.
 (b) The circle $|z - ir_1| = r_1$ is mapped onto the line $v = 0$.
 (c) The circle $|z - ir_2| = r_2$ is mapped onto the line $v = 1$.

- (d) The line $x = 0$ is mapped onto the line $u = 0$.
- (e) The line $y = 0$ is mapped onto the line $v = \frac{r_2}{r_2 - r_1}$.
- (f) The point $z_3 = ir_1(1 - e^{i\theta})$ is mapped onto the point $w_3 = \frac{r_2}{r_2 - r_1} \cot \frac{\theta}{2}$.
- (g) The small circle touching both circles and passing through z_3 is mapped onto the circle $|w + w_3 - \frac{1}{2}i| = \frac{1}{2}$.

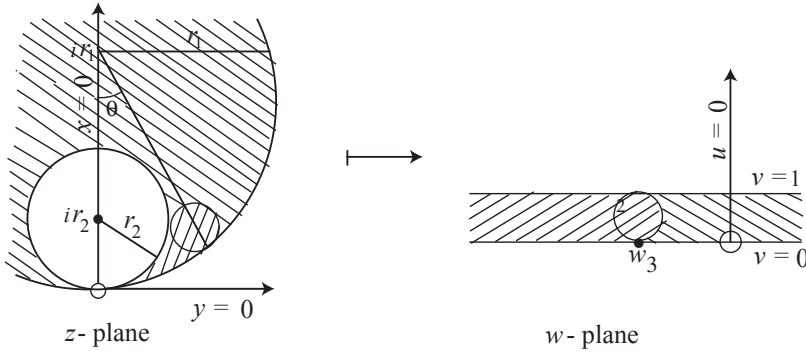


Figure 3.32 Two circles in outer contact.

Map 3.39. The transformation

$$w = \frac{ir_2}{r_1 + r_2} \left(1 - \frac{2r_1}{z}\right) \quad (3.3.27)$$

is presented in Figure 3.33.

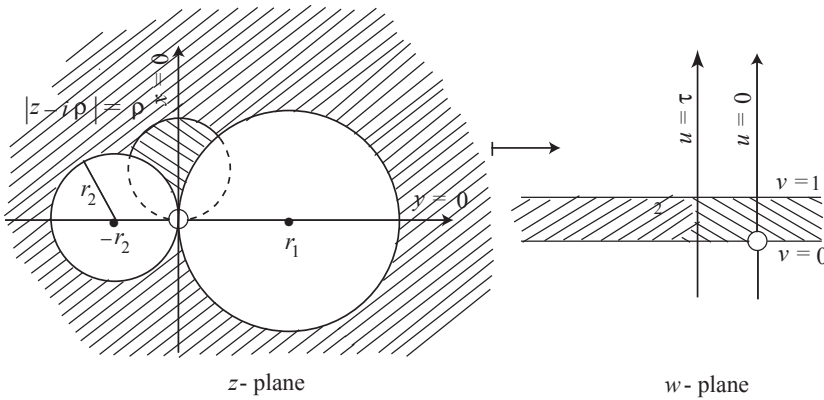


Figure 3.33 Two circles in outer contact.

The details of the transformation for this case with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The points $z = 0; \infty; 2r_1$ are mapped onto the points $w = \infty; \frac{ir_2}{r_1 + r_2}; 0$.
- (b) The circle $|z - r_1| = r_1$ is mapped onto the line $v = 0$.
- (c) The circle $|z + r_2| = r_2$ is mapped onto the line $v = 1$.

(d) The circle $|z - i\rho| = r_2$ is mapped onto the line $u = \tau$, where $\tau = -\frac{r_1 r_2}{\rho(r_1 + r_2)}$.

(e) The lines $x = 0$; $y = 0$ are mapped onto the lines $v = \frac{r_2}{r_2 - r_1}$; $u = 0$.

Map 3.40. The transformation that maps the upper-half of the unit circle in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is defined by

$$w = \left(\frac{1+z}{1-z} \right)^2,$$

and is presented in Figure 3.34.

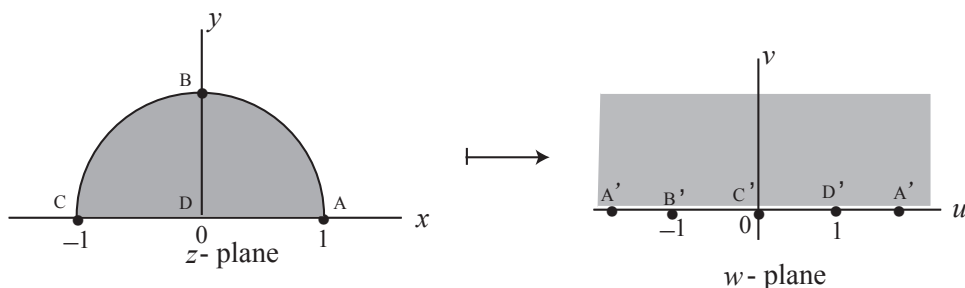


Figure 3.34 Upper-half of the unit circle onto the upper half-plane.

Map 3.41. The transformation that maps the upper half plane $\Im\{z\} > 0$ onto a round corner in the w -plane is $w = (z + 1)^\alpha + (z - 1)^\alpha$, $1 < \alpha < 2$; it is presented in Figure 3.35. The transformation maps the line segment $y = 0$, $-1 \leq x \leq 1$ in the z -plane onto the part $-2^\alpha \cos(\alpha\pi) \leq u \leq 2^\alpha$, $-2^\alpha \sin(\alpha - 1)\pi \leq v \leq 0$, i.e., the part DCB, of $(v \cos(\alpha\pi) - u \sin(\alpha\pi))^{1/\alpha} + (-v)^{1/\alpha} = 2(-\sin(\alpha\pi))^{1/\alpha}$. Note that for $\alpha = 3\pi/2$, the curve is the asteroide $u^{2/3} + (-v)^{2/3} = 2$.

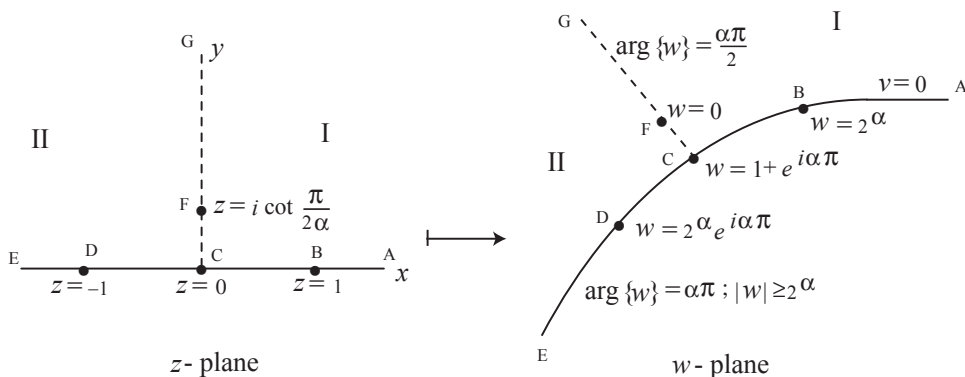


Figure 3.35 $\Im\{z\} > 0$ onto a round corner.

Map 3.42. The transformation that maps the sector of angle π/m of a circle in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is defined by

$$w = \left(\frac{1+z^m}{1-z^m} \right)^2,$$

and is presented in Figure 3.36.

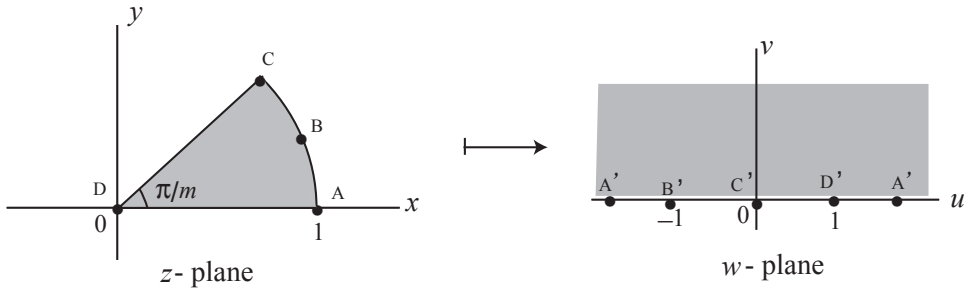


Figure 3.36 Sector of angle π/m of a circle onto the upper half-plane.

Map 3.43. The transformation

$$w = \frac{iz^2 + 1}{iz^2 - 1}, \quad (3.3.28)$$

conformally maps the upper half-plane $U = \Im\{z > 0\}$ onto the unit disk $C(0, 1)$. To prove it, first note that the transformation $z = g(w) = \frac{w-1}{w+1}$ (Map 3.17) maps the right half-plane $\Re\{w\} > 0$ onto $B(0, 1)$. Also, multiplication by $i = e^{i\pi/2}$, with $z = h(w) = iw$ rotates the complex plane by $\pi/2$ and thus maps the right half-plane $C(0, \frac{1}{2})$ onto the upper half-plane U , while its inverse $w = h^{-1}(z) = -iz$ will map U to the right half-plane. Hence, to map the upper half-plane U onto the unit disk, the composition of these two mappings gives the required map:

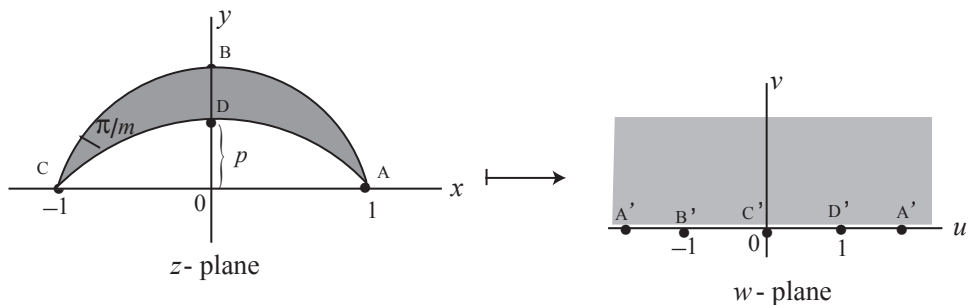
$$w = g \circ h^{-1}(z) = \frac{-iz - 1}{-iz + 1} = \frac{iz + 1}{iz - 1}. \quad (3.3.29)$$

Similarly, we already know (see Maps 4.6) that the square map $w = z^2$ maps the upper right quadrant $\{0 < \arg\{z\} < \pi/2\}$ onto the upper half-plane U . Thus, composition of this map with the previously constructed map, where we replace z by w in (3.3.29), we obtain the final required mapping (3.3.28).

Map 3.44. The transformation that maps the lens-shaped region of angle π/m , where this region is bounded by two circular arcs, in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is defined by

$$w = e^{2mi \operatorname{arccot}(p)} \left(\frac{z+1}{z-1} \right)^m, \quad (3.3.30)$$

and is presented in Figure 3.37.

Figure 3.37 Lens-shaped region of angle π/m onto the upper half-plane.

Map 3.45. The transformation $w = \frac{z-a}{az-1}$, where $a > 1$, $-1 < x_2 < x_1 < 1$, $R_0 > 1$, and

$$a = \frac{1 + x_1x_2 + \sqrt{(1-x_1^2)(1-x_2^2)}}{x_1 + x_2},$$

$$R_0 = \frac{1 - x_1x_2 + \sqrt{(1-x_1^2)(1-x_2^2)}}{x_1 + x_2}, \quad (3.3.31)$$

maps the region between the two circles in the z -plane onto the annular region in the w -plane, as shown in Figure 3.38.

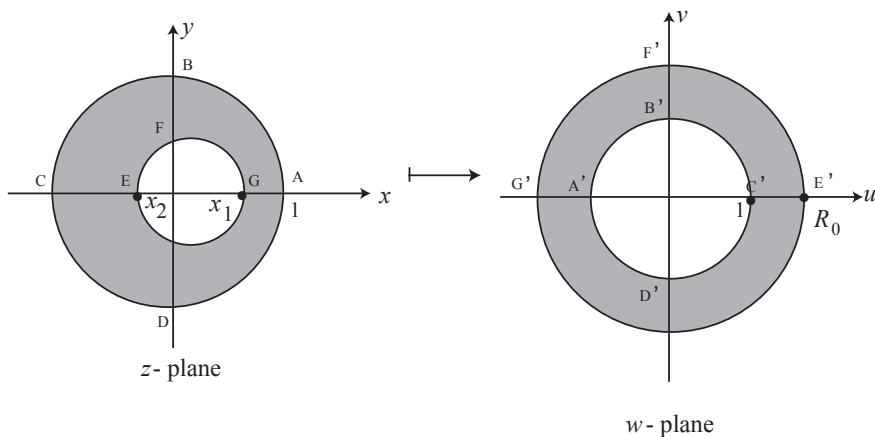


Figure 3.38 Map 3.45.

Map 3.46. The transformation $w = \frac{z-a}{az-1}$, where a and R_0 are the same as in (3.3.31), with $x_2 < a < x_1$, and $0 < R_0 < 1$ when $1 < x_2 < x_1$, maps the region exterior to the two circles in the z -plane onto the annular region in the w -plane, as shown in Figure 3.39.

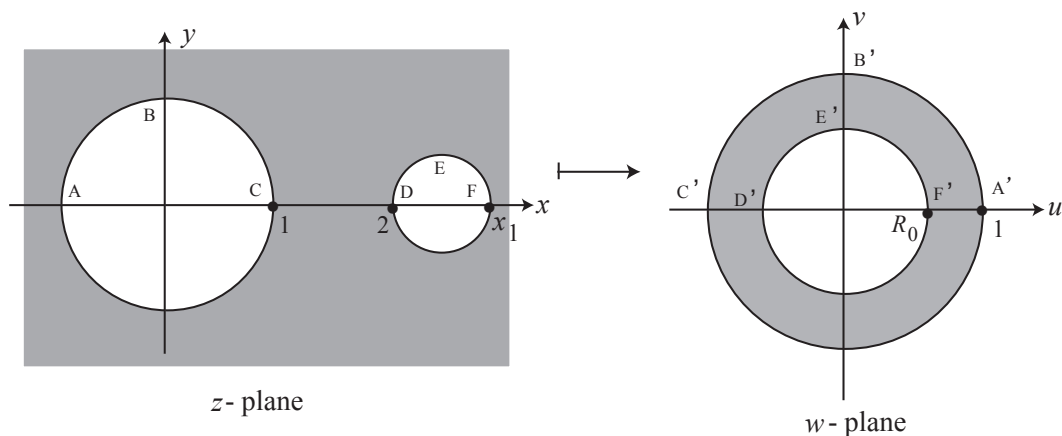


Figure 3.39 Map 3.46.

Map 3.47. The transformation $w = e^{i\lambda} \frac{z - z_0}{z + \bar{z}_0}$, where λ is a real number, maps the upper half-plane $-\infty < x < \infty, 0 \leq y < \infty$, in the z -plane onto the unit disk $|w| \leq 1$ in the w -plane.

Map 3.48. The transformation $w = e^{i\lambda} \frac{z - z_0}{1 - \bar{z}_0 z}$, where λ is a real number, maps the unit disk $x^2 + y^2 \leq 1$, in the z -plane onto the unit disk $|w| \leq 1$ in the w -plane.

Map 3.49. The transformation $w = e^{i\beta} \frac{z - \alpha}{\bar{\alpha}z - 1}$ for some $|\alpha| < 1$, $-\pi < \beta < \pi$, maps the unit disk onto itself.

Map 3.50. CASSINI'S OVALS. There are four cases:

Map 3.50a. Either part of the interior of Cassini's ovals $|z - z_1||z - z_2| = C$, with foci $z_1 = \pm\sqrt{w_1}$, is mapped onto the interior of the circle $|w - w_1| = C$, $C \leq |w_1|$. This case is presented in Figure 3.40, where w_1 is taken as a *real* and positive number.

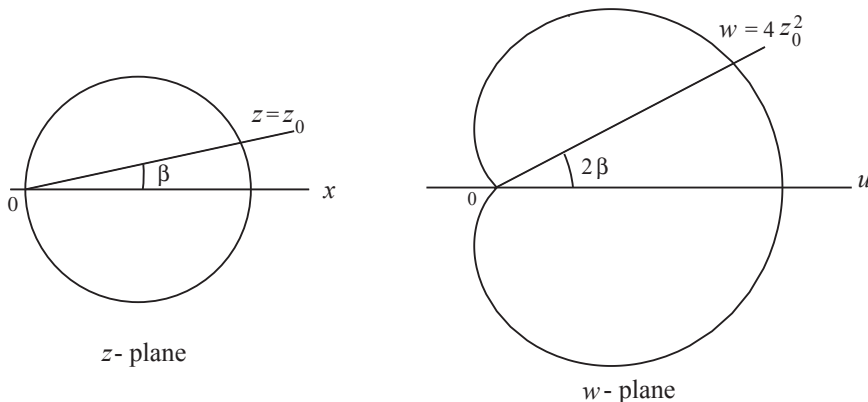


Figure 3.40 Cassini's ovals, Map 3.50a.

Map 3.50b. Cassini's ovals $|z - z_1||z - z_2| = C$, with foci $z_1 = \pm\sqrt{w_1}$, is mapped onto the circle $|w - w_1| = C$, $C > |w_1|$, counted twice.

Map 3.50c. The part $x > 0$ of the interior of Cassini's ovals, and the part $x < 0$ of the interior of Cassini's ovals, both are mapped onto the interior of the circle cut from $w = 0$ to $w' = \Re\{w_1\} = (C^2 - \Im\{w_1\}^2)^{1/2}$, i.e., point a . This case is presented in Figure 3.41.

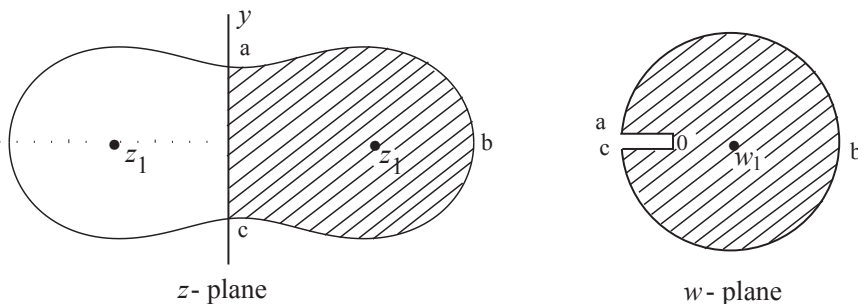


Figure 3.41 Cassini's ovals, Map 3.50c.

Map 3.50d. (Generalized Cassini's Ovals) The details of the transformation are as follows:

- (a) Each part of $|z - z_1||z - z_2| \cdots |z - z_n| = C$, with foci $z_j = e^{2i\pi j/n} \sqrt[n]{w_0}$, $j = 1, 2, \dots, n$, is mapped onto the circle $|w - w_0| = C$, $0 < C \leq |w_0|$.
- (b) Closed curve $|z - z_1||z - z_2| \cdots |z - z_n| = C$ (which is approximately circle for large C) is mapped onto the circle $|w - w_0| = C$, counted n times, $0 < |w_0| < C$.
- (c) Region bounded by this curve and the lines $z = r e^{i\theta}$, $z = r e^{i(\theta+2\theta/n)}$, θ fixed, is mapped onto the interior of $|w - w_0| = C$, $C > |w_0|$, cut from 0 to the point at which the line $w = r^n e^{in\theta}$ meets the circle.

This transformation for $n = 4$ is presented in Figure 3.42.

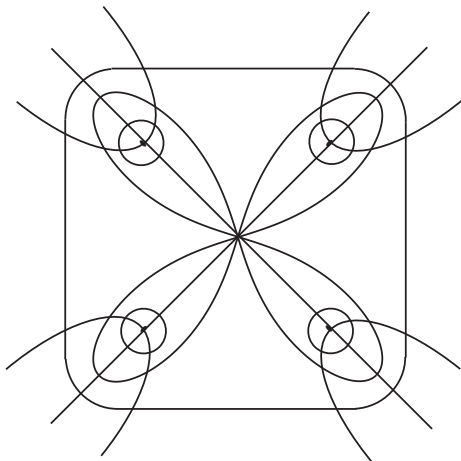


Figure 3.42 Cassini's ovals.

Map 3.51. CARDIOID AND LIMAÇON. The mapping from z -plane onto the w -plane is

presented in Figure 3.43.

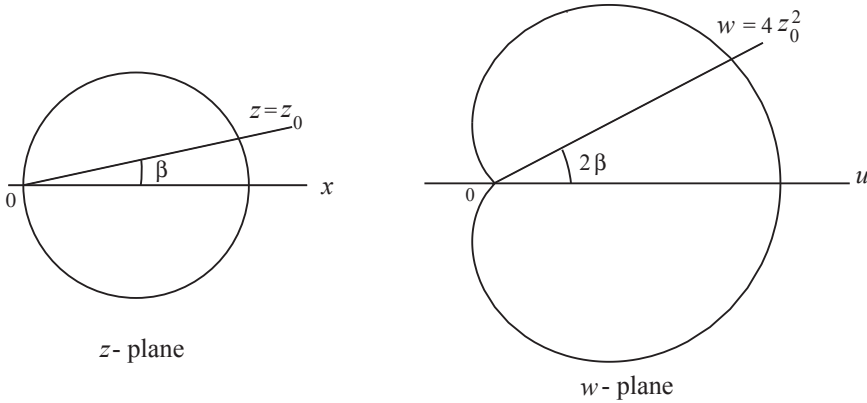


Figure 3.43 Circle onto cardioid.

The details of this transformation are as follows:

- (a) The circles $\left\{ \begin{array}{l} |z - z_0| = |z_0| \\ |z + z_0| = |z_0| \end{array} \right\}$ are mapped onto the cardioid $w = 2z_0^2(1 + \cos \theta)e^{i\theta}$, $-\pi \leq \theta < \pi$.
- (b) The circles $\left\{ \begin{array}{l} |z - z_0| = c \\ |z + z_0| = c \end{array} \right\}$, $c > 0, z_0 \neq 0$, are mapped onto the limaçon $w - w_1 = 2c(z_0 + c \cos \theta)e^{i\theta}$, $w_1 = z_0^2 - c^2$, $-\pi \leq \theta < \pi$.

Map 3.52. CARDIOID AND GENERALIZED CARDIOIDS ($\alpha > 1$), AND LEMNISCATES ($0 < \alpha < 1$) (Figure 3.44).

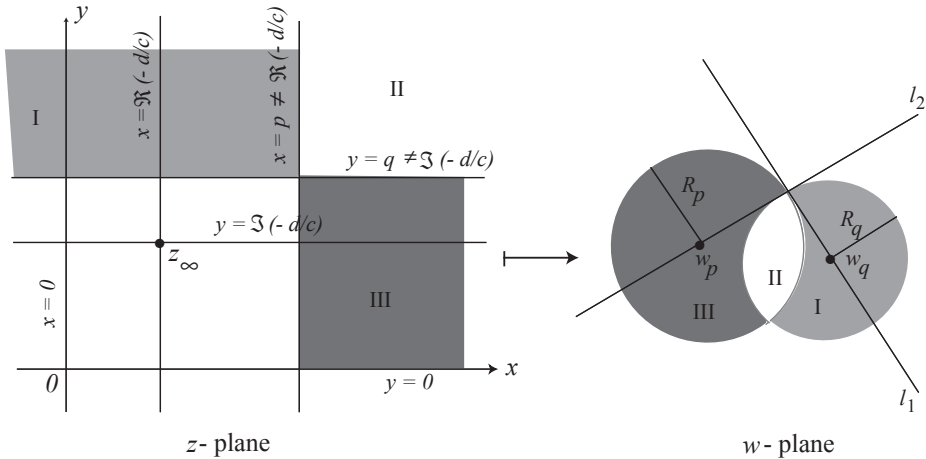


Figure 3.44 Generalized cardioid.

The details of this transformation are as follows:

The line $\Re\{e^{-i\theta} z\} = s$, $s > 0$ is mapped onto each part of $R^\beta = s^{-1} \cos(\beta\theta + \phi)$, $|\theta - \alpha(2k\pi - \phi)| \leq \alpha\pi/2$, $k = 0, \pm 1, \pm 2, \dots$; the curves are closed, with a cusp at 0, with tangents $w = R e^{i\alpha(k\pi + \frac{\pi}{2} - \phi)}$, $k = 0, \pm 1, \pm 2, \dots$; there are n parts if $\alpha^{-1} = n \in \mathbb{Z}$.

3.4 Straight Lines and Circles

We will consider different cases of linear and bilinear transformations that map given elements, like points, lines, circles, onto prescribed points, lines, and circles, using the cross-ratio. This classification of linear and bilinear transformations is based on Kober [1957:2-32]. A consequence of the Riemann mapping theorem is that simply connected domains D_z and D_w can always be mapped conformally onto each other. Three parameters may be chosen at will, based on the elements in w corresponding to similar elements in z , such as points, circles, rectangles, and so on. It is often convenient to start with a line, or a circle, or a rectangle (or square) in the w plane and create an image in the z -plane by applying various mapping functions $z = g(w)$, where $g(w) = f^{-1}(w)$ are the inverse of the ‘forward’ mapping function $w = f(z)$. Often a unit circle is used as an intermediate image in constructing sequential mappings. Some examples are as follows.

Map 3.53. LINES PARALLEL TO THE AXES. $z_\infty = -d/c$, $w_\infty = a/c$ (see Figure 3.45).

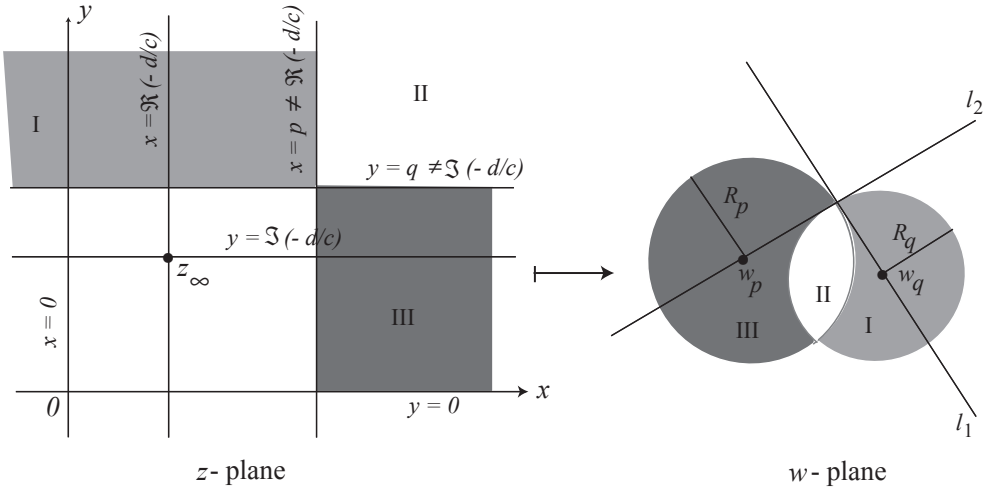


Figure 3.45 Lines parallel to axes (direct map).

Different cases of this transformation, with $z = p, q, r, r_p, r_q$ real and $w = P, Q, R, R_p, R_q$ real, are as follows:

- (a) The line $x = -\Re\{d/c\}$ is mapped onto the line $l_1 : \Re\{c(\bar{a}\bar{d} - \bar{b}\bar{c})(cw - a)\} = 0$.
- (b) The line $x = p, p \neq -\Re\{d/c\}$ is mapped onto the circle $|w - w_p| = R_p$, where $w_p = \frac{a\bar{c}p + \frac{1}{2}(a\bar{d} + b\bar{c})}{p|c|^2 + \Re\{cd\}}$, and $R_p = \left| \frac{ad - bc}{2p|c|^2 + 2\Re\{\bar{c}d\}} \right|$.
- (c) The line $y = -\Im\{d/c\}$ is mapped onto the line $l_2 : \Im\{c(\bar{a}\bar{d} - \bar{b}\bar{c})(cw - a)\} = 0$.
- (d) The line $y = q, q \neq -\Im\{d/c\}$ is mapped onto the circle $|w - w_q| = R_q$, where $w_q = \frac{a\bar{c}p + \frac{1}{2}i(a\bar{d} - b\bar{c})}{q|c|^2 + \Im\{\bar{c}d\}}$, and $R_q = \left| \frac{ad - bc}{2q|c|^2 + 2\Im\{\bar{c}d\}} \right|$.

The inverse mapping is presented in Figure 3.46.

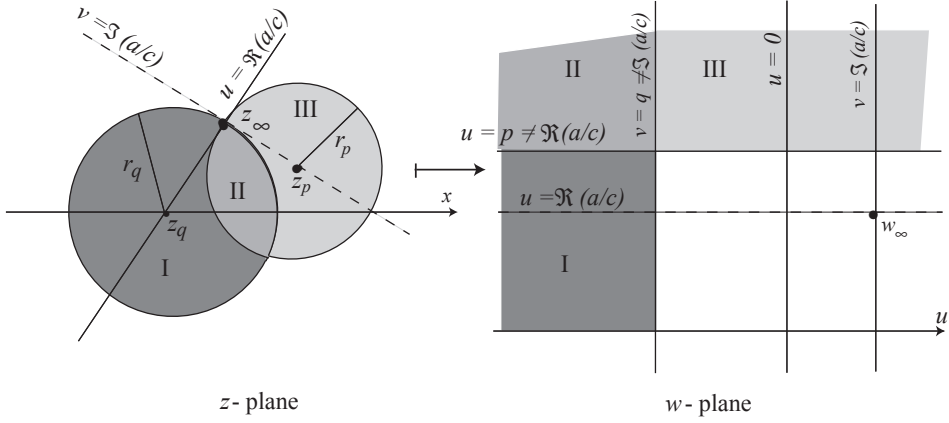


Figure 3.46 Lines parallel to axes (inverse map).

The details of different cases with $z = p, q, r, r_p, r_q$ real and $w = P, Q, R, R_p, R_q$ real, are as follows:

- (a) The line $\Re\{c(\bar{a}\bar{d} - \bar{b}\bar{c})(cz + d)\} = 0$ is mapped onto the line $u = \Re\{a/c\}$.
- (b) The circle with center $z_p = \frac{\frac{1}{2}(\bar{a}d + b\bar{c}) - \bar{c}dp}{p|c|^2 - \Re\{a\bar{c}\}}$ and radius $r_p = \left| \frac{ad - bc}{2P|c|^2 - 2\Re\{a\bar{c}\}} \right|$, is mapped onto the line $u = P, P \neq \Re\{a/c\}$.
- (c) The line $\Im\{c(\bar{a}\bar{d} - \bar{b}\bar{c})(cz + d)\} = 0$ is mapped onto the line $v = \Im\{a/c\}$.
- (d) The circle with center $z_q = \frac{\frac{1}{2}i(\bar{a}d - b\bar{c}) - \bar{c}dq}{q|c|^2 - 2\Im\{a\bar{c}\}}$ and radius $r_q = \left| \frac{ad - bc}{2q|c|^2 - 2\Im\{a\bar{c}\}} \right|$, is mapped onto the line $v = Q, Q \neq \Im\{a/c\}$.

Map 3.54. OTHER LINES AND CIRCLES. $z_\infty = -d/c, w_\infty = a/c$ (see Figure 3.47.)

The details of the transformation with $z = p, q, r, r_p, r_q$ real and $w = P, Q, R, R_p, R_q$ real, are as follows:

- (1a, b.) The line $\Re\{\bar{\lambda}z\} = p$ passing through z_∞ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P$ passing through w_∞ .
- (2.) The line $\Re\{\bar{\lambda}z\} = p$ not passing through z_∞ is mapped onto the circle $|w - w_0| = R$ passing through w_∞ .
- (3.) The circle $|z - z_0| = r$ passing through z_∞ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P$ not passing through w_∞ .

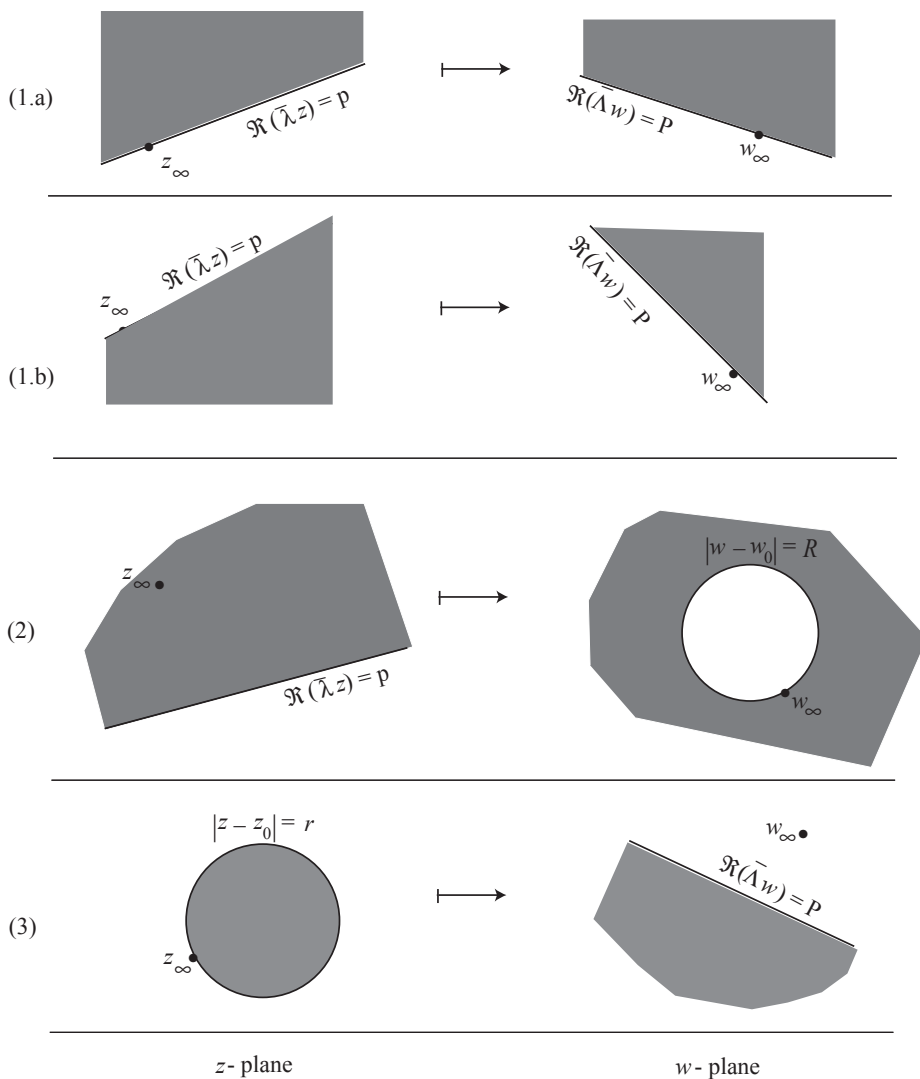
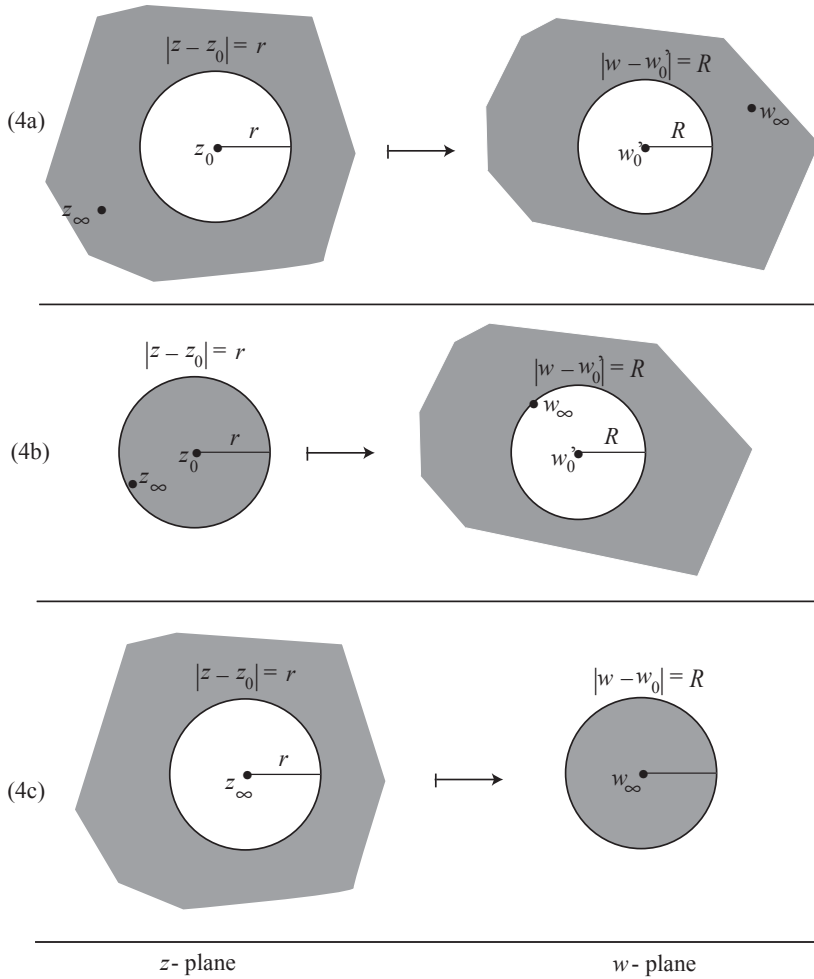


Figure 3.47 Other lines and circles (direct map).

Map 3.55. The circle $|z_0 + d/c| = r$ not passing through z_∞ is mapped onto the circle $|w - w_0| = R$ not passing through w_∞ (see Figure 3.48).

Figure 3.48 Circles not passing through z_∞ .

The details of the transformation with $z = p, q, r, r_p, r_q$ real and $w = P, Q, R, R_p, R_q$ real, are as follows, where

$$w'_0 = \frac{(az_0 + b)(\bar{a}\bar{z}_0 + \bar{d}) - a\bar{c}r^2}{|cz_0 + d|^2 - |c|^2r^2}, \quad R = \frac{r|ad - bd|}{|cz_0 + d|^2 - |c|^2r^2}.$$

(a) The circle $|z - z_0| = r$ not passing through z_∞ is mapped onto the circle $|w - w'_0| = R$ not passing through w_∞ .

(b) The circle $|cz_0 + d| > |c|r$ is mapped onto the circle $|w - w_0| = R$, where $w'_0 = \frac{az_0 + b - \bar{c}R^2s}{cz_0 + d}$, $R = r|s|$, and $s = \frac{ad - bc}{|cz_0 + d|^2 - |cr|^2}$.

(c) The circle $0 < |cz_0 + d| < |c|r$ is mapped onto the circle $|w - w_0| = R$, where w'_0 , and R , as in (4a).

(d) The point $z_0 - d/c = z_\infty$ is mapped onto the point $w_0 = a/c = w_\infty$; $R = \left| \frac{ad - bc}{c^2r} \right|$.

Map 3.56. THREE POINTS ONTO THREE POINTS. There are five cases (a)-(e) discussed below. Using the cross-ratio (3.3.1), the transformation is defined by (3.3.2). Then

Map 56a. This is defined by

$$\frac{w - w_1}{w - w_2} \frac{w_3 - w_2}{w_3 - w_1} = \frac{z - z_1}{z - z_2} \cdot \frac{z_3 - z_2}{z_3 - z_1}, \quad (3.4.1)$$

where z_1, z_2, z_3 are given and finite, and w_1, w_2, w_3 are finite. This is presented in Figure 3.49(a).

Map 3.56b. The transformation is of the form $\{w, w_1, w_2, w_3\} = \{z, z_1, z_2, \infty\}$, i.e.,

$$\frac{w - w_1}{w - w_2} \frac{w_3 - w_2}{w_3 - w_1} = \frac{z - z_1}{z - z_2}, \quad (3.4.2)$$

where the three given points are z_1, z_2, ∞ and w_1, w_2, w_3 . This case is presented in Figure 3.49(b).

Map 3.56c. The transformation is of the form

$$\frac{w - w_3}{w_3 - w_1} = \frac{z_2 - z_1}{z - z_2}, \quad (3.4.3)$$

where the three given points are z_1, z_2, ∞ and w_1, ∞, w_3 . This case is presented in Figure 3.49(c).

Map 3.56d. The transformation is of the form

$$\frac{w - w_1}{w_1 - w_2} = \frac{z - z_1}{z_1 - z_2}, \quad (3.4.4)$$

where the three given points are z_1, z_2, ∞ and w_1, w_2, ∞ . This case is presented in Figure 3.49(d).

Map 3.56e. The transformation is of the form

$$\frac{w - w_1}{w - w_2} = \frac{z - z_1}{z - z_2} \frac{z_3 - z_2}{z_3 - z_1}, \quad (3.4.5)$$

where the three given points are z_1, z_2, z_3 and w_1, w_2, ∞ . This case is presented in Figure 3.49(e).

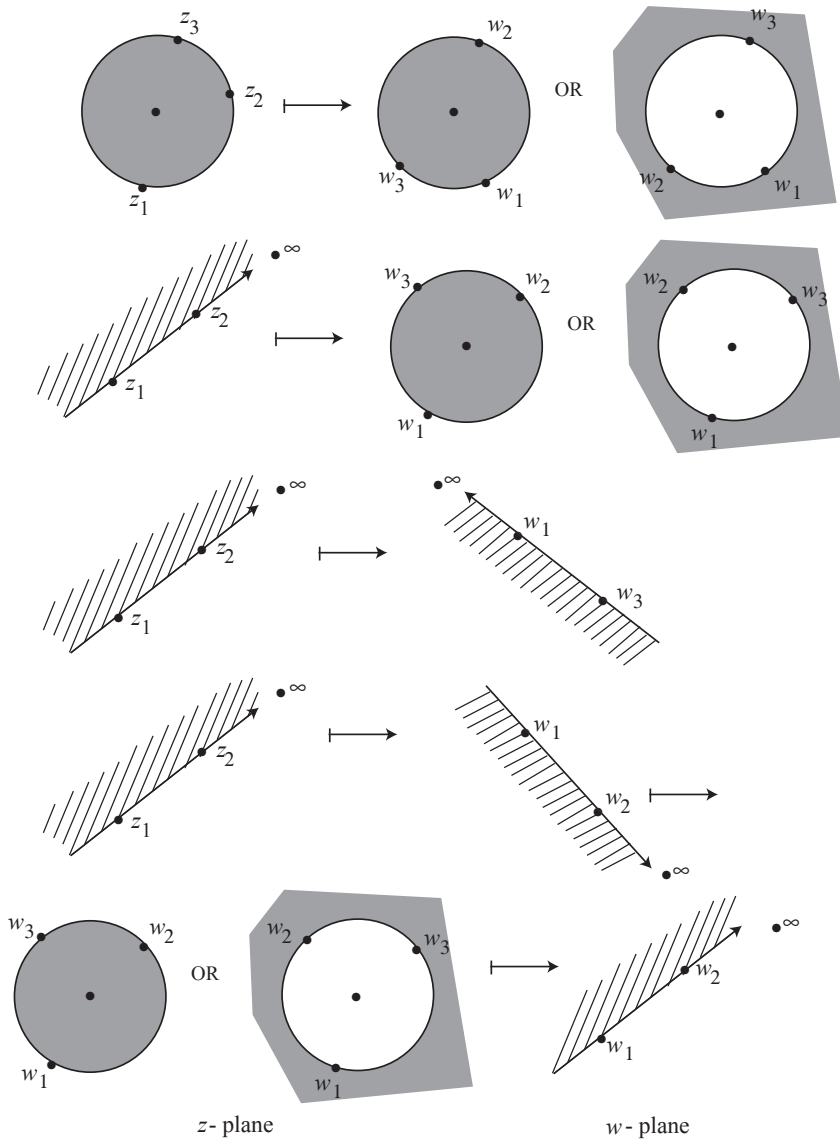


Figure 3.49 Three points onto three points.

Map 3.57. STRAIGHT LINE ONTO STRAIGHT LINE. The transformation is

$$\bar{\lambda}w = P + \frac{a(\bar{\lambda}z - p) - ib}{ic(\bar{\lambda}z - p) + d}, \quad (3.4.6)$$

where λ, p, Λ and P are preassigned. This transformation is presented in Figure 3.50(a). In the case of the mapping of a line onto itself, the above holds except that $\lambda = \Lambda$ and $p = P$, i.e., $\Re\{\bar{\lambda}z\} = p$ is the same as $\Re\{\bar{\Lambda}w\} = P$ (Figure 50(b)).

The details for the case 12a, with $z = p, q$ real and $w = P, Q$ real, are as follows:

If $ad - bd > 0$, then

- (a) the half-plane I is mapped onto the half-plane I; and
- (b) the half-plane II is mapped onto the region II.

If $ad - bc < 0$, then

- (c) the half-plane I is mapped onto the half-plane II.

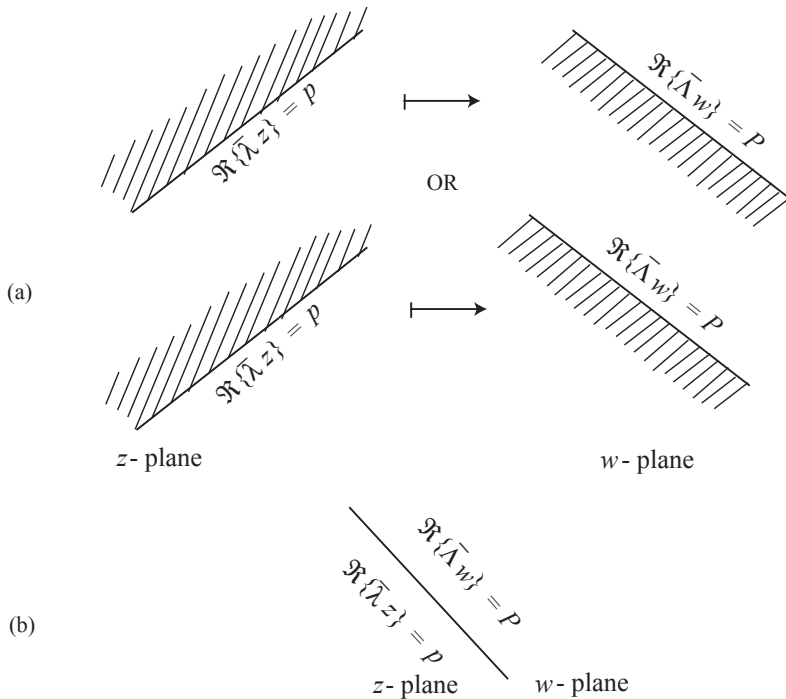


Figure 3.50 Straight line onto straight line.

Map 3.58. ANGLE ONTO ITSELF, WITH ARMS INTERCHANGED. Given, z_0 , $\alpha \neq 0$ ($-\pi < \alpha < \pi$), and τ , where α and τ are real, the case is presented in Figure 3.51. For involutory transformation $(w - w_0)(z - z_0) = b e^{i(\alpha + 2\tau)}$, $b > 0$, the regions I and II are mapped onto the regions I and II, respectively; but for $b < 0$, the region I is mapped onto the region II. Other details for this case, with $z = p, q$ real and $w = P, Q$ real, are as follows:

The points $z_0; z_1 = \infty; z_2 = z_0 + a e^{i\tau}$, where a is real, $a > 0$ in Figure 3.51, are mapped onto the points $w_0 = \infty; w_1 = z_0; w_2 = z_0 + \frac{b}{a} e^{i(\alpha + \tau)}$.

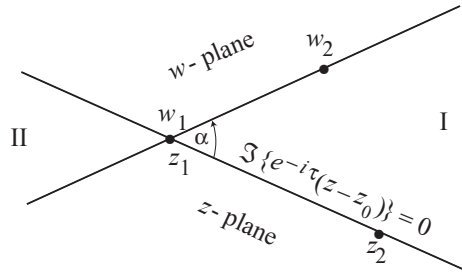


Figure 3.51 Angle onto itself.

Map 3.59. STRAIGHT LINE ONTO CIRCLE. The transformation is

$$w - w_0 = \Re \left\{ e^{i\tau} \frac{\bar{\lambda}z - p + \beta}{\lambda z - p - \beta} \right\}, \quad \tau \text{ real; } \Re\{\beta\} \neq 0, \quad (3.4.7)$$

given in Figure 3.52.

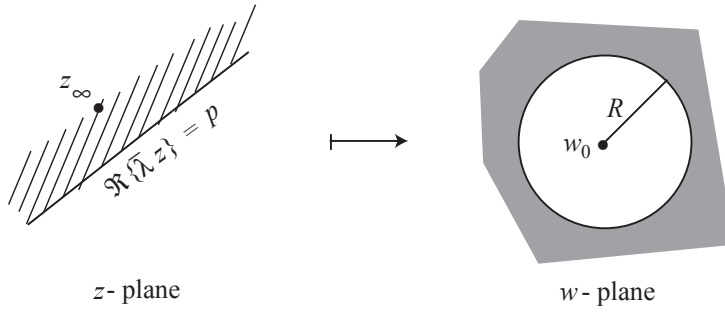


Figure 3.52 Straight line onto circle.

Map 3.60. CIRCLE ONTO STRAIGHT LINE. The transformation is

$$\bar{\Lambda}w = P + ia + b \frac{z - z_0 + r e^{i\tau}}{z - z_0 - r e^{i\tau}}, \quad (3.4.8)$$

where a, b, τ are real, $b \neq 0$, given in Figure 3.53.

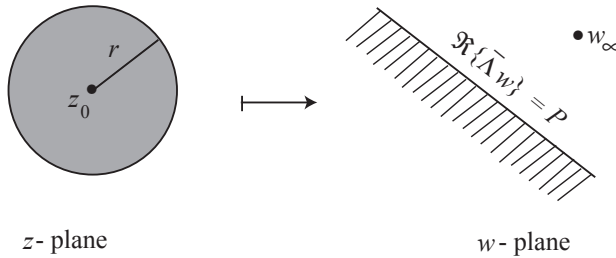


Figure 3.53 Circle onto straight line.

Map 3.61. CIRCLE AND LINE IN-CONTACT MAPPED ONTO TWO PARALLEL LINES. Given $z_0, z_2, \Lambda, P_1, P_2$, the transformation is

$$\bar{\Lambda}w = ia + P_1 + \frac{2(z_2 - z_0)(P_2 - P_1)}{z - z_0}, \quad (3.4.9)$$

where a is real, and $r = |z_2 - z_0|$, $\lambda = z_2 - z_0$, and is presented in Figure 3.54.

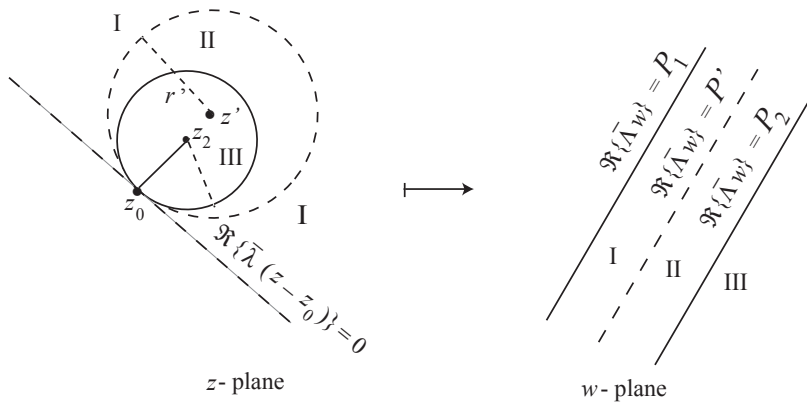


Figure 3.54 Circle onto circle.

The details of the transformation with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The line $\Re\{\bar{\lambda}z\} = \Re\{\bar{\lambda}z_0\}$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_1$.
- (b) The circle $|z - z_2| = r$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_2$.
- (c) The circle $|z - z'| = r'$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P' = P_1 + \frac{r}{r'}(p_2 - p_1)$.

Map 3.62. TWO CIRCLES IN-CONTACT MAPPED ONTO TWO PARALLEL LINES. (Inner Contact) Given $r_1 > 0, r_2 > 0, z_1, z_2$, and a real, the transformation is

$$\bar{\Lambda}w = P_1 + ia + \gamma \frac{z + z_0 - 2z_1}{z_0 - z}, \quad (3.4.10)$$

where $\gamma = r_2 \frac{P_2 - P_1}{r_1 - r_2}$. The details are provided in Figure 3.55 and where $z_0 = \frac{r_1 z_2 - r_2 z_1}{r_1 - r_2}$.

The details of the transformation with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The circle $|z - z_1| = r_1$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_1$.
- (b) The circle $|z - z_2| = r_2$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_2$.
- (c) The circle $|z - z'| = r'$ touching at z_0 is mapped onto the line $\Re\{\bar{\Lambda}w\} = P'$, where

$$P' = P_1 + (P_2 - P_1) \frac{r_2}{r_1} \frac{r' - r_1}{r_2 - r_1}.$$

- (d) The line $\Re\{\bar{\lambda}(z - z_0)\} = 0$ touching at z_0 , where $\lambda = z_1 - z_2$, is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_0$, where $P_0 = \frac{P_1 r_1 - P_2 r_2}{r_1 - r_2}$.

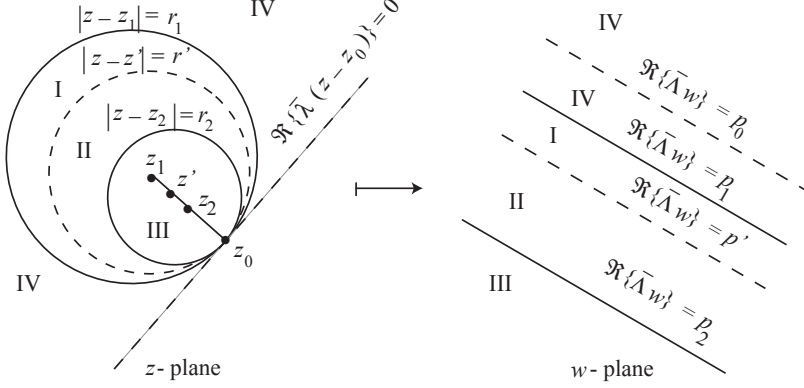


Figure 3.55 Two circles in contact mapped onto two parallel lines (inner contact).

Map 3.63. TWO CIRCLES IN OUTER CONTACT MAPPED ONTO PARALLEL LINES. (Outer Contact) Given $|z_1 - z_2| = r_1 + r_2$, transformation is

$$\bar{\Lambda}w = P_1 + ia + \gamma \frac{z + z_0 - 2z_1}{z_0 - z}, \quad (3.4.11)$$

where a is real, and $\gamma = r_2 \frac{P_1 - P_2}{r_1 + r_2}$. This case is presented in Figure 3.56, where $z_0 = \frac{r_1 z_2 + r_2 z_1}{r_1 + r_2}$.

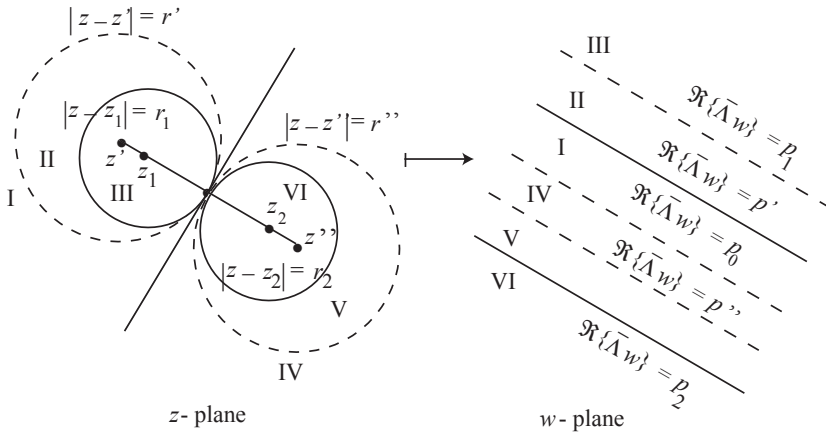


Figure 3.56 Two circles in outer contact mapped onto two parallel lines (outer contact).

The details of the transformation with $z = p, q$ real and $w = P, Q$ real, are as follows:

- The circle $|z - z_1| = r_1$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_1$.
- The circle $|z - z_2| = r_2$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_2$.
- circle $|z - z'| = r'$ touching at z_0 is mapped onto the line $\Re\{\bar{\Lambda}w\} = P'$, where $P' =$

$$P_2 + (P_1 - P_2) \frac{r_1}{r'} \frac{r' + r_2}{r_1 + r_2}.$$

- (d) The circle $|z - z''| = r''$ touching at z_0 , where $\lambda = z_1 - z_2$, is mapped onto the line $\Re\{\bar{\Lambda}w\} + P''$, where $P'' = P_1 + (P_2 - p_1) \frac{r_2}{r''} \frac{r'' + r_1}{r_1 + r_2}$.
- (e) The line $\Re\{\bar{\lambda}(z - z_0)\} = 0$ touching at z_0 , where $\lambda = z_1 - z_2$, is mapped onto the line $\Re\{\bar{\Lambda}w\} = P_0$, where $P_0 = \frac{P_1 r_1 + P_2 r_2}{r_1 + r_2}$.

Map 3.64. TWO ‘CIRCLES’, INTERSECTING AT TWO POINTS, MAPPED ONTO TWO INTERSECTING STRAIGHT LINES. Given $z_1; r_1 > 0; z_2; r_2 > 0$ and $\alpha = \beta$, i.e., $\frac{\bar{\Lambda}_1}{\bar{\Lambda}_2} \frac{z_0 - z_1}{z_0 - z_2}$ is real, the transformation is presented in Figure 3.57, where $|r_1 - r_2| < |z_1 - z_2| < r_1 + r_2$, and z_0, w_0 are the points of intersection of the z - and w -plane, respectively.

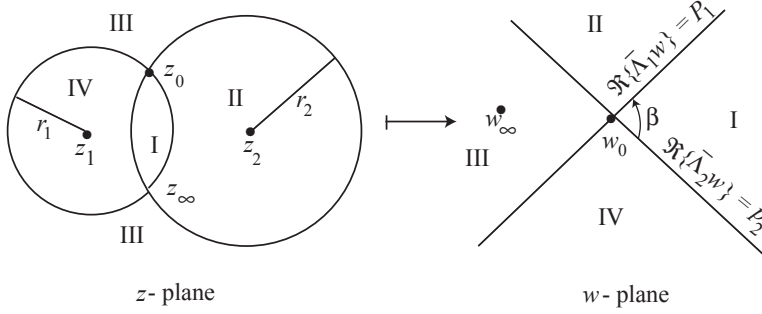


Figure 3.57 Two intersecting ‘circles’.

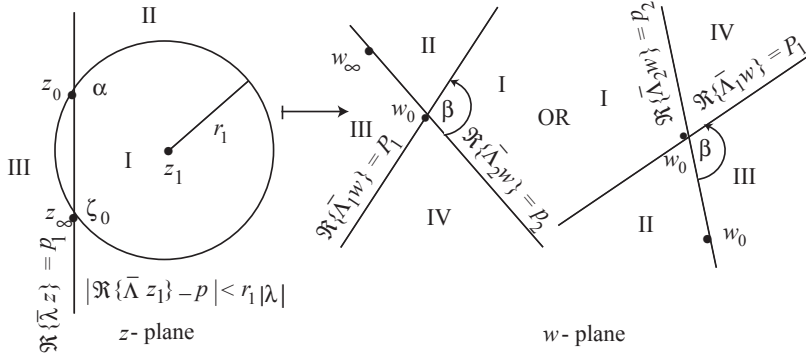


Figure 3.58 Two intersecting ‘circles’, alternative case.

An alternative situation is considered in Figure 3.58, where z_0 and ζ_0 are the points of intersection of the two ‘circles’ in the z -plane; w_0 is the point of intersection of $\Re\{\Lambda_1 w\} = P_1$ and $\Re\{\Lambda_2 w\} = P_2$. The required transformation is

$$w = w_0 + K \Lambda_1 \frac{(z - z_0)(z_1 - \zeta_0)}{(z - \zeta_0)(z_0 - \zeta_0)}, \quad K \neq 0 \text{ real.} \quad (3.4.12)$$

The details of the alternative case with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The points $z = 0; \infty; 2r_1$ is mapped onto the points $w = \infty; \frac{ir_2}{r_1 + r_2}; 0$.
- (b) The circle $|z - z_1| = r_1$ is mapped onto the line $\Re\{\bar{\Lambda}_1 w\} = P_1$.
- (c) The circle $|z - z_2| = r_2$ is mapped onto the line $\Re\{\bar{\Lambda}_2 w\} = P_2$.
- (d) The circle $|z + r_2| = r_2$ is mapped onto the line $\Re\{\bar{\Lambda}_2 w\} = P_2$, or line $\Re\{\bar{\lambda} z\} = p$.
- (e) The points $z = z_0; \zeta_0; z = \infty$ are mapped onto the points $w = w_0; \infty; w_\infty = w_0 + K\Lambda_1 \frac{z_1 - \zeta_0}{z_0 - \zeta_0}$.
- (f) The circle passing through z_0, ζ_0 is mapped onto the lines passing through w_0 .
- (g) The circle orthogonal to $|z - z_0| = r_1$, and to $|z - z_2| = r_2$ or to $\Re\{\bar{\lambda} z\} = p$ is mapped onto the circle with center w_0 .

Map 3.65. The transformation presented in Figure 3.59 is

$$w = \frac{K}{\sigma} \frac{z(i\sigma - z_1)}{z - i\sigma}, \quad \sigma > 0, K > 0. \quad (3.4.13)$$

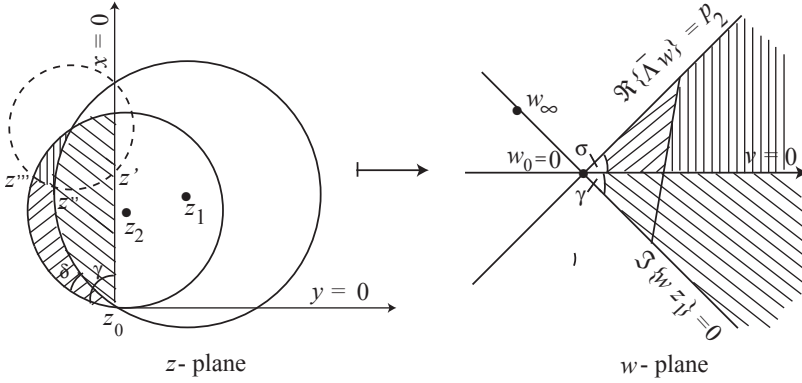


Figure 3.59 Two intersecting 'circles'.

The details of the transformation, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The points $\zeta_0 = i\sigma; z_0 = 0; z'; z''; z'''$ are mapped onto the points $w = \infty; w_0 = 0; w'; w''; w'''$.
- (b) The point $z = \infty$ is mapped onto the $w_\infty = \frac{K}{\sigma}(i\sigma - z_1) = -\frac{K}{\sigma}\bar{z}_1$.
- (c) The circle $|z - z_1| = r_1, \Im\{z_1\} = \frac{1}{2}\sigma$ is mapped onto the line $v = 0$.
- (d) The circle $|z - z_2| = r_2, \Im\{z_2\} = \frac{1}{2}\sigma$ is mapped onto the line $\Re\{\bar{\Lambda}_2 w\} = P_2$.
- (e) The line $x = 0$ is mapped onto the line $\Im\{w z_1\} = 0$.
- (f) The circle (z', z'', z''', ζ_0) is mapped onto the line (w', w'', w''') .
- (g) The curvilinear triangle $\{z_0, z', z''\}$ is mapped onto the right triangle $\{w, w', w''\}$.
- (h) The curvilinear triangle $\{z_0, z'', z'''\}$, each with sum of angles, is mapped onto the right triangle $\{w_0, w'', w'''\}$.

Map 3.66. CIRCLES AND STRAIGHT LINE, WITHOUT COMMON POINT, ONTO TWO CONCENTRIC CIRCLES. The transformation, presented in Figure 3.60, is

$$w - w_1 = R_1 e^{i\tau} \frac{(z - z_0)|\lambda| - \lambda(A + \sigma)}{(z - z_0)|\lambda| - \lambda(A - \sigma)}, \quad (3.4.14)$$

where τ is real, $A = \frac{p - \Re\{\bar{\lambda}z_0\}}{|\lambda|}$, and $\sigma = \sqrt{A^2 - \rho^2} > 0$ or $\sigma = -\sqrt{A^2 - \rho^2}$.

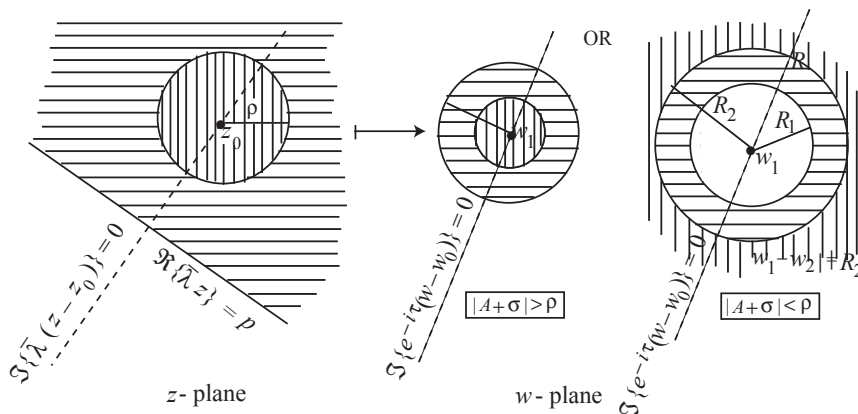


Figure 3.60 Circles and straight lines.

The details of the transformation, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The line $\Re\{\bar{\lambda}z\} = p$ is mapped onto the circle $|w - w_0| = R_1$.
- (b) The circle $|z - z_0| = \rho$ is mapped onto the circle $|w - w_1| = R_2$.
- (c) The line $\Im\{\bar{\lambda}(z - z_0)\} = 0$ is mapped onto the line $\Im\{e^{-i\tau}(w - w_1)\} = 0$.

Map 3.67. TWO CIRCLES, WITHOUT COMMON POINT, ONTO TWO CONCENTRIC CIRCLES. Given $z_1, z_2, r_1 > 0, r_2 > 0, z_1 \neq z_2$, and $w_0, R_1 > 0, R_2 > 0$, and $\frac{R_2}{R_1} = \frac{r_2}{r_1} \left| \frac{t}{s-t} \right|$, where s, t are the real roots of the equations $st = r_1^2$, and $(d-s)(d-t) = r_2^2, d = |z_2 - z_1| > 0$, the transformation is

$$w - w_0 = t \frac{R_1}{r_1} e^{i\theta} \frac{d(z - z_1) - s(z_2 - z_1)}{d(z - z_1) - t(z_2 - z_1)}, \quad \theta \text{ real.} \quad (3.4.15)$$

This transformation is presented Figure 3.61.

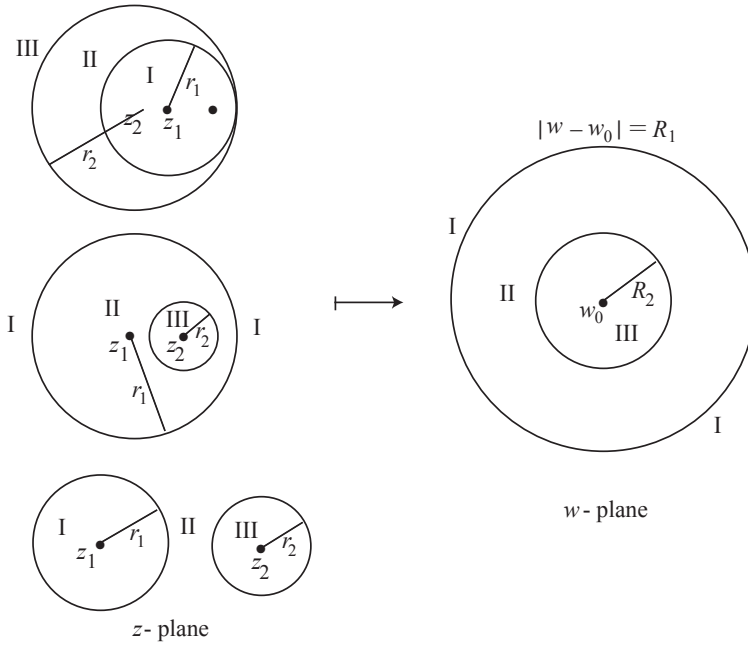


Figure 3.61 Transformation (3.2.35).

The details of this transformation, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The circle $|z - z_1| = r_1$ is mapped onto the circle $|w - w_0| = R_1$.
- (b) The circle $|z - z_2| = r_2$ is mapped onto the circle $|w - w_0| = R_2$.
- (c) The radical axis of these circles is mapped onto the circle $|w - w_0| = R_1|t|/r_1$.
- (d) The point $z_\infty = z_1 + (t/d)(z_2 - z_1)$ is mapped onto the point $w = \infty$.
- (e) The point $z_0 = z_1 + (s/d)(z_2 - z_1)$ is mapped onto the point $w = w_0$.
- (f) Any 'circle' passing through z_∞ and z_0 is mapped onto the line passing through w_0 .

Map 3.68. The transformation, presented in Figure 3.62, is

$$w = R_1 \frac{z - i\sigma}{z + i\sigma}, \quad (3.4.16)$$

where $\sigma = \sqrt{k^2 - \rho^2}$, $0 < \rho < k$.

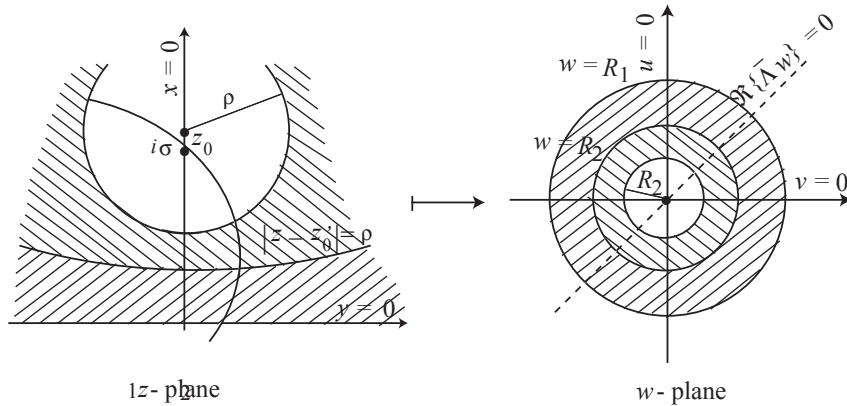


Figure 3.62 Transformation (3.2.33).

The details of the transformation, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The points $z = 0; \infty; i\sigma; -i\sigma$ are mapped onto the points $w = -R_1; R_1; 0; \infty$, respectively.
- (b) The line $x = 0$ is mapped onto the line $v = 0$.
- (c) The line $y = 0$ is mapped onto the circle $|w| = R_1$.
- (d) The circle $|z - ik| = \rho'$, where $z'_0 = i\sigma \frac{R_1^2 + R^2}{R_1^2 - R^2}$, $\rho' = \frac{2\sigma R_1 R}{|R_1^2 - R^2|}$, and $R_1 \neq R$, is mapped onto the circle $|w| = R_2 = \frac{R_1 \rho}{|k + \sigma|}$.
- (e) The circle $|z| = \sigma$ is mapped onto the line $u = 0$.
- (f) The circle $|z - \sigma \frac{\Im\{\Lambda\}}{\Re\{\Lambda\}}| = \sigma \left| \frac{\Lambda}{\Re\{\lambda\}} \right|$, where $\Re\{\Lambda\} \neq 0$, intersecting $x = 0$ at $\pm i\sigma$, is mapped onto the line $\Re\{\bar{\Lambda}w\} = 0$.

Map 3.69. The transformation

$$w = \frac{tR_1}{r_1} \frac{z - s}{z - t}, \quad (3.4.17)$$

where s and t are roots of the equations $st = r_1^2$ and $(z_2 - s)(z_2 - t) = r_2^2$, is presented in Figure 3.63.

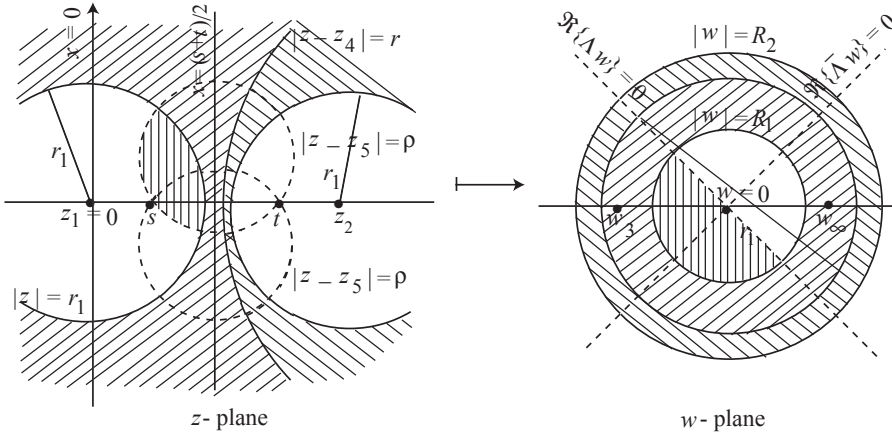


Figure 3.63 Transformation (3.4.17).

The details of this transformation, with $z = p, q$ real and $w = P, Q$ real, are as follows:

- (a) The circles $|z - z_{1,2}| = r_{1,2}$, $z_1 = 0, z_2 > 0$ are mapped onto the circles $|w| = R_{1,2}$; $\frac{R_2}{R_1} = \frac{r_2}{r_1} \frac{t}{r_2 - t}$.
- (b) The points $z_0 = s; z_\infty = t; z = \infty$; and $z_3 = \frac{s+t}{2}$ are mapped onto the points $w_0 = 0, \infty; w_\infty = \frac{tR_1}{r_1}$; and $w_3 = -\frac{tR_1}{r_1}$.
- (c) The circle $|z - z_0| = r$, where $z_4 = r_1^2 \frac{R_1^2 - R_2^2}{tR_1^2 - sR_1^2}$ and $r = \frac{r_1 R_1 R |t - s|}{|sR_1^2 - tR_1^2|}$, is mapped onto the circle $|w| = R \neq \infty$.
- (d) The line $x = \frac{s+t}{2}$ is mapped onto the circle $|w| = \frac{R_1 t}{r_1} = w_\infty$.
- (e) The circle $|z - z_5| = \rho$, where $z_5 = \frac{t\Lambda + s\bar{\Lambda}}{2\Re\{\Lambda\}}$ and $\rho = \left| \frac{\Lambda(s-t)}{2\Re\{\Lambda\}} \right|$, $\Re\{\Lambda\} \neq 0$ is mapped onto the line $\Re\{\bar{\Lambda}w\} = 0$.
- (f) The circle $|z - \bar{z}_5| = \rho$ is mapped onto the line $\Re\{\Lambda w\} = 0$.
- (g) The circle $y = 0$ is mapped onto the line $v = 0$.
- (h) The curvilinear rectangular quadrilateral formed by $|z| = r_1, |z - z_2| = r_2$ between $|w| = R_1, |w| = R$ and $|z - z_5| \leq \rho$, is mapped onto the half-ring left of $\Re\{\bar{\Lambda}\} = 0$.
- (i) The region $|z - z_2| \leq r_2$ is mapped onto the region $|w| \geq R_2$.

3.5 Ellipses and Hyperbolas

For ellipses and hyperbolas, we will set t, t' , and τ fixed, $0 < t < t' < 1$, and $0 < \tau < \pi/2$; also $l = t + 1/t$, $l' = t' + 1/t'$, and $\sigma = \arg\{k/\alpha\}$. The ellipse has been defined in §3.1.3, and the hyperbolas in §3.1.4.

Map 3.70. In the case of ellipses, we find that (i) the regions $|z| < |k/\alpha|t$ and $|z| > |k/\alpha|t^{-1}$ are both mapped onto the region $|w + 2k| + |w - 2k| > 2l|k|$ which is the exterior

of the ellipses; (ii) the annular regions $t|k/\alpha| < |z| < |k/\alpha|$ and $t^{-1}|k/\alpha| > |z| > |k/\alpha|$ are both mapped onto the region $2l|k| > |w + 2k| + |w - 2k| > 4|k|$ which is the exterior of the same ellipses, excluding the segment joining the foci; and (iii) the annular regions $t|k/\alpha| < |z| < t'|k/\alpha|$ and $t^{-1}|k/\alpha| > |z| > (1/t')|k/\alpha|$ are both mapped onto the region $2l|k| > |w + 2k| + |w - 2k| > 2l'|k|$ which is the annular region bounded by two confocal ellipses.

Map 3.71. The transformation $z = \frac{1}{2} \left[(a-b)w + \frac{a+b}{w} \right]$ maps the exterior of an ellipse with semi-axes a and b : $(x/a)^2 + (y/b)^2 \geq 1$, in the z -plane onto the unit disk $|w| \leq 1$ in the w -plane.

Map 3.72. In the case of hyperbolas, we find that (i) the half-lines $\arg\{z\} = \sigma + \tau$ and $\arg\{z\} = \sigma - \tau$ are both mapped onto the line $|w + 2k| - |w - 2k| = 4|k| \cos \tau$ which is one branch of the hyperbola, with asymptotes $\arg\{w\} = \arg\{k \pm \tau\}$; (ii) the half-lines $\arg\{z\} = \sigma + \tau + \pi$ and $\arg\{z\} = \sigma - \tau + \pi$ are both mapped onto the line $|w + 2k| - |w - 2k| = -4|k| \cos \tau$ which is the other branch of the hyperbola; (iii) the half-lines $\arg\{z\} = \sigma + \pi/2$ and $\arg\{z\} = \sigma - \pi/2$ are both mapped onto the line $\Re\{\bar{k}w\} = 0$; (iv) the half-line $\arg\{z\} = \sigma$ is mapped onto the half-line $|w| > 2|k|$, $\arg\{w/k\} = 0$, counted twice; and (v) the half-line $\arg\{z\} = \sigma + \pi$ is mapped onto the half-line $|w| > 2|k|$, $\arg\{w/k\} = \pi$, counted twice. These properties are presented in Figure 3.64.

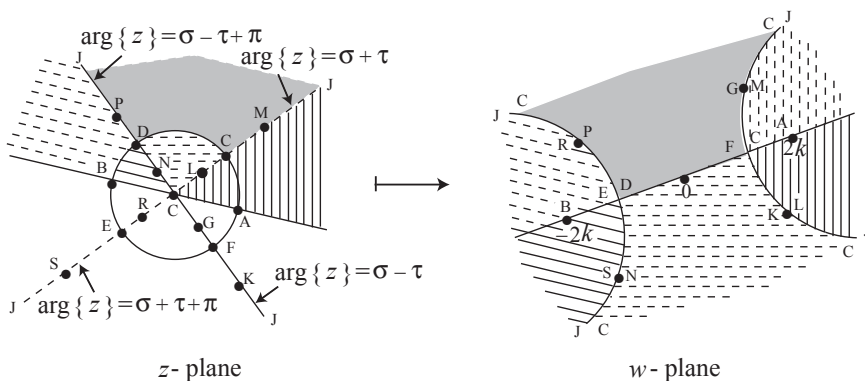


Figure 3.64 Mapping of different regions onto different regions of a hyperbola.

REFERENCES USED: Ahlfors [1966], Boas [1987], Carrier, Krook and Pearson [1966], Ivanov and Trubetskov [1995], Kantorovich and Krylov [1958], Kober [1957], Kythe [1996], Nehari [1952], Nevanlinna and Paatero [1969], Pennisi et al. [1963], Papamichael, Warby and Hough [1986], Saff and Snider [1976], Schinzinger and Laura [2003], Silverman [1967], Wen [1992].

7

Schwarz-Christoffel Transformations

As mentioned earlier, in conformal mapping we always try to determine a function f which maps a given region $D \subset \mathbb{C}$ conformally onto another region $G \subset \mathbb{C}$. However such a function f does not always exist, and it is not always uniquely determined. Recall that the Riemann mapping theorem guarantees the existence and uniqueness of a conformal map of D onto the unit disk U under certain specific conditions. In this chapter the Schwarz-Christoffel transformations are presented. There are three types of equivalent definitions for this transformation, and they all lead to integral equations which are then solved as boundary value problems.

7.1 Schwarz-Christoffel Transformations

The *Schwarz-Christoffel formula* helps solve an important class of boundary value problems that involve regions with polygonal boundaries. This transformation has an integral representation and is used to conformally map the interior of a given polygon of n sides ($n \geq 2$) in the w -plane onto the upper half-plane $\Im\{z\} > 0$ or the unit disk U in the z -plane.

Map 7.1. Let $\Gamma = \bigcup_{j=1}^n \Gamma'_j$ denote the sum of the sides Γ_j of the polygon $G = \text{Int}(\Gamma)$ in the w -plane with vertices w_j and **exterior angles $\pi\alpha_j$** , $|\alpha_j| < 1$, $\sum_{j=1}^n \alpha_j = 2$ for $j = 1, \dots, n$ (Figure 7.1). This transformation is defined by

$$\frac{dw}{dz} = C_1 (z - z_1)^{-\alpha_1} (z - z_2)^{-\alpha_2} \cdots (z - z_n)^{-\alpha_n}, \quad (7.1.1)$$

or

$$\begin{aligned} w(z) &= C_1 \int_{z_0}^z (t - z_1)^{-\alpha_1} (t - z_2)^{-\alpha_2} \cdots (t - z_n)^{-\alpha_n} dt + C_2 \\ &= C_1 \int_{z_0}^z \prod_{j=1}^n (t - z_j)^{-\alpha_j} dt + C_2, \end{aligned} \quad (7.1.2)$$

where the complex constant C_1 provides the scaling, and the complex constant C_2 gives the position of the orientation of the polygon, the integration is carried out along any path in D that joins $z_0 \in D$ to z , and the principal branch is used for the multiple-valued function $(t - z_j)^{-\alpha_j}$ in the integrand such that $0 < \arg\{t - z_j\} < \pi$, $j = 1, \dots, n$, for $\Im\{z\} > 0$. These branches are a direct analytic continuation into the upper half-plane of the real-valued functions $(x - x_j)^{-\alpha_j}$, where $x > x_j$. Then the integral (7.1.2) is a single-valued analytic function in the upper half-plane $\Im\{z\} > 0$, and the points x_j lying on the x -axis are the singularities of the Schwarz-Christoffel integral (7.1.2). This function $w = f(z)$ defines a conformal mapping of G onto D provided the points x_i are suitably chosen.

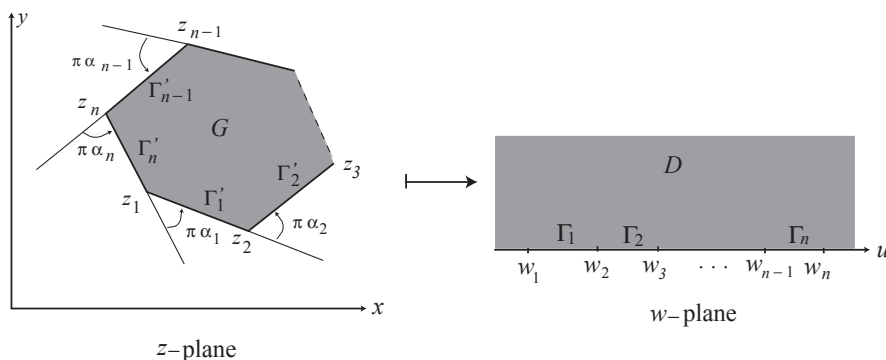


Figure 7.1 Schwarz-Christoffel transformation.

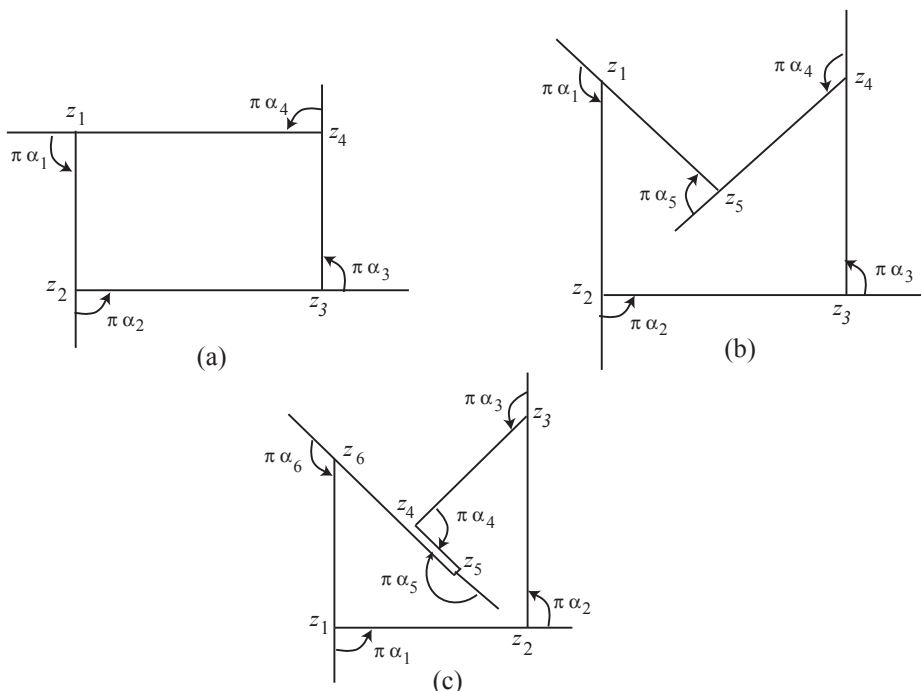


Figure 7.2 Signs of external angles.

The following facts about this transformation must be noted.

- (i) The sides of the polygon G must not intersect one another; self-contacts are permitted

and the vertices may lie at infinity. Thus, the sides must form the boundary of a simply-connected region G which is the interior of the polygon. If this region lies to the left when we traverse the sides from one vertex to another in order, then it is mapped onto the upper half-plane $y > 0$. The angles $\pi\alpha_j$ denote the change of direction when we pass through the vertex z_j to z_{j+1} . If $\alpha_{n+1} = 0$, then the polygon has n vertices $z_1, z_2, \dots, z_n$. Some cases are presented in Figure 7.2.

(ii) The angles $\pi\alpha_j$, which represent the change of direction at z_j (taken counterclockwise positive), are measured by traversing the sides in order so that the interior of the polygon lies to the right.

(iii) For change of direction when passing through a vertex at infinity, draw a sufficiently large circle that contains all the finite vertices in its interior. Instead of passing through a vertex, we pass from the 'side' p to the 'side' q along that arc which, joining p to q , lies in the interior of the polygon, and find the total change of direction.

(iv) Certain transformations are equivalent from the topological point of view. Thus, when $w = f(z)$ maps the interior G of the polygon conformally onto the upper half of the z -plane, then $\tilde{w} = \tilde{f}(z)$ has the same property iff it can be represented in the form

$$\tilde{w} = f\left(\frac{az + b}{cz + d}\right),$$

where a, b, c, d are real such that $ad - bc \neq 0$. For example, the transformation $w = e^z$ maps the strip $0 < v < \pi$ (i.e., a polygon with two vertices both at infinity, with $\alpha_1 = \alpha_2 = 1$, also called the *trigonal region*, see Map 7.3 below) onto $y > 0$.

(v) Not every differential equation of the form (7.1.1) defines a Schwarz-Christoffel transformation. For example, $\frac{dz}{dw} = w^{-1}(w-2)^{-\alpha_2}(w-3)^{-\alpha_3}$ does not, whenever $-2 \leq \alpha_2 + \alpha_3 < -1$. However, for every polygon defined as above, a Schwarz-Christoffel transformation can always be constructed.

In general, there are five major steps that can be used to determine the Schwarz-Christoffel transformation for a given mapping problem.

STEP 1. Label the vertices of the polygon in the z -plane as $A, B, \dots$, or $1, 2, \dots$. To each vertex j , assign the corresponding exterior angle $\pi\alpha_j$. Note that we have to turn at vertex j on the boundary (or to get back on the boundary if the polygon opens at infinity) while traversing in the counterclockwise direction. Check that the sum of all α_j equals 2.

STEP 2. Select a suitable location on the x -axis to correspond to any three or less vertices in the z -plane. The sets of three points, viz., $-1, 0, 1$; $0, 1, \infty$; or $-\infty, 0, 1$, are often chosen. Assign the vertices with exterior angles 2π or π to $-\infty$, and 0.

STEP 3. Formulate and solve the Schwarz-Christoffel integral equation (7.1.2) or (7.1.5), leaving the term corresponding to $\pm\infty$. Use boundary conditions on the polygon in the z -plane to determine the integration constants A and B as well as any x_j which were not specified in Step 2. The remaining points x_j (which total $n - 3$) can be determined from $n - 3$ ratios between the sides of the polygon which determine its shape. The length of a side is given by the Schwarz-Christoffel integral evaluated between the endpoints of the corresponding section of the x -axis.

STEP 4. The specific rules for the parallel lines are as follows: (i) If the transition between two parallel lines in the z -plane occurs at a location corresponding to $x_k = 0$ on the x -axis, then the constant B is determined from

$$d = -i\pi B \prod_{j \neq k} (0 - x_j)^{-\alpha_j}, \quad (7.1.3)$$

where d is the complex separation between the lines, excluding $x_j = \infty$. But if the transition is made to occur at $x = \infty$, then $d = i\pi B$.

STEP 5. Transform the x -axis configuration into the w -plane.

7.1.1 Triangles. Before we establish formulas (7.1.1) and (7.1.2), we will consider the case of a triangular region G . The mapping of the interior of a polygon onto the upper half-plane is defined by the Schwarz-Christoffel transformation which uses the angular change encountered at corners when one traces the boundary of the polygon counterclockwise. The domain D in the z -plane is mapped onto the domain G in the w -plane. The polygonal boundary of G is restricted to straight line segments Γ_i , $i = 1, 2, \dots, n$, in the w -plane.

Map 7.2. Let us choose an arbitrary point $A_1 : x = x_1$ on the x -axis in the segment marked Γ_1 (Figure 7.3(a), where the region G is a three-sided polygon). For $x < x_1$, we have $(x - x_1) = |x - x_0|e^{i\pi}$ since x is always smaller than x_1 and $\arg\{x - x_1\} = \pi$. As we move in the direction of increasing x , we find that $(x - x_1) = |x - x_0|e^{i0}$ as soon as we cross $x = x_1$ and enter the segment Γ_2 . Note that $\arg\{x - x_1\}$ has undergone a step change of $-\pi$ at the point A_2 , since we are tracing the polygonal contour in anticlockwise direction. Then the function $w - w_1 = (z - x_1)^{1/2}$ will map the upper half of the z -plane onto the interior of the right angle sector in the w -plane, and the mapping function $w = (z - x_1)^{1/2}$ is analytic at all values of z . Similarly, a function $w = (z - x_1)^{1/\rho}$ would produce a sector of angle π/ρ . The function $w = (z - x_1)^{1/\rho}$ produces a one-to-one mapping except at the values of w for which the derivative $\frac{dw}{dz} = \frac{1}{\rho}(z - x_1)^{(1/\rho)-1} = 0$. This can occur at $z = x_1$ or $x = \infty$ depending on the value of ρ . Problems arising from $x = \infty$ are handled by a procedure that will be explained later (see Eq (7.1.6)).

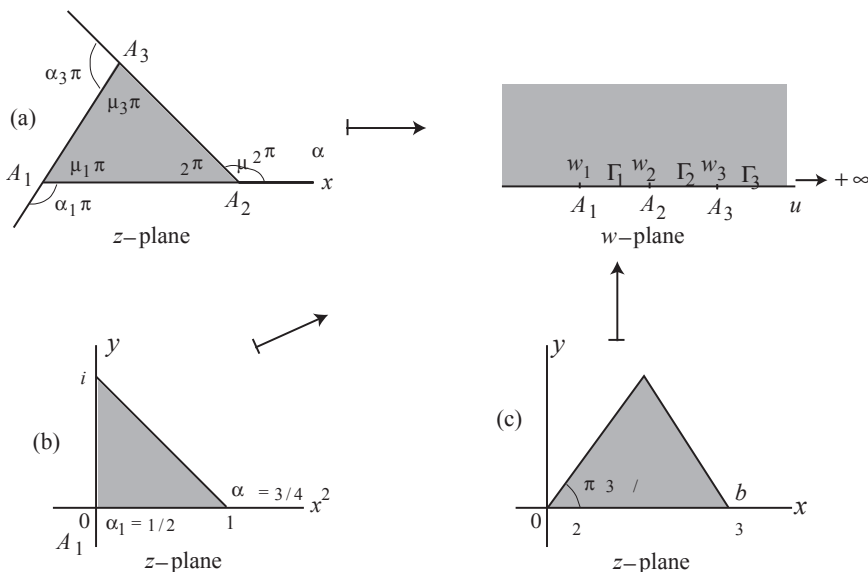


Figure 7.3 Mapping of triangles.

Since the Riemann mapping theorem permits us to choose three points, we take the

points A_1, A_2, A_3 along with segments $\Gamma_1, \Gamma_2, \Gamma_3$ on the x -axis. Thus, in general, we have the exterior angles $\pi\alpha_1, \pi\alpha_2, \pi\alpha_3$ with the corresponding interior angles $\pi - \pi\alpha_1 = \pi\mu_1, \pi - \pi\alpha_2 = \pi\mu_2, \pi - \pi\alpha_3 = \pi\mu_3$, respectively, at the points A_1, A_2, A_3 . As we trace the contour and pass into the segment Γ_2 , approaching the point A_2 , any increment dw has an argument equal to π/ρ . As we pass around A_2 into the segment Γ_3 , the dw -increments change their slopes to $\rho + \alpha_2$. Thus, the change in the argument of dw at A_2 is equal to $\pi\alpha_2$ and the interior angle π/ρ is equal to $\pi - \pi\alpha_2$. Hence $1/\rho - 1 = -\alpha_2$, which gives

$$\frac{dw}{dz} = \frac{1}{\rho} (z - z_2)^{(1/\rho)-1} = C_0 (z - z_2)^{-\alpha_2},$$

where $C_0 (= 1/\rho)$ is a complex constant that gives scaling. Following the above process for the third vertex, we find that the mapping is given by

$$\frac{dw}{dz} = C_0 (z - z_1)^{-\alpha_1} (z - z_2)^{-\alpha_2} (z - z_3)^{-\alpha_3}. \quad (7.1.4)$$

This is the required mapping of a regular triangular region onto the upper half-plane. The above argument can be continued to the n vertices of a regular polygon and establish the transformation given by (7.1.1).

Note that the sum of the exterior angles is 2π , while the sum of all interior angles is $(n - 2)\pi$.

If all the numbers x_j are finite, then the function $f(z)$ defined by (7.1.2) remains bounded in the neighborhood of the singularities x_i . To see that the Schwarz-Christoffel integral (7.1.2) remains bounded as $z \rightarrow \infty$, we rewrite this integral as

$$\begin{aligned} w = f(z) &= C_1 \int_{z_0}^z t^{-(\alpha_1 + \dots + \alpha_n)} \left(1 - \frac{x_1}{t}\right)^{-\alpha_1} \dots \left(1 - \frac{x_n}{t}\right)^{-\alpha_n} dt + C_2 \\ &= C_1 \int_{z_0}^z \frac{1}{t^2} \prod_{j=1}^n \left(1 - \frac{x_j}{t}\right)^{-\alpha_j} dt + C_2, \end{aligned} \quad (7.1.5)$$

which shows that the integral is convergent as $z \rightarrow \infty$. Hence, the Schwarz-Christoffel integral (7.1.2) defines a univalent function of z in $D = \Im\{z\} > 0$ that maps D conformally onto the bounded polygon G in the w -plane. If G is convex, then $\alpha_j > 0$. Thus, there are $(n - 1)$ independent numbers α_j , and $(n - 3)$ independent numbers x_j .

To see the structure of the boundary $\Gamma = \bigcup_{j=1}^n \Gamma_j$ of the polygon G , consider the derivative of the Schwarz-Christoffel integral (7.1.2), which is (7.1.1), i.e.,

$$f'(z) = C_1 (z - x_1)^{-\alpha_1} \dots (z - x_n)^{-\alpha_n}.$$

Notice that $f' \neq 0$ everywhere in $\Im\{z\} \geq 0$ except at the singularities x_i where it vanishes or becomes unbounded. As z varies along every interval $x_k < x < x_{k+1}$, $k = 1, \dots, n - 1$, the value of $\arg\{f'(z)\}$ does not change, since

$$\arg\{(x - x_k)^{-\alpha_k}\} = \begin{cases} -\pi\alpha_k, & \text{if } x < x_k, \\ 0, & \text{if } x > x_k. \end{cases}$$

Geometrically, $\arg\{f'(z)\}$ determines the size of the angle through which the tangent to a Jordan curve passing through x_k must be rotated in order to obtain the tangent of the image

of this curve at the point $w_k = f(x_k)$. Hence the segments $x_k < x < x_{k+1}$, $k = 1, \dots, n-1$, of the real axis are mapped by $f(z)$ into rectilinear segments in the w -plane. The points x_k of the real axis are transformed into the vertices w_k in the w -plane, where the polygon Γ is made up of the polygonal lines through $w_1, w_2, \dots, w_n$ with straight line segments as sides. As the point z traverses the entire real x -axis in the positive sense, the corresponding point w travels completely counterclockwise through the polygonal lines of Γ .

7.1.2 Size of the Angles. The size of the angle between adjacent segments of the polygon Γ can be determined as follows: Consider the variation of $\arg\{f'(z)\}$ as z passes through the point x_i in the positive direction (Figure 7.4). Then the angle between the vectors $\overrightarrow{w_{i-1}w_i}$ and $\overrightarrow{w_iw_{i+1}}$ is equal to $\pi\alpha_i$. For $\mu_i < 1$, where $\mu_i = 1 - \alpha_i$, the transition from the vector $\overrightarrow{w_{i-1}w_i}$ to the direction of the vector $\overrightarrow{w_iw_{i+1}}$ occurs in the positive sense (Figure 7.4(a)), whereas for $\mu_i > 1$ it occurs in the negative sense (Figure 7.4(b)), although the angle of transition in the positive sense in both cases from the direction of the vector $\overrightarrow{w_{i-1}w_i}$ to the direction of the vector $\overrightarrow{w_iw_{i+1}}$ is $\pi\alpha_i$. If a polygonal line does not intersect itself, it becomes the boundary of a closed polygon. Also the sum of all interior angles of this closed polygon is equal to $\sum_{i=1}^n \pi\mu_i = \sum_{i=1}^n \pi(1 - \alpha_i) = (n - 2)\pi$.

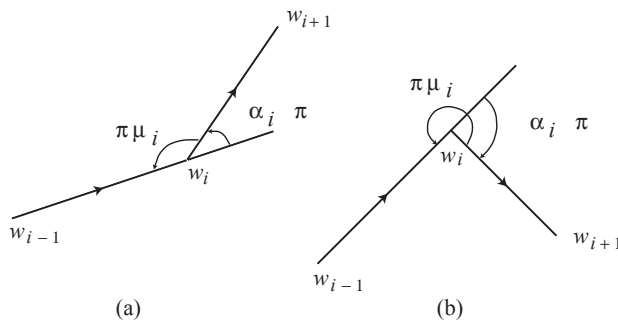


Figure 7.4 Interior and exterior angles.

For practical purposes, while constructing a conformal map of D onto G one can specify any three points x_i, x_j, x_k of the real axis that go into the three prescribed vertices w_i, w_j, w_k of the polygon Γ . When one of the points x_i , say x_n , coincides with the point at infinity, the vertices of the polygon Γ correspond to the points $x'_1, \dots, x'_{n-1}, \infty$, and the formula (7.1.2) becomes

$$w = f(z) = C'_1 \int_{z_0}^z \prod_{i=1}^{n-1} (t - x'_i)^{-\alpha_i} dt + C_2. \quad (7.1.6)$$

Note that formula (7.1.6) is similar to (7.1.2) except that the term corresponding to the point $x_n = \infty$ has been dropped. The inverse Schwarz-Christoffel transformation is given by

$$z(w) = C \int_{w_0}^w \prod_{i=1}^n (t - w_i)^{-\mu_i} dt + C_0. \quad (7.1.7)$$

Note that the mapping of the upper half z -plane onto the exterior of the polygon G is given by

$$z(w) = C_1 \int_{z_0}^z \prod_{j=1}^n (t - w_j)^{\alpha_j} \frac{dt}{(1 + t^2)} + C_2. \quad (7.1.8)$$

The exterior of the unit circle is mapped onto the exterior of the polygon G by

$$z(w) = C_1 \int_{w_0}^w \prod_{j=1}^n (t - z_j)^{\alpha_j} \frac{dt}{t^2} + C_2. \quad (7.1.9)$$

So far we have presented the mapping of the interior of a polygon onto the upper half z -plane. But if we want to map the interior of a polygon onto the interior of the unit circle, we will use the additional transformation that maps the upper half-plane onto the unit disk $|w| < 1$. This transformation is given by

$$w(z) = \frac{z - i}{z + i}, \quad \text{or} \quad z(w) = i \frac{1 + w}{1 - w} \quad (\text{Map 3.22}). \quad (7.1.10)$$

Similarly, for mapping onto a regular strip in the w -plane we have the transformation

$$w(z) = \log z, \quad \text{or} \quad z(w) = e^w \quad (\text{Map 5.8}). \quad (7.1.11)$$

If the n -sided polygon is regular, then the mapping of $|w| < 1$ onto the interior of such a regular polygon is

$$z(w) = C_1 \int_{w_0}^w \frac{dt}{(1 - t^n)^{2/n}} + C_2, \quad n \geq 3, \quad (7.1.12)$$

where in case $w = z = 0$, we get $C_2 = 0$, and C_1 is determined from

$$\frac{z(w)}{C_1} = \int_0^w \frac{dt}{(1 - t^n)^{2/n}} = w + \sum_{j=1}^{\infty} \frac{2(2+n)(2+2n) \cdots (2+(j-1)n)}{w^{jn+1}}. \quad (7.1.13)$$

For the series expression, see Kober [1957:183]. The length L of each side of the regular polygon is given by (Sansone and Gerretson [1960, vol. II:153])

$$L = \frac{1}{n} 2^{1-4/n} \frac{\Gamma^2\left(\frac{1}{2} - \frac{1}{n}\right)}{\Gamma\left(1 - \frac{2}{n}\right)}, \quad (7.1.14)$$

which for $n = 3$ (triangle) is $L = \frac{1}{2\pi} \Gamma^3\left(\frac{1}{3}\right) \approx 3.059908$, and for $n = 4$ (square) $L = \frac{1}{4\sqrt{\pi}} \Gamma^2\left(\frac{1}{4}\right) \approx 1.854075$; the values of C_1 are as follows: $C_1 = 1.0788$ for $n = 4$; $C_1 = 1.0515$ for $n = 5$; $C_1 = 1.0376$ for $n = 6$; $C_1 = 1.0279$ for $n = 7$; and $C_1 = 1.0220$ for $n = 8$.

Map 7.3. The inverse Schwarz-Christoffel transformation of the polygonal region G in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is given by

$$\frac{dz}{dw} = C_1 (w - w_1)^{-\alpha_1} (w - w_2)^{-\alpha_2} \cdots (w - w_n)^{-\alpha_n}, \quad (7.1.15)$$

or

$$z(w) = C_1 \int_{w_0}^w \prod_{j=1}^n (t - w_j)^{-\alpha_j} dt + C_2, \quad (7.1.16)$$

which are obtained from (7.1.2) and (7.1.3) by interchanging z and w .

Map 7.4. The Schwarz-Christoffel transformation of the unit disk in the z -plane onto a polygonal region G with exterior angles $\alpha_k \pi$ at vertices $w_k = f(z_k)$ for each $k = 1, 2, \dots, n$, is defined by

$$\frac{dw}{dz} = C_1 \left(1 - \frac{z}{z_1}\right)^{-\alpha_1} \left(1 - \frac{z}{z_2}\right)^{-\alpha_2} \cdots \left(1 - \frac{z}{z_n}\right)^{-\alpha_n}, \quad (7.1.17)$$

or

$$w(z) = C_1 \int_{z_0}^z \prod_{k=1}^n \left(1 - \frac{z'}{z_k}\right)^{-\alpha_k} dz' + C_2. \quad (7.1.18)$$

Some authors, e.g., Papamichael and Stylianopoulos [2010: 25], take the interior angles at each vertex w_k as $\alpha_k\pi$, in which case the exterior angles are $(1-\alpha_k)\pi$ for each $k = 1, 2, \dots, n$. In this situation the quantity $-\alpha_k$ in the above transformations must be replaced by $\alpha_k - 1$. Then, for example, the map (7.1.8) becomes

$$w(z) = C_1 \int_{z_0}^z \prod_{k=1}^n \left(1 - \frac{z'}{z_k}\right)^{\alpha_k-1} dz' + C_2. \quad (7.1.19)$$

Note that the Maps 7.3 and 7.4 are distinct cases of Map 7.1; they are provided here separately for convenience.

Note that formulas (7.1.2), (7.1.15), and (7.1.18) apply to polygons with vertices at infinity and polygons with slits, i.e., vertices with angle 2π . They are also used in mapping of the exterior of the unit circle onto exterior of the polygon. The images $w_j, j = 1, 2, \dots, n$, of the vertices z_j of the polygon, where $w_j = f(z_k)$, are not known *a priori*, and their determination is known as the *parameter problem* which is discussed in §8.1.

7.1.3 Transition between Two Parallel Lines in the z -plane. Such a transition is always chosen to occur at a location corresponding to $\xi_k = 0$ in the ζ -plane. Then the constant C_1 can be found from formula

$$d = -i\pi \prod_{j \neq k} (\xi_k - \xi_j)^{\mu_j} = -i\pi \prod_{j \neq k} (0 - \xi_j)^{\mu_j},$$

where d is the complex separation of the two parallel lines. But if this transition is made at $\xi = \infty$, then

$$d = i\pi C_1. \quad (7.1.20)$$

7.2 Specific Transformations

Some specific cases of Schwarz-Christoffel Transformations are as follows.

Map 7.5. To map the upper half-plane $\Im\{z\} > 0$ onto a triangle $A_1A_2A_3$ with interior angles $\mu_1\pi, \mu_2\pi$ and $\mu_3\pi$, respectively (Figure 7.3(a)), such that the vertex A_1 corresponds to $x_1 = 0$, A_2 to $x_2 = 1$ and A_3 to $x_3 = \infty$, we get, by using (7.1.2), the mapping function

$$w = C_1 \int_0^z t^{-\alpha_1} (1-t)^{-\alpha_2} dt + C_2. \quad (7.2.1)$$

If the triangle is *right-angled isosceles* (Figure 7.3(b)) such that the vertices A_1, A_2, A_3 are at $w = 0$, $w = 1$, and $w = i$, respectively, then from (7.2.1) the transformation becomes

$$w = C_1 \int_0^z t^{-1/2} (1-t)^{-3/4} dt + C_2. \quad (7.2.2)$$

Since $w = 0$ corresponds to $z = 0$, we get $C_2 = 0$, and since $w = 1$ corresponds to $z = 1$, we find from (7.2.2) that

$$C_1 = \frac{1}{\int_0^1 t^{-1/2} (1-t)^{-3/4} dt}. \quad (7.2.3)$$

For an *equilateral* triangle of side b (Figure 7.3(c)), let $x_1 = -1$ correspond to the vertex w_1 , $x_2 = 1$ to $w = 0$, and $x_3 = \infty$ to $w_3 = b$. Since $\alpha_1 = \alpha_2 = \alpha_3 = 2/3$, the transformation is given by

$$w = \int_1^z (t+1)^{-2/3} (t-1)^{-2/3} dt. \quad (7.2.4)$$

When $z = -1$, we set $t = x$. Then for $-1 < x < 1$, we have $x+1 > 0$ and $\arg\{x+1\} = 0$, but $|x-1| = 1-x$, and $\arg\{x-1\} = \pi$. Thus,

$$\begin{aligned} w_1 &= \int_{-1}^1 (x+1)^{-2/3} e^{-2i\pi/3} (1-x)^{-2/3} dx = e^{i\pi/3} \int_{-1}^1 \frac{dx}{(1-x^2)^{2/3}} \\ &= 2e^{i\pi/3} C_1 B\left(\frac{1}{2}, \frac{1}{3}\right) = b e^{i\pi/3}, \end{aligned} \quad (7.2.5)$$

where $b = 2B\left(\frac{1}{2}, \frac{1}{3}\right)$, and B denotes the beta function of its arguments. For the vertex w_3 , note that it is on the positive u -axis, i.e.,

$$w_3 = \int_1^\infty (x+1)^{-2/3} (x-1)^{-2/3} dx = \int_1^\infty \frac{dx}{(x^2-1)^{2/3}}.$$

But w_3 is also defined by (7.2.4) when z goes to ∞ along the negative u -axis, i.e.,

$$\begin{aligned} w_3 &= \int_1^\infty (x+1)^{-2/3} (x-1)^{-2/3} dx \\ &= w_1 + e^{-4i\pi/3} \int_{-1}^{-\infty} [|x+1| |x-1|]^{-2/3} dx \quad \text{using (7.2.5)} \\ &= b e^{i\pi/3} + e^{-i\pi/3} \int_1^\infty \frac{dx}{(x^2-1)^{2/3}} \\ &= b e^{i\pi/3} + w_3 e^{-i\pi/3}, \end{aligned}$$

which yields $w_3 = b$.

Map 7.6. (Trigonal region) A trigon (Zweieck in German) is an open polygon of three sides with three vertices. Unlike a regular triangle, its two sides are parallel and extend to infinity, so it looks like an infinitely deep slot. We solve two problems: (a) We will determine the Schwarz-Christoffel transformation that maps a trigon in the z -plane onto the upper-half ζ -plane; and (b) we will determine the transformation that maps the region between two parallel lines $v = 0$ and $v = pi$ in the w -plane onto the upper half-plane $\Im\{\zeta\} > 0$, as

presented in Figure 7.5(a)-(b).

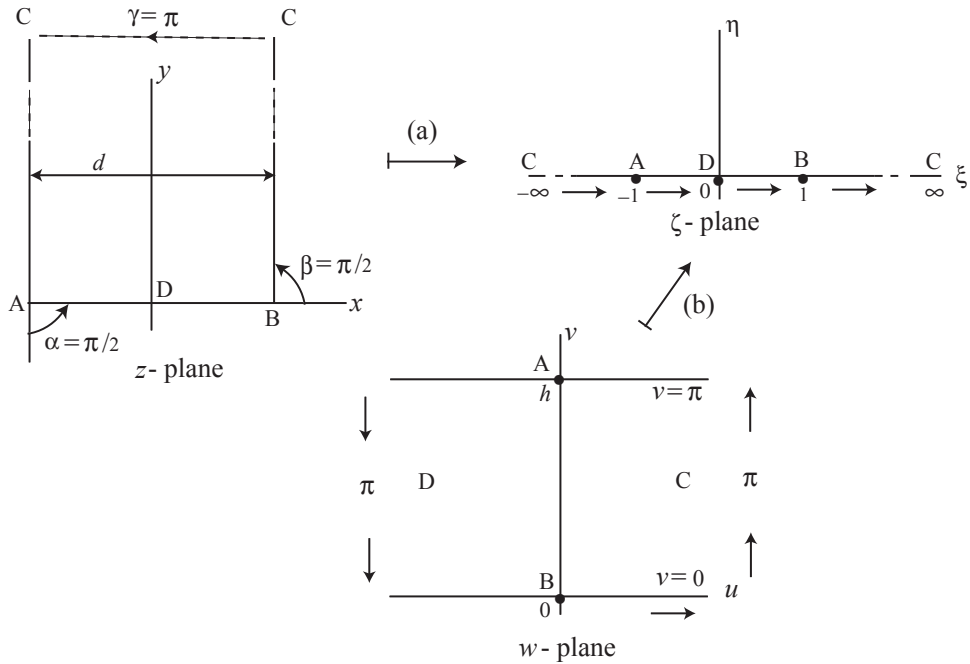


Figure 7.5 Trigonol region.

(a) Let the exterior angle at the vertex A be $\pi\alpha$ and at the vertex B be $\pi\beta$; however, the exterior angle at the vertex C is π , since this vertex is at infinity. Then using formula (7.2.1) we have

$$w(\zeta) = C_1 \int_{\zeta_0}^{\zeta} (t - \zeta_1)^{-\alpha} (t - \zeta_2)^{-\beta} dt + C_2. \quad (7.2.6)$$

Let us choose $\zeta_1 = 0, \zeta_2 = 1$. Then the above integral becomes

$$w(\zeta) = C_1 \int_{\zeta_0}^{\zeta} \frac{dt}{t^{\alpha}(t-1)^{\beta}} + C_2.$$

Since $w = 0$ at $\zeta = 0$ and $w = 1$ at $\zeta = 1$, we get $C_2 = 0$, and then we have

$$1 = C_1 \int_0^1 \frac{dt}{t^{\alpha}(t-1)^{\beta}} = C_1 \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)},$$

or

$$C_1 = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} = \frac{1}{B(\alpha, \beta)},$$

where $B(\alpha, \beta)$ is the beta function of the arguments. However, this does not complete the integration because C_1 is unknown. However, for the trigonol region in Figure 7.5 with $\alpha = \beta = \pi/2$, we find that

$$w(\zeta) = C_1 \int_0^1 \frac{dt}{\sqrt{t^2-1}} = C \int_0^1 \frac{dt}{\sqrt{1-t^2}} = C \arcsin(\zeta), \quad (7.2.7)$$

where $C = -C_1$. Using the boundary conditions $z_1 = -d/2$ at $\zeta_1 = -1$, and $z_2 = d/2$ at $\zeta_2 = 1$, we find that $-\frac{d}{2} = C(-\frac{1}{2}\pi)$ and $\frac{d}{2} = C(\frac{1}{2}\pi)$ yield $C = d/\pi$, which gives

$$w(\zeta) = \frac{d}{\pi} \arcsin(\zeta), \quad \text{or} \quad \zeta(z) = \sin \frac{\pi z}{d}. \quad (7.2.8)$$

(b) For the mapping from the ζ -plane to the w -plane, let $\zeta_1 = -1$ and $\zeta_2 = 1$ correspond to $w = 0$ and $w_2 = ih$, respectively. Since all the exterior angles at w_1 and w_2 are zero and at C at both ends it is π , the Schwarz-Christoffel transformation is

$$w(\zeta) = B_1 \int_{\zeta_0}^{\zeta} \frac{dt}{(t+1)^0(t-1)^0(t-0)^1} + B_2 = B_1 \log(\zeta) + B_2,$$

where $\zeta_0 = 1$ (point B). We use the following conditions: at B, $\zeta = 1$ and $w = 0$, which gives $B_2 = 0$; at A, $\zeta = -1$ and $w = ih$, which gives $B_1 = h/\pi$. Hence, the transformation is

$$w(\zeta) = \frac{h}{\pi} \log(\zeta). \quad (7.2.9)$$

Map 7.7. To map the upper half-plane $\Im\{z\} > 0$ onto a rectangle $A_1 A_2 A_3 A_4$ with vertices at the points $w = \pm a, \pm a + ib$, where $2a$ and b are the width and the height of the rectangle (Figure 7.6), note that in the formula (7.1.2) with $n = 4$, only three of the four points x_1, x_2, x_3, x_4 may be chosen arbitrarily. Since the rectangle is symmetric about the v -axis, we can choose the x 's symmetrically. Thus, for the right-half rectangle $OA_1 A_2 B$ let $w = 0, a, ib$ correspond to $z = 0, 1, \infty$, respectively, and let the preimage of A_2 be $z = 1/k$, $0 < k < 1$. Similarly, for the left-half rectangle $OB A_3 A_4$ let $w = 0, -a, ib$ correspond to $z = 0, -1, -\infty$, respectively, with the preimage of A_3 as $-1/k$. Then the formula (7.1.2) yields

$$\begin{aligned} w &= C_1 \int_{\zeta_0}^z \prod_{i=1}^4 (t - x_i)^{-\alpha_i} dt + C_2 \\ &= C_1 \int_0^z \frac{dt}{\sqrt{t-1} \sqrt{t+1} \sqrt{t-1/k} \sqrt{t+1/k}} \\ &= C_1 \int_0^z \frac{dt}{\sqrt{t^2-1} \sqrt{t^2-1/k^2}} = C_1 \int_0^z \frac{dt}{\sqrt{t^2-1} \sqrt{1-k^2t^2}}, \end{aligned} \quad (7.2.10)$$

where $C_2 = 0$ since $z = 0$ goes into $w = 0$, and C_1 is an arbitrary constant. Now, to evaluate the constants C_1 and k , note that since $z = 1$ goes into $w = a$, we get from (7.2.10)

$$a = C_1 \int_0^1 \frac{dt}{\sqrt{1-t^2} \sqrt{1-k^2t^2}}. \quad (7.2.11)$$

Also, since $z = 1/k$ goes into $w = a + ib$, using (7.2.11) we find that

$$\begin{aligned} a + ib &= C_1 \int_0^{1/k} \frac{dt}{\sqrt{1-t^2} \sqrt{1-k^2t^2}} \\ &= C_1 \left(\int_0^1 + \int_1^{1/k} \right) \frac{dt}{\sqrt{1-t^2} \sqrt{1-k^2t^2}} \\ &= a + i C_1 \int_1^{1/k} \frac{dt}{\sqrt{1-t^2} \sqrt{1-k^2t^2}}, \end{aligned}$$

where C_1 is an arbitrary complex constant. Thus, with $C_1 = 1$

$$b = \int_1^{1/k} \frac{dt}{\sqrt{1-t^2} \sqrt{1-k^2 t^2}}. \quad (7.2.12)$$

Hence, we can determine C_1 and k from (7.2.11) and (7.2.12) if a and b are prescribed. But if k is preassigned and we take $C_1 = 1$, then the values of a and b are determined from (7.2.11) and (7.2.12). Then the mapping function becomes

$$w = \int_0^z \frac{dt}{\sqrt{1-t^2} \sqrt{1-k^2 t^2}}, \quad (7.2.13)$$

which is known as the elliptic integral of the first kind. The inverse function is called the elliptic sine function $z = \operatorname{sn} w = \operatorname{sn}(w; k)$ which is a Jacobian elliptic function. The function sn is a $2a$ - and $2ib$ -periodic function. For more material on the function sn and other Jacobian elliptic functions, see Appendix F. If we denote the value of a , given by (7.2.11), by $K(k)$, where $0 < k < 1$, then the ratio a/b of the sides of the rectangle is given by

$$\frac{a}{b} = 2 \frac{K(k)}{K(\sqrt{1-k^2})}. \quad (7.2.14)$$

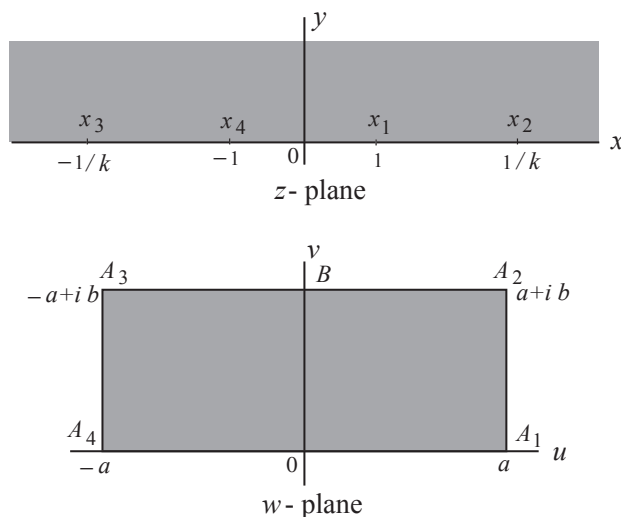


Figure 7.6 $\Im\{z\} > 0$ onto a rectangle.

An evaluation of $K(k)$ is accomplished by the Landen transformation: Set $k_1 = \sqrt{1-k^2}$, and $k' = \frac{1-k_1}{1+k_1}$. Then

$$K(k) = \frac{2}{1+k_1} K(k'). \quad (7.2.15)$$

In fact, by separating the integral (7.2.11) into two parts, we get

$$K(k) = \left(\int_0^{1/\sqrt{1+k_1}} + \int_{1/\sqrt{1+k_1}}^1 \right) \frac{dt}{\sqrt{(1-t^2)(1-k^2 t^2)}} \equiv I_1 + I_2.$$

If we set $t = \sqrt{\frac{1-t^2}{1-k^2t^2}}$, where $1/\sqrt{1+k_1} \leq t \leq 1$, and $t = \sqrt{\frac{1-t^2}{1-k^2t^2}}$, $0 \leq t \leq 1/\sqrt{1+k_1}$, then

$$I_2 = \int_{1/\sqrt{1+k_1}}^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}} = \int_0^{1/\sqrt{1+k_1}} \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}.$$

Moreover, if we set $\tau = (1+k_1) \frac{t\sqrt{1-t^2}}{\sqrt{1-k^2t^2}}$, $0 \leq t \leq 1/\sqrt{1+k_1}$, then $0 \leq \tau \leq 1$, and

$$I_1 = \frac{1}{1+k_1} K\left(\frac{1-k_1}{1+k_1}\right).$$

Hence,

$$I_1 + I_2 = 2I_1 = \frac{2}{1+k_1} K\left(\frac{1-k_1}{1+k_1}\right),$$

which proves (7.2.15). Since

$$k' = \frac{1-k_1}{1+k_1} = \frac{1-k_1^2}{(1+k_1)^2} = \frac{k^2}{(1+k^2)^2} < k^2 < 1,$$

after n applications of the Landen transformation we find that the quantities $k'^{(n)} \rightarrow 0$ very rapidly as $n \rightarrow \infty$. This means that we can use the Landen transformation to evaluate the integral in (7.2.11) step-by-step with smaller and smaller values of k , so that after a finite number of steps the value of the integral (7.2.11) becomes equal to $\arcsin(1) = \pi/2$. However, for values of k closer to 1, the convergence becomes very slow. In that case we can use the inverse Landen transformation

$$K(k') = \frac{1}{2} (1+k_1) K(k), \quad (7.2.16)$$

where $k_1 = \frac{1-k'}{1+k'}$, $k = \sqrt{1-k_1^2} = \frac{2\sqrt{k'}}{1+k'}$, and then we can evaluate $K(k)$ approximately with sufficient accuracy from the asymptotic formula

$$K(k) \approx \ln \frac{4}{k_1} = \frac{1}{2} \ln \frac{16}{1-k^2} \quad \text{for } k \text{ close to } 1. \quad (7.2.17)$$

Further, if we set $t = \sin \phi$ in (7.2.11), then

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1-k^2 \sin^2 \phi}} = \frac{2}{1+k_1} \int_0^{\pi/2} \frac{d\phi}{\sqrt{1-k'^2 \sin^2 \phi}},$$

or

$$\begin{aligned} K(k) &= \int_0^{\pi/2} \frac{d\phi}{\sqrt{\cos^2 \phi + k_1^2 \sin^2 \phi}} \\ &= \frac{2}{1+k_1} \int_0^{\pi/2} \frac{d\phi}{\sqrt{\cos^2 \phi + (1+k'^2) \sin^2 \phi}}. \end{aligned}$$

Let $k_1 = \frac{d_0}{c_0}$, where $c_0 = 1$ and $d_0 = k_1$. Then

$$1 - k'^2 = 1 - \left(\frac{1 - k_1}{1 + k_1} \right)^2 = \left(\frac{2\sqrt{k_1}}{1 + k_1} \right)^2 = \left(\frac{2\sqrt{c_0 d_0}}{c_0 + d_0} \right)^2 = \left(\frac{d_1}{c_1} \right)^2,$$

where $d_1 = \sqrt{c_0 d_0}$, and $c_1 = (c_0 + d_0)/2$, which yields

$$\begin{aligned} K(k) &= \int_0^{\pi/2} \frac{d\phi}{\sqrt{c_0^2 \cos^2 \phi + d_0^2 \sin^2 \phi}} \\ &= \frac{2}{1 + d_0/c_0} \int_0^{\pi/2} \frac{d\phi}{\sqrt{\cos^2 \phi + (d_1/c_1)^2 \sin^2 \phi}} \\ &= \int_0^{\pi/2} \frac{d\phi}{\sqrt{c_1^2 \cos^2 \phi + d_1^2 \sin^2 \phi}}. \end{aligned}$$

Repeating this process n -times, we get

$$K(k) = \int_0^{\pi/2} \frac{d\phi}{\sqrt{c_n^2 \cos^2 \phi + d_n^2 \sin^2 \phi}}, \quad (7.2.18)$$

where c_n and d_n are determined recursively from

$$c_n = \frac{1}{2}(c_{n-1} + d_{n-1}), \quad d_n = \sqrt{c_{n-1} d_{n-1}}.$$

Hence, as $n \rightarrow \infty$, the sequences $\{c_j\}$ and $\{d_j\}$ for $j = 0, 1, 2, \dots$, and $c_0 > d_0$, converge to the same limit, i.e.,

$$\lim_{n \rightarrow \infty} c_n = \lim_{n \rightarrow \infty} d_n = M(k), \quad (7.2.19)$$

which yields

$$K(k) = \frac{\pi}{2M(k)}.$$

Proof of this result is given in Andersen et al. [1962:163]. Hence, from (7.2.14) the ratio between the sides of the rectangle is given by

$$\frac{a}{b} = 2 \frac{K(k)}{K(k_1)} = 2 \frac{M(k_1)}{M(k)}, \quad (7.2.20)$$

where $M(k_1)$ is obtained in the same manner as (7.2.19) by applying the above recursion to $k_1 = \sqrt{1 - k^2}$. For more on elliptic integrals and Jacobian elliptic functions, see Phillips [1943, Ch. 1 and 2]. ■

Map 7.8. To map the trapezoid $A_1 A_2 A_3 A_4$ in the z -plane onto the upper half-plane (Figure 7.7), the mapping function from (7.2.14) is given by

$$w = C_2 + C_1 \int_0^z (\zeta + 1)^{-1/6} (\zeta - 1)^{-1/3} (\zeta + k)^{-2/3} (\zeta - 3)^{-5/6} d\zeta. \quad (7.2.21)$$

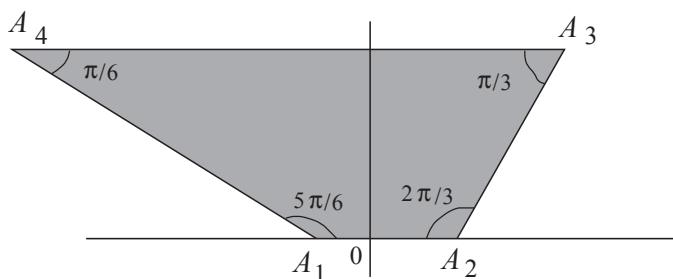


Figure 7.7 Trapezoid onto the upper half-plane.

Map 7.9. A special case of the above transformation is the mapping of the upper half-plane $\Im\{z\} > 0$ onto the interior of a square of side a . We start with the unit disk $|w| < 1$ in the w -plane, and use the transformation (7.1.11) which gives

$$z(w) = \int_0^w \frac{dt}{\sqrt{1-t^4}}, \quad (7.2.22)$$

with the length of the square's side equal to $\sqrt{2}$ in the z -plane.

Map 7.10. To map the upper half-plane $\Im\{z\} > 0$ onto an arbitrary quadrilateral $A_1A_2A_3A_4$ with interior angles $\mu_1\pi$, $\mu_2\pi$, $\mu_3\pi$, and $\mu_4\pi$, respectively, such that the angle $\mu_1\pi$ at A_1 is the smallest and the ratio of the side A_4A_1 to the side A_1A_2 is λ . Without loss of generality, let the vertices A_1, A_2, A_3, A_4 correspond to the points $x_1 = -1$, $x_2 = 1$, $x_3 = k$, $x_4 = 3$. Then, by (7.1.2), the transformation is given by

$$w = f(z) = C_1 \int_1^z (t+1)^{-\alpha_1} (t-1)^{-\alpha_2} (t-k)^{-\alpha_3} (t-3)^{-\alpha_4} dt + C_2. \quad (7.2.23)$$

Newton's method for numerical evaluation of improper integrals in (7.2.23) and an approximate value of k is given in §8.2. ■

Map 7.11. To map a horizontal parallel strip with a rectilinear horizontal cut onto the upper half-plane (Figure 7.8), note that the given strip is equivalent to a quadrilateral with vertices at A_1, A_2 and A_3 , all at infinity, and A_4 at the origin. For the vertices at A_1, A_2, A_3 we have $\alpha_1 = \alpha_2 = \alpha_3 = 1$, and for A_4 we have $\alpha_4 = -1$. We choose $x_1 = 1$, $x_2 = \infty$, $x_4 = 0$, and let $w = 0$ correspond to $z = 0$. Then C_2 in (7.1.2) is zero. With this choice the point A_3 would correspond to some point $x_3 = -k$ on the negative x -axis ($k > 0$). Then from (7.1.2) the required transformation with proper choice of principal values for the integrand is given by

$$\begin{aligned} w &= C_1 \int_0^z \frac{t dt}{(1-t)(k+t)} \\ &= C_1 \int_0^z \frac{1}{1+k} \left(\frac{1}{1-t} - \frac{k}{k+t} \right) dt \\ &= C_1 \left[\log(1-z) + k \log \left(1 + \frac{z}{k} \right) \right], \end{aligned} \quad (7.2.24)$$

where C_1 is an arbitrary constant. To determine C_1 and k if a and b are given, let $z = 1 - \varepsilon e^{i\theta}$, $-\pi \leq \theta < 0$, $\varepsilon > 0$ and small, and define a half-circle C_ε . Then, as z moves along

C_ε , its image point w varies from the ray $A_4 A_1$ to the line $A_1 A_2$. Thus, for the end point of the image curve the difference Δw in the values of w on $A_4 A_1$ and $A_1 A_2$ is

$$\Delta w = -ia + o(1). \quad (7.2.25)$$

But the difference Δw given by the right side of (7.2.24) is given by

$$\Delta w = C_1 (-i\pi) + o(1). \quad (7.2.26)$$

Hence, comparing (7.2.25) and (7.2.26), we find that $C_1 = a/\pi$. Similarly, as z moves on the other half-circle C'_ε with center at $z = -k$, i.e., $z + k = \varepsilon e^{i\theta}$, the value of w varies from its value on $A_2 A_3$ to $A_3 A_4$, and in this case the difference is $\Delta w = -ib + o(1)$, which from (7.2.24) is equal to $-C_1 k i \pi + o(1)$. This yields $b = C_1 k \pi = ka$, or $k = b/a$. Thus, the required mapping is given by

$$w = \frac{a}{\pi} \log(1 - z) + \frac{b}{\pi} \log\left(1 + \frac{a}{b} z\right). \blacksquare \quad (7.2.27)$$

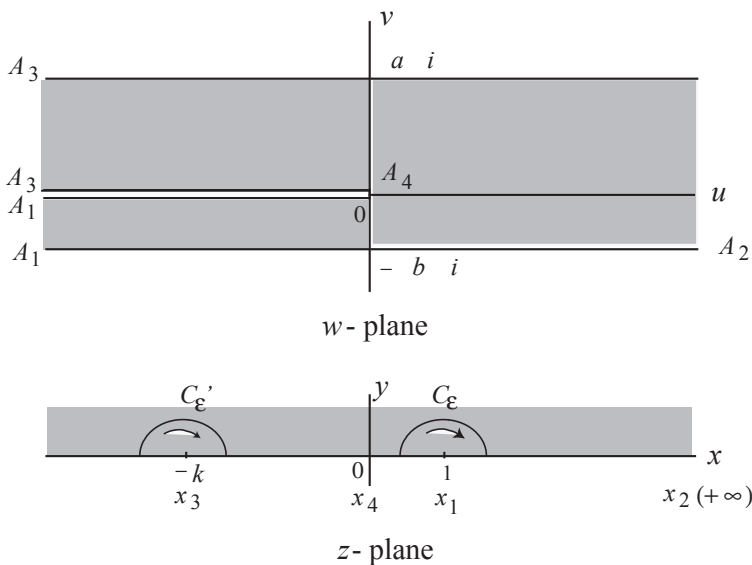


Figure 7.8 Map 7.11.

Map 7.12. To map the upper half-plane $\Im\{z\} > 0$ onto a set of two parallel lines, we use Map 7.11 by treating the w -plane as an intermediate ζ -plane ($\zeta = \xi + i\eta$), and the set of two parallel lines in the w -plane which are h distance apart (Figure 7.9). Let the images of the two parallel lines be at $\xi_1 = -1$ and $\xi_2 = 1$, which correspond to $w = 0$ and $w = ih$. The exterior angles in the w -plane are $\theta_1 = 0 = \theta_2$ at A and B, and $\theta_3 = \theta_4 = \pi$ at C and D. The infinitely distant point C is equally distant in the w -plane, while D translates into $\Re\{\xi\} = \infty$. Thus, the Schwarz-Christoffel transformation is

$$\begin{aligned} w(\zeta) &= B_1 \int_{\zeta_0}^{\zeta} \frac{dt}{(t - \zeta_1)^{\theta_1/\pi} (t - \zeta_2)^{\theta_2/\pi} (t - \zeta_4)^{\theta_4/\pi}} + B_2 \\ &= B_1 \int_{\zeta_0}^{\zeta} \frac{dt}{(t + 1)^0 (t - 1)^0 (t - 0)^1} + B_2 \\ &= B_1 \log \zeta + B_2, \end{aligned} \quad (7.2.28)$$

where $\xi_0 = \zeta_0 = 1$ (point B). To determine the integration constants, $\xi = \zeta = 1$ and $w = 0$ at B, which gives $B_2 = 0$; and $\xi = \zeta = -1$ and $w = ih$ at A, which gives $ih = B_1 \log(-1) = B_1 \log(e^{i\pi})$, thus $B_1 = h/\pi$. Hence, $w(\zeta) = \frac{h}{\pi} \log \zeta$.

Map 7.13. A simple reorganizing of the above case yields the following result: The Schwarz-Christoffel transformation $w = f(z) = \frac{2}{\pi} \arcsin z$ maps the upper half-plane $\Im\{z\} > 0$ onto the semi-infinite strip $|\Re\{w\}| < 1, \Im\{w\} > 0$, such that $f(0) = 0$ and $f(1) = 1$.

Map 7.14. Map the semi-infinite strip $u > 0, 0 < v < a$, as the limit of the triangle OCA as $\theta \rightarrow \pi/2$ onto the upper half-plane $\Im\{z\} > 0$ (Figure 7.9), such that $z = -1, 1$ correspond to $w = ia, 0$, respectively.

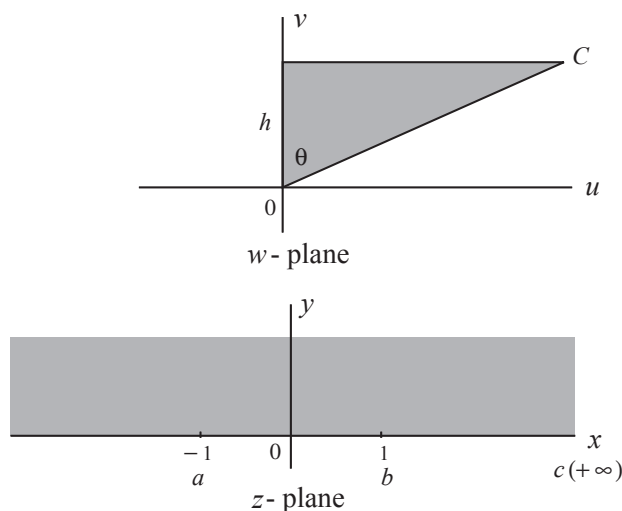


Figure 7.9 Triangle as limit of the upper half-plane.

The transformation (7.1.2) yields

$$\begin{aligned} w = f(z) &= C_2 + \lim_{\theta \rightarrow \pi/2} C_1 \int^z \frac{dt}{(t+1)^{1/2}(t-1)^{1-\theta/\pi}} \\ &= C_1 \int_1^z \frac{dt}{\sqrt{t^2-1}} = B \cosh^{-1} z, \end{aligned} \quad (7.2.29)$$

where $C_2 = 0$ since $w = 0$ corresponds to $z = 1$. From (7.2.29) we have $w/C_1 = \cosh^{-1} z$, which gives $z = \cosh(w/C_1)$, and since $w = ia$ corresponds to $z = -1 = \cosh i\pi$, we get $C_1 = a/\pi$. Hence, from (7.2.29)

$$w = \frac{a}{\pi} \cosh^{-1} z,$$

and the required transformation is given by

$$z = \cosh \frac{\pi w}{a}. \blacksquare \quad (7.2.30)$$

Map 7.15. To map the trapezoidal region in the z -plane, shown in Figure 7.10, onto the upper half-plane $\Im\{w\} > 0$, formula (7.2.1) gives

$$z = C \int_1^w \frac{dt}{t^{1-k}(t-1)^k}, \quad 0 < k < 1. \quad (7.2.31)$$

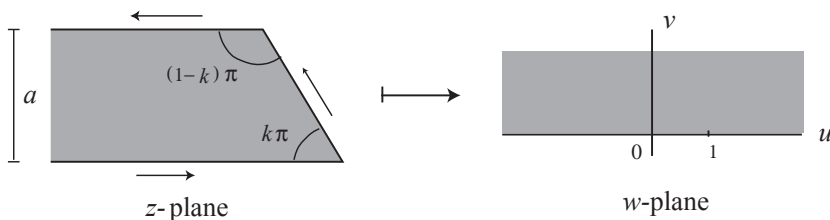


Figure 7.10 Semi-infinite strip onto $\Im\{w\} > 0$.

To determine the constant C , we integrate along the half-circle $w = Re^{i\theta}$, $0 < \theta < \pi$. Since the residue of the integrand at $t = \infty$ is -1 , we find from (7.2.31), as $R \rightarrow \infty$, that $ai = -Ci\pi$, i.e., $C = -a/\pi$ (see Figure 7.9). The integral in (7.2.31) can be evaluated in terms of known functions if k is rational. Let $k = p/q$, $p < q$, where $p, q \in \mathbb{R}^+$. Set $\left(\frac{\zeta-1}{\zeta}\right)^{1/q} = t$. Then, $d\zeta = \frac{qt^{q-1}}{(1-t^q)^2} dt$, and (7.2.31) becomes

$$z = \frac{aq}{\pi} \int_0^t \frac{t^{-p+q-1}}{t^q - 1} dt. \quad (7.2.32)$$

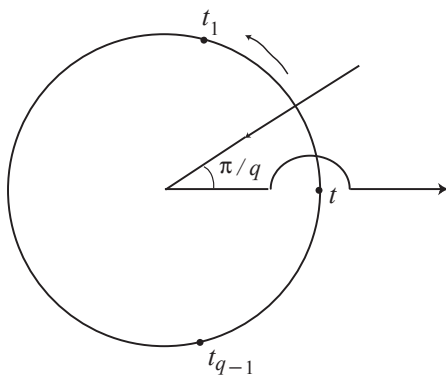


Figure 7.11 Evaluation of C .

Now, the q poles of the integrand in (7.2.35) are the q -th zeros of $(t^q - 1)$, i.e., they are at $t_n = e^{2ni\pi/q}$, $n = 0, 1, \dots, q-1$. Thus,

$$\frac{t^{-p+q-1}}{t^q - 1} = \sum_{n=0}^{q-1} \frac{t_n^{-p+q-1}}{q t_n^q - 1} \frac{1}{t - t_n} = \frac{1}{q} \sum_{n=0}^{q-1} \frac{1}{t_n^p} \frac{1}{t - t_n},$$

and

$$\int_0^t \frac{1}{t - t_n} dt = \ln |t - t_n| + \text{const} = \ln \left(1 - \frac{t}{t_n}\right),$$

where the constant is zero because the integrand is zero at $t = 0$ and the principal value of the logarithm is taken. Thus, the required transformation becomes

$$z = \frac{a}{\pi} \sum_{n=0}^{q-1} \frac{1}{t_n^q} \ln \left(1 - \frac{t}{t_n} \right). \quad (7.2.33)$$

As a special case, if $\mu = 1/2$, i.e., if $p = 1$, $q = 2$, the transformation (7.2.33) reduces to

$$z = \frac{a}{\pi} \ln \frac{1-t}{1+t}, \quad t = \left(\frac{t-1}{t} \right)^{1/2}. \blacksquare \quad (7.2.34)$$

In the next chapter we will discuss the problem of approximately computing the values of the $(2n+2)$ parameters involved in the Schwarz-Christoffel formula (7.1.2). This discussion involves a numerical solution of improper integrals known as Schwarz-Christoffel integrals.

Map 7.16. The Schwarz-Christoffel transformation

$$w = f(z) = \frac{1}{\pi} \sqrt{2(1-z)} + \frac{1}{\pi} \log \frac{\sqrt{1-z} - \sqrt{2}}{\sqrt{1-z} + \sqrt{2}} - i \quad (7.2.35)$$

maps the upper half-plane $\Im\{z\} > 0$ onto the region consisting of the fourth quadrant plus the strip $0 < v < 1$ in the w -plane, $w = u + iv$, such that $f(1) = 0$.

Map 7.17. The mapping of the exterior of a thin straight-line slit in the z -plane onto the exterior of the unit circle $|w| > 1$ can be carried out using the Schwarz-Christoffel transformation. Let the segment AB of length $4a$ be symmetrical with respect to the y -axis. The exterior angles at its endpoints A and B are each equal to π . Let $w_1 = -1$ and $w_2 = 1$. Then, using the transformation (7.1.9) of the required mapping is

$$\begin{aligned} z(w) &= C_1 \int_{w_0}^w (t+1)(t-1) \frac{dt}{t^2} \\ &= C_1 \int_{w_0}^w \left(1 - \frac{1}{t^2} \right) dt + C'_2 = C_1 \left(w + \frac{1}{w} \right) + C'_2. \end{aligned}$$

Since $z = \pm 2a$ correspond to $w = \pm 1$, respectively, we have $C_1 = a$ and $C'_2 = 0$. Hence, the required transformation is

$$z(w) = a \left(w + \frac{1}{w} \right), \quad \text{or} \quad w(z) = \frac{1}{2a} [z \pm \sqrt{z^2 - 4a^2}], \quad (7.2.36)$$

where the positive square-root is used so that $z = \infty$ corresponds to $w = \infty$. Note that the *negative* square-root in (7.2.36) will map the exterior of the slit onto the interior of the

unit circle $|w| < 1$.

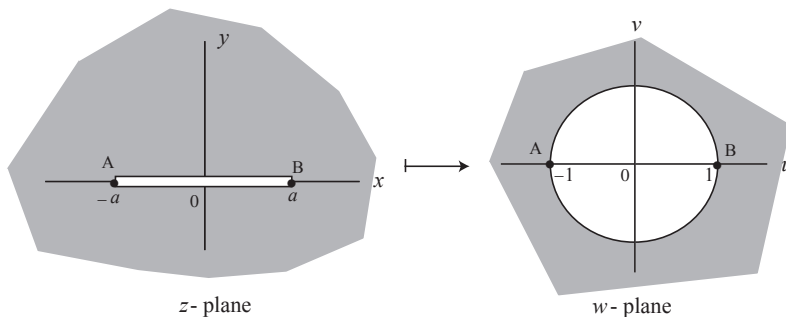


Figure 7.12 Exterior of a slit onto the exterior of unit circle.

7.3 Regions Exterior to and between Rectangles

We will consider cases of mapping the regions exterior to rectangles and between rectangles.

Map 7.18. Consider the exterior of a rectangle as in Figure 7.13, where the mapping from the z -plane onto the ζ -plane is defined by

$$z(\zeta) = \int_0^\zeta t^{1/2}(t^2 - 1)^{-1/2} dt, \quad (7.3.1)$$

where, according to Bieberbach [1914] and von Koppenfels and Stallman [1959], the semi-infinite lines BA and DE in Figure 7.13 can be taken as axes of symmetry.

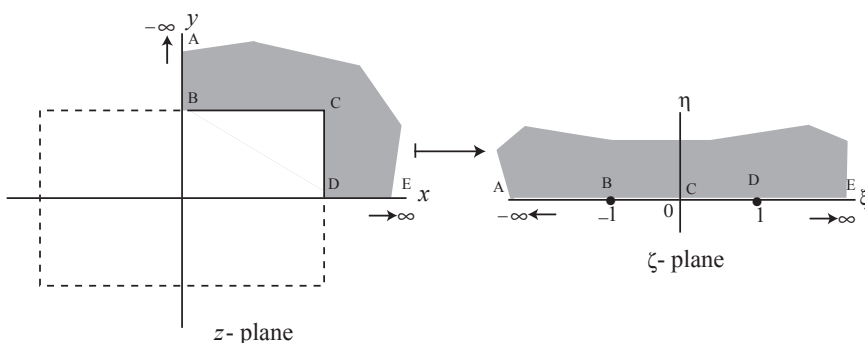


Figure 7.13 Exterior of a rectangle onto the upper half-plane.

Map 7.19. The symmetry mentioned in the above case can also be used to determine the transformation that maps the region between two rectangles (one an inner and the other an outer rectangle) with a common center at $z = 0$ onto the upper half-plane $\Im\{\zeta\} > 0$, as shown in Figure 7.14. The Schwarz-Christoffel transformation is

$$z(\zeta) = C_1 \int_0^\zeta \left(\frac{t^2 - \xi_2^2}{(t^2 - 1)(t^2 - k^2)(t^2 - \xi_1^2)} \right)^{1/2} dt, \quad (7.3.2)$$

where the values of the constant C_1 and the modulus k can be found from the given

dimensions of the rectangles.

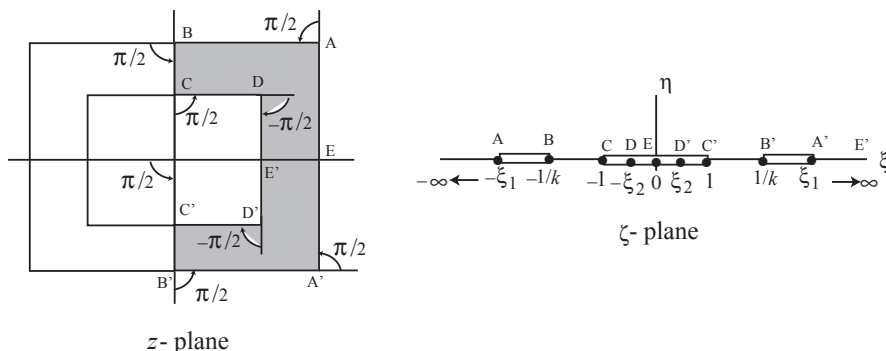


Figure 7.14 Region between two rectangles onto the upper half-plane.

Let the value of the integral (7.3.2) be taken as

$$I(\zeta_m, \zeta_n) = \int_{\zeta_m}^{\zeta_n} \left(\frac{|t^2 - \xi_2^2|}{|(t^2 - 1)(t^2 - k^2)(t^2 - \xi_1^2)|} \right)^{1/2} dt. \quad (7.3.3)$$

Then, using Figure 7.14, we have: $a/2 = C_1 I(\xi_2, 1)$; $b/2 = C_1 I(0, \xi_2)$; $d/2 = \text{segment (B-C)} + \text{segment (D-E)}$; $C_1 = I(0, \xi_2) + I(1, 1/k)$; and $c/2 = \text{segment (B-A)} = C_1 I(1/k, \xi_2)$. These values can be now used to determine the three unknowns ξ_1, ξ_2 and k from the known ratios of $a/d, b/d$ and c/d . Moreover, we can determine C_1 using the relation $I(0, 1) = a/2 + b/2 = K(k)$, a complete elliptic integral of the first kind. Finally, the transformation $w = \log(\zeta)$ will map the upper half-plane $\Im\{\zeta\} > 0$ onto a parallel strip of length $K(k)$ and height $K'(k)$.

As an application, this method has been widely used to determine the characteristic impedance of various thin sheet registers and transmission line cross-sections (see parameter problem, and a numerical method in §22.3).

Map 7.20. To map two concentric squares symmetric about a diagonal axis as onto a parallel strip in the w -plane shown in Figure 7.15, we consider two cases:

Map 7.20a. For the mapping shown in Figure 7.15, the Schwarz-Christoffel transformation is

$$\begin{aligned} z(\zeta) &= C_1 \int_0^\zeta \frac{dt}{(t-0)^{3/4}(t-1)^{1/2}(t-1/k)^{1/2}} \\ &= C \int_0^\zeta \frac{dt}{t^{3/4}(1-t)^{1/2}(1-kt)^{1/2}}, \end{aligned} \quad (7.3.4)$$

where $C = C_1/k$. Using Bowman's method (Bowman [1933]), we carry out one more intermediate mapping $\zeta = (w^*)^2$, and then set $w^* = \sin w$. This will provide a simple

parallel strip of length of the sides K and K' in the w -plane.

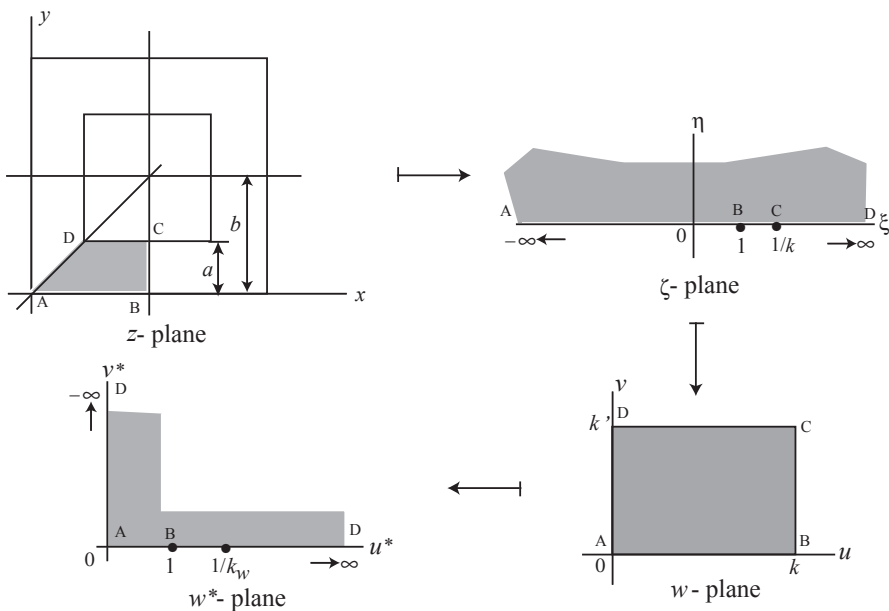


Figure 7.15 Two concentric squares, Map 7.20a.

Map 7.20b. For the mapping of the inner square set like a diamond inside a corner square as shown in Figure 7.16, the Schwarz-Christoffel transformation is

$$z(\zeta) = C_1 \int_0^\zeta \frac{dt}{(t-0)^{1/2}(t-1)^{1/4}(t-i/k)^{1/2}}. \quad (7.3.5)$$

Using the same Bowman's method, the transformation to map it onto the parallel strip of sides K and K' in the w -plane is given by

$$\zeta = \operatorname{sn}^2(w). \quad (7.3.6)$$

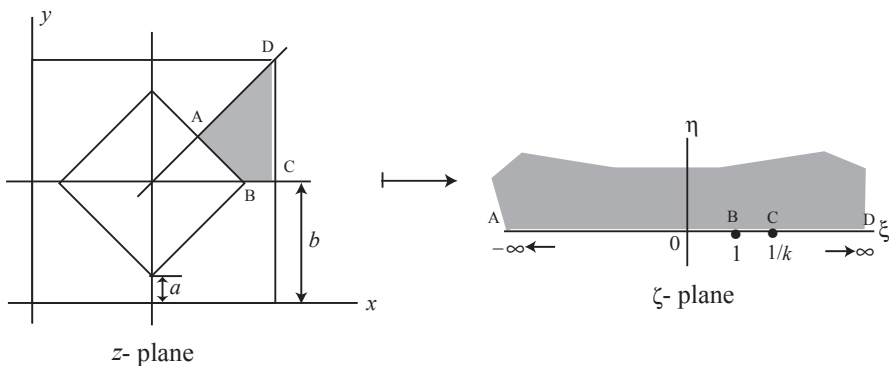


Figure 7.16 Two concentric squares, Map 7.20b.

7.4 Polygons with Round Corners

Map 7.21. In the real world problems, one usually finds round corners in manufactured items. Sharp corners are idealized or mathematically simplified cases. Some of the causes of round corners are: (i) wear and tear by constant use of the physical object; and (ii) production of round cornered objects on purpose to reduce the high potential gradients which can result in physical failures especially in electrically operated objects. We will consider an example of polygons with round corners as presented in Figure 7.17, and use Schwarz-Christoffel transformation to determine the field between the right angle (unrounded corner at C) and the y -axis. First, for the right angle corner at C we have the transformation

$$z(w) = C_1 \int \frac{dt}{(t-0)^1(t-1)^{-1/2}} + C_2 = C_1 \int \frac{\sqrt{t-1}}{t} dt + C_2, \quad (7.4.1)$$

which corresponds to the points A, B, C as marked in the z -plane and w -plane in Figure 7.17. Note that $z = 0$ does *not* correspond to $u = 0$, and, thus, C_2 need not be zero. Also, the angle $3\pi/2$ at the point A in the z -plane is avoided by moving this point to infinity in the w -plane. Thus, the solution for the unrounded or right-angle corner at C is

$$z(w) = 2C_1 \left(\sqrt{t-1} + \frac{i}{2} \log \frac{i - \sqrt{t-1}}{i + \sqrt{t-1}} \right) + C_2. \quad (7.4.2)$$

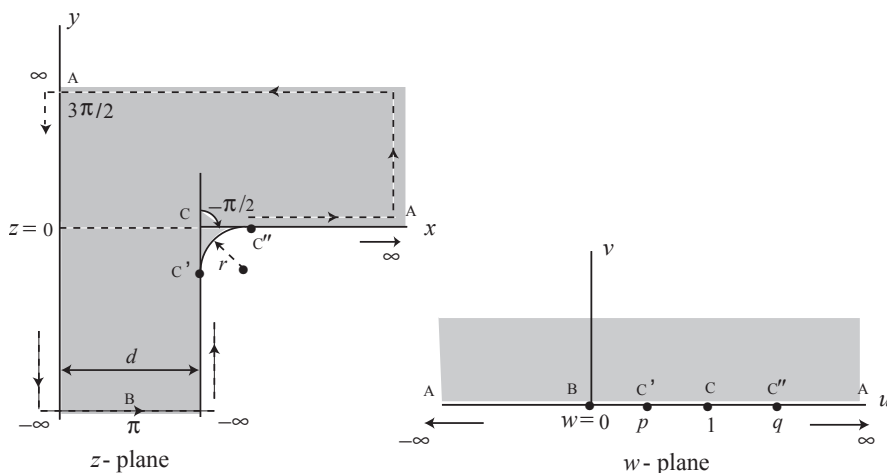


Figure 7.17 Polygon with rounded corner.

The jump across the gap of width d between $z = 0$ and $z = d$ (corner C) occurs at B, which corresponds to $w = 0$. Using formula (7.1.18) we get

$$d = -i\pi C_1(0-1)^{1/2}, \quad \text{or} \quad C_1 = d/k. \quad (7.4.3)$$

Since the point $z = d$ (point C) in the z -plane is mapped onto the point $w = 1$ (C in the w -plane), we have the boundary condition $z(1) = d$, using which we find that $C_2 = 2d$, thus, giving the solution for the right-angle corner at C as

$$z(w) = \frac{2\pi}{d} \left(\sqrt{e^w - 1} + \frac{i}{2} \log \frac{i - \sqrt{e^w - 1}}{i + \sqrt{e^w - 1}} \right) + 2d. \quad (7.4.4)$$

Next we resolve the problem for the round corner at C in two steps: Firstly, we solve the problem with the sharp (unrounded corner); then we find an equipotential line near this corner which has the shape nearly equal to the rounding. It then becomes the new boundary.

Secondly, in a general setting we may replace the right corner by a circular arc. In the case of Figure 7.17 the point C is replaced by a quarter arc of a circle of radius r such that this arc is tangent to the polygon at C' and C'' with coordinates $z = d - ir$ and $z = d + r - i0$, respectively. In the w -plane, the point C' corresponds to $u = p$ and the point C'' to $u = q$.

Thus, the factor of the type $(w - u_j)^{-\alpha_j}$ is replaced by $(u - p)^{-\alpha_j} + \lambda(u - q)^{\alpha_j}$. Hence, the transformation is

$$z(w) = C_1 \int \frac{\sqrt{t-p} + \lambda\sqrt{t-q}}{t} dt + C_2, \quad (7.4.5)$$

which after integrating yields

$$z(w) = 2C_1 [\sqrt{u-p} - \sqrt{p} \arctan(\sqrt{(u-p)/p})] + \lambda [\sqrt{u-q} - \sqrt{q} \arctan(\sqrt{(u-q)/q})] + C_2. \quad (7.4.6)$$

Although the weight factor λ is not known, the constant C_1 can be evaluated using (7.1.18) as $C_1 = \frac{d}{\pi} (\sqrt{p} + \lambda\sqrt{q})^{-1/2}$. Also, using the boundary condition $z = d - ir$ at $u = p$, we find that $C_2 = d$.

The weight λ can be evaluated in two ways: (i) choose a value for either p or q on the xi -axis and use optimization method to determine which λ will provide the best fit; (ii) the other choice is simply take $q = 1 - p$, which by using the ratio r/d leads to a function for λ which can be found in a graphical form in Weber [1950:375].

Note that in the case of a polygon that is not regular, we may have to rescale the field to reestablish the original dimensions of the polygon. Note that such a rounding does not run sufficiently parallel to the sides of the polygon away from the corner.

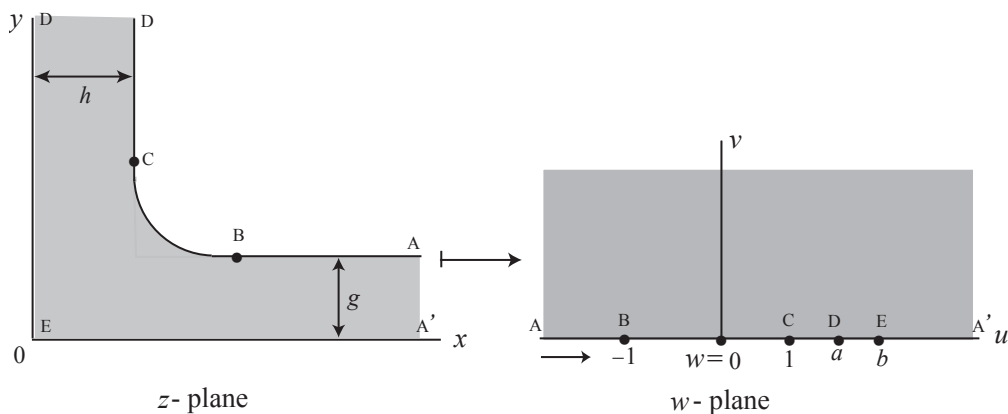


Figure 7.18 Region with a curved boundary-line onto $\Im\{z\} > 0$.

Map 7.22. The transformation $\frac{dz}{dw} = \frac{\sqrt{w+1} + c\sqrt{w-1}}{(w-a)\sqrt{w-b}}$, where $b > a > 1$, and $c = \sqrt{\frac{b+1}{b-1}} > 0$, maps the region with a curved boundary-line in the z -plane onto the

upper half-plane $\Im\{w\} > 0$. This mapping is shown in Figure 7.18, where the curve is almost a quarter of a circle, $g = (1+c)\pi$, and $h = \left(\sqrt{\frac{a+1}{b-a}} + c\sqrt{\frac{a-1}{b-a}}\right)\pi$.

Map 7.23. The above transformation (Map 7.22), when combined with $w = b + e^{-i\pi} \zeta^2$, maps the region bounded by $x = 0$ and the two curved boundary-lines onto the upper half-plane $\Im\{\zeta\} > 0$ (i.e., $\eta > 0$), where $\zeta = \xi + i\eta$, and g and h are defined as above. This mapping is presented in Figure 7.19.

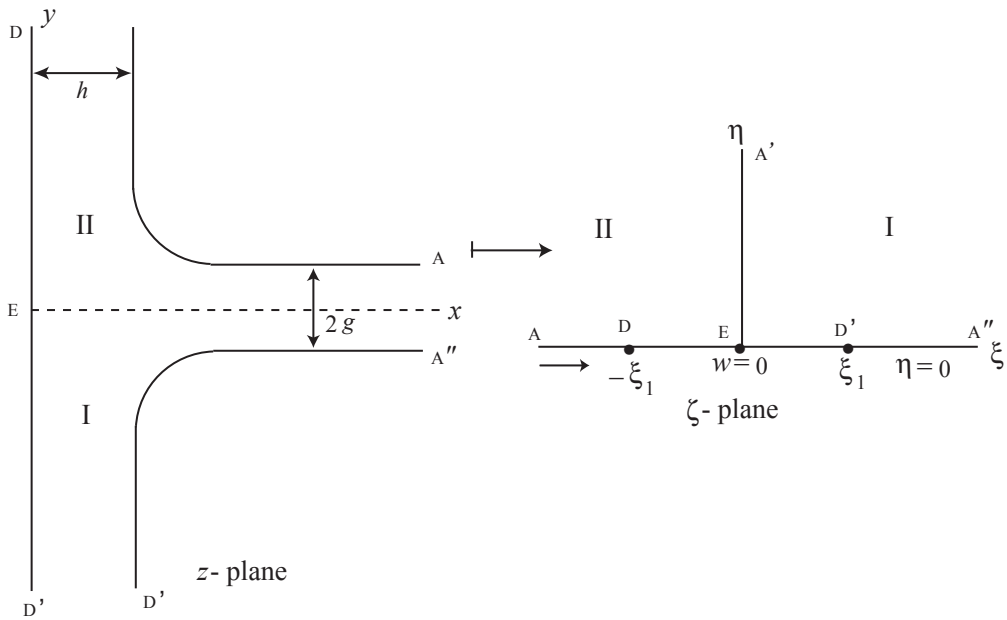


Figure 7.19 Region with two curved boundary-lines onto $\Im\{\zeta\} > 0$.

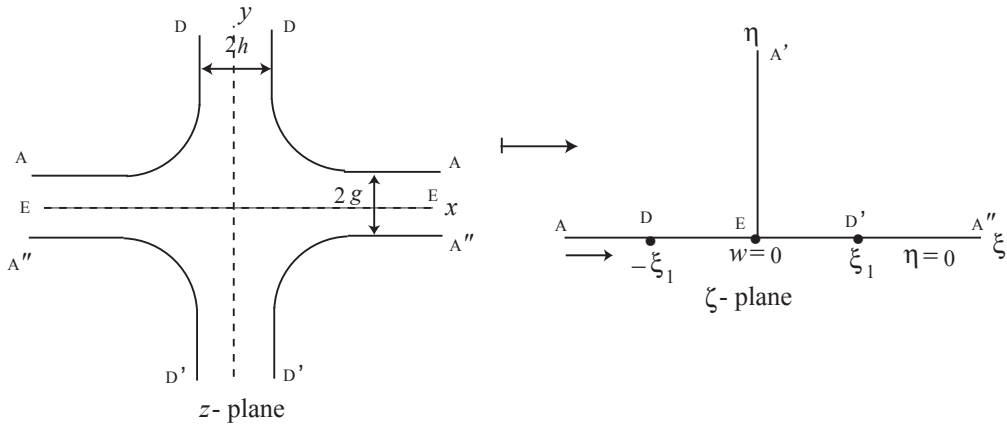


Figure 7.20 Region with four curved boundary-lines onto $\Im\{\zeta\} > 0$.

Map 7.24. The above transformation (Map 7.23), when combined with $w = b + \frac{a-b}{4}(\zeta + 1/\zeta)^2$, maps the region bounded by the four curved boundary-lines onto the upper half-plane $\Im\{\zeta\} > 0$ (i.e., $\eta > 0$), where $\zeta = \xi + i\eta$, and g and h are defined as above. This mapping is presented in Figure 7.20.

7.5 Weierstrass Integral Equation

Map 7.25. The Weierstrass elliptic integral equation is another method to map a rectangle onto the upper half-plane or onto the interior of a circle. This approach is different from that of the Jacobi elliptic integral only in the location of the zero point and specification of the scale factor. Such parameters are available in Sansone and Garretson [1969:132], Betz [1964:352], and Abramowitz and Stegun [1972: Ch 18]. The Weierstrass elliptic integral and its inverse are

$$z(\xi) = \int_{\xi}^{\infty} \frac{d\xi}{\sqrt{4\xi^3 - g_2\xi - g_3}}, \quad \xi = \wp(z; g_2, g_3). \quad (7.5.1)$$

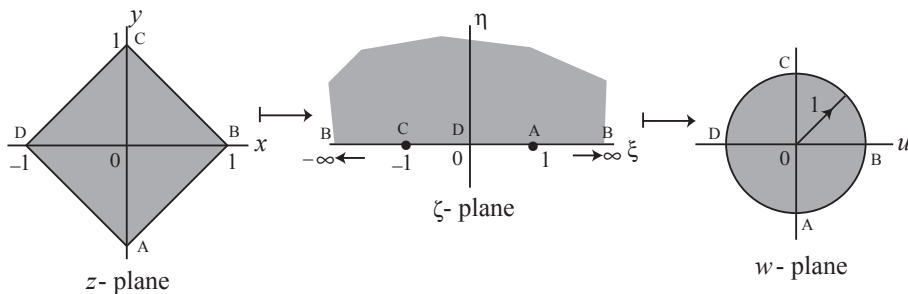


Figure 7.21 Mapping of the square onto a circle.

The function that maps the special case of the square of side length L in the z -plane onto the upper half of the ζ -plane is

$$\zeta = \wp(z; 4, 0), \quad L = \frac{1}{4\sqrt{2\pi}} \Gamma^2\left(\frac{1}{4}\right), \quad (7.5.2)$$

where $\wp(z; 4, 0)$ denotes the Weierstrass elliptic function of order 2 with periods 4 and 0. The inverse mapping is the function $z(\xi)$ given by (7.5.1) with $g_2 = 4$ and $g_3 = 0$ (see Figure 7.21).

We will use Schwarz's symmetry and provide mappings of a square into the circle in the w -plane using $w^2 = \wp^{-1}(z; 4, 0)$. The details are shown in Figure 7.22. Note that the mapping in Figure 7.22(a) is $\xi = \wp(z; 4, 0)$; the mapping in Figure 7.22(b) is $w = \wp^2(z; 4, 0)$; and the mapping in Figure 7.22(c) is $w = 1/\wp(z; 4, 0)$.

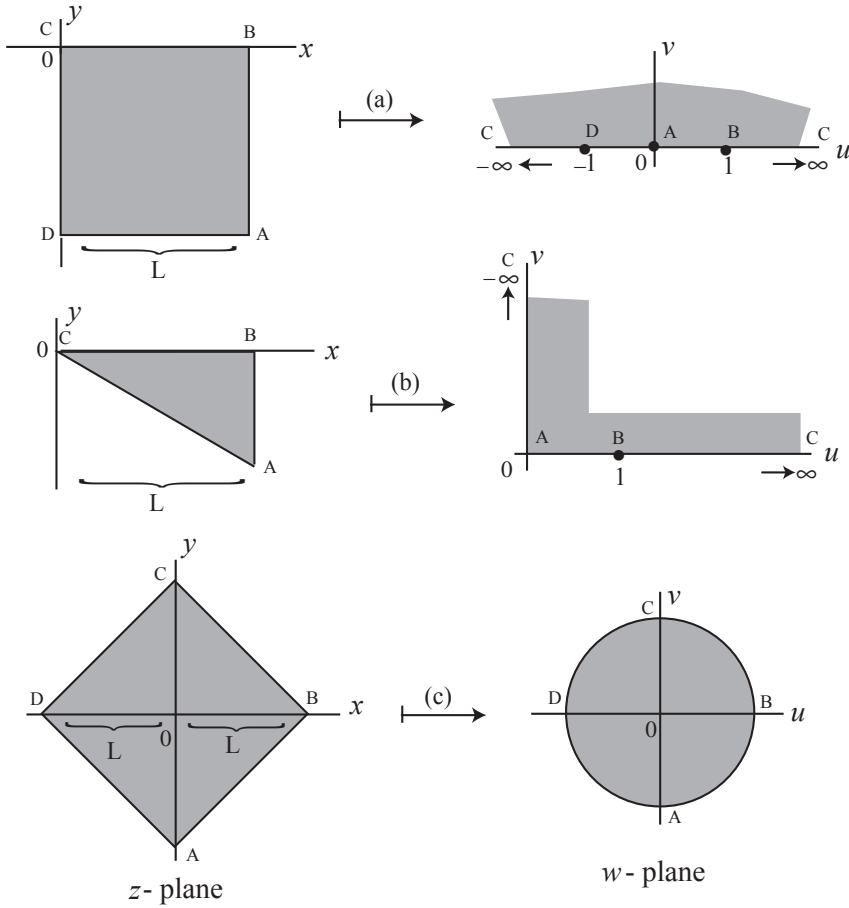


Figure 7.22 Details of the mapping of the square onto a circle.

An application of these mappings is

Map 7.26. Consider the transformation of the interior of a rectangle onto the half-plane, as presented in Figure 7.23. The Schwarz-Christoffel transformation is

$$z = \int_{-\infty}^w \frac{dt}{\sqrt{4(t-e_1)(t-e_2)(t-e_3)}} = \int_{-\infty}^w \frac{dt}{\sqrt{4t^3 - g_2t - g_3}}, \quad (7.5.3)$$

where

$$\begin{aligned} e_1 &= \frac{1}{3} \left(\frac{\pi}{2\omega_1} \right)^2 (e_3^4(\tau) + e_0^4(\tau)), \quad e_2 = \frac{1}{3} \left(\frac{\pi}{2\omega_1} \right)^2 (e_2^4(\tau) + e_0^4(\tau)), \\ e_3 &= -\frac{1}{3} \left(\frac{\pi}{2\omega_1} \right)^2 (\Theta_2^4(\tau) + \Theta_3^4(\tau)), \quad \tau = \frac{\omega_3}{\omega_1}, \quad i\tau < 0; -\infty < e_3 < e_2 < e_1 < \infty \\ g_2 &= \frac{2}{3} \left(\frac{\pi}{2\omega_1} \right)^2 (\Theta_2^8(\tau) + \Theta_3^8(\tau) + \Theta_0^8(\tau)), \quad g_3 = 4e_1e_2e_3. \end{aligned} \quad (7.5.4)$$

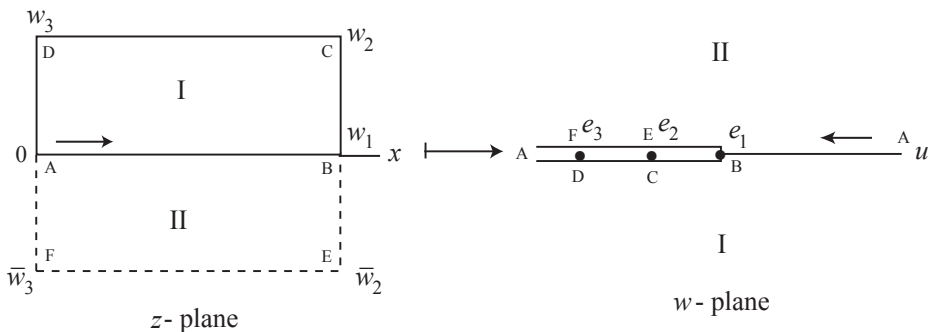


Figure 7.23 Interior of a rectangle onto the upper half-plane.

The mapping function (7.5.4) can be expressed in terms of the Weierstrass elliptical integral function as

$$w = \wp(z) = e_3 + \frac{e_1 - e_3}{\operatorname{sn}^2(z\sqrt{e_1 - e_3}, k)}, \quad (7.5.5)$$

where $k^2 = \frac{e_2 - e_3}{e_1 - e_3}$.

The mapping function (7.5.5) can also be represented in terms of the θ -series as

$$w = \wp(z) = e_1 + (\mu\vartheta_2(\xi|\tau)^2) = e_2 + (\mu\vartheta_3(\xi|\tau)^2) = e_3 + \left(\mu \frac{\Theta_2(\tau)}{\Theta_0(\tau)} \vartheta_0(\xi|\tau)\right)^2, \quad (7.5.6)$$

where $\tau = \frac{\omega_3}{\omega_1}$, $\mu = \frac{\vartheta_1'(\tau)}{2\omega_1\Theta_2(\tau)\vartheta_1(\xi|\tau)}$, and $\nu = \frac{z}{2\omega_1}$.

The transformation (7.5.4) can be defined alternatively in terms of e_1, e_2, e_3 , all real, with $e_3 < e_2 < e_1$, $e_1 + e_2 + e_3 = 0$, and

$$k^2 = \frac{e_2 - e_3}{e_1 - e_3}, \quad k'^2 = \frac{e_1 - e_2}{e_1 - e_3},$$

$$\omega_1 = \frac{K}{\sqrt{e_1 - e_3}} \text{ for } \omega_1 > 0; \quad \omega_3 = \frac{iK}{\sqrt{e_1 - e_3}} \text{ for } \omega_3/i > 0,$$

where

$$K = \int_0^1 \frac{dt}{\sqrt{1-t^2}\sqrt{1-k^2t^2}} = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1-k^2\sin^2\phi}}$$

$$= \frac{\pi}{2} \left\{ 1 + \sum_{n=1}^{\infty} \left(\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} \right)^2 k^{2n} \right\} = F\left(k, \frac{\pi}{2}\right);$$

$$K' = \int_0^{1/k} \frac{dt}{\sqrt{-1+t^2}\sqrt{1-k^2t^2}} = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1-k'^2\sin^2\phi}}$$

$$= \int_0^1 \frac{dt}{\sqrt{-1+t^2}\sqrt{1-k^2t^2}} = F\left(k', \frac{\pi}{2}\right).$$
(7.5.7)

7.6 Rectangles and Other Regions

Map 7.27. The transformation that maps a square onto the lower half-plane, such that the four right triangles of angles $\pi/2, \pi/4, \pi/4$ within the square marked I, II, III, and IV, are mapped onto the four regions of the w -plane separated by the arcs of a semicircle in the w -plane (Figure 7.24), is given by

$$w = \wp(z; 4, 0), \quad (7.6.1)$$

where $g_2 = 4, g_3 = 0, e_1 = -e_3 = 1, e_2 = 0$, and

$$\omega_1 = \int_0^1 \frac{dt}{\sqrt{1-t^4}} = \frac{\pi}{2\sqrt{2}} \left\{ 1 + \sum_{n=1}^{\infty} \left(\frac{1 \cdot 3 \cdot 5 \cdots (2n-1)}{2 \cdot 4 \cdot 6 \cdots 2n} \right)^2 2^{-n} \right\},$$

and $\omega_3 = i\omega_1$.

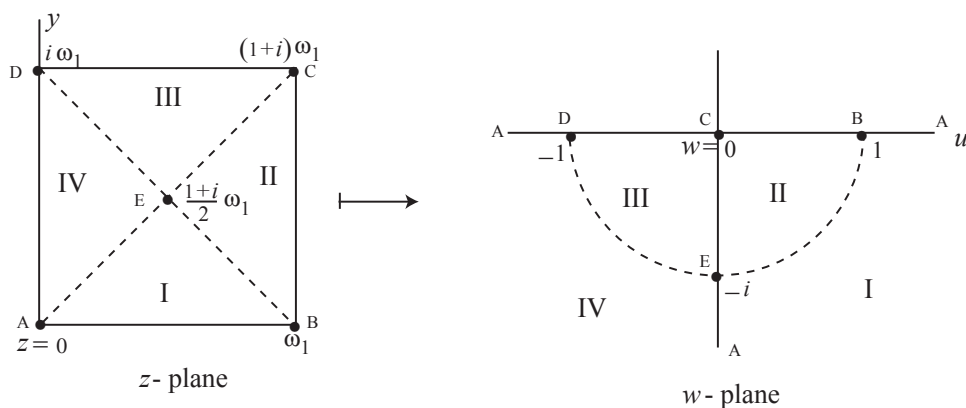


Figure 7.24 Square onto the lower half-plane.

Note that triangles with angles $\pi/2, \pi/4, \pi/4; \pi/3, \pi/3, \pi/3; 2\pi/3, \pi/6, \pi/6; \pi/2, \pi/3, \pi/6$ are the only four cases of finite triangles in the z -plane that are mapped onto a half-plane under a conformal mapping function $w = f(z)$; see Map 7.5.

Map 7.28. To map a rectangle onto a half-plane or onto a quarter of a plane, as shown in Figure 7.25, is given by

$$z = \alpha \int_0^w \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad (7.6.2)$$

or

$$w = \operatorname{sn} \left(\frac{z}{\alpha}, k \right) = \operatorname{sn} \frac{z}{\alpha}, \quad (7.6.3)$$

where

$$\alpha = \frac{a}{K}, \quad K = \int_0^1 \int_0^w \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}. \quad (7.6.4)$$

In this case both a, b are positive, so that

$$\tau = \frac{ib}{a}, \quad k = \frac{\Theta_2(\tau)}{\Theta_3(\tau)} = \sqrt{\lambda(\tau)}, \quad 0 < k < 1.$$

As seen from Figure 7.25, the interior of the rectangle ABDE (marked I, II) is mapped onto the quadrant $u > 0, v > 0$ (first quadrant) in the w -plane; the interior of the rectangle ABB'A' (marked II, II') is mapped onto the upper half-plane $v > 0$ in the w -plane; and the line-segment $y = b/2, -a \leq x \leq a$ in the z -plane is mapped onto the semi-circle $|w| = \sqrt{e}, v \geq 0$ in the w -plane.

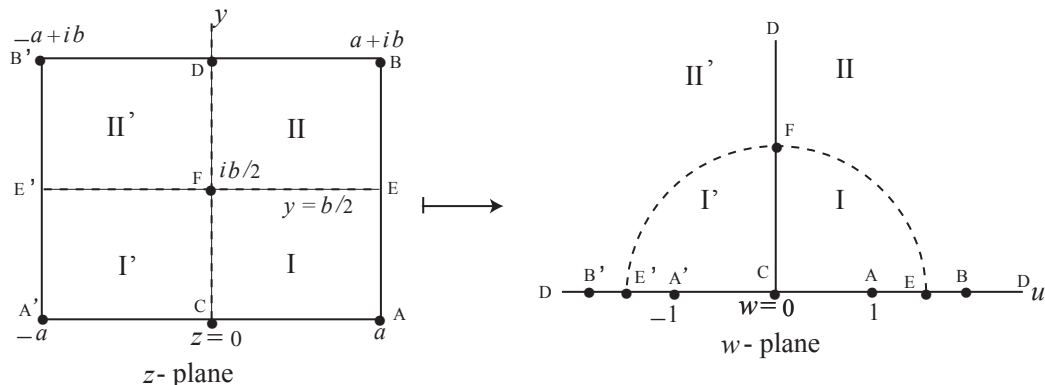


Figure 7.25 Rectangle onto the upper half-plane.

If $k < 1$, then the transformation is defined by

$$w = \operatorname{sn}(z, k) = \operatorname{sn}(z), \quad \text{or} \quad w = \operatorname{sn}^2(z, k), \quad (7.6.5)$$

where

$$a = K, \quad b = K' = \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-k'^2 t^2)}}, \quad k'^2 = 1 - k^2,$$

and K is the same as defined in (7.6.1). In this case, the interior of the rectangle with vertices at $z = 0, K, K + iK', iK'$ is mapped onto the upper half-plane $v > 0$.

Map 7.29. To map a square onto a quarter of a plane, the transformation is

$$w = \operatorname{sn}\left(z, \frac{1}{\sqrt{2}}\right), \quad (7.6.6)$$

with $K = K' = 1$. This mapping is described in Figure 7.26 as follows:

- (i) The points $z = 0$ (marked A); K (marked B); $(1+i)K$ (marked C); iK (marked D); $\frac{1+i}{2}K$ (marked E); and $\frac{iK}{2}$ (marked F) in the z -plane are mapped onto the points $w = 0$ (A); 1 (B); $\sqrt{2}$ (C); ∞ (D); $\sqrt[4]{2}e^{i\pi/8}$ (E); and $i\sqrt[4]{2}$ (F), respectively, in the w -plane.
- (ii) The line-segment $y + \frac{1}{2}K, -K \leq x \leq K$, in the z -plane is mapped onto the semi-circle $|w| = \sqrt[4]{2}v \geq 0$ in the w -plane.
- (iii) The segment $x > 0, y > 0$ of the line $x + y = 1$ (BED) in the z -plane is mapped onto part $u > 0, v > 0$ of the hyperbola $u^2 - v^2 = 1$.
- (iv) The segment $x < 0, y > 0$ of the line $y - x = 1$ (B'E'D) in the z -plane is mapped onto part $u < 0, v > 0$ of the same hyperbola.

- (v) The segment $0 < x < K$ of the line $y = x$ (AEC) in the z -plane is mapped onto part $u > 0, v > 0$ of the lemniscate $|w - 1||w + 1| = 1$.
- (vi) The segment $-K < x < 0$ of the line $y = -x$ (AE'C') in the z -plane is mapped onto part $u < 0, v > 0$ of the above lemniscate.

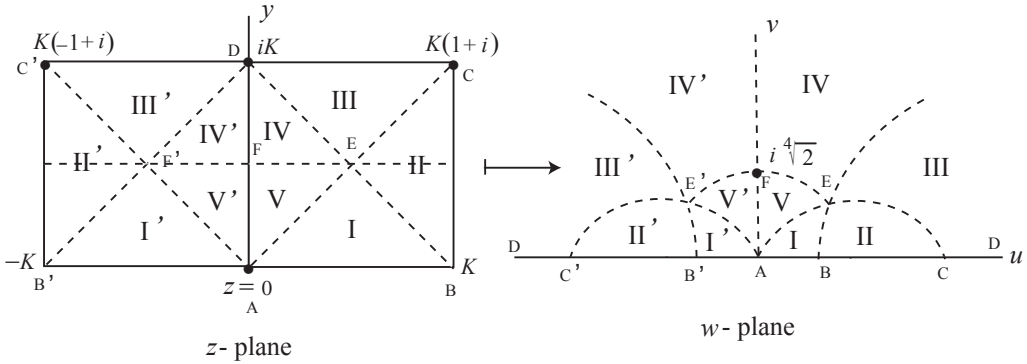


Figure 7.26 Square onto a quarter plane.

Map 7.30. The transformation of a rectangle, with four sub-rectangles marked I, II, III, and IV, in the z -plane onto the right-half of a Cassini's oval with a slit from $0 \leq w \leq 1$ is given by

$$w = \operatorname{cn} z = \sqrt{1 - \operatorname{sn}^2 z}, \quad z = \int_w^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad k^2 + k'^2 = 1. \quad (7.6.7)$$

This transformation is presented in Figure 7.27.

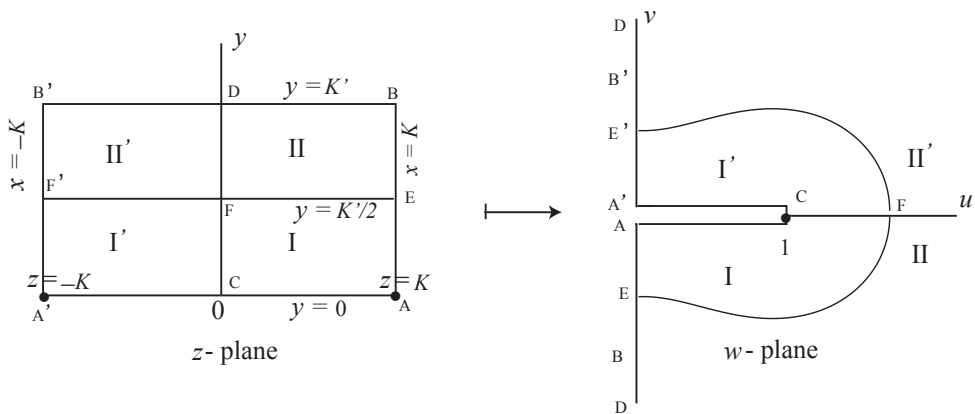


Figure 7.27 Rectangle onto a Cassini's oval.

The details of this mapping are as follows:

- (i) The points $z = 0$ (C); K (A); $-K$ (A'); iK' (D); $K + iK'$ (B); $-K + iK'$ (B'); $iK'/2$ (F); $K + iK'/2$ (E); and $-K + iK'/2$ (E'), in the z -plane are mapped onto the points $w = 1$ (C); 0 (A); 0 (A') ∞ (D); $-ik'/k$ (B); ik'/k (B'); $(k+1)^{1/2}k^{-1/2}$ (F); $-i(1-k)^{1/2}k^{-1/2}$ (E); and $i(1-k)^{1/2}k^{-1/2}$ (E'), respectively, in the w -plane.

- (ii) The line-segment $y = K'/2, -K \leq x \leq K$ in the z -plane is mapped onto the part $u \geq 0$ of Cassini's oval $|w + 1||w - 1| = k^{-1}$.

Map 7.31. The transformation of the same rectangle as in Map 7.29 onto the right-half of a closed Cassini's oval with a slit from $w = 0$ to $w = 1$ (B to C in Figure 7.28) is given by

$$w = \operatorname{dn}(z) = \sqrt{1 - k^2 \operatorname{sn}^2(z)}, \quad \text{or} \quad z = \int_w^1 \frac{dt}{\sqrt{(1-t^2)(t^2-k'^2)}}, \quad (7.6.8)$$

where $k^2 + k'^2 = 1$. The details of this mapping are as follows:

- (i) The points $z = 0$ (C); K (A); $K + iK'/2$ (E); $K + iK'$ (B); iK' (D); $-K + iK'$ (B'); $-K + iK'/2$ (E'); $-K$ (A'); and $iK'/2$ (F) in the z -plane are mapped onto the points $w = 1$ (C); k' (A); $\sqrt{1-k}$ (E); 0 (B); ∞ (D); 0 (B'); $\sqrt{1+k}$ (E'); k' (A'); and $\sqrt{1+k}$ (F), respectively, in the w -plane.
- (ii) The line-segment $y = K', -K \leq x \leq K$, in the z -plane are mapped onto the part $u \geq 0$ of Cassini's oval $|w + 1||w - 1| = k$.

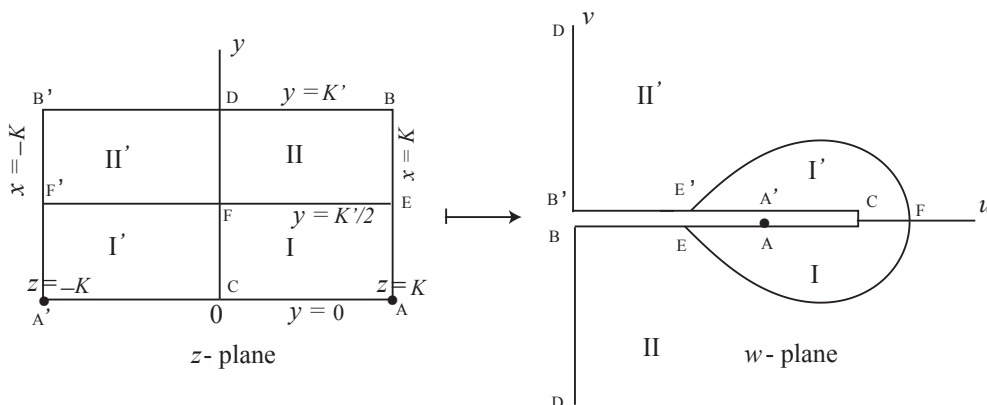


Figure 7.28 Rectangle onto a closed Cassini's oval.

Map 7.32. The transformation of an ellipse with semi-axes a and b , $a > b$, onto the interior of the unit circle is defined by

$$w = \sqrt{k} \operatorname{sn} \left(\frac{2K}{\pi} \arcsin \frac{z}{\sqrt{a^2 - b^2}} \right), \quad (7.6.9)$$

where

$$k = \left(\frac{\Theta_2(\tau)}{\Theta_3(\tau)} \right)^2, \quad \tau = \frac{2i}{\pi} \log \frac{a+b}{a-b}, \quad e^{i\pi\tau} = \left(\frac{a-b}{a+b} \right)^2,$$

$$K = \int_0^{\pi/2} \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}} = F \left(k, \frac{\pi}{2} \right).$$

The transformation is presented in Figure 7.29, and the details are as follows:

- (i) The points $z = 0; \sqrt{a^2 - b^2}; -\sqrt{a^2 - b^2}; a$; and ib in the z -plane are mapped onto the points $w = 0; \sqrt{k}; -\sqrt{k}; 1$; and i , respectively, in the w -plane.

- (ii) The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in the z -plane is mapped onto the unit circle $|w| = 1$.

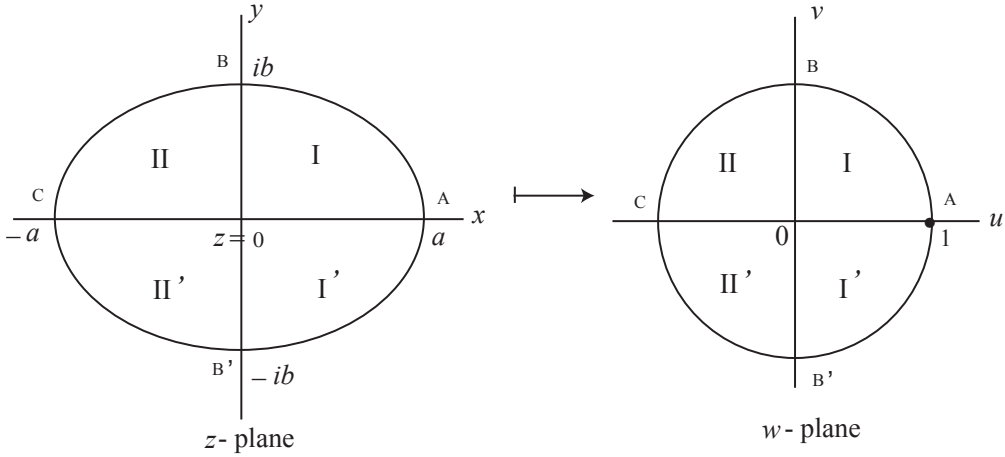


Figure 7.29 Ellipse onto the unit circle.

Map 7.33. The Schwarz-Christoffel transformation that maps the exterior of a rectangle of sides a, b , $a > b$, and the vertices at $\pm a \pm ib$ onto the upper half of the w -plane such that the boundary of the rectangle is mapped onto the boundary of the upper half of the unit circle is given by

$$\frac{dz}{dw} = C (w^2 - c^2)^{1/2} (w^2 - c^{-2})^{1/2} (w^2 + 1)^{-2}, \quad (7.6.10)$$

or

$$z = \frac{c}{k} \left\{ \operatorname{sn} \xi + \left(\frac{E}{K} - k'^2 \right) \xi \right\}, \quad w = \frac{1 - \operatorname{dn}(\xi, k)}{k \operatorname{sn}(\xi, k)} = \frac{1 - \operatorname{dn} \xi}{k \operatorname{sn} \xi}, \quad (7.6.11)$$

where

$$E = \int_0^{\pi/2} \sqrt{1 - k^2 \sin^2 \phi} d\phi, \quad k'^2 = 1 - k^2,$$

$$E' = \int_0^{\pi/2} \sqrt{1 - k'^2 \sin^2 \phi} d\phi, \quad C = -\frac{ibK}{E - k'^2 K};$$

ξ is an auxiliary variable, and $\operatorname{sn}(\xi)$ is the Jacobian zeta function

$$\operatorname{sn}(\xi) = \frac{\vartheta'_0\left(\frac{\xi}{2K} \middle| \tau\right)}{2K \vartheta_0\left(\frac{\xi}{2K} \middle| \tau\right)}, \quad \tau = \frac{iK'}{K},$$

K and K' are defined by (7.6.1); k depends on b/a and is the root of $(E - k'^2 K) a = (E' - k^2 K) b$; and $c = \frac{1 - \sqrt{1 - k^2}}{k}$, $0 < c < 1$. If $a = b$, then the rectangle reduces to a square of side a ; then $k = 1/\sqrt{2}$, and $c = \tan \pi/8 = \sqrt{2} - 1$. This transformation from the

z -plane is presented in Figure 7.30.

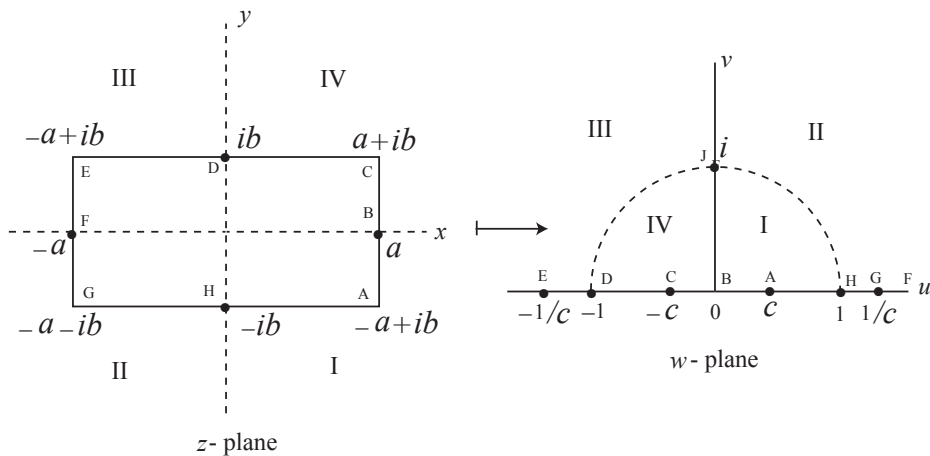


Figure 7.30 Exterior of a rectangle onto the upper half-plane.

This mapping from the ξ -plane is shown in Figure 7.31.

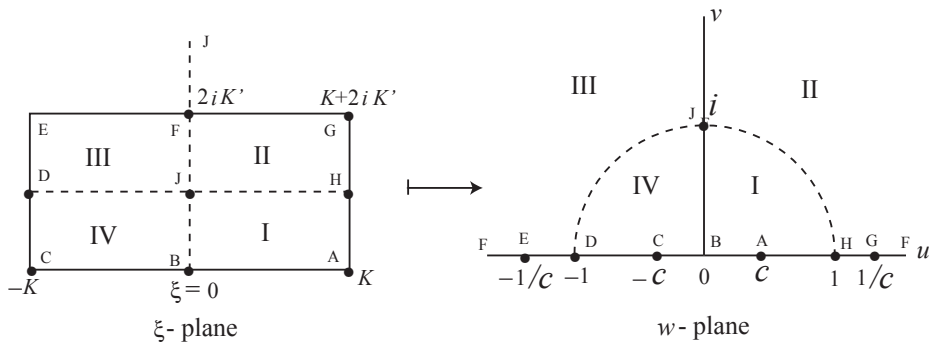


Figure 7.31 Rectangle of Figure 7.30 in the ξ -plane.

Map 7.34. The Schwarz-Christoffel transformation that maps the region exterior to two semi-infinite strips (trigonal regions) of width $2a$ and separated by $2b$, $a > 0, b > 0$, in the z -plane onto the lower half of the w -plane is given by

$$\frac{dz}{dw} = C (w^2 - k^2)^{1/2} (w^2 - 1)^{1/2} w^{-2}, \quad (7.6.12)$$

or

$$z = -C \left\{ \xi \left(k'^2 - \frac{2E}{K} \right) - 2 \operatorname{sn}(\xi) - \frac{\operatorname{cn}(\xi) \operatorname{dn}(\xi)}{\operatorname{sn}(\xi)} \right\} - b, \quad (7.6.13)$$

where $w = 1/\operatorname{sn}(\xi)$ and $C = -\frac{ia}{2E - k'^2 K}$; and depending on b/a only, k is a root of $\frac{b}{a} = \frac{K' k'^2 - 2K' + 2E'}{2(Kk'^2 - 2E)}$. This transformation is presented in Figure 7.32.

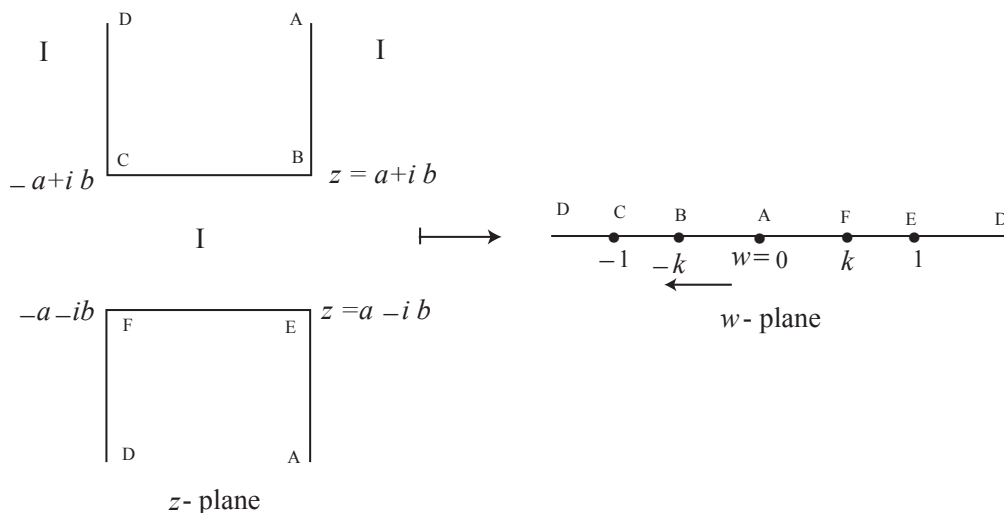


Figure 7.32 Region exterior to two semi-strips onto the lower half-plane.

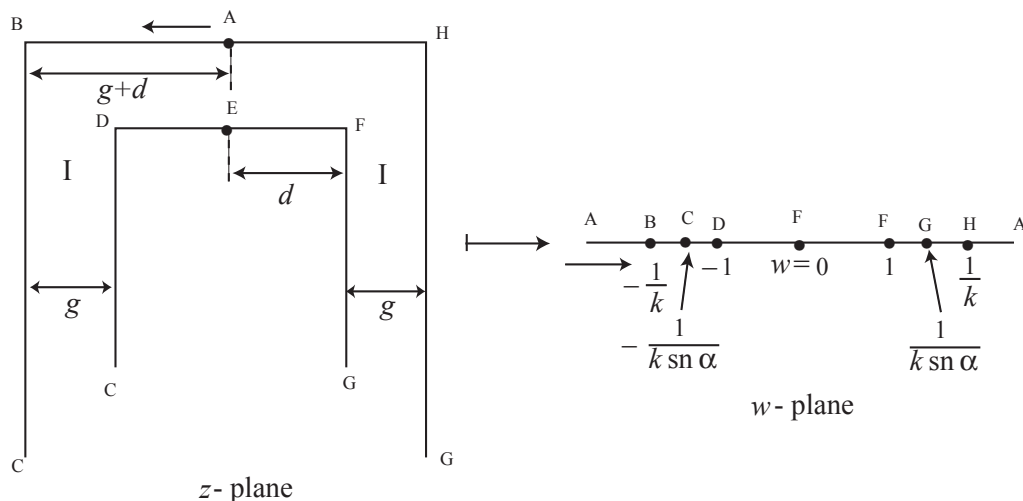


Figure 7.33 Region interior to a semi-infinite strip, but exterior to another one.

Map 7.35. The Schwarz-Christoffel transformation that maps the region interior to a semi-infinite strip, but exterior to another one in the z -plane onto the upper half of the w -plane is defined by

$$\frac{dz}{dw} = \left(\frac{1-w^2}{1-k^2w^2} \right)^{1/2} (1-k^2w^2 \operatorname{sn}^2(\alpha))^{-1}, \quad (7.6.14)$$

or

$$w = \operatorname{sn}(\xi), \quad z = \xi - \frac{\operatorname{dn}(\alpha)}{k^2 \operatorname{sn}(\alpha) \operatorname{cn}(\alpha)} \Pi(\xi, \alpha), \quad 0 < k < 1, \quad (7.6.15)$$

where

$$\alpha = K - K'/2, \quad \Pi(\xi, \alpha) = \frac{\xi}{2K} \frac{\vartheta'_0\left(\frac{\alpha}{2K}|\tau\right)}{\vartheta_0\left(\frac{\alpha}{2K}|\tau\right)} + \frac{1}{2} \log \frac{\vartheta_0\left(\frac{\xi-\alpha}{2K}\right)}{\vartheta_0\left(\frac{\xi+\alpha}{2K}\right)},$$

$$\tau = \frac{iK'}{K}, \quad d = K\left(\frac{1+k}{2k}\right), \quad g = \frac{\pi}{2k}, \quad h = K'\frac{1+K}{2k}.$$

This mapping is shown in Figure 7.33.

Map 7.36. A transformation similar to the one in Map 7.34 and defined by

$$\frac{dz}{dw} = \left(\frac{1-k^2w^2}{1-w^2}\right)^{1/2}, \quad \text{or} \quad w = \operatorname{sn}(\xi), \quad z = \xi - \frac{\operatorname{cn}(\alpha)}{\operatorname{sn}(\alpha)\operatorname{dn}(\alpha)} \prod(\xi, \alpha), \quad (7.6.16)$$

maps the region in the z -plane shown in Figure 7.34 onto the upper half of the w -plane.

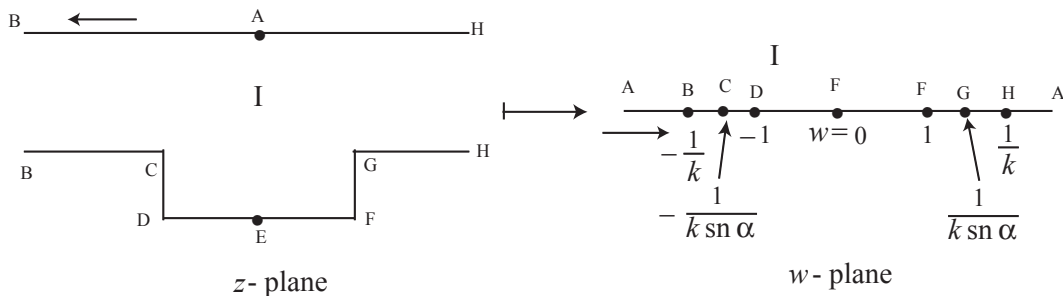


Figure 7.34 Region in the z -plane onto the upper half w -plane.

Map 7.37. The transformation that maps the region inside a (diamond-shaped) square in the z -plane onto the unit disk is defined by

$$z = \int_0^w \frac{dt}{1-t^4}, \quad \text{or} \quad w = \frac{1}{\sqrt{2}} \frac{\operatorname{sn}(z\sqrt{2}, k)}{\operatorname{dn}(z\sqrt{2}, k)}, \quad (7.6.17)$$

and presented in Figure 7.35, where $k = 1/\sqrt{2}$, and

$$\omega_1 = \int_0^1 \frac{dt}{1-t^4} = \frac{K}{\sqrt{2}}, \quad K = \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}} \approx 1.8541. \quad (7.6.18)$$

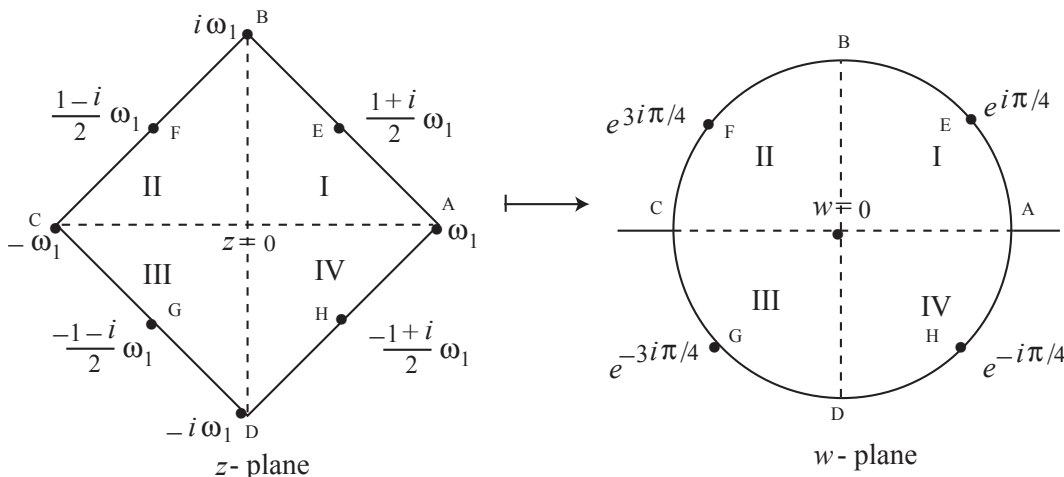


Figure 7.35 Region inside a square onto the unit disk in the w -plane.

Another form of the transformation (7.6.17) is

$$w^2 = \frac{1}{\wp(z; 4, 0)}, \quad (7.6.19)$$

which means that $g_2 = 4, g_3 = 0, e_1 = -e_3 = 1, e_2 = 0$; and $\omega_3 = i\omega_1$, where ω_1 is defined in (7.6.18).

Map 7.38. The transformation that maps a triangle with angles $\pi/2, \pi/4, \pi/4$ onto the half-plane is defined by

$$w = \sqrt{2} \operatorname{sn}(z\sqrt{2}) \operatorname{dn}(z\sqrt{2}). \quad (7.6.20)$$

The details of this transformation are: (i) The points $z = -\omega_1; \omega_1; i\omega_1$ in the z -plane are mapped onto the points $w = -1; 1; \infty$, respectively; and (ii) the triangle (CAB in Figure 7.35) is mapped onto the half-plane $v > 0$, where ω_1 and k are defined in Map 7.36. Compare this transformation with (7.2.2).

Map 7.39. The transformation that maps a regular polygon with n vertices ($n \geq 3$) in the z -plane onto the unit circle in the w -plane is defined by

$$z = \int_0^w \frac{dt}{(1-t^n)^{2/n}} = w + \sum_{j=1}^{\infty} \frac{2 \cdot (2+n) \cdot (2+2n) \cdots (2+(j-1)n)}{j! n^j (jn+1)}. \quad (7.6.21)$$

The details of this transformations are: (i) The points $z = \omega e^{2mi\pi/n}; \frac{\omega}{\cos(\pi/n)} e^{(2m+1)i\pi/n}$, $m = 0, 1, \dots, n-1$, in the z -plane, where

$$\omega = \int_0^1 (1-t^n)^{-2/n} dt, \quad (7.6.22)$$

are mapped onto the points $w = e^{2mi\pi/n}; e^{(2m+1)i\pi/n}$, respectively, in the w -plane; (ii) the line-segment $\arg\{z\} = 2m\pi/n, 0 < |z| < \omega$, in the z -plane is mapped onto the segment $\arg\{w\} = 2m\pi/n, 0 < |w| < 1$; and (iii) the line-segment $\arg\{z\} = (2m+1)\pi/n, 0 < |z| < \omega/\cos(\pi/n)$ in the z -plane is mapped onto the segment $\arg\{w\} = (2m+1)\pi/n, 0 < |w| < 1$. Compare it with (7.1.3).

Map 7.40. The transformation that maps an equilateral triangle in the z -plane onto the half-plane is defined by

$$\begin{aligned} w &= \int_0^z \frac{dt}{(t-t^2)^{2/3}} = f(z), \\ \text{or } z - z^2 &= \wp^3\left(i\frac{w+w_2}{3}; 0, 1\right), \\ \text{or } z &= \frac{1}{2} + \frac{A \operatorname{sn}(\zeta, k) \operatorname{dn}(\zeta, k)}{(1 + \operatorname{cn}(\zeta, k))^2}, \end{aligned} \quad (7.6.23)$$

where

$$\begin{aligned} k &= \frac{\sqrt{3}}{2\sqrt{2}}; \quad \zeta = \frac{2w-w_2}{\sqrt[4]{27}\sqrt[3]{2}}; \quad A = \sqrt[4]{27}; \\ w_1 &= 0; \quad w_2 = \int_0^1 \frac{dt}{(t-t^2)^{2/3}} \approx 5.298, \quad w_3 = e^{i\pi/3}w_2. \end{aligned} \quad (7.6.24)$$

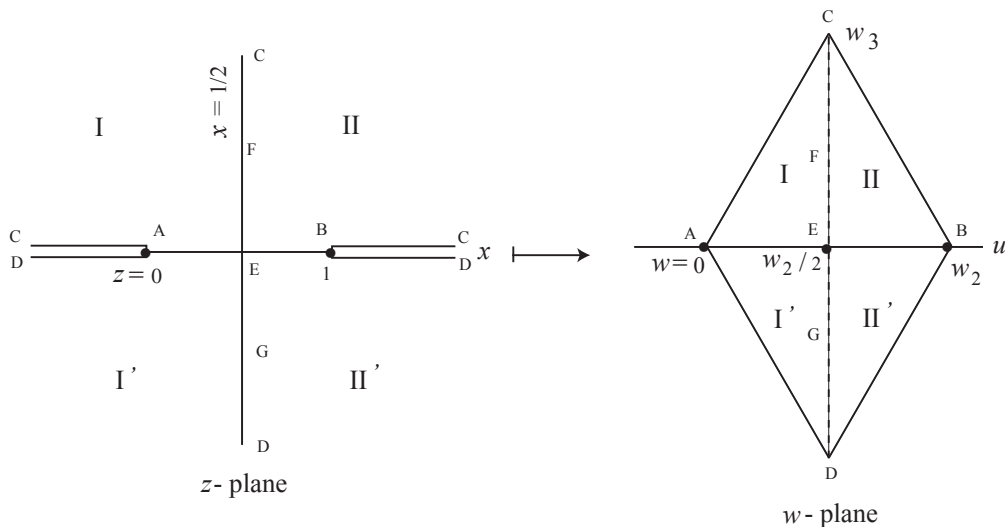


Figure 7.36 Equilateral triangles onto half-planes.

This transformation is presented in Figure 7.36, where the two equilateral triangles are each in the upper and lower half-planes joined at their common base at the u -axis. Compare this transformation with (7.2.4).

When z is real and $0 \leq z \leq 1$,

$$f(z) = \frac{w_2}{2} \mp \frac{\sqrt[4]{27}}{\sqrt[3]{4}} \int_0^\gamma \frac{d\phi}{\sqrt{1 - k^2 \sin^2 \phi}}, \quad (7.6.25)$$

where the minus or plus sign is taken according as $0 \leq z \leq \frac{1}{2}$ or $\frac{1}{2} \leq z \leq 1$; and

$$k = \sin \frac{5\pi}{12} = \frac{\sqrt{6} + \sqrt{2}}{4}; \quad \tan \gamma = 3^{-1/4} \{1 - (4z - 4z^2)^{1/3}\}, \quad 0 \leq \gamma \leq 1.842.$$

Map 7.41. Another case of Schwarz-Christoffel transformation of two triangles, one each in the upper and lower half-planes and joined at their common base on the u -axis onto the upper and lower half-planes with a slit $-\infty < x \leq 1$ is shown in Figure 7.37. The transformation in this case is given by

$$w = \frac{2}{\sqrt[4]{27}} \int_z^\infty \frac{dt}{(t^3 - t^2)^{2/3}}, \quad \text{or} \quad z = c \frac{1 + \operatorname{cn}(w^2)}{\operatorname{sn}(w) \operatorname{dn}(w)}, \quad (7.6.26)$$

where $k = \frac{1 + \sqrt{3}}{2\sqrt{2}}$, and $c = \frac{\sqrt[4]{3}}{6}$.

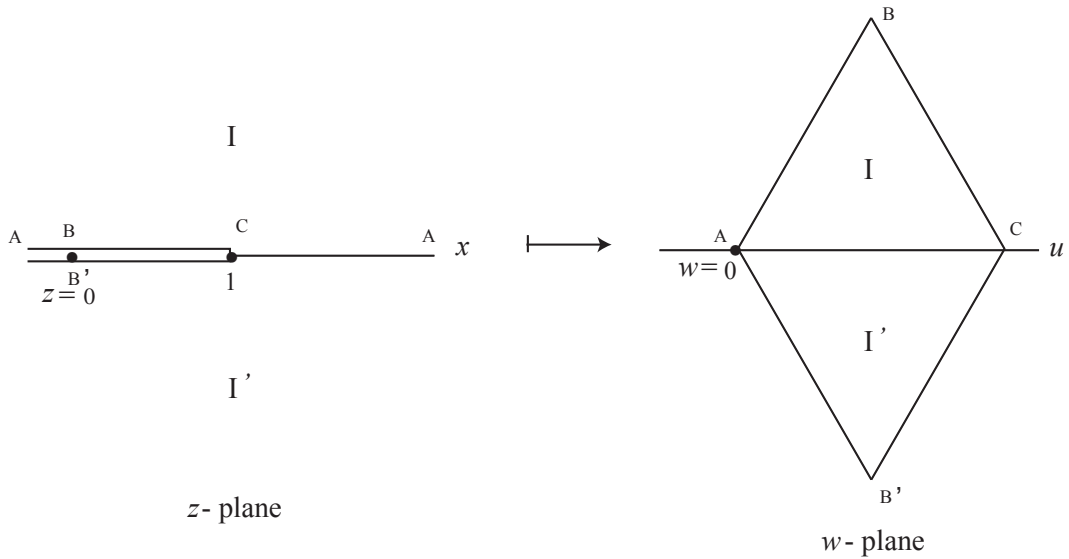
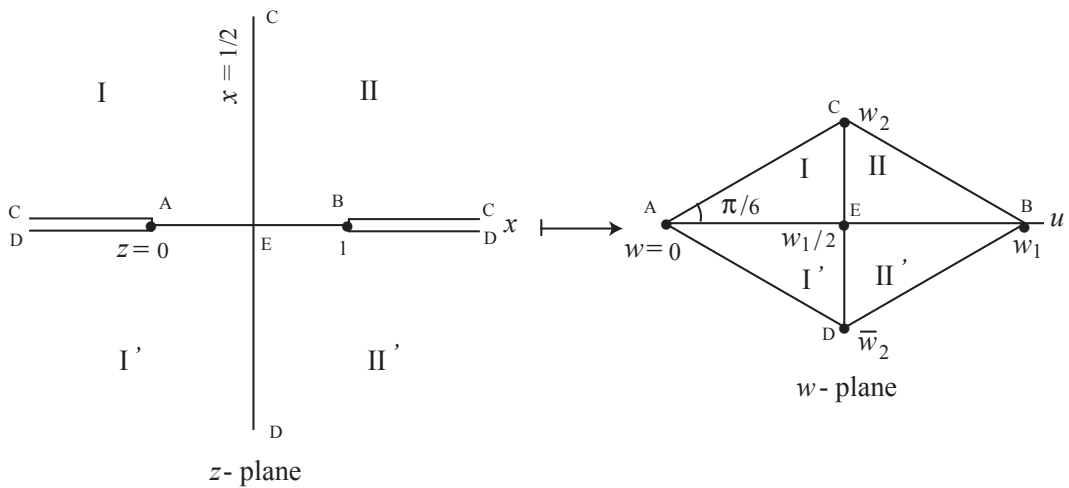


Figure 7.37 Equilateral triangles onto half-planes.

Figure 7.38 Triangle with angles $2\pi/3, \pi/6, \pi/6$.

Map 7.42. The Schwarz-Christoffel transformation of a triangle with angles $\frac{2\pi}{3}, \frac{\pi}{6}, \frac{\pi}{6}$ onto the z -plane with slits $-\infty < x \leq 0$ and $1 \leq x < \infty$, quarter-by-quarter in both planes as presented in Figure 7.38, is given by

$$w = \int_0^z \frac{dt}{(t-t^2)^{5/6}}, \quad \text{or} \quad z - z^2 = \frac{1}{16} \wp^{-3} \left(\frac{w}{3\sqrt[3]{2}}; 0, 1 \right), \quad (7.6.27)$$

$$w_1 = \int_0^1 \frac{dt}{(t-t^2)^{5/6}}, \quad w_2 = \frac{w_1 e^{i\pi/6}}{\sqrt{3}}. \quad (7.6.28)$$

This transformation maps the quarter of the z -plane onto the interior of a triangle with

angles $\pi/2, \pi/3, \pi/6$; and the half-plane $y > 0$ onto the interior of the triangle ABC with angles $\pi/6, \pi/6, 2\pi/3$.

Map 7.43. The Schwarz-Christoffel transformation of a triangle with angles $\frac{\pi}{2}, \frac{\pi}{3}, \frac{\pi}{6}$ onto the z -plane with slits $-\infty < x \leq 0$ and $1 \leq x < \infty$, as shown in Figure 7.39, is defined by

$$w = \frac{1}{\sqrt[4]{108}} \int_0^z \frac{dt}{t^{1/2}(1-t)^{2/3}(1+t)^{5/6}}, \quad \text{or} \quad \frac{1-z}{1+z} = 1 - \sqrt{3} \frac{\operatorname{sn}^2(w)}{\operatorname{cn}^2(w)}, \quad (7.6.29)$$

where $k = \frac{1 + \sqrt{3}}{2\sqrt{2}}$.

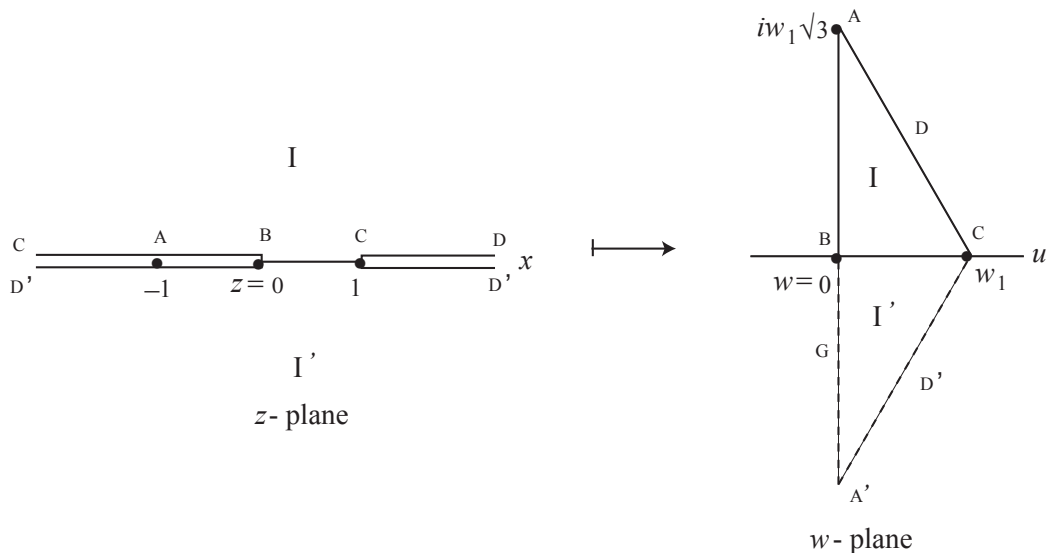


Figure 7.39 Triangle with angles $\pi/2, \pi/3, \pi/6$.

Map 7.44. The Schwarz-Christoffel transformation of the interior of a rectangle onto the exterior of another rectangle is defined by

$$w = \zeta(z) + e_2 z, \quad \text{or} \quad \frac{dw}{ds} = -\frac{1}{2} (s - e_1)^{-1/2} (s - e_2)^{1/2} (s - e_3)^{-1/2}, \quad s = \wp(z), \quad (7.6.30)$$

where

$$\begin{aligned} \omega_1 > 0, \quad \frac{\omega_3}{i} > 0; \quad e_1 > e_2 > e_3; \\ \zeta(z) &= \frac{1}{z} - \int_0^z (\wp(t) - t^{-2}) dt, \quad \zeta'(z) = -\wp(z); \\ \tau &= \omega_3/\omega_1; \quad \zeta(\omega_1) = \eta, \quad \zeta(\omega_3) = \eta'; \quad \eta\omega_3 - \eta'\omega_1 = i\pi/2. \end{aligned} \quad (7.6.31)$$

This transformation maps the points $z = \omega_1$ (A); $-\omega_1$ (E); ω_3 (C); $-\omega_3$ (G); $\omega_1 + \omega_3$ (B); $\omega_3 - \omega_1$ (D); $-\omega_1 - \omega_3$ (F); $\omega_1 - \omega_3$ (H); and 0 (K) in the z -plane onto $w = \eta + e_2\omega_1$ (A); $-\eta - e_2\omega_1$ (E); $\eta' + e_2\omega_3$ (C); $-\eta' - e_2\omega_3$ (G); $\eta + \eta' + e_2(\omega_1 + \omega_3)$ (B); $\eta' - \eta + e_2(\omega_3 - \omega_1)$ (D); $-\eta + \eta' - e_2(\omega_1 - \omega_3)$ (F); $\eta - \eta' + e_2(\omega_1 - \omega_3)$ (H); and 0 (K) in the w -plane.

(D); $-\eta - \eta' - e_2(\omega_1 + \omega_3)$ (F); $\eta - \eta' + e_2(\omega_1 - \omega_3)$ (H); and ∞ (C), respectively, in the w -plane.

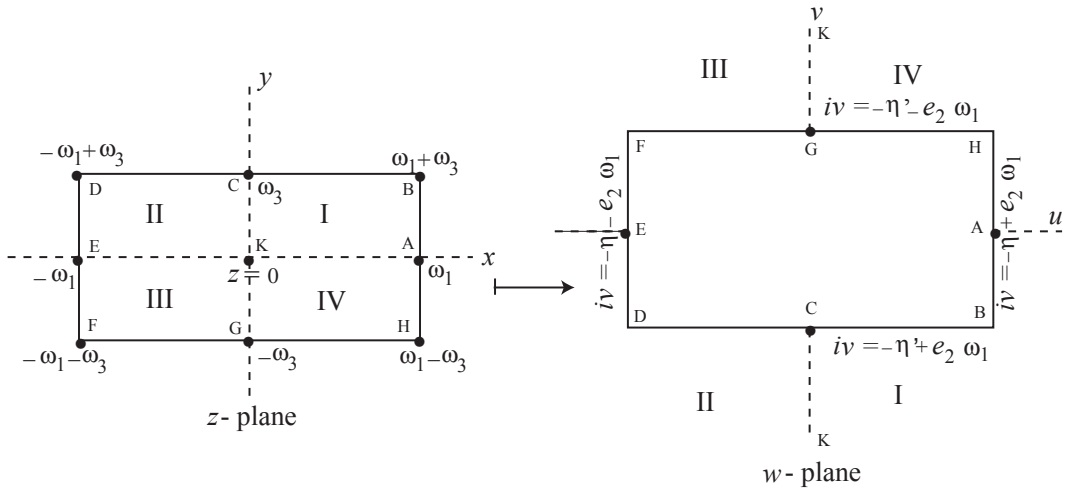


Figure 7.40 Interior of a rectangle onto exterior of another rectangle.

Note that if $\omega_3 = i\omega_1$, then $e_2 = e_1 + e_3 = 0$, $w = \zeta(z)$, and the rectangle becomes a square.

Map 7.45. The Schwarz-Christoffel transformation of the interior of a rectangle onto the region exterior to two semi-infinite strips, one opening in the right-half and the other in the left-half of the w -plane, is defined by

$$w = \zeta(z) + e_1 z, \quad \text{or} \quad \frac{dw}{ds} = -\frac{1}{2} (s - e_1)^{1/2} (s - e_2)^{-1/2} (s - e_3)^{-1/2}, \quad s = \wp(z), \quad (7.6.32)$$

where $\omega_1, \omega_3; e_1, e_2, e_3; \eta, \eta'$ are the same as in (7.6.31).

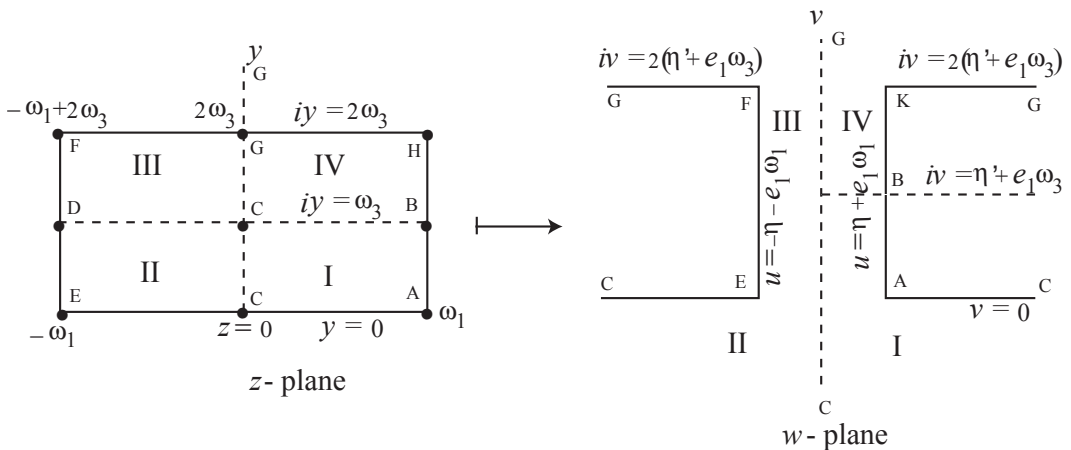


Figure 7.41 Interior of a rectangle onto region of two semi-infinite strips.

This transformation, presented in Figure 7.41, maps the points $z = \omega_1$ (A); $\omega_1 + \omega_3$ (B);

$\omega_1 + 2\omega_3$ (K); ω_3 (C); $2\omega_3$ (G); 0 (C); $-\omega_1$ (E); $-\omega_1 + \omega_3$ (D); and $-\omega_1 + \omega_3$ (F) in the z -plane onto the points $w = \eta + e_1\omega_1$ (A); $\eta + \eta' + e_1(\omega_1 + \omega_3)$ (B); $\eta + 2\eta' + e_1(\omega_1 + 2\omega_3)$ (K); $\eta' + e_1\omega_3$ (C); ∞ (G); ∞ (C); $-\eta - e_1\omega_1$ (E); $-\eta + \eta' + e_1(\omega_3 - \omega_1)$ (D); and $-\eta + 2\eta' + e_1(2\omega_3 - \omega_1)$ (F), respectively, in the w -plane.

Map 7.46. The Schwarz-Christoffel transformation of the interior of a rectangle onto the region exterior to two semi-infinite strips, one opening in the upper-half and the other in the lower-half of the w -plane, is defined by

$$w = \zeta(z) + e_3 z, \quad (7.6.33)$$

where $\omega_1, \omega_3; e_1, e_2, e_3; \eta, \eta'$ are the same as in (7.6.31).

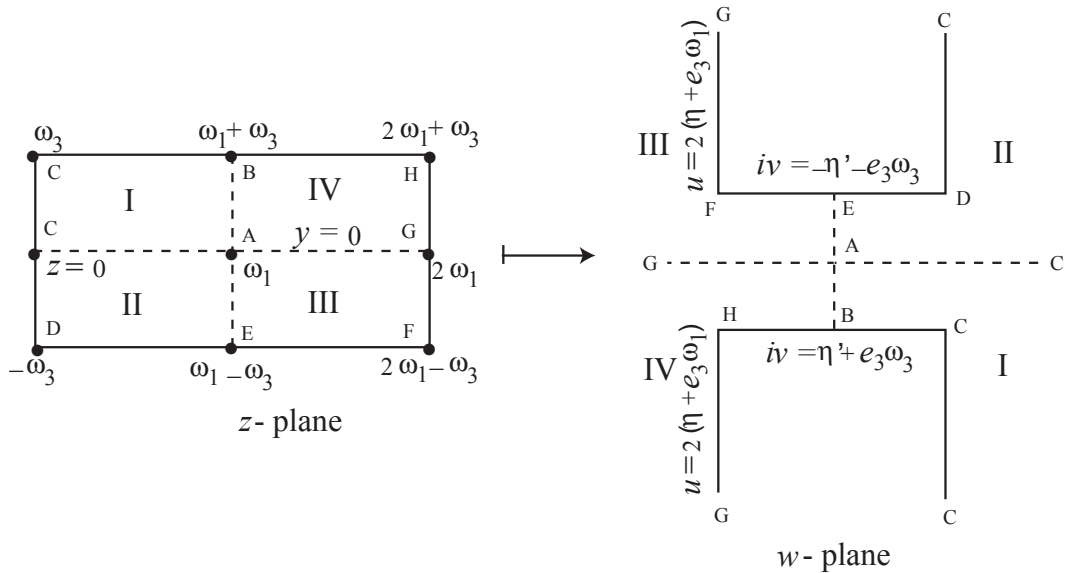


Figure 7.42 Interior of a rectangle onto region of two semi-infinite strips.

This transformation, presented in Figure 7.42, maps the points $z = \omega_1 - \omega_3$ (E); ω_1 (A); $\omega_1 + \omega_3$ (B); ω_3 (C); 0 (C); $-\omega_3$ (D); $2\omega_1 - \omega_3$ (F); $2\omega_1$ (G); and $2\omega_1 + \omega_3$ (K) in the z -plane onto the points $w = \eta - \eta' + e_3(\omega_1 - \omega_3)$ (E); $\eta + e_3\omega_1$ (A); $\eta' + e_3\omega_3$ (C); ∞ (C); $-\eta' - e_3\omega_3$ (D); $2\eta - \eta' + e_3(2\omega_1 - \omega_3)$ (F); ∞ (G); and $2\eta + \eta' + e_3(2\omega_1 + \omega_3)$ (K), respectively, in the w -plane.

Map 7.47. The Schwarz-Christoffel transformation of the interior of a rectangle onto a cut plane, presented in Figure 7.43, is defined by

$$w = \zeta(z) - \frac{\eta}{\omega_1} z = \frac{1}{2\omega_1} \frac{\vartheta'_1\left(\frac{z}{2\omega_1} \middle| \tau\right)}{\vartheta_1\left(\frac{z}{2\omega_1} \middle| \tau\right)}, \quad (7.6.34)$$

where $\omega_1, \omega_3; e_1, e_2, e_3; \eta, \eta'$ are the same as in (7.6.31). The transformation is a composite of

$$\frac{dw}{ds} = -\frac{s + \eta/\omega_1}{\sqrt{4(s - e_1)(s - e_2)(s - e_3)}}, \quad \text{and} \quad s = \wp(z). \quad (7.6.35)$$

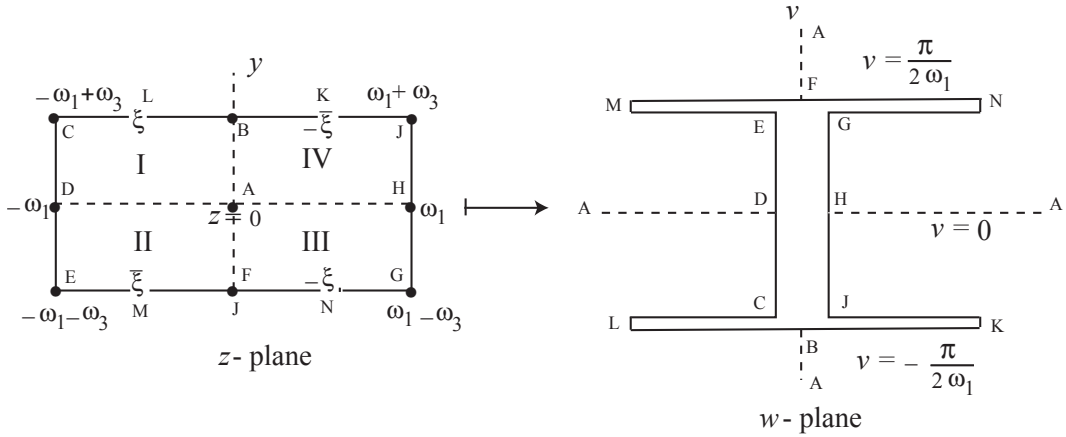


Figure 7.43 Interior of a rectangle onto a cut plane.

This transformation maps (i) the points $z = \omega_3 - \omega_1$ (C); $\omega_1 + \omega_3$ (J); ω_3 (B) onto $w = -\frac{i\pi}{2\omega_1}$ (C, J, B); the points $z = -\omega_3 - \omega_1$ (E); $\omega_1 - \omega_3$ (G); $-\omega_3$ (F) onto the points $w = \frac{i\pi}{2\omega_1}$ (E, G, F); the points $z = -\omega_1$ (D); ω_1 (K); 0 (A) onto the points $w = 0$ (D); 0 (K); ∞ (A); and the points $x = \xi$ (L); $\bar{\xi}$ (M); $-\bar{\xi}$ (K); and $-\xi$ (N) onto the points $w_P = \zeta(\xi)$; $\bar{w}_P$ (M); $-\bar{w}_P$ (K); and $-w_P$ (N) in the w -plane; and (ii) the line-segments $y = 0$, $\omega_1 < x < 0$ or $0 < x < \omega_1$, are mapped onto the half-lines $v = 0$, $-\infty < u < 0$ or $0 < u < \infty$ in the w -plane.

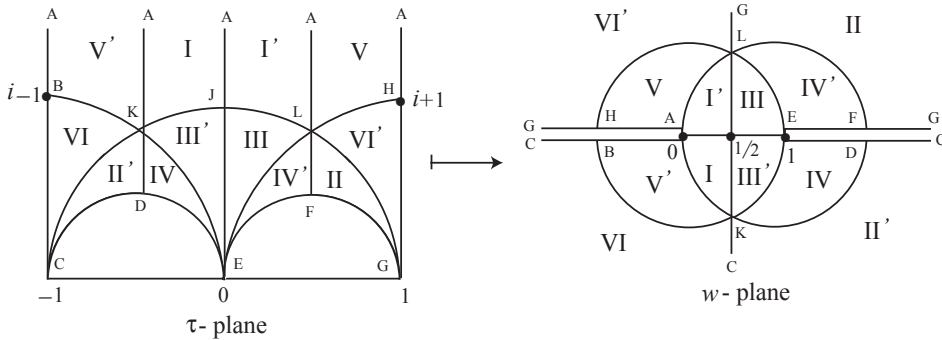


Figure 7.44 Curvilinear triangles onto two intersecting unit circles.

Map 7.48. We will consider curvilinear triangles with angles (i) $0, 0, 0$; (ii) $\pi/3, \pi/3, 0$; (iii) $\pi/2, \pi/3, 0$; (iv) $\pi/2, 0, 0$; and (v) $2\pi/3, 0, 0$ in the τ -plane, and their mapping onto two intersecting unit circles $|w| = 1$ and $|w - 1| = 1$. The transformation for this mapping is defined by

$$w = \lambda(\tau) = \frac{\Theta_2^4(\tau)}{\Theta_3^4(\tau)} = k^2, \quad (7.6.36)$$

where $\lambda(\tau)$ is the modular function (see Appendix F for definition), and $\Im\{\tau\} > 0$ ($\tau = x + iy$). Let $\alpha, \beta, \gamma, \delta$ be integers, where α and δ are odd, and β and γ are even, such that

$\alpha\delta - \gamma\beta = 1$. The critical points of the mapping are $\tau = \infty$ and any point on $y = 0$. A typical mapping is presented in Figure 7.44.

The details of the mapping are as follows:

- (i) The points $\frac{\gamma + \delta\tau}{\alpha + \beta\tau}; \frac{\tau}{\tau + 1}; \frac{1}{\tau}; \tau + 1; \frac{1}{\tau_1}; \frac{\tau + 1}{\tau}$ in the z -plane are mapped onto the points $w = \lambda(\tau); \frac{1}{\lambda(\tau)}; 1 - \lambda(\tau); \frac{\lambda(\tau)}{\lambda(\tau) - 1}; \frac{1}{1 - \lambda(\tau)}; \frac{\lambda(\tau) - 1}{\lambda(\tau)}$, respectively.
- (ii) The points $\tau = 1$ (J); $i - 1$ (B); $-\frac{1}{2} + \frac{i}{2}\sqrt{3}$ (K); $\frac{1}{2} + \frac{i}{2}\sqrt{3}$ (L); $\pm\frac{1}{2} + \frac{i}{2}$ (F, D) in the z -plane are mapped onto the points $w = \frac{1}{2}$ (J); -1 (B); $\frac{1}{2} - \frac{i}{2}\sqrt{3}$ (K); $\frac{1}{2} + \frac{i}{2}\sqrt{3}$ (L); 2 (F, D), respectively, in the w -plane.
- (iii) The area of the curvilinear quadrilateral, bounded by half-lines $x = \pm 1, y \geq 0$ and by semi-circles $|\tau \pm \frac{1}{2}| = \frac{1}{2}, y \geq 0$, is mapped onto the whole w -plane with two slits $-\infty < x \leq 0$ and $1 \leq x < \infty$.
- (iv) The area of the curvilinear triangles ACJGA, with angles $0, 0, 0$ in the z -plane is mapped onto the cut half-plane $u < 0$ (i.e., VI+V'+I+I'+V+VI').
- (v) The area of the curvilinear triangles AKJLA, with angles $\pi/3, \pi/3, 0$ in the z -plane is mapped onto the region (I+I').
- (vi) The area of the curvilinear triangles AJLA, with angles $\pi/2, \pi/3, 0$ in the z -plane is mapped onto the region I'.
- (vii) The area of the curvilinear triangles AJLGA, with angles $\pi/2, 0, 0$ in the z -plane is mapped onto the quadrant $u < 0, v > 0$ (i.e., I'+V+VI').
- (viii) The area of the curvilinear triangles AJELA, with angles $0, 2\pi/3, 0$ in the z -plane is mapped onto the region (I'+III).

Note that the transformation $\tau' = \frac{\gamma + \delta\tau}{\alpha + \beta\tau}$ maps the area of the quadrilateral, mentioned in (iii) above, onto another curvilinear quadrilateral with angles $0, 0, 0, 0$. Next, the transformation $w = \lambda(\tau')$ maps both of these quadrilaterals onto the whole w -plane, with two slits as in Figure 7.44, where all quadrilaterals together cover the half-plane $y > 0$.

Map 7.49. The transformation that maps two intersecting unit circles in the ζ -plane, shown in Figure 7.45, onto the whole w -plane with three slits in line is defined by

$$\begin{aligned}
 w = J(\tau) &= \frac{1}{54} \frac{\Theta_0^8(\tau) + \Theta_2^8(\tau) + \Theta_3^8(\tau)}{[\pi\Theta_0(\tau)\Theta_2(\tau)\Theta_3(\tau)]^8} \\
 &= \frac{4}{27} \frac{(1 - \lambda(\tau) + \lambda^2(\tau))^3}{(\lambda^2(\tau) - \lambda(\tau))^2} \\
 &= \frac{27g_2^3}{27g_2^3 - g_3^2}, \quad y > 0, \tau = x + iy; \quad \pi\Theta_0(\tau)\Theta_2(\tau)\Theta_3(\tau) = \theta'_1.
 \end{aligned} \tag{7.6.37}$$

The critical points of this transformation are: $\tau = \infty$, and any point on the axis $y = 0$.

The transformation (7.6.37) is a composition of $\zeta = \lambda(\tau)$ and $w = \frac{4}{27} \frac{(1 - \zeta + \zeta^2)^3}{(\zeta^2 - \zeta)^2}$.

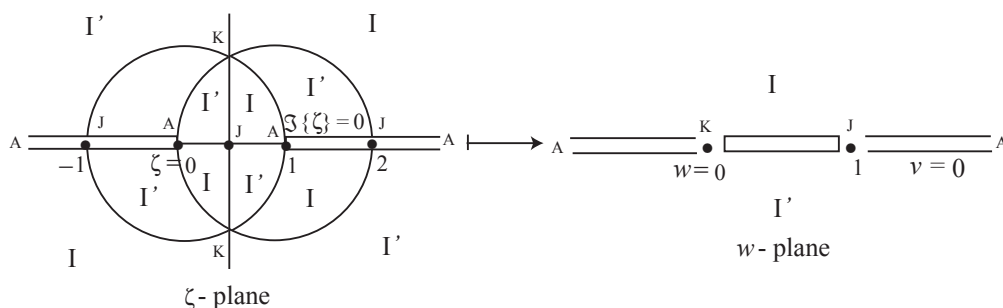


Figure 7.45 Two intersecting unit circles onto the whole plane with three slits in line.

Map 7.50. The transformation that maps the regions over the upper half of the unit circle $|\tau| = 1$ in the τ -plane onto the whole $w = J(\tau)$ -plane with one slit $-\infty < u \leq 0$ is presented in Figure 7.46; it has the following properties: (i) The points $\tau = i$ (J); $\frac{1}{2} + \frac{i}{2}\sqrt{3}$ (L); $-\frac{1}{2} + \frac{i}{2}\sqrt{3}$ (K) are mapped onto the points $w = 1$ (J); 0 (K); 0 (L), respectively, in the w -plane; and (ii) the point $\tau' = \frac{\gamma + \delta\tau}{\alpha + \beta\tau}$ is mapped onto the point $w = J(\tau) = J(\tau')$.

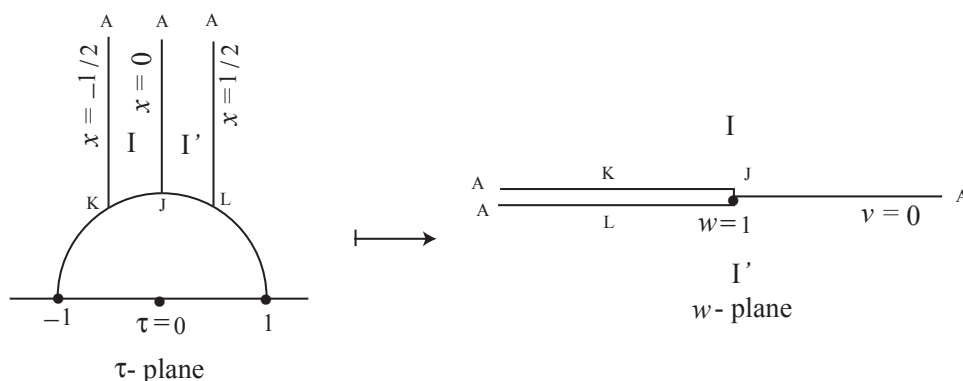


Figure 7.46 Upper half of the unit circle onto the whole plane with one slit.

Map 7.51. The transformation that maps the equilateral equiangular circular triangle, with angle $\alpha, 0 \leq \alpha < \pi$, in the z -plane onto a circle in the w -plane, as shown in Figure 7.47, is defined by

$$w = z \frac{\Gamma(\frac{2}{3}) \Gamma(\frac{5}{6} + \beta) F(\frac{5}{6} - \beta, \frac{1}{2} - \beta, \frac{4}{3}; z^3)}{\Gamma(\frac{4}{3}) \Gamma(\frac{1}{6} + \beta) F(\frac{1}{6} - \beta, \frac{1}{2} - \beta, \frac{2}{3}; z^3)}, \quad (7.6.38)$$

where $\beta = \alpha/(2\pi)$, and

$$F(a, b, c; z) = 1 + \frac{ab}{1!c}z + \frac{a(a+1)b(b+1)}{2!c(c+1)}z^2 + \frac{a(a+1)(a+2)b(b+1)(b+2)}{3!c(c+1)(c+2)}z^3 + \cdots, \quad |z| < 1, \quad (7.6.39)$$

is the Gaussian hypergeometric series.

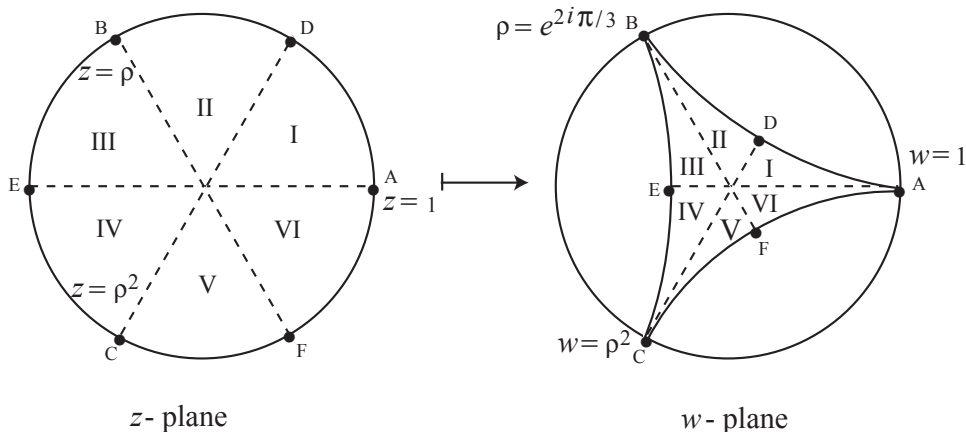


Figure 7.47 Equilateral equiangular circular triangle onto a circle.

Notice that in Figure 7.47, the radius ADB is equal to $\frac{\sqrt{3}}{2} \left| \cos \left(\frac{\alpha}{2} + \frac{\pi}{3} \right) \right|^{-1}$.

For $\alpha = \pi/3$, see Map 7.39.

Map 7.52. $\alpha = 0$: The transformation that maps the interior of the circular triangle ABC, with angles $0, 0, 0$, in the z -plane onto the half-plane $y > 0$ is

$$z = \lambda \left(\frac{\rho^2 - w\rho}{w - 1} \right), \quad (7.6.40)$$

where $\lambda(\tau), \tau = \frac{\rho^2 - w\rho}{w - 1}$ and $\rho = e^{2i\pi/3}$, is the modular function defined in Appendix F. This transformation maps (i) the points $z = 0; 1; \infty$ in the z -plane onto the points $\tau = \infty; 0; 1$ in the τ -plane, and the points $w = 1$ (A); ρ (B); ρ^2 (C), respectively, in the w -plane; (ii) the points $z = 2; \frac{1}{2}$ in the z -plane onto the points $\tau = \frac{1+i}{2}; i$ in the τ -plane, and onto the points $w = \sqrt{3} - 2$ (E); $(2 - \sqrt{3}) e^{i\pi/3}$ (D), respectively, in the w -plane; and (iii) the points $z = -1; e^{i\pi/3}$ in the z -plane onto the points $\tau = i + 1; -\rho^2 = e^{i\pi/3}$ in the τ -plane, and onto the points $w = (2 - \sqrt{3}) e^{-i\pi/3}$ (F); 0 (C), respectively, in the w -plane.

Map 7.53. The transformation $w = \left(\frac{\zeta^{\pi/\alpha} + 1}{\zeta^{\pi/\alpha} - 1} \right)^2$, where $\zeta = \frac{z_1 - \zeta_0}{z_1 - z_0} \frac{z - z_0}{z - \zeta_0}$, with given z_0, z_1, ζ_0 and the angle α ; z_0, ζ_0 finite, and $0 < \alpha < 2\pi$, maps the curvilinear triangle with two right angles, and the third angle $\alpha \neq \pi$, shown in Figure 7.48, in the z -plane, onto the upper half-plane $\Im\{w\} > 0$. For other cases, see Map 4.50 and 4.53.

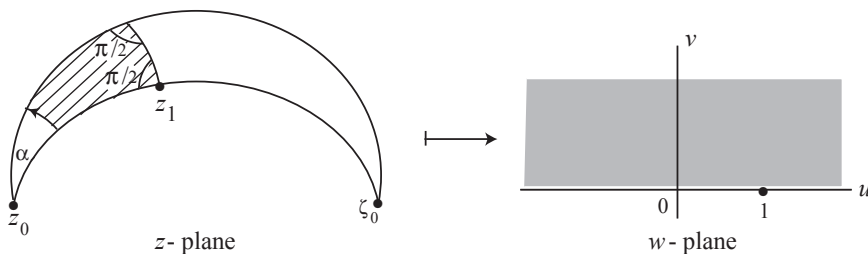


Figure 7.48 Curvilinear triangle with two right angles onto the upper half-plane.

Map 7.54. The transformation that maps the interior of a triangle in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is defined by

$$w = \int_0^z t^{\alpha/\pi-1} (1-t)^{\beta/\pi-1} dt, \quad (7.7.1)$$

and is presented in Figure 7.49.

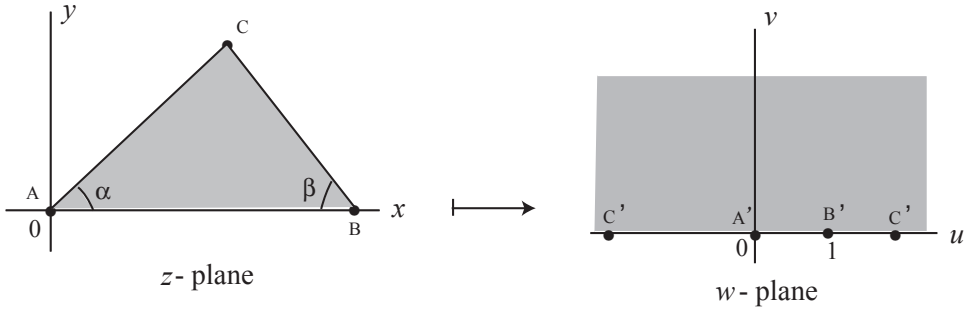


Figure 7.49 Interior of a triangle onto the upper half-plane.

Map 7.55. The transformation that maps the interior of a rectangle in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is defined by

$$w = \int_0^z \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}}, \quad (7.7.2)$$

and is presented in Figure 7.50.

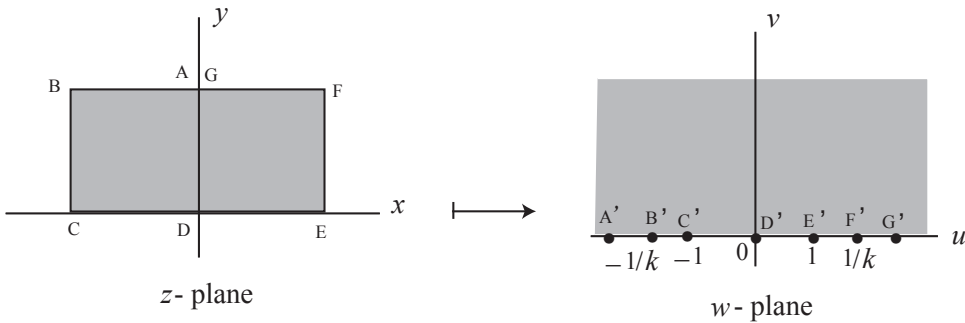


Figure 7.50 Interior of a rectangle onto the upper half-plane.

Map 7.56. The Schwarz-Christoffel transformation that maps the upper half of the z -plane with the path AOCB, where C is at $w = ia$, onto the upper half-plane $\Im\{w\} > 0$, as shown in Figure 7.51, is defined by

$$\frac{dw}{dz} = C_1(z-0)^{-1/2}(z-1)^{-3\pi/2}C_1\sqrt{\frac{1-z}{z}} = K\sqrt{\frac{1-z}{z}},$$

which gives

$$w = K \int \sqrt{\frac{1-z}{z}} dz.$$

To integrate, take $z = \sin^2 \theta$, which leads to $w = K \left(\arcsin \sqrt{z} + \sqrt{z(1-z)} \right) + C_2$. The constants are evaluated using the boundary conditions: $w(0) = 0$ and $w(1) = ib$, which yields

$$w = \frac{2ib}{\pi} \left(\arcsin \sqrt{z} + \sqrt{z(1-z)} \right). \quad (7.7.3)$$

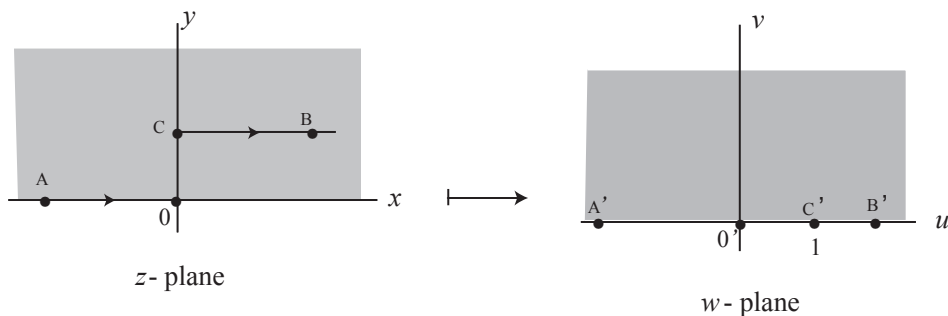


Figure 7.51 Upper half-plane onto upper half-plane.

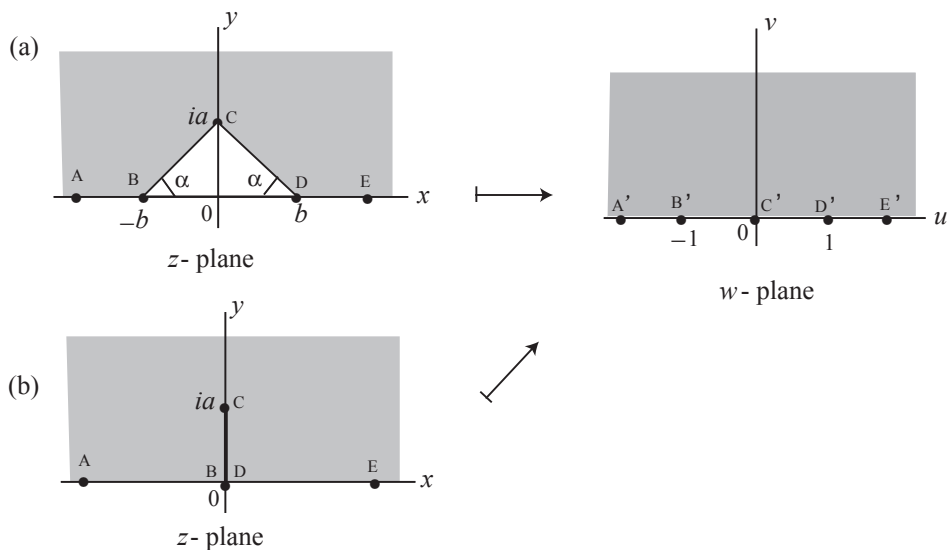


Figure 7.52 Upper half-plane with a triangle removed onto upper half-plane.

Map 7.57. The Schwarz-Christoffel transformation that maps the upper half of the z -plane with an isosceles triangle of angle α removed onto the upper half-plane $\Im\{w\} > 0$, as presented in Figure 7.52, is defined by

$$w = K \int_0^z \frac{t^{2\alpha/\pi}}{(1-t^2)^{\alpha/\pi}} dt + C_2,$$

where using the boundary condition $w(0) = ia$ we obtain $C_2 = ia$, and using the boundary

condition $w(1) = b$ the value of K can be expressed in terms of the gamma function as

$$K = \frac{(b - ia)\sqrt{\pi}}{\Gamma\left(\frac{\alpha}{\pi} + \frac{1}{2}\right)\gamma\left(1 - \frac{\alpha}{\pi}\right)}.$$

The case $b = 0$ is shown in Figure 7.52(b). In this case $\alpha \rightarrow \pi/2$, and the transformation from part (a) reduces to

$$w = ia - ia \int_0^z \frac{t}{\sqrt{1-t^2}} dt = ia\sqrt{1-z^2} = a\sqrt{z^2-1}.$$

Map 7.58. The function

$$w = f(z) = \sqrt{k(q)} \operatorname{sn}\left(\frac{2K}{\pi} \arcsin z, q\right), \quad q = \left(\frac{a-b}{a+b}\right)^2, \quad (7.7.4)$$

maps the interior of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ onto the unit disk $|w| < 1$, such that the foci of the ellipse go into the points $w = \pm\sqrt{k(q)}$, and

$$\begin{aligned} f'(0) &= \frac{2}{a+b} \left(\sum_{n=0}^{\infty} q^{n(n+1)} \right) \cdot \left(1 + 2 \sum_{n=1}^{\infty} q^n \right) \\ &= \begin{cases} 1.0165984 & \text{for } a/b = 1.2, \\ 1.2376223 & \text{for } a/b = 2, \\ 2.372368 & \text{for } a/b = 5. \end{cases} \end{aligned}$$

Map 7.59. Let the field of an infinite two-dimensional capacitor be as shown in the Figure 7.53, with the values of the potential on the curves Γ_1 and Γ_2 as 0 and 1. Then the function $w = f(z)$ that maps the upper half-plane onto the triangle $A_1A_2A_3$ is given by

$$w = C_0 \int_{-1}^z \zeta^{-1} (1 + \zeta)^\alpha d\zeta + C_1,$$

where $C_1 = ih$ and $C_0 = h/\pi$. (Sveshnikov and Tikhonov [1978:217-218].)

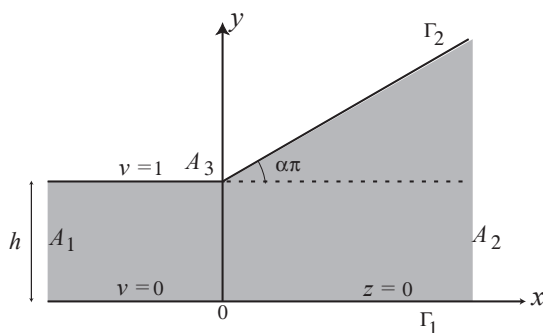


Figure 7.53 Two-dimensional capacitor.

Map 7.60. The function $w = \int_0^z \frac{dt}{\sqrt{t(1-t^2)}}$ maps the upper half-plane $\Im\{z\} > 0$ onto the interior of the square of side $\frac{\Gamma^2(1/4)}{2\sqrt{2}\pi}$. [(Phillips 1966:65).]

Map 7.61. The Schwarz-Christoffel transformation that maps the region (shaded) in Figure 7.54 onto the upper half-plane $\Im\{w\} > 0$ such that $z = 0$ goes into $w = 0$ is given by

$$z = \frac{a}{\mu\pi} \int_1^w \left(\frac{\zeta-1}{\zeta} \right)^\mu d\zeta.$$

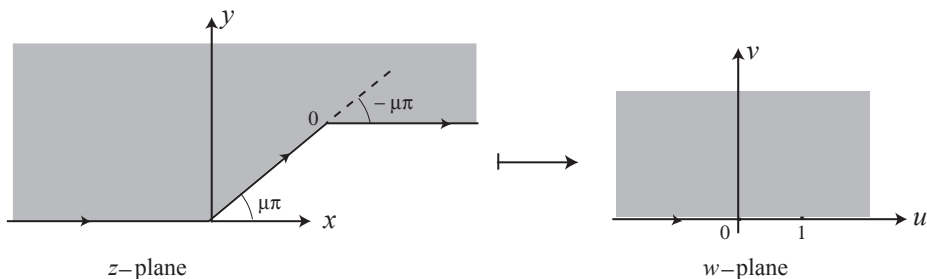


Figure 7.54 The shaded region.

Let $\mu = p/q$, $p < q$ ($p, q \in \mathbb{R}^+$), and set $t = \left(\frac{\zeta-1}{\zeta} \right)^{1/q}$, as in Map 7.15. Show that the mapping function becomes

$$z = \frac{a}{\pi} \left[-\frac{1}{\mu} \frac{t^p}{t^q - 1} + \sum_{n=0}^{q-1} t_n^p \log \left(1 - \frac{t}{t_n} \right) \right],$$

where $t_n = e^{2i\pi n/q}$, $n = 0, 1, \dots, q-1$, are the q zeros of $t^q - 1$. [Use $\frac{t^{p-1}}{t^q - 1} = \frac{1}{q} \sum_{n=0}^{q-1} \frac{t_n^p}{t - t_n}$.]

In particular, for $\mu = 1/2$, $p = 1$, $q = 2$, show that the mapping function is given by $z = \frac{a}{\pi} \left[\frac{2t}{1-t^2} + \log \frac{1-t}{1+t} \right]$ (von Koppenfels [1939: 210-211]).

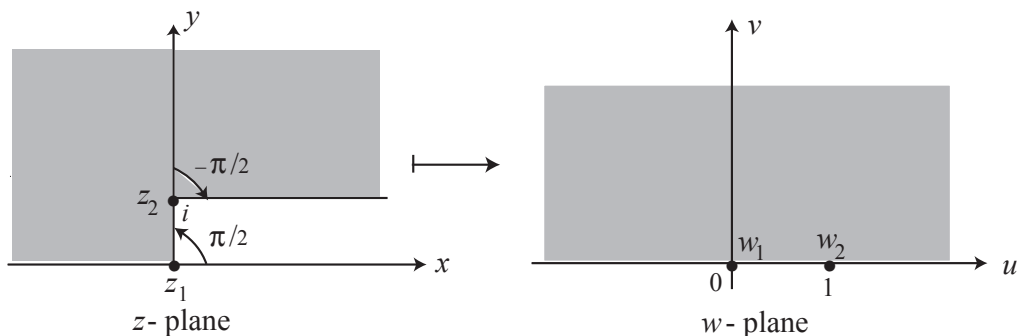


Figure 7.55 For Map 7.62.

Map 7.62. This is a special case of Map 7.61 when $\mu = \frac{1}{2}$. The region shown in Figure 7.55 is mapped onto the upper half-plane $\Im\{w\} > 0$, such that the three vertices z_1, z_2, z_3 are mapped onto the points $w_1 = 0, w_2 = 1$, and $w_3 = \infty$, respectively. Then the Schwarz-Christoffel transformation is given by

$$w = A' \int_{z_0}^z t^{-1/2}(t-1)^{1/2} dt + B. \quad (7.7.5)$$

If we write $A'(z-1)^{1/2} = A(1-z)^{1/2}$, then the mapping (7.7.5) becomes

$$w = A \int_{z_0}^z \sqrt{\frac{1-z}{z}} dz + B. \quad (7.7.6)$$

If we set $z = \sin^2 \theta$, then $\frac{1-z}{z} = \frac{1-\sin^2 \theta}{\sin^2 \theta}$, so that the mapping (7.7.6) becomes

$$w = A \int_{z_0}^z (1 + \cos 2\theta) d\theta + B = \theta + \sin \theta \cos \theta.$$

Again, since $\sin \theta = \sqrt{z}$, the map (7.7.6) becomes

$$w = A \int_{z_0}^z \sqrt{\frac{1-z}{z}} dz + B = A \left(\arcsin \sqrt{z} + \sqrt{z(1-z)} \right) + B,$$

which after applying the conditions that $z_1 = 0$ corresponds to $w_1 = 0$ and $z_2 = i$ corresponds to $w_2 = 1$, gives $B = 0$, and $A = 2i/\pi$. Hence, the Schwarz-Christoffel transformation for this problem is

$$w = \frac{2i}{\pi} \left(\arcsin \sqrt{z} + \sqrt{z(1-z)} \right). \quad (7.7.7)$$

Map 7.63. The Schwarz-Christoffel transformation that maps the rectangular region in the z -plane onto the upper half-plane $\Im\{w\} > 0$ (see Figure 7.56) is given by

$$w(z) = A \int_{z_0}^z \frac{dt}{\sqrt{(t-z_1)(t-z_2)(t-z_3)}} + B, \quad (7.7.8)$$

where the points z_1, z_2, z_3 are mapped onto the points w_1, w_2, w_3 respectively, and the point z_4 is mapped onto the point at infinity.

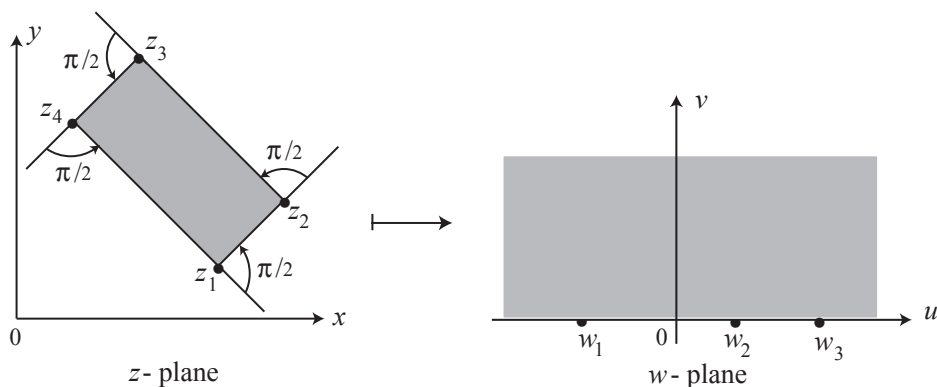


Figure 7.56 For Map 7.63.

Map 7.64. Consider the region in the z -plane as shown in Figure 7.57. There are three vertices at infinity and two are finite at the points z_2 and z_4 , with exterior angles as marked in the figure. Notice that the sum of the exterior angles is $\pi - 3\pi/4 + \pi - \pi/4 + \pi = 2\pi$. The Schwarz-Christoffel transformation that maps the region in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is given by

$$w = A \int_{z_0}^z \frac{(t - z_2)^{3/4}(t - z_4)^{1/4}}{(t - z_1)(t - z_3)} dt + B. \quad (7.7.9)$$

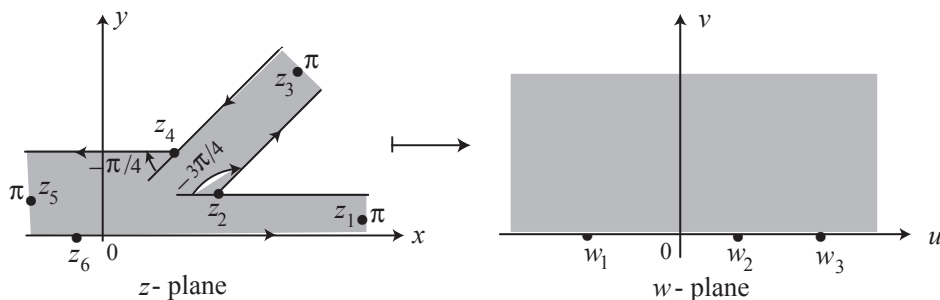


Figure 7.57 For Map 7.64.

Map 7.65. The function $\pi w = \cosh^{-1} z - \arcsin(1/z) + \pi/2$ maps the region in the positive quadrant of the w -plane bounded by the lines $u = 0$, $v = 0$, $u = 1$ ($v > 1$), and $v = 1$ ($u > 1$), onto the upper half-plane $\Im\{z\} > 0$. (Phillips [1966:66].)

Map 7.66. We will determine the Schwarz-Christoffel transformation that maps the shaded region with a slit from $z = 0$ to $z = i$ in the z -plane onto the upper half-plane $\Im\{w\} > 0$ (Figure 58).

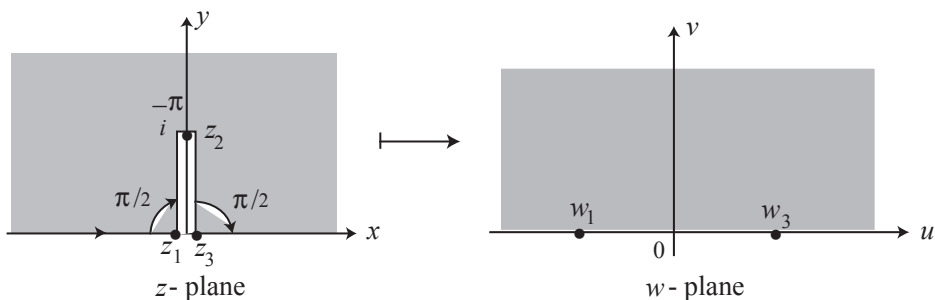


Figure 7.58 For Map 7.66.

The region in the z -plane has four vertices, namely, $z = 0$, $z_2 = i$, $z_3 = 0$, and $z_4 = \infty$, which are mapped onto the points $w_1 = -1$, $w_3 = 1$, and $w_4 = \infty$, but the point w_2 cannot be specified arbitrarily since it is the fourth point. All we know is that $-1 < w_2 < 1$. The exterior angles at z_1, z_2, z_3, z_4 are, respectively, $\pi/2, -\pi, \pi/2, 2\pi$, and the sum of these

angles is 2π . Then the mapping function is

$$\begin{aligned} w &= A' \int_{z_0}^z \frac{t - x_2}{\sqrt{t^2 - 1}} dt + B = A \int_{z_0}^z \frac{t - x_2}{\sqrt{1 - t^2}} dt + B \\ &= A \int_{z_0}^z \frac{t}{\sqrt{1 - t^2}} dt - \int_{z_0}^z \frac{x_2}{\sqrt{1 - t^2}} dt + B \\ &= -A\sqrt{1 - z^2} - x_2 A \arcsin z + B, \end{aligned} \quad (7.7.10)$$

which, after using the conditions that $w = -1$ when $z = 0$, yields $B = \frac{\pi x_2 A}{2}$, and then using (7.7.10), we get $B = 0$, which gives $x_2 A = 0$. Since $A \neq 0$ (otherwise the mapping becomes constant), we use the condition that $w = i$ when $z = x_2 = 0$ in (7.7.10) and find that that $A = -i$, so that the mapping is

$$w = i\sqrt{1 - z^2}.$$

Map 7.67. The Schwarz-Christoffel transformation that maps the region (shaded) in the z -plane onto the upper half-plane $\Im\{w\} > 0$ is given by

$$w = \frac{1-b}{2} \log(z-1) + \frac{1+b}{2} \log(z+1),$$

where b satisfies the equation

$$\frac{i\pi(1-b)}{2} + \frac{1-b}{2} \log(1-b) + \frac{1+b}{2} \log(1+b) = a i.$$

Note that Map 7.11 is a particular case of this mapping.

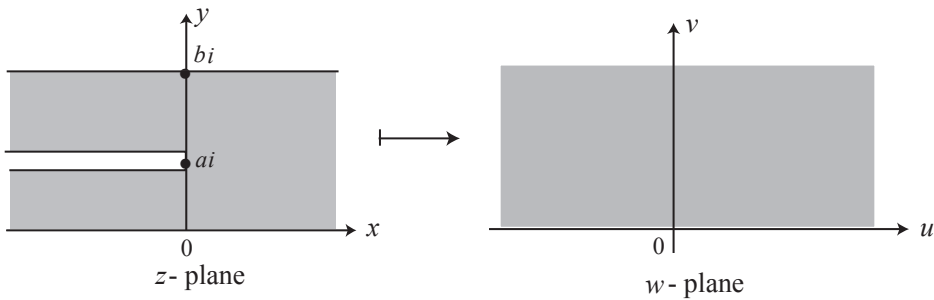


Figure 7.59 For Map 7.67.

7.7 Inverse Schwarz-Christoffel Mapping

The inverse of the Schwarz-Christoffel mapping is defined by Eq (7.1.7). However, if this inverse is defined by $z = g(w)$, it can be found, in principle, by solving the first order, nonlinear differential equation

$$g'(w) = \frac{1}{f'(z)} \Big|_{z=g(w)} = \frac{1}{C_1} (g(w) - z_1)^{\alpha_1} (g(w) - z_2)^{\alpha_2} \cdots (g(w) - z_n)^{\alpha_n}. \quad (7.7.1)$$

In case $g'(w)$ contains a single factor, Eq (7.7.1) becomes

$$g'(w) = \frac{1}{C_1} \left(g(w) - z_1 \right)^{\alpha_1},$$

which, when divided by $(g - z_1)_1^\alpha$ and integrated, yields

$$g(w) = \begin{cases} z_1 + \left(\frac{1 - \alpha_1}{C_1} \right)^{1/(1-\alpha_1)} w^{1/(1-\alpha_1)} & \text{if } \alpha_1 \neq 1, \\ z_1 + e^{w/C_1} & \text{if } \alpha_1 = 1. \end{cases} \quad (7.7.2)$$

If $g'(w)$ contains two or more factors, Eq (7.7.1) becomes a difficult, and more likely an impossible, nonlinear differential equation to solve in closed form.

REFERENCES USED: Ahlfors [1966], Andersen et al. [1962], Betz [1964], Boas [1987], Carrier, Krook and Pearson [1966], Ivanov and Trubetskov [1995], Kantorovich and Krylov [1958], Kober [1957], Kythe [1996], Nehari [1952], Phillips [1943; 1966], Polyanin and Nazaikinskii [2016], Sansone and Gerretsen [1969], Schinzinger and Laura [2003], Silverman [1967], von Koppenfels [1959], Wen [1992], Whittaker and Watson [1962].